ASYMPTOTICS OF LOCAL VOLUMES NEAR LOG CANONICAL THRESHOLDS

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Abstract. We prove that, when the coefficient of a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor added to the boundary of a klt singularity increases to the log canonical threshold, the local volume is asymptotic to a positive constant times the distance to the threshold raised to the codimension of the minimal log canonical center of the pair at the threshold. When the minimal log canonical center is not a point, we compute the constant in terms of the log canonical places over the center and the generalized pair induced on the center by adjunction. We also answer Question 6.10 of Li–Liu–Xu's survey on normalized volume positively.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

C. Li [Li18] introduced the normalized volume of a valuation centered at a closed point of a klt variety. For a klt singularity $(X \ni x, \Delta)$ of dimension n, [LLX20, Definition 2.3] defines the normalized volume $a(v, X, \Delta)^n \operatorname{vol}(v)$ of a valuation v on X with $\operatorname{center}_X v = x$, and the local volume

$$\widehat{\operatorname{vol}}(X \ni x, \Delta) := \inf_v a(v, X, \Delta)^n \operatorname{vol}(v),$$

where v runs over all such valuations (see Definition 2.2, where $\widehat{\operatorname{vol}}$ is also defined at non-closed points).

Li, Liu, and Xu asked the following question in their survey [LLX20].

Question 1.1 ([LLX20, Question 6.10])**.** Let $(X \ni x, \Delta)$ be a klt singularity of dimension n and $D \geq 0$ a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X with $x \in \operatorname{Supp} D$. Let $c := \operatorname{lct}(X \ni x, \Delta; D)$, let W be the minimal lc center of $(X, \Delta + cD)$ containing x, and let w be the generic point of W. By

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Kawamata's subadjunction, $(K_X+\Delta+cD)|_W = K_W+\Delta_W+J_W$, where (W,Δ_W+J_W) is a generalized klt pair. Let $k := n-\dim W$. Is it true that

$$\lim_{\epsilon\to 0^+}\frac{\widehat{\mathrm{vol}}(X\ni x,\Delta+(1-\epsilon)cD)}{n^n\epsilon^k}\geq\frac{\widehat{\mathrm{vol}}(X\ni w,\Delta)}{k^k}\cdot\frac{\widehat{\mathrm{vol}}(W\ni x,\Delta_W+J_W)}{(n-k)^{n-k}},$$

where $\widehat{\mathrm{vol}}(W\ni x,\Delta_W+J_W)$ is defined as for klt pairs in [LLX20, Definition 2.3]?

The local volume of a generalized klt pair is defined in Definition 2.6. We refer the reader to [Kaw98, Theorem 1] for Kawamata's subadjunction. See also [FG12, Theorem 7.2] and [FH23, Theorem 1.2].

In the notation of Question 1.1, the limit was known in the following two cases.

(1) When D is a normal prime divisor and $(X,\Delta+D)$ is plt, so that $W=D$ and $k=1$ [LLX20, Proposition 6.8]. The proof of the lower bound for the limit in [LLX20, Proposition 6.8] is based on [LZ21, Theorem 4.6].
(2) When $W=\{x\}$ and Δ is a $\mathbb{Q}$-divisor [LLX20, Proposition 6.9].

In the other cases, in particular when $2\leq k\leq n-1$, Question 1.1 was open.

Let $(X\ni x,\Delta)$ be a klt pair of dimension n, where $x\in X$ is a closed point, and let $D\geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. A valuation v on X with $\mathrm{center}_X v = x$ *computes* $\mathrm{lct}(X\ni x,\Delta;D)$ if

$$\mathrm{lct}(X\ni x,\Delta;D)=\frac{a(v,X,\Delta)}{v(D)},$$

and *computes* $\widehat{\mathrm{vol}}(X\ni x,\Delta)$ if $\widehat{\mathrm{vol}}(X\ni x,\Delta)=a(v,X,\Delta)^n\,\mathrm{vol}(v)$. The value $v(D)$ is defined in Definition 2.2(4), and quasi-monomial valuations are recalled in Definition 2.2(6).

Theorem 1.2 ([LLX20, Proposition 6.9])**.** *Let $(X\ni x,\Delta)$ be a klt singularity of dimension n, where Δ is a $\mathbb{Q}$-divisor, and let $D\geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X such that $(X,\Delta+D)$ is lc and $\{x\}$ is the minimal lc center of $(X,\Delta+D)$ containing x. Then there exist a real number $\epsilon_0>0$ and a quasi-monomial valuation v on X with $\mathrm{center}_X v=x$ such that v computes $\mathrm{lct}(X\ni x,\Delta;D)$ and $\widehat{\mathrm{vol}}(X\ni x,\Delta+(1-\epsilon)D)$ for every $\epsilon\in(0,\epsilon_0)$. In particular,*

$$\widehat{\mathrm{vol}}(X\ni x,\Delta+(1-\epsilon)D)=a(v,X,\Delta)^n\,\mathrm{vol}(v)\cdot\epsilon^n\quad\text{for every }\epsilon\in(0,\epsilon_0).$$

Li and Xu [LX20, Remark 2.5] expected a formula relating the local volume at a not necessarily closed point ξ to the local volume at a general closed point of $\overline{\{\xi\}}$, and [LLX20, Conjecture 6.7] proposed the formula. Kim proved [LLX20, Conjecture 6.7].

Theorem 1.3 ([Kim26, Theorem 1.1])**.** *Let (X,Δ) be a klt pair of dimension n, where Δ is a $\mathbb{Q}$-divisor, let $\xi\in X$ be a point which is not necessarily closed, $d:=\dim\overline{\{\xi\}}$, and let x' be a general closed point of $\overline{\{\xi\}}$. Then*

$$\widehat{\mathrm{vol}}(X\ni x',\Delta)=\frac{n^n}{(n-d)^{n-d}}\cdot\widehat{\mathrm{vol}}(X\ni\xi,\Delta).$$

The goal of this paper is to answer Question 1.1 for any $1\leq k\leq n$ and any $\mathbb{R}$-divisor Δ: we show that the limit exists and is a positive real number, and we compute the limit when $k\leq n-1$.

Theorem 1.4. *Let $(X\ni x,\Delta)$ be a klt pair of dimension n, where $x\in X$ is a closed point, and let $D\geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X with $x\in\mathrm{Supp}\,D$. Let $c:=\mathrm{lct}(X\ni x,\Delta;D)$ and $B:=\Delta+cD$. Possibly shrinking X to a neighborhood of x, we may assume that (X,B) is lc and that every lc center of (X,B) contains x. Let W be the minimal lc center of (X,B) containing x, let w be the generic point of W, and let $\ell:=\dim W$ and $k:=n-\ell$. If $\ell\geq 1$, let $(W,\Delta_W,\mathbf{M})$ be the generalized pair induced by adjunction from (X,B) to W (see Definition-Theorem 2.7), and let*

$$\widehat{\mathrm{vol}}^B(X\ni w,\Delta):=\inf\{a(E,X,\Delta)^k\,\mathrm{vol}(E)\mid E\text{ is an lc place of }(X,B)\text{ with }\mathrm{center}_X E=W\}$$

(see Definition 2.2 for volumes and local volumes, also at the non-closed point w, and Definition 2.6 for local volumes of generalized pairs). Then:

(1) the limit

$$\lim_{\epsilon\to 0^+}\frac{\widehat{\mathrm{vol}}(X\ni x,\Delta+(1-\epsilon)cD)}{n^n\epsilon^k}$$

is a positive real number;

(2) if $\ell\geq 1$, then the limit in (1) is equal to

$$\frac{\widehat{\mathrm{vol}}^B(X\ni w,\Delta)}{k^k}\cdot\frac{\widehat{\mathrm{vol}}(W\ni x,\Delta_W,\mathbf{M})}{\ell^\ell},$$

and $\widehat{\mathrm{vol}}^B(X\ni w,\Delta)=\widehat{\mathrm{vol}}(X\ni w,\Delta)$ if $k=1$;

(3) the limit in (1) is at least

$$\frac{\widehat{\mathrm{vol}}(X\ni w,\Delta)}{k^k}\cdot\frac{\widehat{\mathrm{vol}}(W\ni x,\Delta_W,\mathbf{M})}{\ell^\ell},$$

where the second factor is defined to be 1 if $\ell=0$.

With $J_W:=\mathbf{M}_W$ (Remark 1.5), (3) answers Question 1.1 positively, where $n-k=\ell$.

Remark 1.5. Question 1.1 does not specify the moduli part J_W beyond Kawamata's subadjunction, and defines $\widehat{\mathrm{vol}}(W\ni x,\Delta_W+J_W)$ only by reference to [LLX20, Definition 2.3]. In Theorem 1.4, $(W,\Delta_W,\mathbf{M})$ is the generalized pair of [FH23, Theorem 1.2]. The generalized log discrepancy of $(W,\Delta_W,\mathbf{M})$ at any prime divisor on any higher model of W is an lc threshold on an lc place of (X,B) with center W (Definition-Theorem 2.7(2)). The local volume $\widehat{\mathrm{vol}}(W\ni x,\Delta_W,\mathbf{M})$ is the infimum in [LLX20, Definition 2.3] with the log discrepancy replaced by the generalized log discrepancy (Definition 2.6).

When $\ell=0$ and Δ is not a $\mathbb{Q}$-divisor, Theorem 1.4 does not determine the value of the limit in (1). When $\ell=0$ and Δ is a $\mathbb{Q}$-divisor, Theorem 1.2 determines it.

Sketch of the proof. Let B_ϵ, $\epsilon\in[0,1]$, be as in Set-up 2.8. Thus $B_0=B$. If $\ell=0$, then $W=\{x\}$, and (1) and (3) follow from Theorem 1.2 for the rational approximations $(X\ni x,\Delta_j)$ and c_jD below, as the comparison factors $(1\pm\delta_j)^n$ tend to 1. Assume that $\ell\geq 1$. The proof of Theorem 1.4 in Section 9 combines the following three results.

- The upper bound of Theorem 5.1:

$$\limsup_{\epsilon\to 0^+}\frac{\widehat{\mathrm{vol}}(X\ni x,B_\epsilon)}{n^n\epsilon^k}\leq\frac{\widehat{\mathrm{vol}}^B(X\ni w,\Delta)\cdot\widehat{\mathrm{vol}}(W\ni x,\Delta_W,\mathbf{M})}{k^k\ell^\ell}.$$

- The lower bound of Theorem 7.1:

$$\liminf_{\epsilon\to 0^+}\frac{\widehat{\mathrm{vol}}(X\ni x,B_\epsilon)}{n^n\epsilon^k}\geq\frac{\widehat{\mathrm{vol}}(X\ni w,\Delta)\cdot\widehat{\mathrm{vol}}(W\ni x,\Delta_W,\mathbf{M})}{k^k\ell^\ell}.$$

- The limit at the generic point w:

$$\lim_{\sigma\to 0^+}\frac{\widehat{\mathrm{vol}}(X\ni w,B_\sigma)}{\sigma^k}=\widehat{\mathrm{vol}}^B(X\ni w,\Delta),$$

which is Proposition 8.1 if $k\geq 2$. If $k=1$, then W is the only prime divisor over X with center W, $a(W,X,B)=0$, and

$$\widehat{\mathrm{vol}}(X\ni w,B_\sigma)=a(W,X,B_\sigma)=\sigma a(W,X,\Delta)=\sigma\widehat{\mathrm{vol}}^B(X\ni w,\Delta)\quad\text{for any }\sigma\in(0,1].$$

For $k \geq 2$, the proof of Proposition 8.1 uses the ACC for lc thresholds [HMX14, Theorem 1.1], as in [LLX20, Proof of Proposition 6.9] (Theorem 1.2). By [Kim26, Lemma 2.2], [Xu21, Lemma 2.11], and the ACC, if Δ is a $\mathbb{Q}$-divisor and $K_X + \Delta$ is $\mathbb{Q}$-Cartier, then there exists a real number $\sigma_0 > 0$ such that, for any rational number $\sigma \in (0, \sigma_0)$, $\widehat{\mathrm{vol}}(X \ni w, B_\sigma)$ is the infimum of $a(E, X, B_\sigma)^k \mathrm{vol}(E)$ over the lc places E of (X, B) with $\mathrm{center}_X E = W$. As $a(E, X, B_\sigma) = \sigma a(E, X, \Delta)$ for any such E, we obtain, also for irrational σ by the monotonicity of $\widehat{\mathrm{vol}}(X \ni w, B_\sigma)$ in σ,

$$\widehat{\mathrm{vol}}(X \ni w, B_\sigma) = \sigma^k \widehat{\mathrm{vol}}^B(X \ni w, \Delta) \quad \text{for any } \sigma \in (0, \sigma_0).$$

Since

$$\widehat{\mathrm{vol}}(X \ni w, \Delta) \leq \widehat{\mathrm{vol}}^B(X \ni w, \Delta),$$

the upper and lower bounds do not determine the limit. For $\sigma \in (0, 1]$, the pair (X, B_σ) is klt and $\mathrm{lct}(X \ni x, B_\sigma; D) = \sigma c$ (Lemma 2.9). Since $B_\sigma + \sigma c D = B$, replacing Δ with B_σ replaces c with σc and B_ϵ with $B_{\sigma\epsilon}$, and does not change B, W, w, ℓ, k, and $(W, \Delta_W, \mathbf{M})$. Since $(\sigma\epsilon)^k = \sigma^k \epsilon^k$, Theorem 7.1 for B_σ in place of Δ implies

$$\liminf_{\epsilon \to 0^+} \frac{\widehat{\mathrm{vol}}(X \ni x, B_\epsilon)}{n^n \epsilon^k} \geq \frac{\widehat{\mathrm{vol}}(X \ni w, B_\sigma)}{\sigma^k} \cdot \frac{\widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{k^k \ell^\ell} \quad \text{for any } \sigma \in (0, 1].$$

As $\sigma \to 0^+$, the right-hand side tends to the right-hand side of the upper bound, which implies the existence of the limit in (1) and the formula in (2). Then (3) is the lower bound of Theorem 7.1.

For example, let $\ell \geq 1$, $X = \mathbb{A}^2_{u,v} \times \mathbb{A}^\ell$, $\Delta = 0$, $D = \mathrm{div}(v^2 - u^3)$, and x the origin. Then $c = 5/6$, $W = \{u = v = 0\}$, $k = 2$, and the only lc place of (X, B) is the exceptional divisor E of the weighted blow-up of W with weights 2 and 3 for u and v. Let E' be the exceptional divisor of the blow-up of W. Then

$$\widehat{\mathrm{vol}}^B(X \ni w, 0) = a(E, X, 0)^2 \mathrm{vol}(E) = \frac{5^2}{6} > 4 = a(E', X, 0)^2 \mathrm{vol}(E') \geq \widehat{\mathrm{vol}}(X \ni w, 0),$$

so the inequality in Theorem 1.4(3) is strict.

Possibly shrinking X to a neighborhood of x, we let $\Delta_j \geq 0$ and $\delta_j > 0$, $j \geq 1$, be the $\mathbb{Q}$-divisors and real numbers of Construction-Proposition 3.1: $\lim_{j \to +\infty} \delta_j = 0$, and, for $c_j := \mathrm{lct}(X \ni x, \Delta_j; D)$, the lc places of $(X, \Delta_j + c_j D)$ are exactly the lc places of (X, B), and

$$(1 - \delta_j) a(v, X, B_\epsilon) \leq a(v, X, \Delta_j + (1 - \epsilon) c_j D) \leq (1 + \delta_j) a(v, X, B_\epsilon)$$

for any $\epsilon \in [0, 1]$ and any valuation v on X with a center on X. By Corollary 3.2, the local volumes at x, at w, and, if $\ell \geq 1$, on W for Δ_j differ from the corresponding local volumes for Δ by factors in $[(1 - \delta_j)^n, (1 + \delta_j)^n]$, uniformly in $\epsilon \in (0, 1]$. In the proofs of Theorem 7.1 and Proposition 8.1, Corollary 3.2 allows us to assume that Δ is a $\mathbb{Q}$-divisor and $K_X + \Delta$ is $\mathbb{Q}$-Cartier.

The proof of Theorem 5.1 uses test valuations as in [LLX20, Proof of Proposition 6.8]. Let E be an lc place of (X, B) with $\mathrm{center}_X E = W$, and P a prime divisor over W with $\mathrm{center}_W P = \{x\}$. Lemma 5.5 gives valuations v_λ, $\lambda > 0$, with $\mathrm{center}_X(v_\lambda) = x$ and

$$a(v_\lambda, X, B) = \lambda \cdot a(P, W, \Delta_W, \mathbf{M}), \qquad \lim_{\lambda \to 0^+} c v_\lambda(D) = a(E, X, \Delta),$$
$$\limsup_{\lambda \to 0^+} \lambda^\ell \mathrm{vol}(v_\lambda) \leq \mathrm{vol}(E) \cdot \mathrm{vol}(P),$$

where the volume bound uses the graded pieces I_j / I_{j+1}, $j \in \mathbb{N}$, of the filtration

$$I_j := \{f \in \mathcal{O}_{X,x} \mid \mathrm{mult}_E \mathrm{div}(f) \geq j\}$$

in place of Okounkov bodies [KK12, Definition 1.10] (Definition-Proposition 5.4). For $\lambda > 0$ with

$$a(v_\lambda, X, B_\epsilon) = \frac{n a(E, X, \Delta) \epsilon}{k},$$

we obtain

$$\limsup_{\epsilon\to 0^+} \frac{\widehat{\mathrm{vol}}(X\ni x, B_\epsilon)}{n^n\epsilon^k} \le \frac{a(E,X,\Delta)^k \,\mathrm{vol}(E)\cdot a(P,W,\Delta_W,\mathbf{M})^\ell \,\mathrm{vol}(P)}{k^k\ell^\ell},$$

and Theorem 5.1 follows by taking the infimum over P (Lemma 4.2) and then over E.

The proof of Theorem 7.1 (Subsection 7.5) takes rational numbers $\epsilon_i \to 0$ which realize the $\liminf$, and divisorial valuations v_i with $\mathrm{center}_X v_i = x$, $cv_i(D) = 1$, and

$$a(v_i, X, B_{\epsilon_i})^n \,\mathrm{vol}(v_i) \le (1+1/i)\widehat{\mathrm{vol}}(X\ni x, B_{\epsilon_i}).$$

Then $a(v_i, X, B_{\epsilon_i}) = a(v_i, X, B) + \epsilon_i$. By [Zhu24, Lemma 3.4] and Lemma 7.9, there exists a real number $\gamma > 0$ with $a(v_i, X, B) \le \gamma\epsilon_i$ for any i, and the values of the v_i on the ideal of W in $\mathcal{O}_{X,x}$ are bounded from below by a positive constant. Let $h : Y \to X$, $K_Y + B_Y := h^*(K_X + B)$, and the reduced snc divisor E on Y be as in Set-up 7.2(2)(3), and let $v_i(E_j)$ be the value of v_i at a local equation of a component E_j of E at the center of v_i on Y. Passing to a subsequence, the components $E_1, \dots, E_m$ of E with $\mathrm{mult}_{E_j} B_Y = 1$ and $\lim_{i\to+\infty} v_i(E_j) > 0$ cut out an irreducible component S of $E_1 \cap \cdots \cap E_m$ which contains the centers of the v_i on Y and satisfies $h(S) = W$. Let $K_S + B_S := (K_Y + B_Y)|_S$, let $S \xrightarrow{\pi_S} T \xrightarrow{\tau} W$ be the Stein factorization of $h|_S$, and let $r := \deg\tau$. The proof shows

$$\liminf_{i\to+\infty} a(\mu_i, X, B)^\ell \,\mathrm{vol}(\mu_i) \ge \widehat{\mathrm{vol}}(X\ni w, \Delta)\cdot \widehat{\mathrm{vol}}(W\ni x, \Delta_W, \mathbf{M})$$

for the Gauss valuations μ_i along S defined in the next paragraph, which satisfy $\mathrm{vol}(v_i) \ge (1 + O(\epsilon_i))^{-n}\,\mathrm{vol}(\mu_i)$, where the constant in $O(\epsilon_i)$ does not depend on i, and $a(\mu_i, X, B) \le a(v_i, X, B)$. Since $a(v_i, X, B_{\epsilon_i}) \ge a(\mu_i, X, B) + \epsilon_i$, and the function $t \to (t+\epsilon_i)^n/t^\ell$ on $(0, +\infty)$ attains its minimum $n^n\epsilon_i^k/(k^k\ell^\ell)$ at $t = \ell\epsilon_i/k$, Theorem 7.1 follows.

The Gauss valuations are defined in Construction-Proposition 7.3. Near a point of S, take étale coordinates on Y in which $E_j = \{z_j = 0\}$ for $1 \le j \le m$ and $S = \{z_1 = \cdots = z_m = 0\}$. Any regular function f near the point has a unique expansion $f = \sum_{\beta\in\mathbb{N}^m} \iota(f_\beta) z^\beta$ in the completion along S, where the f_β are regular functions on S and ι lifts regular functions on S to the completion. For a valuation θ on S with a center on S and $\boldsymbol{\alpha} \in \mathbb{R}^m_{>0}$, let

$$\mu_{\theta,\boldsymbol{\alpha}}(f) := \min_\beta\{\theta(f_\beta) + \boldsymbol{\alpha}\cdot\beta\}.$$

For $v = v_i$, Proposition 7.5 takes $\theta := v\circ\iota$ and $\boldsymbol{\alpha} := (v(z_1), \dots, v(z_m))$, and shows

$$\mu_{\theta,\boldsymbol{\alpha}} \le v \le \Big(1 + \frac{ma(v,Y,E)}{\min_j v(z_j)}\Big)\mu_{\theta,\boldsymbol{\alpha}} \quad\text{and}\quad a(\mu_{\theta,\boldsymbol{\alpha}}, X, B) = a(\theta, S, B_S) \le a(v, X, B).$$

Here $a(v_i, Y, E) \le a(v_i, X, B) \le \gamma\epsilon_i$ as $B_Y \le E$, and $\min_j v_i(z_j)$ is bounded from below by a positive constant. We let θ_i and $\boldsymbol{\alpha}_i$ be θ and $\boldsymbol{\alpha}$ for $v = v_i$, and $\mu_i := \mu_{\theta_i,\boldsymbol{\alpha}_i}$. The first inequality holds term by term. For the second, if the minimum defining $\mu_{\theta,\boldsymbol{\alpha}}(f)$ is attained at β, then the product of the operators $z_j\partial_{z_j} - \beta'_j$, over $1 \le j \le m$ and the integers $0 \le \beta'_j \le \mu_{\theta,\boldsymbol{\alpha}}(f)/v(z_j)$ with $\beta'_j \ne \beta_j$, maps f to an element f' with $v(f') = \mu_{\theta,\boldsymbol{\alpha}}(f)$. Each of the at most $m\mu_{\theta,\boldsymbol{\alpha}}(f)/\min_j v(z_j)$ factors lowers v by at most $a(v, Y, E)$, as $z_j\partial_{z_j}$ is a logarithmic derivation along E (Lemma 7.4).

Assume that $k \ge 2$, and let $(T, \Delta_T, \tau^*\mathbf{M})$ be the pullback of $(W, \Delta_W, \mathbf{M})$ to T (Construction 7.7). By Lemma 7.6 and Proposition 7.8(1), for the push-forwards $(\pi_S)_*\theta_i$ and $(h|_S)_*\theta_i$ of θ_i to T and W (Subsection 6.2),

$$a((\pi_S)_*\theta_i, T, \Delta_T, \tau^*\mathbf{M}) = a((h|_S)_*\theta_i, W, \Delta_W, \mathbf{M}) \le a(\theta_i, S, B_S) = a(\mu_i, X, B).$$

Passing to a subsequence, the center $p \in T$ of $(\pi_S)_*\theta_i$ does not depend on i. After an étale base change of X near x, which by Lemma 4.4 and Theorem 1.3 changes neither $\widehat{\mathrm{vol}}(X\ni w, \Delta)$ nor $\widehat{\mathrm{vol}}(W\ni x, \Delta_W, \mathbf{M})$, an open subset of the spring of W [Kol16, Proposition 19] is a Galois cover of W and of T which has only one point over x and only one point over p, and on which the

pullback of $(W, \Delta_W, \mathbf{M})$ is a generalized pair (Proposition 7.8(2)). Thus Proposition 4.1, which extends the degree formula [XZ21, Theorem 1.3] to generalized pairs, implies

$$\widehat{\mathrm{vol}}(T \ni p, \Delta_T, \tau^*\mathbf{M}) = r \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M}).$$

Thus, by Lemma 4.2,

$$a(\mu_i, X, B)^\ell \,\mathrm{vol}((\pi_S)_*\theta_i) \geq r \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M}).$$

Passing to a subsequence, $\boldsymbol{\alpha}_i \to \boldsymbol{\alpha}_\infty \in \mathbb{R}^m_{>0}$. Fix $\rho \in (0,1)$ and a vector $\mathbf{b} \in \mathbb{R}^m_{>0}$ with $\mathbb{Q}$-linearly independent coordinates such that $(1-\rho)\boldsymbol{\alpha}_i \leq \mathbf{b} \leq (1+\rho)\boldsymbol{\alpha}_i$ for any $i \gg 0$, and let $u_\mathbf{b}$ be the monomial valuation with weights $\mathbf{b}$ along $E_1, \dots, E_m$ at the generic point of S. Then $(1-\rho)\mu_i \leq \mu_{\theta_i,\mathbf{b}} \leq (1+\rho)\mu_i$ for any $i \gg 0$, $\mathrm{center}_X u_\mathbf{b} = w$, and, as $\mathrm{mult}_{E_j} B_Y = 1$ for $1 \leq j \leq m$,

$$a(u_\mathbf{b}, X, \Delta) = cu_\mathbf{b}(D) \leq (1+\rho)cv_i(D) = 1+\rho,$$

so $(1+\rho)^k \,\mathrm{vol}(u_\mathbf{b}) \geq \widehat{\mathrm{vol}}(X \ni w, \Delta)$. Let B'_S be as in Construction 7.7. Then (S, B'_S) is klt, and $a(\theta_i, S, B'_S) \to 0$ as $i \to +\infty$, since $a(\theta_i, S, B_S) \leq \gamma\epsilon_i$ and $v_i(E_j) \to 0$ for the components $E_j \neq E_1, \dots, E_m$ of E with $\mathrm{mult}_{E_j} B_Y = 1$. For any $\zeta \in (0,1)$, Proposition 7.10 constructs a real number $C > 0$ and a continuous function $\varphi : \mathbb{R}^m_{>0} \to \mathbb{R}_{>0}$ with $r \cdot \varphi(\mathbf{b}) \geq (1-\zeta)\,\mathrm{vol}(u_\mathbf{b})$ such that

$$\mathrm{vol}(\mu_{\theta_i,\mathbf{b}}) \geq \varphi\big(\mathbf{b} + Ca(\theta_i, S, B'_S) \cdot (1, \dots, 1)\big) \cdot \mathrm{vol}((\pi_S)_*\theta_i) \quad \text{for any } i.$$

Since $\mathrm{vol}(\mu_i) \geq (1-\rho)^n \,\mathrm{vol}(\mu_{\theta_i,\mathbf{b}})$ and $a(\theta_i, S, B'_S) \to 0$, the last inequality, the lower bound for $a(\mu_i, X, B)^\ell \,\mathrm{vol}((\pi_S)_*\theta_i)$, and

$$r \cdot \varphi(\mathbf{b}) \geq (1-\zeta)\,\mathrm{vol}(u_\mathbf{b}) \geq (1-\zeta)(1+\rho)^{-k}\widehat{\mathrm{vol}}(X \ni w, \Delta)$$

give

$$\liminf_{i \to +\infty} a(\mu_i, X, B)^\ell \,\mathrm{vol}(\mu_i) \geq \frac{(1-\rho)^n(1-\zeta)}{(1+\rho)^k} \cdot \widehat{\mathrm{vol}}(X \ni w, \Delta) \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M}),$$

and we let $\zeta, \rho \to 0$. If $k = 1$, then $m = 1$, $S = h_*^{-1}W$, $T = W$, and $r = 1$. In this case, the lower bound for $a(\mu_i, X, B)^\ell \,\mathrm{vol}((h|_S)_*\theta_i)$ follows from Lemmas 7.6 and 4.2, and $\mathrm{vol}(\mu_i)$ is bounded from below directly by the filtration of $\mathcal{O}_{X,x}$ by the order of vanishing along W, without Proposition 7.10 and Theorem 6.1.

In the proof of Proposition 7.10, the graded algebra of $u_\mathbf{b}$ over the generic point of T is approximated by finitely generated subalgebras [KK12, Theorems 1.6 and 1.18], whose normalizations embed into section rings of line bundles $\mathcal{L}$ on $S \times \mathbb{P}^{m-1}$ over T, and the shift of each coordinate of $\mathbf{b}$ is the error term of Theorem 6.1 for the morphism $S \times \mathbb{P}^{m-1} \to T$, the klt pair $(S \times \mathbb{P}^{m-1}, \mathrm{pr}_S^* B'_S)$, and the extension θ'_i of θ_i with weight 0 along $\mathbb{P}^{m-1}$, divided by the degree N:

$$\theta'_i(s/e^N) \leq \min_\beta \theta_i(b_\beta) + CNa(\theta_i, S, B'_S)$$

for any integer $N \geq N_0$ and any rational section $s = \sum_\beta b_\beta s_\beta$ of $\mathcal{L}^{\otimes N}$, where $(s_\beta)_\beta$ is a fixed basis of standard monomials (Section 6) over the generic point of T, the b_β are rational functions on T, e is a local generator of $\mathcal{L}$ at the center of θ'_i, and C and N_0 do not depend on i.

Theorem 6.1 holds for any projective surjective morphism from a klt pair to a variety of positive dimension whose generic fiber is geometrically integral of positive dimension. By [dJ97, Theorem 5.9] and a base change to a resolution of the base of the tower constructed there, the total space is the image of a generically finite morphism from the top of a tower of flat families of nodal curves over a smooth variety. At the top, the log discrepancy of any divisorial valuation is at most a constant times the log discrepancy, with respect to the klt pair, of the push-forward of the valuation to the original total space, since the crepant pullback of the klt pair to the top is sub-klt. Along the tower, the total spaces are klt and the log discrepancies do not increase toward the base (Lemma 6.4), and on a family of nodal curves, the error caused by the nodes is at most a constant times N times the log discrepancy (Lemma 6.5).

Structure of the paper. Section 2 recalls valuations, local volumes at not necessarily closed points, local volumes of generalized pairs, and the generalized pair induced by adjunction to a minimal lc center. In Section 3 we approximate Δ by $\mathbb{Q}$-divisors. In Section 4 we study local volumes of generalized pairs. In Section 5 we bound $\widehat{\mathrm{vol}}(X \ni x, B_\epsilon)/\epsilon^k$ from above as $\epsilon \to 0^+$. In Section 6 we prove a uniform estimate for the values of sections on projective klt families. In Section 7 we bound $\widehat{\mathrm{vol}}(X \ni x, B_\epsilon)/\epsilon^k$ from below as $\epsilon \to 0^+$. In Section 8 we study the local volume at the generic point w of W. In Section 9 we prove the main theorem.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and valuations.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c\,\mathrm{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X. We say that a pair $(X \ni z, B)$ is lc (resp. klt) if (X, B) is lc (resp. klt) near z.

For an lc pair $(X \ni z, B)$ and an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $D \geq 0$ on X, we define

$$\mathrm{lct}(X \ni z, B; D) := \sup\{t \geq 0 \mid (X \ni z, B + tD) \text{ is lc}\}$$

as the *lc threshold* of D with respect to $(X \ni z, B)$. If (X, B) is lc, then

$$\mathrm{lct}(X, B; D) := \sup\{t \geq 0 \mid (X, B + tD) \text{ is lc}\}.$$

Definition 2.2. Let X be a normal variety, $\xi \in X$ a point which is not necessarily closed, and $d_\xi := \dim \mathcal{O}_{X,\xi}$.

(1) A *valuation* on X is a map v from the set of non-zero rational functions on X to $\mathbb{R}$ which is not identically zero, such that $v(fg) = v(f) + v(g)$, $v(f + g) \geq \min\{v(f), v(g)\}$, and $v(\mathbb{C}^\times) = 0$. We set $v(0) := +\infty$. We say that v has *center* ξ on X, and write $\mathrm{center}_X v = \xi$, if $v \geq 0$ on $\mathcal{O}_{X,\xi}$ and $v > 0$ on $\mathfrak{m}_\xi$. We say that v has a center on X if $\mathrm{center}_X v = \xi'$ for some point $\xi' \in X$. For a prime divisor E over X, let v_E be the valuation $f \to \mathrm{mult}_E \mathrm{div}(f)$ on X. The divisorial valuations in Definition 2.1 are the cv_E with $c > 0$. The closure of $\mathrm{center}_X v_E$ is called the *center* of E on X and is denoted by $\mathrm{center}_X E$.

(2) For a valuation v with $\operatorname{center}_X v = \xi$ and a real number $\lambda > 0$, let

$$\mathfrak{a}_\lambda(v) := \{f \in \mathcal{O}_{X,\xi} \mid v(f) \geq \lambda\}.$$

The *volume* of v is

$$\operatorname{vol}(v) := \lim_{\lambda \to +\infty} \frac{d_\xi! \cdot \operatorname{length}(\mathcal{O}_{X,\xi}/\mathfrak{a}_\lambda(v))}{\lambda^{d_\xi}}.$$

For a prime divisor E over X with $\operatorname{center}_X E = \overline{\{\xi\}}$, let $\operatorname{vol}(E) := \operatorname{vol}(v_E)$.

(3) For an $\mathfrak{m}_\xi$-primary ideal $\mathfrak{b} \subset \mathcal{O}_{X,\xi}$, the *Hilbert–Samuel multiplicity* of $\mathfrak{b}$ is

$$e(\mathfrak{b}) := \lim_{m \to +\infty} \frac{d_\xi! \cdot \operatorname{length}(\mathcal{O}_{X,\xi}/\mathfrak{b}^m)}{m^{d_\xi}}.$$

(4) Let (X, B) be a sub-pair, $h : X' \to X$ a log resolution of (X, B), $K_{X'} + B' := h^*(K_X + B)$, and $B' = \sum_i b_i E_i$. For a valuation v on X with a center on X, let

$$a(v, X, B) := A_{X'}(v) - \sum_i b_i v(E_i),$$

where $A_{X'}$ is the log discrepancy of [JM12, Section 5] and $v(E_i)$ is the value of a local equation of E_i at $\operatorname{center}_{X'} v$. This does not depend on h and coincides with Definition 2.1 for divisorial valuations. For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor G on X and a valuation v on X with $\operatorname{center}_X v = \zeta$, let $v(G) := v(g)/N$, where N is a positive integer such that NG is Cartier and g is a local equation of NG at ζ.

(5) If (X, B) is a klt pair, then the *local volume* of $(X \ni \xi, B)$ is

$$\widehat{\operatorname{vol}}(X \ni \xi, B) := \inf\{a(v, X, B)^{d_\xi} \operatorname{vol}(v) \mid \operatorname{center}_X v = \xi\},$$

where $a(v, X, B)^{d_\xi} \operatorname{vol}(v) := +\infty$ if $a(v, X, B) = +\infty$.

(6) A valuation v on X is *quasi-monomial* if there exist a projective birational morphism $X' \to X$ from a smooth variety X', a point $\eta \in X'$, a regular system of parameters $y_1, \ldots, y_N$ of $\mathcal{O}_{X',\eta}$, and $\alpha \in \mathbb{R}^N_{\geq 0}$, such that $v(f) = \min\{\alpha \cdot \beta \mid c_\beta \neq 0\}$ for any non-zero $f \in \mathcal{O}_{X',\eta}$, where $f = \sum_{\beta \in \mathbb{N}^N} c_\beta y^\beta$ in the completion of $\mathcal{O}_{X',\eta}$ with all c_β in a coefficient field (see [JM12, Section 3]).

For a closed point ξ, Definition 2.2(5) is [LLX20, Definition 2.3]. For a point ξ which is not necessarily closed and a $\mathbb{Q}$-divisor B, it is [Kim26, Definition 2.1].

Lemma 2.3 ([Cut14, Theorem 6.5]). *Let X be a normal variety, $\xi \in X$ a point which is not necessarily closed, $d_\xi := \dim \mathcal{O}_{X,\xi}$, and v a valuation on X with $\operatorname{center}_X v = \xi$. Then $\mathfrak{a}_\lambda(v)$ is $\mathfrak{m}_\xi$-primary for any real number $\lambda > 0$, the limit defining $\operatorname{vol}(v)$ in Definition 2.2(2) exists, and*

$$\operatorname{vol}(v) = \lim_{m \to +\infty} \frac{e(\mathfrak{a}_m(v))}{m^{d_\xi}}.$$

Proof. For any real number $\lambda \geq 1$,

$$\mathfrak{a}_{\lceil \lambda \rceil}(v) \subset \mathfrak{a}_\lambda(v) \subset \mathfrak{a}_{\lfloor \lambda \rfloor}(v).$$

Since $\mathcal{O}_{X,\xi}$ is analytically unramified [Cut14, Section 2] and $\{\mathfrak{a}_m(v)\}_{m \in \mathbb{N}^+}$ is a graded family of $\mathfrak{m}_\xi$-primary ideals, the lemma follows from [Cut14, Theorem 6.5]. □

Lemma 2.4 ([JM12, Proposition 5.1, Corollary 5.4]). *Let X be a smooth variety, $E = \sum_{i=1}^m E_i$ a reduced snc divisor on X, $b_1, \ldots, b_m$ real numbers, and $B := \sum_{i=1}^m b_i E_i$. Then for any valuation v on X with a center on X,*

$$a(v, X, B) = a(v, X, E) + \sum_{i=1}^m (1 - b_i) v(E_i) \quad \text{and} \quad a(v, X, E) \geq 0.$$

Proof. The equality follows from Definition 2.2(4) with $h = \mathrm{id}_X$, and

$$a(v, X, E) = A_X(v) - \sum_{i=1}^{m} v(E_i) \geq 0$$

by [JM12, Proposition 5.1, Corollary 5.4]. □

Lemma 2.5. *Let X be a smooth variety, $E = \sum_{i=1}^m E_i$ a reduced snc divisor on X, $\delta \in (0,1)$, and $b_1, \dots, b_m, b'_1, \dots, b'_m$ real numbers such that, for any i, $b_i \leq 1$ and*

$$(1-\delta)(1-b_i) \leq 1 - b'_i \leq (1+\delta)(1-b_i).$$

Let $B := \sum_{i=1}^m b_i E_i$ and $B' := \sum_{i=1}^m b'_i E_i$. Then

$$0 \leq (1-\delta)a(v, X, B) \leq a(v, X, B') \leq (1+\delta)a(v, X, B) \leq +\infty$$

for any valuation v on X with a center on X.

Proof. By Lemma 2.4, $a(v, X, B) = a(v, X, E) + \sum_{i=1}^m (1-b_i)v(E_i)$ and $a(v, X, B') = a(v, X, E) + \sum_{i=1}^m (1-b'_i)v(E_i)$, where $a(v, X, E) \geq 0$ and $v(E_i) \geq 0$ for any i. Since $1 - b_i \geq 0$ and $1 - \delta \leq 1 \leq 1 + \delta$, the lemma follows by comparing the right-hand sides term by term. □

2.2. **Generalized pairs and adjunction.**

Definition 2.6. (1) A ***b***-divisor $\mathbf{M}$ of a normal variety X is a choice of an $\mathbb{R}$-divisor $\mathbf{M}_{X'}$ on each normal variety X' with a projective birational morphism $X' \to X$, compatible with pushforward. We call $\mathbf{M}_{X'}$ the *trace* of $\mathbf{M}$ on X'. If every $\mathbf{M}_{X'}$ is a $\mathbb{Q}$-divisor, then we say that $\mathbf{M}$ is a $\mathbb{Q}$-***b***-divisor. We say that $\mathbf{M}$ *descends* to X' if $\mathbf{M}_{X'}$ is $\mathbb{R}$-Cartier and $\mathbf{M}_{X''} = g^*\mathbf{M}_{X'}$ for any projective birational morphism $g : X'' \to X'$ from a normal variety X''.

(2) A *generalized pair* $(X, B, \mathbf{M})$ consists of a normal quasi-projective variety X, an $\mathbb{R}$-divisor $B \geq 0$ on X, and a ***b***-divisor $\mathbf{M}$ of X which descends to some X' with $\mathbf{M}_{X'}$ nef over X, such that $K_X + B + \mathbf{M}_X$ is $\mathbb{R}$-Cartier. A pair (X, B) is regarded as the generalized pair $(X, B, \mathbf{M})$ with $\mathbf{M} = 0$.

(3) Let $(X, B, \mathbf{M})$ be a generalized pair, $h : X' \to X$ a log resolution of (X, B) such that $\mathbf{M}$ descends to X', and $K_{X'} + B' + \mathbf{M}_{X'} = h^*(K_X + B + \mathbf{M}_X)$. For a valuation v on X with a center on X we let $a(v, X, B, \mathbf{M}) := a(v, X', B')$, which does not depend on h. For a prime divisor E over X we let $a(E, X, B, \mathbf{M}) := a(E, X', B')$. We say that $(X, B, \mathbf{M})$ is *generalized klt* if $a(E, X, B, \mathbf{M}) > 0$ for any prime divisor E over X.

(4) If $(X, B, \mathbf{M})$ is generalized klt near a closed point $x \in X$, then the *local volume* of $(X, B, \mathbf{M})$ at x is

$$\widehat{\mathrm{vol}}(X \ni x, B, \mathbf{M}) := \inf\{a(v, X, B, \mathbf{M})^{\dim X} \mathrm{vol}(v) \mid \mathrm{center}_X v = x\},$$

where $a(v, X, B, \mathbf{M})^{\dim X} \mathrm{vol}(v) := +\infty$ if $a(v, X, B, \mathbf{M}) = +\infty$. If $\mathbf{M} = 0$, then

$$\widehat{\mathrm{vol}}(X \ni x, B, \mathbf{M}) = \widehat{\mathrm{vol}}(X \ni x, B),$$

where the right-hand side is as in Definition 2.2(5) for $\xi = x$.

Generalized pairs are introduced in [BZ16, Definition 1.4].

Definition-Theorem 2.7 ([Fuj11, Theorem 9.1], [FH23, Theorem 1.2, Definition 1.3], [FH22, Theorem 1.2])**.** Let (X, B) be an lc pair and W a minimal lc center of (X, B). Then W is normal, and there exist an $\mathbb{R}$-divisor $\Delta_W \geq 0$ on W and a ***b***-divisor $\mathbf{M}$ of W with the following properties.

(1) $(W, \Delta_W, \mathbf{M})$ is a generalized pair which is generalized klt, and

$$K_W + \Delta_W + \mathbf{M}_W = (K_X + B)|_W.$$

(2) Let S be an lc place of (X, B) with $\operatorname{center}_X S = W$, $h : Y \to X$ a log resolution of (X, B) on which S is a divisor, $K_Y + B_Y := h^*(K_X + B)$ and $K_S + B_S := (K_Y + B_Y)|_S$. Let P be a prime divisor on a normal variety W' with a projective birational morphism $W' \to W$ such that the induced map $h_S : S \dashrightarrow W'$ is a morphism. Then

$$a(P, W, \Delta_W, \mathbf{M}) = \sup\{\lambda \in \mathbb{R} \mid (S, B_S + \lambda h_S^* P) \text{ is sub-lc over the generic point of } P\},$$

where, as W' is normal, P is Cartier on an open neighborhood $U \subset W'$ of the generic point of P, and *sub-lc over the generic point of* P means that $(h_S^{-1}(U), (B_S + \lambda h_S^* P)|_{h_S^{-1}(U)})$ is sub-lc for some such U.

(3) If B is a $\mathbb{Q}$-divisor and $K_X + B$ is $\mathbb{Q}$-Cartier, then Δ_W is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is a $\mathbb{Q}$-$\boldsymbol{b}$-divisor.

We call $(W, \Delta_W, \mathbf{M})$ the *generalized pair induced by adjunction* from (X, B) to W.

Proof. By [Fuj11, Theorem 9.1], W is normal. In (1), $\Delta_W \geq 0$, the generalized klt property, and the descent of $\mathbf{M}$ to a smooth variety $\widetilde{W}$ with a projective birational morphism $\widetilde{W} \to W$ are [FH23, Theorem 1.2], and $\mathbf{M}_{\widetilde{W}}$ is nef over W by [FH23, Remark 2.3]. For the morphism $\rho : W' \to W$ in (2), $K_{W'} + B_{W'} + \mathbf{M}_{W'} = \rho^*(K_W + \Delta_W + \mathbf{M}_W)$ by [FH23, Definition 1.3], where $B_{W'}$ is the trace on W' of the discriminant, so $a(P, W, \Delta_W, \mathbf{M}) = 1 - \operatorname{mult}_P B_{W'}$, and $1 - \operatorname{mult}_P B_{W'}$ is the supremum in (2) by [FH22, Definition 2.1, Theorem 1.2]. This proves (2). Item (3) is the last sentence of [FH23, Theorem 1.2]. □

By Definition-Theorem 2.7(2), $a(P, W, \Delta_W, \mathbf{M})$ does not depend on the choice of Δ_W and $\mathbf{M}$ for any prime divisor P over W. Thus the boundaries in Definition 2.6(3) on a log resolution of W on which the $\boldsymbol{b}$-divisors of two choices descend coincide, and $\widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M})$ does not depend on the choice for any closed point $x \in W$.

2.3. **Set-up.**

Set-up 2.8. Let X, x, Δ, D, c, B, and W be as follows.

(1) X is a normal quasi-projective variety of dimension n, and $x \in X$ is a closed point.
(2) $\Delta \geq 0$ is an $\mathbb{R}$-divisor on X such that (X, Δ) is klt.
(3) $D \geq 0$ is a non-zero $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X with $x \in \operatorname{Supp} D$.
(4) $c := \operatorname{lct}(X \ni x, \Delta; D)$, $B := \Delta + cD$, and (X, B) is lc.
(5) Every lc center of (X, B) contains x.
(6) W is the minimal lc center of (X, B), which is unique by (4), (5), and [Fuj11, Theorem 9.1(4)].

Let w be the generic point of W, $\mathcal{I}_W \subset \mathcal{O}_X$ the ideal sheaf of W, $\ell := \dim W$, and $k := n - \ell$. For any $\epsilon \in [0, 1]$, let $B_\epsilon := \Delta + (1 - \epsilon)cD$. If $\ell \geq 1$, let $(W, \Delta_W, \mathbf{M})$ be the generalized pair induced by adjunction from (X, B) to W (Definition-Theorem 2.7). Let

$$\widehat{\operatorname{vol}}^B(X \ni w, \Delta) := \inf\{a(E, X, \Delta)^k \operatorname{vol}(E) \mid E \text{ is an lc place of } (X, B) \text{ with } \operatorname{center}_X E = W\}.$$

Lemma 2.9. *Notation and conditions as in Set-up 2.8. For any $\epsilon \in (0, 1]$ and any prime divisor E over X,*

$$a(E, X, B_\epsilon) = \epsilon a(E, X, \Delta) + (1 - \epsilon)a(E, X, B) > 0.$$

In particular, (X, B_ϵ) is klt and $\operatorname{lct}(X \ni x, B_\epsilon; D) = \epsilon c$.

Proof. Since $B_\epsilon = \epsilon\Delta + (1 - \epsilon)B$, the equality holds, and the right-hand side is positive as (X, Δ) is klt and (X, B) is lc. Since $B_\epsilon + tD = \Delta + ((1 - \epsilon)c + t)D$ for any $t \geq 0$ and $c = \operatorname{lct}(X \ni x, \Delta; D)$, we have $\operatorname{lct}(X \ni x, B_\epsilon; D) = \epsilon c$. □

3. Rational approximation of the boundary

Construction-Proposition 3.1. Notation and conditions as in Set-up 2.8. Possibly shrinking X to a neighborhood of x, there exist $\mathbb{Q}$-divisors $\Delta_j \geq 0$ on X and real numbers $\delta_j \in (0, 1/2)$, $j \geq 1$, with $\lim_{j\to+\infty} \delta_j = 0$, satisfying the following. For every j, let $c_j := \operatorname{lct}(X \ni x, \Delta_j; D)$, $B_j := \Delta_j + c_j D$, and $B_{j,\epsilon} := \Delta_j + (1-\epsilon)c_j D$ for $\epsilon \in [0,1]$. Then:

(1) (a) $K_X + \Delta_j$ is $\mathbb{Q}$-Cartier and (X, Δ_j) is klt;
(b) $c_j \in \mathbb{Q}$;
(c) (X, B_j) is lc;
(d) the lc places of (X, B_j) are exactly the lc places of (X, B);
(e) W is the minimal lc center of (X, B_j).

(2) For every $\epsilon \in [0,1]$ and every valuation v on X with a center on X,

$$(1-\delta_j)a(v, X, B_\epsilon) \leq a(v, X, B_{j,\epsilon}) \leq (1+\delta_j)a(v, X, B_\epsilon).$$

(3) If $\ell \geq 1$ and $(W, \Delta_{W,j}, \mathbf{M}_j)$ is the generalized pair induced by adjunction from (X, B_j) to W, then for every valuation ν on W with a center on W,

$$(1-\delta_j)a(\nu, W, \Delta_W, \mathbf{M}) \leq a(\nu, W, \Delta_{W,j}, \mathbf{M}_j) \leq (1+\delta_j)a(\nu, W, \Delta_W, \mathbf{M}).$$

Proof. Since W is an lc center of (X, B), there exists an lc place S of (X, B) such that $\operatorname{center}_X S = W$. We let $\operatorname{mult}_S D$ be the coefficient of S in the pullback of D to a higher model of X on which S is a divisor. Since (X, Δ) is klt and $a(S, X, B) = 0$,

$$0 < a(S, X, \Delta) = a(S, X, B) + c \operatorname{mult}_S D = c \operatorname{mult}_S D.$$

Thus $c > 0$ and $\operatorname{mult}_S D > 0$.

Step 1. Write $\Delta = \sum_{i=1}^m b_{0,i} R_i$, where $m \in \mathbb{N}$, $R_1, \dots, R_m$ are distinct prime divisors, and $b_{0,i} > 0$ for any i. Fix a canonical divisor K_X. Since $K_X + \Delta$ is $\mathbb{R}$-Cartier, possibly shrinking X to a neighborhood of x, we may assume that there exist Cartier divisors $L_1, \dots, L_N$ on X and real numbers $z_{0,1}, \dots, z_{0,N}$ such that

$$K_X + \sum_{i=1}^m b_{0,i} R_i = \sum_{i=1}^N z_{0,i} L_i.$$

Let $f : X' \to X$ be a log resolution of $(X, \Delta + D)$, let

$$E_{X'} := \operatorname{Exc}(f) + f_*^{-1} \operatorname{Supp}(\Delta + D),$$

which is a reduced snc divisor, and fix a canonical divisor $K_{X'}$ such that $f_* K_{X'} = K_X$. We write points of $\mathbb{R}^{m+N+1}$ as $\mathbf{u} = (b_1, \dots, b_m, z_1, \dots, z_N, t)$. For any prime divisor F on X' and any $\mathbf{u} \in \mathbb{R}^{m+N+1}$, let

$$\begin{aligned} \alpha_F(\mathbf{u}) &:= 1 + \operatorname{mult}_F K_{X'} - \sum_{i=1}^N z_i \operatorname{mult}_F f^* L_i - t \operatorname{mult}_F f^* D, \\ \beta_F(\mathbf{u}) &:= \alpha_F(\mathbf{u}) + t \operatorname{mult}_F f^* D. \end{aligned}$$

Since $\operatorname{mult}_F K_{X'} \in \mathbb{Z}$, $\operatorname{mult}_F f^* L_i \in \mathbb{Z}$ and $\operatorname{mult}_F f^* D \in \mathbb{Q}$, α_F and β_F are affine functions on $\mathbb{R}^{m+N+1}$ with rational coefficients. We let

$$\Delta(\mathbf{u}) := \sum_{i=1}^m b_i R_i \quad \text{and} \quad B(\mathbf{u}) := \Delta(\mathbf{u}) + tD,$$

and let $\Pi \subset \mathbb{R}^{m+N+1}$ be the set of points $\mathbf{u} = (b_1, \dots, b_m, z_1, \dots, z_N, t)$ such that

$$K_X + \sum_{i=1}^m b_i R_i = \sum_{i=1}^N z_i L_i \tag{3.1}$$

as $\mathbb{R}$-divisors, and

$$\alpha_F(\mathbf{u}) = 0 \text{ for any prime component } F \text{ of } E_{X'} \text{ such that } a(F, X, B) = 0. \tag{3.2}$$

For any $\mathbf{u} \in \Pi$, $K_X + \Delta(\mathbf{u}) = \sum_{i=1}^N z_i L_i$ is $\mathbb{R}$-Cartier by (3.1), and

$$f^*(K_X + B(\mathbf{u})) = \sum_{i=1}^N z_i f^* L_i + t f^* D,$$

so for any prime divisor F on X',

$$a(F, X, B(\mathbf{u})) = \alpha_F(\mathbf{u}) \quad \text{and} \quad a(F, X, \Delta(\mathbf{u})) = \beta_F(\mathbf{u}).$$

Let

$$\mathbf{u}_0 := (b_{0,1}, \dots, b_{0,m}, z_{0,1}, \dots, z_{0,N}, c).$$

Then $\Delta(\mathbf{u}_0) = \Delta$, $B(\mathbf{u}_0) = B$, and $\mathbf{u}_0 \in \Pi$.

Since (3.1) and (3.2) are finitely many linear equations with rational coefficients and $\mathbf{u}_0 \in \Pi$, $\Pi \cap \mathbb{Q}^{m+N+1}$ is dense in Π.

Step 2. Fix $\eta \in (0, 1/2)$. Since α_F and β_F are continuous, $E_{X'}$ has finitely many components, $a(F, X, \Delta) > 0$ and $a(F, X, B) \geq 0$ for any prime divisor F on X', and $\alpha_F|_\Pi = 0 = a(F, X, B)$ for any prime component F of $E_{X'}$ with $a(F, X, B) = 0$ by (3.2), there exists an open neighborhood Π_η of $\mathbf{u}_0$ in Π such that

$$\begin{aligned} (1-\eta)a(F, X, B) \leq \alpha_F(\mathbf{u}) \leq (1+\eta)a(F, X, B), \\ (1-\eta)a(F, X, \Delta) \leq \beta_F(\mathbf{u}) \leq (1+\eta)a(F, X, \Delta) \end{aligned} \tag{3.3}$$

for any $\mathbf{u} \in \Pi_\eta$ and any prime component F of $E_{X'}$. Since $b_{0,i} > 0$ for any i and $c > 0$, possibly shrinking Π_η, we may assume that $b_i > 0$ for any i and $t > 0$ for any $\mathbf{u} = (b_1, \dots, b_m, z_1, \dots, z_N, t) \in \Pi_\eta$.

Let $\mathbf{u} = (b_1, \dots, b_m, z_1, \dots, z_N, t) \in \Pi_\eta$ and $\epsilon \in [0, 1]$. Since $b_i > 0$ and $t > 0$, $\operatorname{Supp} \Delta(\mathbf{u}) = \operatorname{Supp} \Delta$ and $\operatorname{Supp} B(\mathbf{u}) = \operatorname{Supp}(\Delta + D)$, so f is a log resolution of $(X, \Delta(\mathbf{u}) + (1-\epsilon)tD)$. Since

$$\Delta(\mathbf{u}) + (1-\epsilon)tD = \epsilon\Delta(\mathbf{u}) + (1-\epsilon)B(\mathbf{u})$$

and, for any prime divisor F on X' which is not a component of $E_{X'}$, $\alpha_F(\mathbf{u}) = \beta_F(\mathbf{u}) = 1$ by (3.1) and $f_* K_{X'} = K_X$, we have

$$f^*(K_X + \Delta(\mathbf{u}) + (1-\epsilon)tD) = K_{X'} + B'_{\mathbf{u},\epsilon},$$

where

$$B'_{\mathbf{u},\epsilon} := \sum_F \big(1 - \epsilon\beta_F(\mathbf{u}) - (1-\epsilon)\alpha_F(\mathbf{u})\big)F$$

and F runs over the prime components of $E_{X'}$. By Definition 2.2(4), for any valuation v on X with a center on X,

$$a(v, X, \Delta(\mathbf{u}) + (1-\epsilon)tD) = a(v, X', B'_{\mathbf{u},\epsilon}) \quad \text{and} \quad a(v, X, B_\epsilon) = a(v, X', B'_{\mathbf{u}_0,\epsilon}).$$

Since $\epsilon \geq 0$, $1 - \epsilon \geq 0$, $\alpha_F(\mathbf{u}_0) = a(F, X, B) \geq 0$, and $\beta_F(\mathbf{u}_0) = a(F, X, \Delta) > 0$, $\operatorname{mult}_F B'_{\mathbf{u}_0,\epsilon} \leq 1$ for any prime component F of $E_{X'}$, and $B'_{\mathbf{u}_0,\epsilon}$ and $B'_{\mathbf{u},\epsilon}$ satisfy the conditions of Lemma 2.5 with $E = E_{X'}$ and $\delta = \eta$ by (3.3). Thus

$$(1-\eta)a(v, X, B_\epsilon) \leq a(v, X, \Delta(\mathbf{u}) + (1-\epsilon)tD) \leq (1+\eta)a(v, X, B_\epsilon) \tag{3.4}$$

in $[0, +\infty]$, for any $\epsilon \in [0, 1]$ and any valuation v on X with a center on X.

Step 3. Let $\mathbf{u} = (b_1, \dots, b_m, z_1, \dots, z_N, t) \in \Pi_\eta$. By (3.4) with $\epsilon = 1$, $a(F, X, \Delta(\mathbf{u})) \geq (1-\eta)a(F, X, \Delta) > 0$ for any prime divisor F over X, so $(X, \Delta(\mathbf{u}))$ is klt. By (3.4) with $\epsilon = 0$,

$$(1-\eta)a(F, X, B) \leq a(F, X, B(\mathbf{u})) \leq (1+\eta)a(F, X, B)$$

for any prime divisor F over X. Thus $(X, B(\mathbf{u}))$ is lc, and the lc places of $(X, B(\mathbf{u}))$ are exactly the lc places of (X, B). In particular, the lc centers of $(X, B(\mathbf{u}))$ are exactly the lc centers of (X, B), so W is the minimal lc center of $(X, B(\mathbf{u}))$.

Since $(X, B(\mathbf{u}))$ is lc, S is an lc place of $(X, B(\mathbf{u}))$, $x \in \operatorname{center}_X S$, and $\operatorname{mult}_S D > 0$, $t = \operatorname{lct}(X \ni x, \Delta(\mathbf{u}); D)$.

For any integer $j \geq 1$, let $\delta_j := \frac{1}{j+2} \in (0, 1/2)$. Since $\Pi \cap \mathbb{Q}^{m+N+1}$ is dense in Π, there exists $\mathbf{u}_j \in \Pi_{\delta_j} \cap \mathbb{Q}^{m+N+1}$. Write

$$\mathbf{u}_j = (b_{j,1}, \dots, b_{j,m}, z_{j,1}, \dots, z_{j,N}, t_j).$$

Let $\Delta_j := \Delta(\mathbf{u}_j)$. Then $\Delta_j \geq 0$ is a $\mathbb{Q}$-divisor, $\operatorname{Supp} \Delta_j = \operatorname{Supp} \Delta$, and $\lim_{j \to +\infty} \delta_j = 0$. By the previous paragraph with $\eta = \delta_j$,

$$c_j = t_j \in \mathbb{Q}_{>0}, \quad B_j = B(\mathbf{u}_j), \quad \text{and} \quad B_{j,\epsilon} = \Delta(\mathbf{u}_j) + (1-\epsilon) t_j D$$

for any $\epsilon \in [0, 1]$. Since $z_{j,i} \in \mathbb{Q}$ for any i, $K_X + \Delta_j = \sum_{i=1}^N z_{j,i} L_i$ is $\mathbb{Q}$-Cartier. Thus the first two paragraphs of this step imply (1), and (3.4) with $\eta = \delta_j$ and $\mathbf{u} = \mathbf{u}_j$ is (2). Moreover, since $\operatorname{Supp} \Delta_j = \operatorname{Supp} \Delta$ and $c_j > 0$,

$$\operatorname{Supp} B_j = \operatorname{Supp}(\Delta + D) = \operatorname{Supp} B. \tag{3.5}$$

Step 4. Assume that $\ell \geq 1$ and fix j. By (1), (X, B_j) is lc, W is the minimal lc center of (X, B_j), and S is an lc place of (X, B_j), so $(W, \Delta_{W,j}, \mathbf{M}_j)$ is defined by Definition-Theorem 2.7. We first show that

$$(1-\delta_j) a(P, W, \Delta_W, \mathbf{M}) \leq a(P, W, \Delta_{W,j}, \mathbf{M}_j) \leq (1+\delta_j) a(P, W, \Delta_W, \mathbf{M}) \tag{3.6}$$

for any projective birational morphism $W' \to W$ from a normal variety W' and any prime divisor P on W'.

Let $h : Y \to X$ be a log resolution of (X, B) such that S is a divisor on Y and the induced map $h_S : S \dashrightarrow W'$ is a morphism. By (3.5), h is also a log resolution of (X, B_j). Let

$$E_Y := \operatorname{Exc}(h) + h_*^{-1} \operatorname{Supp} B,$$

which is a reduced snc divisor with S as a component, and let

$$K_Y + B_Y := h^*(K_X + B), \qquad K_Y + B_{Y,j} := h^*(K_X + B_j),$$
$$K_S + B_S := (K_Y + B_Y)|_S, \qquad K_S + B_{S,j} := (K_Y + B_{Y,j})|_S.$$

Then B_Y and $B_{Y,j}$ are supported in E_Y, and $\operatorname{mult}_S B_Y = \operatorname{mult}_S B_{Y,j} = 1$. Thus $B_S = (B_Y - S)|_S$ and $B_{S,j} = (B_{Y,j} - S)|_S$ are supported in the reduced snc divisor $E_S := (E_Y - S)|_S$. Since E_Y is snc, for any prime component F' of E_S there exists exactly one prime component F of $E_Y - S$ such that $F' \subset F$, and

$$1 - \operatorname{mult}_{F'} B_S = a(F, X, B) \geq 0 \quad \text{and} \quad 1 - \operatorname{mult}_{F'} B_{S,j} = a(F, X, B_j).$$

By (2) with $\epsilon = 0$ and $v = v_F$,

$$(1-\delta_j)(1 - \operatorname{mult}_{F'} B_S) \leq 1 - \operatorname{mult}_{F'} B_{S,j} \leq (1+\delta_j)(1 - \operatorname{mult}_{F'} B_S)$$

for any prime component F' of E_S. Since S is smooth, E_S is a reduced snc divisor on S, and $\operatorname{mult}_{F'} B_S \leq 1$ for any prime component F' of E_S, Lemma 2.5 with $E = E_S$ and $\delta = \delta_j$ implies that, for any valuation μ on S with a center on S, $a(\mu, S, B_S) \geq 0$ and

$$(1-\delta_j) a(\mu, S, B_S) \leq a(\mu, S, B_{S,j}) \leq (1+\delta_j) a(\mu, S, B_S). \tag{3.7}$$

Let $\mathcal{G}_P$ be the set of prime divisors G over S such that $P \subset h_S(\operatorname{center}_S G)$. Since W' is normal, P is Cartier near the generic point of P, so $\operatorname{mult}_G h_S^* P \geq 0$ is defined for any $G \in \mathcal{G}_P$, and $\operatorname{mult}_G h_S^* P > 0$ if and only if $h_S(\operatorname{center}_S G) = P$. For any $\lambda \in \mathbb{R}$, $(S, B_S + \lambda h_S^* P)$ is sub-lc over the generic point of P if and only if

$$a(G, S, B_S + \lambda h_S^* P) = a(G, S, B_S) - \lambda \operatorname{mult}_G h_S^* P \geq 0 \text{ for any } G \in \mathcal{G}_P,$$

as one sees on a log resolution of $(S, E_S + h_S^* P)$ over a neighborhood of the generic point of P. Since $a(G, S, B_S) \geq 0$ for any G, the inequality holds for any λ when $\operatorname{mult}_G h_S^* P = 0$.

Let $\mathcal{G}_P^+ := \{G \in \mathcal{G}_P \mid \operatorname{mult}_G h_S^* P > 0\}$. Since S and h satisfy the conditions of Definition-Theorem 2.7(2) both for (X, B) and for (X, B_j),

$$a(P, W, \Delta_W, \mathbf{M}) = \inf_{G \in \mathcal{G}_P^+} \frac{a(G, S, B_S)}{\operatorname{mult}_G h_S^* P} \quad \text{and} \quad a(P, W, \Delta_{W,j}, \mathbf{M}_j) = \inf_{G \in \mathcal{G}_P^+} \frac{a(G, S, B_{S,j})}{\operatorname{mult}_G h_S^* P}.$$

By (3.7) with $\mu = v_G$,

$$(1 - \delta_j) \frac{a(G, S, B_S)}{\operatorname{mult}_G h_S^* P} \leq \frac{a(G, S, B_{S,j})}{\operatorname{mult}_G h_S^* P} \leq (1 + \delta_j) \frac{a(G, S, B_S)}{\operatorname{mult}_G h_S^* P}$$

for any $G \in \mathcal{G}_P^+$, and taking infima implies (3.6).

Step 5. By Definition 2.6(2), $\mathbf{M}$ and $\mathbf{M}_j$ descend to some higher models of W, hence to any higher model dominating both. Let $g : W'' \to W$ be a log resolution of $(W, \Delta_W + \Delta_{W,j})$ such that $\mathbf{M}$ and $\mathbf{M}_j$ descend to W'', let

$$E_{W''} := \operatorname{Exc}(g) + g_*^{-1} \operatorname{Supp}(\Delta_W + \Delta_{W,j}),$$

which is a reduced snc divisor, and let

$$K_{W''} + \Delta_{W''} + \mathbf{M}_{W''} := g^*(K_W + \Delta_W + \mathbf{M}_W),$$
$$K_{W''} + \Delta_{W'',j} + \mathbf{M}_{W'',j} := g^*(K_W + \Delta_{W,j} + \mathbf{M}_{W,j}).$$

Then $\Delta_{W''}$ and $\Delta_{W'',j}$ are supported in $E_{W''}$, since $g_* \Delta_{W''} = \Delta_W$ and $g_* \Delta_{W'',j} = \Delta_{W,j}$. By Definition 2.6(3), for any prime component P of $E_{W''}$,

$$a(P, W, \Delta_W, \mathbf{M}) = 1 - \operatorname{mult}_P \Delta_{W''} > 0 \quad \text{and} \quad a(P, W, \Delta_{W,j}, \mathbf{M}_j) = 1 - \operatorname{mult}_P \Delta_{W'',j},$$

where the positivity holds since $(W, \Delta_W, \mathbf{M})$ is generalized klt (Definition-Theorem 2.7(1)). Thus, by (3.6) with $W' = W''$, for any prime component P of $E_{W''}$,

$$(1 - \delta_j)(1 - \operatorname{mult}_P \Delta_{W''}) \leq 1 - \operatorname{mult}_P \Delta_{W'',j} \leq (1 + \delta_j)(1 - \operatorname{mult}_P \Delta_{W''}).$$

By Definition 2.6(3), $a(\nu, W, \Delta_W, \mathbf{M}) = a(\nu, W'', \Delta_{W''})$ and $a(\nu, W, \Delta_{W,j}, \mathbf{M}_j) = a(\nu, W'', \Delta_{W'',j})$ for any valuation ν on W with a center on W. Since W'' is smooth and $\operatorname{mult}_P \Delta_{W''} < 1$ for any prime component P of $E_{W''}$, Lemma 2.5 with $E = E_{W''}$ and $\delta = \delta_j$ implies (3). □

Corollary 3.2. *Notation as in Construction-Proposition 3.1. Then for every j:*

(1) for every $\epsilon \in (0, 1]$,

$$(1 - \delta_j)^n \widehat{\operatorname{vol}}(X \ni x, B_\epsilon) \leq \widehat{\operatorname{vol}}(X \ni x, B_{j,\epsilon}) \leq (1 + \delta_j)^n \widehat{\operatorname{vol}}(X \ni x, B_\epsilon);$$

(2) for every $\epsilon \in (0, 1]$,

$$(1 - \delta_j)^k \widehat{\operatorname{vol}}(X \ni w, B_\epsilon) \leq \widehat{\operatorname{vol}}(X \ni w, B_{j,\epsilon}) \leq (1 + \delta_j)^k \widehat{\operatorname{vol}}(X \ni w, B_\epsilon);$$

(3) if $\ell \geq 1$, then

$$(1 - \delta_j)^\ell \widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M}) \leq \widehat{\operatorname{vol}}(W \ni x, \Delta_{W,j}, \mathbf{M}_j) \leq (1 + \delta_j)^\ell \widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M}).$$

Proof. Fix j and $\epsilon \in (0, 1]$. By Lemma 2.9, (X, B_ϵ) is klt, and $(X, B_{j,\epsilon})$ is klt by Construction-Proposition 3.1(2). Thus all local volumes in (1) and (2) are defined. By Construction-Proposition 3.1(2), $a(v, X, B_\epsilon) = +\infty$ if and only if $a(v, X, B_{j,\epsilon}) = +\infty$, for any valuation v on X with a center on X.

If $\ell \geq 1$, then, since (X, B) and (X, B_j) are lc with minimal lc center W (Set-up 2.8 and Construction-Proposition 3.1(1)), Definition-Theorem 2.7(1) implies that the generalized pairs $(W, \Delta_W, \mathbf{M})$ and $(W, \Delta_{W,j}, \mathbf{M}_j)$ are generalized klt, so both local volumes in (3) are defined.

Since $\operatorname{vol}(v)$ does not depend on the boundary, raising the inequalities of Construction-Proposition 3.1(2) and (3) to the powers $n = \dim \mathcal{O}_{X,x}$, $k = \dim \mathcal{O}_{X,w}$, and $\ell = \dim W$ and taking infima imply (1), (2), and (3). □

4. Local volumes of generalized pairs

The main result of this section is the following degree formula for Galois covers, which we prove at the end of the section.

Proposition 4.1. *Let $(X, B, \mathbf{M})$ be a generalized pair such that B is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is a $\mathbb{Q}$-$\boldsymbol{b}$-divisor, let $x \in X$ be a closed point such that $(X, B, \mathbf{M})$ is generalized klt near x, and let $f : X' \to X$ be a finite Galois morphism of degree r from a normal variety X' such that $f^{-1}(x)$ is a single point x'. Let $Y \to X$ be a projective birational morphism such that $\mathbf{M}$ descends to Y. Let Y' be the normalization of the irreducible component of $Y \times_X X'$ dominating X', and let $f^*\mathbf{M}$ be the $\mathbb{Q}$-$\boldsymbol{b}$-divisor of X' which descends to Y' with $(f^*\mathbf{M})_{Y'}$ the pullback of $\mathbf{M}_Y$. Let B' be the $\mathbb{Q}$-divisor with*

$$K_{X'} + B' + (f^*\mathbf{M})_{X'} = f^*(K_X + B + \mathbf{M}_X).$$

Assume that $B' \geq 0$. Then $(X', B', f^*\mathbf{M})$ is generalized klt near x' and

$$\widehat{\mathrm{vol}}(X' \ni x', B', f^*\mathbf{M}) = r \cdot \widehat{\mathrm{vol}}(X \ni x, B, \mathbf{M}).$$

Lemma 4.2 (cf. [LLX20, Theorem 2.6])**.** *Let $(X, B, \mathbf{M})$ be a generalized pair which is generalized klt near a closed point $x \in X$. Then*

$$\widehat{\mathrm{vol}}(X \ni x, B, \mathbf{M}) = \inf_E a(E, X, B, \mathbf{M})^{\dim X} \mathrm{vol}(E),$$

where E runs over the prime divisors over X with $\mathrm{center}_X E = \{x\}$.

Proof. The case $\dim X = 0$ is trivial. Let $d := \dim X \geq 1$, and let Λ be the infimum on the right-hand side. If E is a prime divisor over X with $\mathrm{center}_X E = \{x\}$, then $\mathrm{center}_X v_E = x$, $a(v_E, X, B, \mathbf{M}) = a(E, X, B, \mathbf{M})$, and $\mathrm{vol}(v_E) = \mathrm{vol}(E)$. Thus $\widehat{\mathrm{vol}}(X \ni x, B, \mathbf{M}) \leq \Lambda$.

Possibly shrinking X to a neighborhood of x, we may assume that $(X, B, \mathbf{M})$ is generalized klt. For a valuation v on X with $\mathrm{center}_X v = x$ and an ideal $J \subset \mathcal{O}_{X,x}$, let $v(J) := \min\{v(f) \mid f \in J\}$.

Let $J \subset \mathcal{O}_{X,x}$ be an $\mathfrak{m}_x$-primary ideal and $\mathcal{J} \subset \mathcal{O}_X$ the ideal sheaf such that $\mathcal{J}_x = J$ and $\mathrm{Supp}(\mathcal{O}_X/\mathcal{J}) = \{x\}$. Let $h : Y \to X$ be a log resolution of (X, B) such that $\mathbf{M}$ descends to Y and

$$\mathcal{J}\mathcal{O}_Y = \mathcal{O}_Y\Big(-\sum_{i=1}^N \alpha_i F_i\Big),$$

where $\alpha_i \in \mathbb{N}$, $F_1, \dots, F_N$ are distinct prime divisors on Y, and $\sum_{i=1}^N F_i$ is an snc divisor whose support contains $\mathrm{Exc}(h) \cup \mathrm{Supp}\, h_*^{-1}B$. Write

$$K_Y + B_Y + \mathbf{M}_Y = h^*(K_X + B + \mathbf{M}_X) \quad \text{and} \quad B_Y = \sum_{i=1}^N b_i F_i.$$

Then $a(F_i, X, B, \mathbf{M}) = 1 - b_i > 0$ for any i. Since $J \neq \mathcal{O}_{X,x}$, we have $\alpha_i > 0$ for some i, and $\mathrm{center}_X F_i = \{x\}$ for any i with $\alpha_i > 0$ as $h(F_i) \subset \mathrm{Supp}(\mathcal{O}_X/\mathcal{J})$. Let

$$\gamma_J := \min\Big\{\frac{a(F_i, X, B, \mathbf{M})}{\alpha_i} \;\Big|\; \alpha_i > 0\Big\}.$$

Possibly reordering indices, we may assume that $\alpha_1 > 0$ and $a(F_1, X, B, \mathbf{M}) = \gamma_J \alpha_1$.

Let v be a valuation on X with $\mathrm{center}_X v = x$. Then v has a center on Y, $v(F_i) \geq 0$ for any i, and $v(J) = \sum_{i=1}^N \alpha_i v(F_i)$. Since $a(v, Y, \sum_{i=1}^N F_i) \geq 0$, by Lemma 2.4,

$$\begin{aligned} a(v, X, B, \mathbf{M}) = a(v, Y, B_Y) &= a\Big(v, Y, \sum_{i=1}^N F_i\Big) + \sum_{i=1}^N (1 - b_i) v(F_i) \\ &\geq \sum_{\alpha_i > 0} a(F_i, X, B, \mathbf{M}) v(F_i) \geq \gamma_J v(J). \end{aligned} \tag{4.1}$$

Since $\operatorname{mult}_{F_1}\operatorname{div}(f) \geq m\alpha_1$ for any positive integer m and any non-zero $f \in J^m$, we have $J^m \subset \mathfrak{a}_{m\alpha_1}(v_{F_1})$. By Lemma 2.3,

$$\operatorname{vol}(F_1) = \lim_{m\to+\infty} \frac{d! \cdot \operatorname{length}(\mathcal{O}_{X,x}/\mathfrak{a}_{m\alpha_1}(v_{F_1}))}{(m\alpha_1)^d} \leq \lim_{m\to+\infty} \frac{d! \cdot \operatorname{length}(\mathcal{O}_{X,x}/J^m)}{(m\alpha_1)^d} = \frac{e(J)}{\alpha_1^d}.$$

Thus

$$\Lambda \leq a(F_1, X, B, \mathbf{M})^d \operatorname{vol}(F_1) = \gamma_J^d \alpha_1^d \operatorname{vol}(F_1) \leq \gamma_J^d e(J). \tag{4.2}$$

Now let v be a valuation on X such that $\operatorname{center}_X v = x$ and $a(v, X, B, \mathbf{M}) < +\infty$, and let m be a positive integer. By Lemma 2.3, $\mathfrak{a}_m(v)$ is $\mathfrak{m}_x$-primary. Since $v(\mathfrak{a}_m(v)) \geq m$, (4.1) for $J = \mathfrak{a}_m(v)$ implies that $\gamma_{\mathfrak{a}_m(v)} \leq a(v, X, B, \mathbf{M})/m$, so by (4.2),

$$\Lambda \leq a(v, X, B, \mathbf{M})^d \cdot \frac{e(\mathfrak{a}_m(v))}{m^d}.$$

Letting $m \to +\infty$, Lemma 2.3 implies that

$$\Lambda \leq a(v, X, B, \mathbf{M})^d \operatorname{vol}(v).$$

Thus $\Lambda \leq \widehat{\operatorname{vol}}(X \ni x, B, \mathbf{M})$. The lemma follows. □

In the proof of Theorem 7.1, we replace X with a pointed étale neighborhood of x.

Lemma 4.3. *Let $\phi : X' \to X$ be an étale morphism between normal quasi-projective varieties, $h : Y \to X$ a projective birational morphism from a normal variety Y, $Y' := Y \times_X X'$, and $h' : Y' \to X'$ and $\phi' : Y' \to Y$ the projections. Then the following hold.*

1. *For any closed point $x' \in X'$ and any $\mathfrak{m}_{x'}$-primary ideal $\mathfrak{b}' \subset \mathcal{O}_{X',x'}$, the inverse image $\mathfrak{b}$ of $\mathfrak{b}'$ in $\mathcal{O}_{X,\phi(x')}$ is $\mathfrak{m}_{\phi(x')}$-primary, $\mathfrak{b}' = \mathfrak{b}\mathcal{O}_{X',x'}$, and $\mathcal{O}_{X,\phi(x')}/\mathfrak{b} \cong \mathcal{O}_{X',x'}/\mathfrak{b}'$.*
2. *The variety Y' is normal, and h' is projective birational. For any prime divisor F on Y and any irreducible component F' of $\phi'^{-1}(F)$, F' is a prime divisor on Y', $\operatorname{mult}_{F'} \phi'^* D = \operatorname{mult}_F D$ for any $\mathbb{R}$-divisor D on Y, and $h'(F')$ is an irreducible component of $\phi^{-1}(h(F))$. Any irreducible component of $\phi^{-1}(h(F))$ is $h'(F'')$ for some irreducible component F'' of $\phi'^{-1}(F)$.*
3. *For any prime divisor E' over X', there exists a prime divisor F over X such that, for any projective birational morphism $h_1 : Y_1 \to X$ from a normal variety Y_1 on which F is a divisor, E' is an irreducible component of the inverse image of F in $Y_1 \times_X X'$.*
4. *If Δ is an $\mathbb{R}$-divisor on X such that $K_X + \Delta$ is $\mathbb{R}$-Cartier and h is a log resolution of (X, Δ), then $a(F', X', \phi^*\Delta) = a(F, X, \Delta)$ for F and F' as in (2).*
5. *If F and F' are as in (2) and $x' \in X'$ is a closed point with $h(F) = \{\phi(x')\}$ and $h'(F') = \{x'\}$, then $\operatorname{vol}(cv_{F'}) = \operatorname{vol}(cv_F)$ for any real number $c > 0$.*
6. *If Δ is as in (4) and $\phi_j : U_j \to X$, $1 \leq j \leq N$, are étale morphisms from normal quasi-projective varieties with $X = \bigcup_{j=1}^N \phi_j(U_j)$, then (X, Δ) is sub-lc if and only if $(U_j, \phi_j^*\Delta)$ is sub-lc for any j.*

Proof. (1) and (2) hold since ϕ is étale: ϕ induces isomorphisms

$$\mathcal{O}_{X,\phi(x')}/\mathfrak{m}_{\phi(x')}^k \cong \mathcal{O}_{X',x'}/\mathfrak{m}_{x'}^k$$

for any positive integer k, and étale base change preserves normality, dimensions of irreducible components and multiplicities along prime divisors. In (2), Y' is irreducible, so h' is projective birational: the connected components of the normal scheme Y' are irreducible, and if $U \subset X$ is a non-empty open subset over which h is an isomorphism, then, since ϕ' is an open map, any connected component of Y' maps onto a non-empty open subset of Y, hence intersects $h'^{-1}(\phi^{-1}(U)) \cong \phi^{-1}(U)$, which is irreducible.

(3) By [Kol13, 2.14(2)], there exist a birational morphism $Y_2 \to X$ from a normal variety and a prime divisor F on Y_2 such that E' is an irreducible component of the inverse image of F in $Y_2 \times_X X'$. Let $h_1 : Y_1 \to X$ be a projective birational morphism from a normal variety Y_1 on

which F is a divisor. Then Y_1 and Y_2 are isomorphic near the generic point of F, hence so are $Y_1 \times_X X'$ and $Y_2 \times_X X'$ near the inverse image of that point, and E' is an irreducible component of the inverse image of F in $Y_1 \times_X X'$.

(4) Since ϕ' is étale and Y is smooth, h' is a log resolution of $(X', \phi^*\Delta)$, and by [Kol13, 2.14(1)],

$$a(F', X', \phi^*\Delta) = a(F, X, \Delta).$$

(5) Let $x := \phi(x')$ and regard $\mathcal{O}_{X,x}$ as a subring of $\mathcal{O}_{X',x'}$. By (2),

$$v_{F'}(f) = \operatorname{mult}_{F'} \phi'^* \operatorname{div}(f) = v_F(f)$$

for any non-zero $f \in \mathcal{O}_{X,x}$, so the inverse image of $\mathfrak{a}_\lambda(cv_{F'})$ in $\mathcal{O}_{X,x}$ is $\mathfrak{a}_\lambda(cv_F)$ for any real number $\lambda > 0$. By Lemma 2.3 and (1),

$$\operatorname{length}(\mathcal{O}_{X',x'}/\mathfrak{a}_\lambda(cv_{F'})) = \operatorname{length}(\mathcal{O}_{X,x}/\mathfrak{a}_\lambda(cv_F))$$

for any $\lambda > 0$. Since $\dim X' = \dim X$, $\operatorname{vol}(cv_{F'}) = \operatorname{vol}(cv_F)$.

(6) Assume that (X, Δ) is sub-lc, and let E' be a prime divisor over U_j for some j. Let F be the prime divisor over X given by (3) for ϕ_j in place of ϕ, and let $h_1 : Y_1 \to X$ be a log resolution of (X, Δ) on which F is a divisor. By (3) and (4) for ϕ_j and h_1 in place of ϕ and h, $a(E', U_j, \phi_j^*\Delta) = a(F, X, \Delta) \geq 0$. Conversely, assume that $(U_j, \phi_j^*\Delta)$ is sub-lc for any j, and let F be a prime divisor over X. Let $h_1 : Y_1 \to X$ be a log resolution of (X, Δ) on which F is a divisor, and choose j such that $\phi_j^{-1}(h_1(F)) \neq \emptyset$. By (2) for ϕ_j and h_1 in place of ϕ and h, $F \times_X U_j$ has an irreducible component F', and by (4), $a(F, X, \Delta) = a(F', U_j, \phi_j^*\Delta) \geq 0$. □

Lemma 4.4 (cf. [HLQ23, Proposition 2.24])**.** *Let $\phi : X' \to X$ be an étale morphism between normal quasi-projective varieties, $x' \in X'$ a closed point, $x := \phi(x')$, B an $\mathbb{R}$-divisor on X, and $B' := \phi^*B$.*

(1) Assume that $(X \ni x, B)$ is a klt pair. Then:
- *(i) $(X' \ni x', B')$ is a klt pair and $\widehat{\operatorname{vol}}(X' \ni x', B') = \widehat{\operatorname{vol}}(X \ni x, B)$;*
- *(ii) for any divisorial valuation v on X with $\operatorname{center}_X v = x$, there exists a divisorial valuation v' on X' such that*
  - *(a) $\operatorname{center}_{X'} v' = x'$,*
  - *(b) $v'(f \circ \phi) = v(f)$ for any non-zero rational function f on X,*
  - *(c) $a(v', X', B') = a(v, X, B)$, and*
  - *(d) $\operatorname{vol}(v') = \operatorname{vol}(v)$.*

(2) Assume that (X, B) is lc, any lc center of (X, B) contains x, and W is the minimal lc center of (X, B). Let $(W, \Delta_W, \mathbf{M})$ be the generalized pair induced by adjunction from (X, B) to W. Then, possibly shrinking X' to a neighborhood of x':
- *(i) any lc center of (X', B') contains x';*
- *(ii) the irreducible component W' of $\phi^{-1}(W)$ containing x' is the minimal lc center of (X', B');*
- *(iii) $\widehat{\operatorname{vol}}(W' \ni x', \Delta_{W'}, \mathbf{M}') = \widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M})$ for the generalized pair $(W', \Delta_{W'}, \mathbf{M}')$ induced by adjunction from (X', B') to W';*
- *(iv) $\Delta_{W'}$ and $\mathbf{M}'$ in (iii) are the pullbacks of Δ_W and $\mathbf{M}$ under the étale morphism $\phi|_{W'} : W' \to W$, where the pullback of $\mathbf{M}$ is the $\mathbf{b}$-divisor of W' which descends to $W_1 \times_W W'$, a normal variety projective and birational over W' by Lemma 4.3(2), with trace the pullback of $\mathbf{M}_{W_1}$, for a projective birational morphism $W_1 \to W$ from a normal variety to which $\mathbf{M}$ descends.*

Proof. **Step 1.** We prove (1). Possibly shrinking X to a neighborhood of x and replacing X' with $\phi^{-1}(X)$, we may assume that (X, B) is klt. Then $B' \geq 0$, and $K_{X'} + B' = \phi^*(K_X + B)$ is $\mathbb{R}$-Cartier as $K_{X'} = \phi^*K_X$. By Lemma 4.3(3)(4) for $\Delta = B$, any prime divisor E' over X' satisfies $a(E', X', B') = a(E, X, B) > 0$ for some prime divisor E over X. Thus $(X' \ni x', B')$ is a klt pair.

Let $v = cv_E$, where $c > 0$ is a real number and E is a prime divisor over X with $\operatorname{center}_X E = \{x\}$. Let $h : Y \to X$ be a log resolution of (X, B) on which E is a divisor, $Y' := Y \times_X X'$, and $h' : Y' \to X'$ and $\phi' : Y' \to Y$ the projections. Since $\{x'\}$ is an irreducible component of $\phi^{-1}(x)$, by Lemma 4.3(2) there exists an irreducible component E' of $\phi'^{-1}(E)$ with $h'(E') = \{x'\}$. Let $v' := cv_{E'}$. By Lemma 4.3(2),

$$v'(f \circ \phi) = c \operatorname{mult}_{E'} \phi'^* \operatorname{div}(f) = c \operatorname{mult}_E \operatorname{div}(f) = v(f)$$

for any non-zero rational function f on X. By Lemma 4.3(4),

$$a(v', X', B') = ca(E', X', B') = ca(E, X, B) = a(v, X, B),$$

and by Lemma 4.3(5), $\operatorname{vol}(v') = \operatorname{vol}(v)$.

Let $d := \dim X$. By Lemma 4.2 (with $\mathbf{M} = 0$), $\widehat{\operatorname{vol}}(X \ni x, B)$ and $\widehat{\operatorname{vol}}(X' \ni x', B')$ are the infima of $a(E, X, B)^d \operatorname{vol}(E)$ and $a(E', X', B')^d \operatorname{vol}(E')$ over the prime divisors E over X with $\operatorname{center}_X E = \{x\}$ and E' over X' with $\operatorname{center}_{X'} E' = \{x'\}$, respectively. By the previous paragraph with $c = 1$, for any such E there exists such an E' with $a(E', X', B')^d \operatorname{vol}(E') = a(E, X, B)^d \operatorname{vol}(E)$. Thus $\widehat{\operatorname{vol}}(X' \ni x', B') \le \widehat{\operatorname{vol}}(X \ni x, B)$. Conversely, let E' be a prime divisor over X' with $\operatorname{center}_{X'} E' = \{x'\}$. By Lemma 4.3(3), there exist a prime divisor E over X and a log resolution $h : Y \to X$ of (X, B) on which E is a divisor, such that E' is an irreducible component of $\phi'^{-1}(E)$. By Lemma 4.3(2), $\{x'\}$ is an irreducible component of $\phi^{-1}(\operatorname{center}_X E)$, so $\operatorname{center}_X E = \{x\}$. By Lemma 4.3(4)(5), $a(E', X', B')^d \operatorname{vol}(E') = a(E, X, B)^d \operatorname{vol}(E)$. Thus $\widehat{\operatorname{vol}}(X' \ni x', B') \ge \widehat{\operatorname{vol}}(X \ni x, B)$, and (1) follows.

Step 2. We prove (2). By Lemma 4.3(3)(4) for $\Delta = B$, (X', B') is lc, and any lc place of (X', B') is an irreducible component of $\phi'^{-1}(E)$ for an lc place E of (X, B) on a log resolution $Y \to X$ of (X, B), with Y' and ϕ' as in **Step 1**. By Lemma 4.3(2)(4), the lc centers of (X', B') are exactly the irreducible components of $\phi^{-1}(V)$ for the lc centers V of (X, B). Since (X, B) has finitely many lc centers, so does (X', B'). Possibly shrinking X' to a neighborhood of x', we may assume that any lc center of (X', B') contains x'. By Definition-Theorem 2.7, W is normal. Since $\phi^{-1}(W) \to W$ is étale, $\phi^{-1}(W)$ is normal, so W' is the only irreducible component of $\phi^{-1}(W)$ containing x'. Since the irreducible components of $\phi^{-1}(W)$ are lc centers of (X', B'), each of them contains x', so $\phi^{-1}(W) = W'$, and W' is an lc center of (X', B').

Let $V' \subset W'$ be an lc center of (X', B'). Then V' is an irreducible component of $\phi^{-1}(V)$ for an lc center V of (X, B). Since $\phi^{-1}(V) \to V$ is étale, $\phi(V')$ is dense in V. Since $\phi(V') \subset \phi(W') \subset W$, we have $V \subset W$, so $V = W$ by the minimality of W. Thus V' is an irreducible component of $\phi^{-1}(W) = W'$, and $V' = W'$. Therefore W' is the minimal lc center of (X', B').

Let $\ell := \dim W$. By Definition-Theorem 2.7(1), $(W, \Delta_W, \mathbf{M})$ and $(W', \Delta_{W'}, \mathbf{M}')$ are generalized klt. By Lemma 4.2,

$$\widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M}) = \inf_P a(P, W, \Delta_W, \mathbf{M})^\ell \operatorname{vol}(P),$$

where P runs over the prime divisors over W with $\operatorname{center}_W P = \{x\}$, and similarly for $(W' \ni x', \Delta_{W'}, \mathbf{M}')$. Since $W' = W \times_X X'$, the restriction $\phi_W := \phi|_{W'} : W' \to W$ is étale. We show the following.

Claim 4.5. *Let $q : W_1 \to W$ be a projective birational morphism from a normal variety W_1, P a prime divisor on W_1, $W_1' := W_1 \times_W W'$, $\phi_W' : W_1' \to W_1$ the projection, and P' an irreducible component of $\phi_W'^{-1}(P)$. Then*

$$a(P', W', \Delta_{W'}, \mathbf{M}') = a(P, W, \Delta_W, \mathbf{M}).$$

Proof. Let S be an lc place of (X, B) with $\operatorname{center}_X S = W$. Since W is quasi-projective, q is the blow-up of an ideal sheaf $\mathcal{I} \subset \mathcal{O}_W$ with $\operatorname{Supp}(\mathcal{O}_W/\mathcal{I}) \ne W$. Let $\mathcal{J} \subset \mathcal{O}_X$ be the inverse image of $\mathcal{I}$, and let $h : Y \to X$ be a log resolution of (X, B) such that S is a divisor on Y and $\mathcal{J}\mathcal{O}_Y$ is invertible. Since $h(S) = W \not\subset \operatorname{Supp}(\mathcal{O}_W/\mathcal{I})$, the ideal sheaf $\mathcal{I}\mathcal{O}_S = \mathcal{J}\mathcal{O}_Y \cdot \mathcal{O}_S$ is invertible,

so $h|_S : S \to W$ factors through a morphism $h_S : S \to W_1$. Let $K_Y + B_Y := h^*(K_X + B)$ and $K_S + B_S := (K_Y + B_Y)|_S$. By Definition-Theorem 2.7(2),

$$a(P, W, \Delta_W, \mathbf{M}) = \sup\{\lambda \in \mathbb{R} \mid (S, B_S + \lambda h_S^* P) \text{ is sub-lc over the generic point of } P\}. \tag{4.3}$$

Let $Y' := Y \times_X X'$, let $h' : Y' \to X'$ and $\phi' : Y' \to Y$ be the projections, and let $S' := \phi'^{-1}(S)$. Since ϕ' is étale and Y is smooth, h' is a log resolution of (X', B'). Since S' is étale over the smooth variety S, the irreducible components of S' are disjoint, and by Lemma 4.3(2)(4), each of them is an lc place of (X', B') whose center on X' is an irreducible component of $\phi^{-1}(W) = W'$, that is, W'. Since $W' = W \times_X X'$,

$$S' = S \times_X X' = S \times_W W' = S \times_{W_1} W'_1,$$

and the second projection $h_{S'} : S' \to W'_1$ is the base change of h_S by ϕ'_W. In particular, $\phi'_S := \phi'|_{S'} : S' \to S$ is étale, and since $\phi'^* S = S'$,

$$K_{S'} + B_{S'} := (K_{Y'} + \phi'^* B_Y)|_{S'} = \phi'^*_S (K_S + B_S).$$

By Definition-Theorem 2.7(2) for (X', B'), h', and any irreducible component of S', the number $a(P', W', \Delta_{W'}, \mathbf{M}')$ is the supremum of the real numbers λ such that the restriction of $(S', B_{S'} + \lambda h_{S'}^* P')$ to that component is sub-lc over the generic point of P'. Since $h_{S'}^* P' \geq 0$, the irreducible components of S' are disjoint, and for each of them the supremum above is $a(P', W', \Delta_{W'}, \mathbf{M}')$,

$$a(P', W', \Delta_{W'}, \mathbf{M}') = \sup\{\lambda \in \mathbb{R} \mid (S', B_{S'} + \lambda h_{S'}^* P') \text{ is sub-lc over the generic point of } P'\}. \tag{4.4}$$

Let $U' \subset W'_1$ be an open neighborhood of the generic point of P' such that $U' \cap \phi'^{-1}_W(P) = U' \cap P'$ and $U := \phi'_W(U')$ is contained in the non-singular locus of W_1. Then U is an open neighborhood of the generic point of P, $U' \to U$ is surjective étale, and $h_{S'}^* P' = h_{S'}^* \phi'^*_W P = \phi'^*_S h_S^* P$ over U'. Let λ be a real number. If $(S, B_S + \lambda h_S^* P)$ is sub-lc over an open neighborhood $U_0 \subset U$ of the generic point of P, then Lemma 4.3(6), for $h_S^{-1}(U_0)$ in place of X and the irreducible components of $h_{S'}^{-1}(U' \cap \phi'^{-1}_W(U_0))$ in place of the U_j, implies that $(S', B_{S'} + \lambda h_{S'}^* P')$ is sub-lc over $U' \cap \phi'^{-1}_W(U_0)$. Conversely, if $(S', B_{S'} + \lambda h_{S'}^* P')$ is sub-lc over an open neighborhood $U'_0 \subset U'$ of the generic point of P', then since $h_{S'}^{-1}(U'_0) \to h_S^{-1}(\phi'_W(U'_0))$ is surjective étale, being the base change of $U'_0 \to \phi'_W(U'_0)$, Lemma 4.3(6), for $h_S^{-1}(\phi'_W(U'_0))$ in place of X and the irreducible components of $h_{S'}^{-1}(U'_0)$ in place of the U_j, implies that $(S, B_S + \lambda h_S^* P)$ is sub-lc over $\phi'_W(U'_0)$. Thus the suprema in (4.3) and (4.4) are equal. □

Let P be a prime divisor over W with $\operatorname{center}_W P = \{x\}$, and let $q : W_1 \to W$ be a projective birational morphism from a normal variety on which P is a divisor. By Lemma 4.3(2) for ϕ_W in place of ϕ, there exists an irreducible component P' of $\phi'^{-1}_W(P)$ with $\operatorname{center}_{W'} P' = \{x'\}$. By Claim 4.5 and Lemma 4.3(5) for ϕ_W in place of ϕ,

$$a(P', W', \Delta_{W'}, \mathbf{M}')^\ell \operatorname{vol}(P') = a(P, W, \Delta_W, \mathbf{M})^\ell \operatorname{vol}(P).$$

Thus

$$\widehat{\operatorname{vol}}(W' \ni x', \Delta_{W'}, \mathbf{M}') \leq \widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M}).$$

Conversely, let P' be a prime divisor over W' with $\operatorname{center}_{W'} P' = \{x'\}$. By Lemma 4.3(3) for ϕ_W in place of ϕ, there exists a prime divisor P over W such that, for a projective birational morphism $q : W_1 \to W$ from a normal variety on which P is a divisor, P' is an irreducible component of $\phi'^{-1}_W(P)$. By Lemma 4.3(2) for ϕ_W in place of ϕ, $\operatorname{center}_W P = \{x\}$, and by Claim 4.5 and Lemma 4.3(5) for ϕ_W in place of ϕ,

$$a(P', W', \Delta_{W'}, \mathbf{M}')^\ell \operatorname{vol}(P') = a(P, W, \Delta_W, \mathbf{M})^\ell \operatorname{vol}(P).$$

Thus

$$\widehat{\operatorname{vol}}(W' \ni x', \Delta_{W'}, \mathbf{M}') \geq \widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M}).$$

Since $K_{W'}+\Delta_{W'}+\mathbf{M}'_{W'}=(K_{X'}+B')|_{W'}$ is the pullback of $K_W+\Delta_W+\mathbf{M}_W$ and, by Claim 4.5 and Lemma 4.3(3) for ϕ_W in place of ϕ, the generalized log discrepancy of $(W',\Delta_{W'},\mathbf{M}')$ at any prime divisor over W' equals that of $(W,\Delta_W,\mathbf{M})$ at the corresponding prime divisor over W, $\Delta_{W'}$ and $\mathbf{M}'$ are the pullbacks of Δ_W and $\mathbf{M}$. Thus (2) follows. □

Proposition 4.6. *Let $(X,B,\mathbf{M})$ be a generalized pair of dimension $d\geq 1$ such that B is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is a $\mathbb{Q}$-$\boldsymbol{b}$-divisor, and let $x\in X$ be a closed point such that $(X,B,\mathbf{M})$ is generalized klt near x. Then for any $\eta\in(0,1)$, possibly shrinking X to a neighborhood of x, there exists a $\mathbb{Q}$-divisor $\Delta_\eta\geq B$ on X such that (X,Δ_η) is klt and*

$$(1-\eta)a(v,X,B,\mathbf{M})\leq a(v,X,\Delta_\eta)\leq a(v,X,B,\mathbf{M})$$

for any valuation v on X with $\operatorname{center}_X v=x$. In particular, $0<\widehat{\operatorname{vol}}(X\ni x,B,\mathbf{M})<+\infty$.

Proof. Fix $\eta\in(0,1)$. Possibly shrinking X to an affine neighborhood of x, we may assume that $(X,B,\mathbf{M})$ is generalized klt. By [Kol13, 1.12.1], there exist a log resolution $h:Y\to X$ of (X,B) such that $\mathbf{M}$ descends to Y, and an h-exceptional divisor $E\geq 0$ on Y such that $-E$ is ample over X. Let $F_1,\dots,F_N$ be the distinct prime divisors on Y whose union is $\operatorname{Exc}(h)\cup\operatorname{Supp}h_*^{-1}B$. Then $\sum_{i=1}^N F_i$ is snc and $\operatorname{Supp}E\subset\operatorname{Supp}\sum_{i=1}^N F_i$. Write

$$K_Y+B_Y+\mathbf{M}_Y=h^*(K_X+B+\mathbf{M}_X),\quad B_Y=\sum_{i=1}^N b_iF_i,\quad\text{and}\quad E=\sum_{i=1}^N e_iF_i.$$

Then $b_i\in\mathbb{Q}$, $1-b_i=a(F_i,X,B,\mathbf{M})>0$, and $e_i\geq 0$ for any i. Moreover, $\mathbf{M}_Y$ is a $\mathbb{Q}$-divisor which is nef over X.

Let $\delta>0$ be a rational number such that $\delta e_i\leq\frac{\eta}{2}(1-b_i)$ for any i. Then $L:=\mathbf{M}_Y-\delta E$ is a $\mathbb{Q}$-divisor which is ample over X. Since X is affine, there exists a positive integer $m\geq\frac{1}{\eta}$ such that mL is a divisor and $\mathcal{O}_Y(mL)$ is globally generated. By Bertini's theorem, a general member H_1 of $|mL|$ is reduced, has no common component with $\sum_{i=1}^N F_i$, and

$$R:=\sum_{i=1}^N F_i+H_1$$

is a reduced snc divisor. Let $H:=\frac{1}{m}H_1$ and $\Delta_\eta:=B+h_*H$. Since $H_1\in|mL|$, there exists a rational function φ on Y such that $H=\mathbf{M}_Y-\delta E+\frac{1}{m}\operatorname{div}(\varphi)$. Since E is h-exceptional and $h_*\mathbf{M}_Y=\mathbf{M}_X$, we have $h_*H=\mathbf{M}_X+\frac{1}{m}\operatorname{div}(\varphi)$. Thus Δ_η is a $\mathbb{Q}$-divisor, $\Delta_\eta\geq B$ as $H\geq 0$, the $\mathbb{Q}$-divisor

$$K_X+\Delta_\eta=K_X+B+\mathbf{M}_X+\frac{1}{m}\operatorname{div}(\varphi)$$

is $\mathbb{Q}$-Cartier, and

$$h^*(K_X+\Delta_\eta)=K_Y+B_Y+\mathbf{M}_Y+\frac{1}{m}\operatorname{div}(\varphi)=K_Y+B_Y+\delta E+H.$$

Moreover, h is a log resolution of (X,Δ_η), as $h_*^{-1}\Delta_\eta=h_*^{-1}B+H$ and

$$\operatorname{Exc}(h)\cup\operatorname{Supp}h_*^{-1}\Delta_\eta=\operatorname{Supp}R.$$

Let v be a valuation on X with a center on X. Then v has a center on Y, and we let $v(H_1)$ be the value of a local equation of H_1 at the center of v on Y. By Lemma 2.4,

$$a(v,X,B,\mathbf{M})=a(v,Y,B_Y)=a(v,Y,R)+\sum_{i=1}^N(1-b_i)v(F_i)+v(H_1)$$

and

$$a(v,X,\Delta_\eta)=a(v,Y,B_Y+\delta E+H)=a(v,Y,R)+\sum_{i=1}^N(1-b_i-\delta e_i)v(F_i)+\Big(1-\frac{1}{m}\Big)v(H_1).$$

Since $a(v,Y,R)\geq 0$, $v(F_i)\geq 0$, $v(H_1)\geq 0$, and

$$(1-\eta)(1-b_i)\leq 1-b_i-\delta e_i\leq 1-b_i \quad \text{and} \quad 1-\eta\leq 1-\frac{1}{m}\leq 1,$$

comparing the expressions for $a(v,X,B,\mathbf{M})$ and $a(v,X,\Delta_\eta)$ term by term, we have

$$(1-\eta)a(v,X,B,\mathbf{M})\leq a(v,X,\Delta_\eta)\leq a(v,X,B,\mathbf{M}), \tag{4.5}$$

where all three terms are $+\infty$ exactly when $a(v,Y,R)=+\infty$. Since $(X,B,\mathbf{M})$ is generalized klt, (4.5) implies that $a(P,X,\Delta_\eta)\geq(1-\eta)a(P,X,B,\mathbf{M})>0$ for any prime divisor P over X. Since $\Delta_\eta\geq 0$, (X,Δ_η) is klt.

By the right inequality of (4.5), $\widehat{\mathrm{vol}}(X\ni x,B,\mathbf{M})\geq\widehat{\mathrm{vol}}(X\ni x,\Delta_\eta)$, which is positive by [LLX20, Theorem 2.5] as $(X\ni x,\Delta_\eta)$ is klt. Let P be a prime divisor over X with $\mathrm{center}_X P=\{x\}$. Since $\mathrm{mult}_P\,\mathrm{div}(f)\geq 1$ for any non-zero $f\in\mathfrak{m}_x$, v_P satisfies $\mathfrak{m}_x^{\lceil\lambda\rceil}\subset\mathfrak{a}_\lambda(v_P)$ for any real number $\lambda>0$, so $\mathrm{vol}(P)\leq e(\mathfrak{m}_x)<+\infty$. Thus $\widehat{\mathrm{vol}}(X\ni x,B,\mathbf{M})\leq a(P,X,B,\mathbf{M})^d\,\mathrm{vol}(P)<+\infty$. □

Proof of Proposition 4.1. The case $\dim X=0$ is trivial. Let $d:=\dim X\geq 1$. Note that $(X',B',f^*\mathbf{M})$ is a generalized pair, as $B'\geq 0$ and $(f^*\mathbf{M})_{Y'}$ is nef over X'.

Let F' be a prime divisor over X'. There exist a prime divisor F over X and an integer $r_{F'}\geq 1$ such that, for a log resolution $g:Z\to X$ of (X,B) which factors through Y and on which F is a divisor, the normalization Z' of the irreducible component of $Z\times_X X'$ dominating X', and the induced morphisms $f_Z:Z'\to Z$ and $g':Z'\to X'$, F' is a prime divisor on Z' with $f_Z(F')=F$ and $\mathrm{mult}_{F'}f_Z^*F=r_{F'}$. Then $\mathrm{mult}_{F'}R(f_Z)=r_{F'}-1$, where $K_{Z'}=f_Z^*K_Z+R(f_Z)$. Therefore, for any $\mathbb{Q}$-divisor D on Z,

$$1-\mathrm{mult}_{F'}\left(f_Z^*D-R(f_Z)\right)=r_{F'}(1-\mathrm{mult}_F D). \tag{4.6}$$

Since $f^*\mathbf{M}$ descends to Y' and Z' factors through Y', we have $(f^*\mathbf{M})_{Z'}=f_Z^*\mathbf{M}_Z$. Let $K_Z+B_Z+\mathbf{M}_Z:=g^*(K_X+B+\mathbf{M}_X)$. Then

$$g'^*(K_{X'}+B'+(f^*\mathbf{M})_{X'})=f_Z^*(K_Z+B_Z+\mathbf{M}_Z)=K_{Z'}+\left(f_Z^*B_Z-R(f_Z)\right)+(f^*\mathbf{M})_{Z'}.$$

Since $f^*\mathbf{M}$ descends to Z', (4.6) for $D=B_Z$ implies that

$$a(F',X',B',f^*\mathbf{M})=1-\mathrm{mult}_{F'}\left(f_Z^*B_Z-R(f_Z)\right)=r_{F'}a(F,X,B,\mathbf{M}). \tag{4.7}$$

Possibly shrinking X to a neighborhood of x and replacing X' with $f^{-1}(X)$, we may assume that $(X,B,\mathbf{M})$ is generalized klt. By (4.7), $(X',B',f^*\mathbf{M})$ is generalized klt.

Fix $\eta\in(0,1)$. By Proposition 4.6, possibly shrinking X to a neighborhood of x and replacing X' with $f^{-1}(X)$, there exists a $\mathbb{Q}$-divisor $\Delta_\eta\geq B$ on X such that (X,Δ_η) is klt and

$$(1-\eta)a(v,X,B,\mathbf{M})\leq a(v,X,\Delta_\eta)\leq a(v,X,B,\mathbf{M}) \tag{4.8}$$

for any valuation v on X with $\mathrm{center}_X v=x$. For a $\mathbb{Q}$-divisor G on X, let f^*G be the $\mathbb{Q}$-divisor on X' defined at the generic points of prime divisors. Let

$$\Delta'_\eta:=B'+f^*(\Delta_\eta-B).$$

Since $B'\geq 0$ and $\Delta_\eta\geq B$, we have $\Delta'_\eta\geq 0$. Let $X^\circ\subset X$ be the largest open subset over which $Y\to X$ is an isomorphism. Then $\dim(X\backslash X^\circ)\leq d-2$, and $Y'\to X'$ is an isomorphism over $f^{-1}(X^\circ)$ as X' is normal. Thus $(f^*\mathbf{M})_{X'}=f^*\mathbf{M}_X$. Since $\Delta_\eta-B-\mathbf{M}_X=(K_X+\Delta_\eta)-(K_X+B+\mathbf{M}_X)$ is $\mathbb{Q}$-Cartier,

$$\begin{aligned}f^*(K_X+\Delta_\eta)&=f^*(K_X+B+\mathbf{M}_X)+f^*(\Delta_\eta-B-\mathbf{M}_X)\\&=K_{X'}+B'+(f^*\mathbf{M})_{X'}+f^*(\Delta_\eta-B)-f^*\mathbf{M}_X=K_{X'}+\Delta'_\eta.\end{aligned}$$

Let $K_Z+\Delta_{\eta,Z}:=g^*(K_X+\Delta_\eta)$. Then $g'^*(K_{X'}+\Delta'_\eta)=f_Z^*(K_Z+\Delta_{\eta,Z})$, so (4.6) for $D=\Delta_{\eta,Z}$ implies that

$$a(F',X',\Delta'_\eta)=r_{F'}a(F,X,\Delta_\eta) \tag{4.9}$$

for any prime divisor F' over X', with F and $r_{F'}$ as in (4.7). In particular, (X', Δ'_η) is klt. By [XZ21, Theorem 1.3],

$$\widehat{\mathrm{vol}}(X' \ni x', \Delta'_\eta) = r \cdot \widehat{\mathrm{vol}}(X \ni x, \Delta_\eta). \tag{4.10}$$

Let F' be a prime divisor over X' with $\mathrm{center}_{X'} F' = \{x'\}$. Then

$$\mathrm{center}_X F = g(F) = f(g'(F')) = \{x\},$$

so (4.8) holds for $v = v_F$. Multiplying it by $r_{F'}$ and applying (4.7) and (4.9), we obtain

$$(1-\eta)a(F', X', B', f^*\mathbf{M}) \le a(F', X', \Delta'_\eta) \le a(F', X', B', f^*\mathbf{M}).$$

By Lemma 4.2 for $(X', B', f^*\mathbf{M})$ and for (X', Δ'_η) (with $\mathbf{M} = 0$),

$$(1-\eta)^d \widehat{\mathrm{vol}}(X' \ni x', B', f^*\mathbf{M}) \le \widehat{\mathrm{vol}}(X' \ni x', \Delta'_\eta) \le \widehat{\mathrm{vol}}(X' \ni x', B', f^*\mathbf{M}).$$

By (4.8),

$$(1-\eta)^d \widehat{\mathrm{vol}}(X \ni x, B, \mathbf{M}) \le \widehat{\mathrm{vol}}(X \ni x, \Delta_\eta) \le \widehat{\mathrm{vol}}(X \ni x, B, \mathbf{M}).$$

The last two estimates and (4.10) imply that

$$(1-\eta)^d \widehat{\mathrm{vol}}(X' \ni x', B', f^*\mathbf{M}) \le r \cdot \widehat{\mathrm{vol}}(X \ni x, B, \mathbf{M}) \le (1-\eta)^{-d} \widehat{\mathrm{vol}}(X' \ni x', B', f^*\mathbf{M}).$$

Letting $\eta \to 0$, we obtain

$$\widehat{\mathrm{vol}}(X' \ni x', B', f^*\mathbf{M}) = r \cdot \widehat{\mathrm{vol}}(X \ni x, B, \mathbf{M}).$$

□

5. The upper bound

Theorem 5.1. *Notation and conditions as in Set-up 2.8, and assume that $\ell \ge 1$. Then*

$$\limsup_{\epsilon \to 0^+} \frac{\widehat{\mathrm{vol}}(X \ni x, B_\epsilon)}{n^n \epsilon^k} \le \frac{\widehat{\mathrm{vol}}^B(X \ni w, \Delta) \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{k^k \ell^\ell}.$$

Theorem 5.1 is proved at the end of Subsection 5.3.

5.1. Extractions of lc places.

Lemma 5.2. *Notation and conditions as in Set-up 2.8, and let E be an lc place of (X, B) with $\mathrm{center}_X E = W$. Then, possibly shrinking X to a neighborhood of x, there exists a projective birational morphism $f : Y \to X$ from a normal variety Y such that*

(1) E is a $\mathbb{Q}$-Cartier prime divisor on Y, $-E$ is ample over X, f is an isomorphism over $X \setminus W$, and $f^{-1}(W) = E$ as sets;
(2) E is the only f-exceptional prime divisor if $k \ge 2$, and f is small if $k = 1$;
(3) $R^i f_* \mathcal{O}_Y(-mE) = 0$ for all integers $i \ge 1$ and $m \ge 0$.

Proof. Possibly shrinking X to a neighborhood of x, we may assume that the conclusion of Construction-Proposition 3.1 holds on X, and that $ND = \mathrm{div}\, g$ for some $N \in \mathbb{N}^+$ and some regular function g on X. We fix an index j and let $a_j := a(E, X, \Delta_j)$. By Construction-Proposition 3.1(1), $K_X + \Delta_j$ is $\mathbb{Q}$-Cartier, (X, Δ_j) is klt, $c_j \in \mathbb{Q}$, (X, B_j) is lc, and E is an lc place of (X, B_j). Thus $a_j > 0$ and

$$a_j = a(E, X, B_j) + c_j \,\mathrm{mult}_E D = c_j \,\mathrm{mult}_E D. \tag{5.1}$$

Step 1: construction of f. Assume that $k \ge 2$. Since $(X, \Delta_j + \frac{c_j}{N} \mathrm{div}\, g) = (X, B_j)$ is lc and $a(E, X, B_j) = 0$, E computes $\mathrm{lct}(X, \Delta_j; \mathrm{div}\, g) = \frac{c_j}{N}$. By [Xu21, Lemma 2.11], there exists a projective birational morphism $f : Y \to X$ from a normal variety Y such that the exceptional locus of f is E and $-E$ is ample over X. In particular, E is a $\mathbb{Q}$-Cartier prime divisor on Y, and E is the only f-exceptional prime divisor.

Assume that $k = 1$. Then W is a prime divisor on X. Since $\mathcal{O}_{X,w}$ is a discrete valuation ring, E is the prime divisor W on X. Thus $\operatorname{mult}_W B_j = 1$, and by (5.1),

$$c_j \operatorname{mult}_W D = a_j = 1 - \operatorname{mult}_W \Delta_j > 0.$$

Let $f_0 : Y_0 \to X$ be a small $\mathbb{Q}$-factorialization of (X, Δ_j) [BCHM10, Corollary 1.4.3], $E_0 := (f_0)_*^{-1}W$, and $D_0 := (f_0)_*^{-1}(D - (\operatorname{mult}_W D)W) \geq 0$. Since f_0 is small, for any rational number t,

$$K_{Y_0} + (f_0)_*^{-1}\Delta_j + tD_0 = f_0^*(K_X + \Delta_j + tD) - t(\operatorname{mult}_W D)E_0 \sim_{\mathbb{Q},X} -t(\operatorname{mult}_W D)E_0.$$

Since Y_0 is $\mathbb{Q}$-factorial and $(Y_0, (f_0)_*^{-1}\Delta_j)$ is klt, we may fix a rational number $t > 0$ such that $(Y_0, (f_0)_*^{-1}\Delta_j + tD_0)$ is klt. Since f_0 is birational, $K_{Y_0} + (f_0)_*^{-1}\Delta_j + tD_0$ is big over X. By [BCHM10, Theorem 1.2], $(Y_0, (f_0)_*^{-1}\Delta_j + tD_0)$ has an lc model $f : Y \to X$ over X. By the definition of the lc model, the induced birational map $Y_0 \dashrightarrow Y$ does not extract any divisor. Since f_0 is small, f is small. Then $E = f_*^{-1}W$ is a prime divisor on Y, and since f is small,

$$K_Y + f_*^{-1}\Delta_j + tf_*^{-1}(D - (\operatorname{mult}_W D)W) = f^*(K_X + \Delta_j + tD) - t(\operatorname{mult}_W D)E.$$

By the definition of the lc model, the left side is $\mathbb{Q}$-Cartier and ample over X. Since $t \operatorname{mult}_W D > 0$, E is $\mathbb{Q}$-Cartier and $-E$ is ample over X.

Step 2: proof of (1) and (2). (2) holds by **Step 1**. Since $-E$ is ample over X, any curve C contracted by f satisfies $E \cdot C < 0$, so $C \subset E$. As $f(E) = W$ and the fibers of f are connected, f is an isomorphism over $X \backslash W$ and $f^{-1}(W) = E$ as sets. This proves (1).

Step 3: proof of (3). Fix a rational number ζ such that $0 < \zeta < \min\{1, a_j\}$, and let

$$D_Y := f^*D - (\operatorname{mult}_E D)E,$$

which is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Y with $D_Y \geq 0$:

$$D_Y = f_*^{-1}D \text{ if } k \geq 2, \qquad D_Y = f_*^{-1}(D - (\operatorname{mult}_W D)W) \text{ if } k = 1.$$

Let $K_Y + \Delta_{j,Y} := f^*(K_X + \Delta_j)$. If $k \geq 2$, then E is the only f-exceptional prime divisor and $\operatorname{mult}_E f_*^{-1}\Delta_j = 0$. If $k = 1$, then f is small and $\operatorname{mult}_E f_*^{-1}\Delta_j = \operatorname{mult}_W \Delta_j = 1 - a_j$. In both cases

$$\Delta_{j,Y} = f_*^{-1}\Delta_j - (\operatorname{mult}_E f_*^{-1}\Delta_j)E + (1 - a_j)E.$$

Since $f^*(c_jD) = c_jD_Y + a_jE$ by (5.1), we have

$$\begin{aligned} K_Y + \Delta_{j,Y} + (a_j - \zeta)E &= f^*(K_X + \Delta_j) + (a_j - \zeta)E \\ &= f^*(K_X + B_j) - \zeta E - c_jD_Y. \end{aligned} \tag{5.2}$$

By the first line of (5.2), the left side of (5.2) is $\mathbb{Q}$-Cartier. By the second line, for any prime divisor F over Y,

$$a\big(F, Y, \Delta_{j,Y} + (a_j - \zeta)E\big) = a(F, X, B_j) + \zeta \operatorname{mult}_F E + c_j \operatorname{mult}_F D_Y.$$

The three terms on the right are non-negative as (X, B_j) is lc. If $\operatorname{center}_Y F \subset E$, then $\operatorname{mult}_F E > 0$. Otherwise $\operatorname{center}_Y F$ intersects $Y \backslash E = f^{-1}(X \backslash W)$, over which f is an isomorphism and the boundary on the left is $f_*^{-1}\Delta_j$. Thus the left side is $a(F, X, \Delta_j) > 0$. Therefore the pair $(Y, \Delta_{j,Y} + (a_j - \zeta)E)$ is klt. For any integer $m \geq 0$, by (5.2),

$$-mE - \big(K_Y + \Delta_{j,Y} + (a_j - \zeta)E\big) \sim_{\mathbb{Q},X} -(m + a_j - \zeta)E,$$

which is ample over X as $m + a_j - \zeta > 0$. Since $-mE$ is a $\mathbb{Q}$-Cartier divisor, [Fuj17, Corollary 5.7.7] implies (3). The lemma follows. □

5.2. **Hilbert functions of graded pieces.**

Lemma 5.3 (cf. [Bha57, Theorem 2]). *Let $R = \bigoplus_{i,j\geq 0} R_{(i,j)}$ be a bigraded $\mathbb{C}$-algebra such that $\dim_{\mathbb{C}} R_{(0,0)} < +\infty$ and R is generated over $R_{(0,0)}$ by finitely many elements of bidegrees $(1,0)$ and $(0,1)$, and let $M = \bigoplus_{i,j\geq 0} M_{(i,j)}$ be a finitely generated bigraded R-module. Then there exist integers $i_0, j_0 \geq 0$ and a polynomial $P \in \mathbb{Q}[u_1, u_2]$ such that $\dim_{\mathbb{C}} M_{(i,j)} = P(i,j)$ for all integers $i \geq i_0$ and $j \geq j_0$.*

Proof. Fix elements $g_1, \ldots, g_{N_1} \in R_{(1,0)}$ and $g'_1, \ldots, g'_{N_2} \in R_{(0,1)}$ which generate R over $R_{(0,0)}$, and let

$$S := \mathbb{C}[U_1, \ldots, U_{N_1}, V_1, \ldots, V_{N_2}]$$

be the polynomial ring, acting on M via $U_{i'} \to g_{i'}$ and $V_{j'} \to g'_{j'}$, bigraded by

$$\deg U_1 = \cdots = \deg U_{N_1} = (1,0), \qquad \deg V_1 = \cdots = \deg V_{N_2} = (0,1).$$

Since $\dim_{\mathbb{C}} R_{(0,0)} < +\infty$, M is generated as an S-module by the products of a $\mathbb{C}$-basis of $R_{(0,0)}$ with finitely many homogeneous generators of M. Thus M is a finitely generated bigraded S-module, and $\dim_{\mathbb{C}} M_{(i,j)} < +\infty$ for all i, j. We prove the conclusion for any finitely generated bigraded S-module M by induction on N_1.

If $N_1 = 0$, then the first degree of any homogeneous element of M is at most the largest first degree d of the generators of M, so $M_{(i,j)} = 0$ for $i > d$, and the lemma holds with $i_0 := d + 1$, $j_0 := 0$, and $P = 0$.

Assume that $N_1 \geq 1$. Let $M'' := \{\alpha \in M \mid U_{N_1}\alpha = 0\}$ and $M' := M/U_{N_1}M$. For all integers $i \geq 1$ and $j \geq 0$, the sequence

$$0 \to M''_{(i-1,j)} \to M_{(i-1,j)} \xrightarrow{\cdot U_{N_1}} M_{(i,j)} \to M'_{(i,j)} \to 0$$

is exact. Since S is Noetherian, M'' and M' are finitely generated bigraded modules over $\mathbb{C}[U_1, \ldots, U_{N_1-1}, V_1, \ldots, V_{N_2}]$, as U_{N_1} annihilates them. By induction, there exist integers $i_1, j_1 \geq 0$ and $Q \in \mathbb{Q}[u_1, u_2]$ such that

$$\dim_{\mathbb{C}} M_{(i,j)} - \dim_{\mathbb{C}} M_{(i-1,j)} = \dim_{\mathbb{C}} M'_{(i,j)} - \dim_{\mathbb{C}} M''_{(i-1,j)} = Q(i,j)$$

for all integers $i > i_1$ and $j \geq j_1$. Moreover, $\bigoplus_{j\geq 0} M_{(i_1,j)}$ is a finitely generated graded $\mathbb{C}[V_1, \ldots, V_{N_2}]$-module: it is generated by the elements $\mu\alpha$, where α runs through the generators of M and μ through the monomials in $U_1, \ldots, U_{N_1}$ whose first degree is i_1 minus the first degree of α. By the Hilbert polynomial theorem if $N_2 \geq 1$, and since $\sum_{j\geq 0} \dim_{\mathbb{C}} M_{(i_1,j)} < +\infty$ if $N_2 = 0$, there exist an integer $j_2 \geq 0$ and $Q_1 \in \mathbb{Q}[u_2]$ such that $\dim_{\mathbb{C}} M_{(i_1,j)} = Q_1(j)$ for all $j \geq j_2$. Thus, for all integers $i \geq i_1$ and $j \geq \max\{j_1, j_2\}$,

$$\dim_{\mathbb{C}} M_{(i,j)} = Q_1(j) + \sum_{i'=i_1+1}^{i} Q(i', j).$$

For any $e \in \mathbb{N}$, $\sum_{i'=i_1+1}^{i} i'^e$ is a polynomial in i with rational coefficients, so the right side is $P(i,j)$ for some $P \in \mathbb{Q}[u_1, u_2]$. Thus the lemma holds with $i_0 := i_1$ and $j_0 := \max\{j_1, j_2\}$. □

Definition-Proposition 5.4. Notation and conditions as in Set-up 2.8, and let E be an lc place of (X, B) with $\operatorname{center}_X E = W$. For any integer $j \geq 0$ let

$$I_j := \{f \in \mathcal{O}_{X,x} \mid \operatorname{mult}_E \operatorname{div}(f) \geq j\} \quad \text{and} \quad \operatorname{gr}_j := I_j/I_{j+1},$$

and let $J \subset \mathcal{O}_{W,x}$ be an $\mathfrak{m}_{W,x}$-primary ideal. Then $I_1 = \ker(\mathcal{O}_{X,x} \to \mathcal{O}_{W,x})$, so that every gr_j is an $\mathcal{O}_{W,x}$-module. Moreover, for any real number $\beta > 0$ the limit

$$\phi_J(\beta) := \lim_{m\to+\infty} \frac{n!}{m^n} \sum_{j=0}^{m} \operatorname{length}\left(\operatorname{gr}_j / J^{\lceil (m-j)/\beta \rceil} \operatorname{gr}_j\right)$$

exists, and

$$\lim_{\beta\to 0^+} \beta^{\ell}\phi_J(\beta) = \mathrm{vol}(E)\cdot e(J).$$

Proof. **Step 1: gr_j as sections on E.** Since $\mathrm{center}_X E = W$, $I_1 = \ker(\mathcal{O}_{X,x} \to \mathcal{O}_{W,x})$, so every gr_j is an $\mathcal{O}_{W,x}$-module. Possibly shrinking X to an affine neighborhood of x, by Lemma 5.2 we may assume that there exists a projective birational morphism $h : Y \to X$ from a normal variety Y satisfying Lemma 5.2(1)–(3) with h in place of f. Then W is affine, and h restricts to a projective surjective morphism $E \to W$, so any $H^0(E,\cdot)$ is an $H^0(W,\mathcal{O}_W)$-module. For $j \geq 0$ let

$$\mathcal{F}_j := \mathcal{O}_Y(-jE)/\mathcal{O}_Y(-(j+1)E).$$

Since Y is normal, $\mathcal{O}_Y(-E)$ is the ideal sheaf of E, so $\mathcal{F}_j$ is a coherent $\mathcal{O}_E$-module. Since Y is normal and $h_*\mathcal{O}_Y = \mathcal{O}_X$,

$$H^0(Y,\mathcal{O}_Y(-jE)) = \{f \in H^0(X,\mathcal{O}_X) \mid \mathrm{mult}_E \,\mathrm{div}(f) \geq j\},$$

and the localization of $H^0(Y,\mathcal{O}_Y(-jE))$ at x is I_j. Since X is affine and $R^1h_*\mathcal{O}_Y(-(j+1)E) = 0$ by Lemma 5.2(3), the sequence

$$0 \to H^0(Y,\mathcal{O}_Y(-(j+1)E)) \to H^0(Y,\mathcal{O}_Y(-jE)) \to H^0(E,\mathcal{F}_j) \to 0$$

is exact. Localizing at x, we obtain isomorphisms

$$\mathrm{gr}_j \cong H^0(E,\mathcal{F}_j) \otimes_{H^0(W,\mathcal{O}_W)} \mathcal{O}_{W,x}, \quad j \geq 0, \tag{5.3}$$

which respect the multiplications

$$\mathrm{gr}_j \times \mathrm{gr}_{j'} \to \mathrm{gr}_{j+j'} \quad \text{and} \quad \mathcal{F}_j \otimes \mathcal{F}_{j'} \to \mathcal{F}_{j+j'}.$$

Thus $\mathrm{gr}_0 = \mathcal{O}_{X,x}/I_1 = \mathcal{O}_{W,x}$.

Step 2: the Veronese decomposition. Since $-E$ is $\mathbb{Q}$-Cartier and ample over X, by Serre's theorems there exists an integer $r \geq 1$ such that rE is Cartier, $\mathcal{H} := \mathcal{O}_Y(-rE)|_E$ is ample over W, and the graded ring

$$\mathcal{S}_W := \bigoplus_{q\geq 0} H^0(E,\mathcal{H}^q)$$

is generated by $H^0(E,\mathcal{H})$ over $H^0(E,\mathcal{O}_E)$. Moreover, $H^0(E,\mathcal{H}^q)$ is a finitely generated $H^0(W,\mathcal{O}_W)$-module for any $q \geq 0$. Since rE is Cartier, $\mathcal{O}_Y(-(j+r)E) = \mathcal{O}_Y(-jE)\otimes\mathcal{O}_Y(-rE)$ for any $j \geq 0$, so

$$\mathcal{F}_{rq+i} \cong \mathcal{F}_i \otimes_{\mathcal{O}_E} \mathcal{H}^q, \quad 0 \leq i \leq r-1,\ q \geq 0.$$

For $0 \leq i \leq r-1$, the graded $\mathcal{S}_W$-module $M_{i,W} := \bigoplus_{q\geq 0} H^0(E,\mathcal{F}_i\otimes\mathcal{H}^q)$ is finitely generated by Serre's theorems. By (5.3), the localizations at x of $\mathcal{S}_W$ and $M_{i,W}$ are

$$\mathcal{S} := \bigoplus_{q\geq 0} \mathrm{gr}_{rq} \quad \text{and} \quad M_i := \bigoplus_{q\geq 0} M_{i,q}, \quad M_{i,q} := \mathrm{gr}_{rq+i}\,.$$

Thus $\mathcal{S}_0 = \mathrm{gr}_0 = \mathcal{O}_{W,x}$, $\mathcal{S}$ is generated by $\mathcal{S}_1$ over $\mathcal{S}_0$, $\mathcal{S}_1$ is a finitely generated $\mathcal{O}_{W,x}$-module, and M_i is a finitely generated graded $\mathcal{S}$-module. In particular, any $M_{i,q}$ is a finitely generated $\mathcal{O}_{W,x}$-module.

Step 3: two-variable Hilbert polynomials. Let

$$G := \bigoplus_{q,b\geq 0} G_{(q,b)}, \quad G_{(q,b)} := J^b\mathcal{S}_q/J^{b+1}\mathcal{S}_q,$$

and for $0 \leq i \leq r-1$ let

$$G_i := \bigoplus_{q,b\geq 0} G_{i,(q,b)}, \quad G_{i,(q,b)} := J^bM_{i,q}/J^{b+1}M_{i,q}.$$

Then G is a bigraded $\mathbb{C}$-algebra with $G_{(0,0)} = \mathcal{O}_{W,x}/J$, so $\dim_{\mathbb{C}} G_{(0,0)} < +\infty$. By **Step 2**, G is generated over $G_{(0,0)}$ by the images of finitely many generators of the $\mathcal{O}_{W,x}$-module $\mathcal{S}_1$, of

bidegree $(1,0)$, and of finitely many generators of J, of bidegree $(0,1)$. Similarly, G_i is a finitely generated bigraded G-module, generated by the images of the homogeneous generators of M_i. For integers $q, t \geq 0$ let

$$\psi_i(q,t) := \operatorname{length}(M_{i,q}/J^t M_{i,q}) = \sum_{b=0}^{t-1} \dim_{\mathbb{C}} G_{i,(q,b)}.$$

Let z be a variable of bidegree $(0,1)$. The bigraded $\mathbb{C}$-algebra $G \otimes_{\mathbb{C}} \mathbb{C}[z]$ satisfies the conditions of Lemma 5.3, $G_i \otimes_{\mathbb{C}} \mathbb{C}[z]$ is a finitely generated bigraded module over it, and

$$\dim_{\mathbb{C}}(G_i \otimes_{\mathbb{C}} \mathbb{C}[z])_{(q,b)} = \sum_{b'=0}^{b} \dim_{\mathbb{C}} G_{i,(q,b')} = \psi_i(q, b+1).$$

By Lemma 5.3 for the modules $G_i \otimes_{\mathbb{C}} \mathbb{C}[z]$ and the substitution $b = t-1$, there exist integers $q_0, t_0 \geq 1$ and polynomials $P_0, \ldots, P_{r-1} \in \mathbb{Q}[u_1, u_2]$ such that

$$\psi_i(q,t) = P_i(q,t) \quad \text{for all integers } 0 \leq i \leq r-1,\ q \geq q_0,\ \text{and } t \geq t_0. \tag{5.4}$$

We show that

$$\deg P_i \leq n-1, \quad 0 \leq i \leq r-1. \tag{5.5}$$

First, we show that $\dim G \leq n$. Let J_W be the inverse image of J in $H^0(W, \mathcal{O}_W)$, y a variable, and $\mathcal{R}_W \subset \mathcal{S}_W[y, y^{-1}]$ the subring generated by $\mathcal{S}_W$, y, and $J_W y^{-1}$, that is,

$$\mathcal{R}_W = \bigoplus_{b \in \mathbb{Z}} J_W^b \mathcal{S}_W \cdot y^{-b}, \quad J_W^b := H^0(W, \mathcal{O}_W) \text{ for } b \leq 0.$$

Since E is integral, $\dim E = n-1$, and $\mathcal{H}$ is ample over W, $\mathcal{S}_W$ is a finitely generated $\mathbb{C}$-domain with $\dim \mathcal{S}_W = n$. Since $\mathcal{S}_W[y] \subset \mathcal{R}_W \subset \mathcal{S}_W[y, y^{-1}]$, $\mathcal{R}_W$ is a finitely generated $\mathbb{C}$-domain with $\dim \mathcal{R}_W = n+1$, so $\dim \mathcal{R}_W/y\mathcal{R}_W = n$. Moreover,

$$\mathcal{R}_W/y\mathcal{R}_W \cong \bigoplus_{q,b \geq 0} J_W^b H^0(E, \mathcal{H}^q)/J_W^{b+1} H^0(E, \mathcal{H}^q).$$

Since J is $\mathfrak{m}_{W,x}$-primary, $H^0(W, \mathcal{O}_W)/J_W \cong \mathcal{O}_{W,x}/J$ is a local ring. The (q,b)-summand on the right is an $H^0(W, \mathcal{O}_W)/J_W$-module, hence equal to the localization of the (q,b)-summand at x, which is $G_{(q,b)}$. Thus $G \cong \mathcal{R}_W/y\mathcal{R}_W$ and $\dim G \leq n$.

Second, we show that for $N \gg 0$, $\sum_{q+b=N} \dim_{\mathbb{C}} G_{i,(q,b)}$ is a polynomial in N of degree at most $n-1$. Let S' be a polynomial ring over $\mathbb{C}$ whose variables map to the generators of G over $G_{(0,0)}$ of total degree 1 fixed above, and $\mathfrak{k}$ the kernel of $S' \to G$. Then G_i is a finitely generated graded S'-module for the total degree, annihilated by $\mathfrak{k}$. Since G is generated as an $S'/\mathfrak{k}$-module by a $\mathbb{C}$-basis of $G_{(0,0)}$, $\dim S'/\mathfrak{k} = \dim G \leq n$, and the claim follows from the Hilbert polynomial theorem.

Thus, for all integers $N \geq 1$,

$$\sum_{q+t=N} \psi_i(q,t) = \sum_{q+b \leq N-1} \dim_{\mathbb{C}} G_{i,(q,b)} = O(N^n),$$

where q, t, b run through $\mathbb{N}$. We may assume that $P_i \neq 0$. Let P_i^{top} be the homogeneous part of P_i of degree $\deg P_i$. For all positive integers a_1, a_2, by (5.4),

$$P_i^{\mathrm{top}}(a_1, a_2) = \lim_{\lambda \to +\infty, \lambda \in \mathbb{N}} \lambda^{-\deg P_i} P_i(\lambda a_1, \lambda a_2) \geq 0,$$

so $P_i^{\mathrm{top}} \geq 0$ on $\mathbb{R}^2_{\geq 0}$ by homogeneity and continuity. Since $P_i^{\mathrm{top}} \neq 0$, there exist real numbers $0 < \alpha_1 < \alpha_2 < 1$ and $\eta > 0$ such that $P_i^{\mathrm{top}}(\tau, 1-\tau) \geq \eta$ for any $\tau \in [\alpha_1, \alpha_2]$. Since

$$P_i(q, N-q) = N^{\deg P_i} P_i^{\mathrm{top}}(q/N, 1-q/N) + O(N^{\deg P_i - 1})$$

uniformly for $0 \le q \le N$, for $N \gg 0$ we have

$$\sum_{q+t=N} \psi_i(q,t) \ge \sum_{\alpha_1 N \le q \le \alpha_2 N} P_i(q, N-q) \ge \frac{\eta(\alpha_2-\alpha_1)}{4} N^{\deg P_i + 1}.$$

Thus $\deg P_i + 1 \le n$, which proves (5.5).

Step 4: the coefficient of u_2^ℓ and $\mathrm{vol}(E)$. For a finitely generated $\mathcal{O}_{W,x}$-module M, let

$$e(J, M) := \lim_{t \to +\infty} \frac{\ell! \cdot \mathrm{length}(M/J^t M)}{t^\ell}.$$

Since $\dim \mathcal{O}_{W,x} = \ell$, there exists a polynomial $Q_M \in \mathbb{Q}[u_2]$ of degree at most ℓ such that $\mathrm{length}(M/J^t M) = Q_M(t)$ for $t \gg 0$, so the limit exists, $e(J, M)$ is $\ell!$ times the coefficient of u_2^ℓ in Q_M, and $e(J, \mathcal{O}_{W,x}) = e(J)$ ([Mat89, Formula 14.1] and Definition 2.2(3)). Since $\mathcal{O}_{W,x}$ is a domain, by [Mat89, Theorem 14.8],

$$e(J, M) = \mathrm{rank}_{\mathcal{O}_{W,x}}(M) \cdot e(J).$$

Write $P_i = \sum_{a \ge 0} P_{i,a}(u_1) u_2^a$ with $P_{i,a} \in \mathbb{Q}[u_1]$. Fix an integer $q \ge q_0$. By (5.4), $P_i(q,t) = \mathrm{length}(M_{i,q}/J^t M_{i,q})$ for all integers $t \ge t_0$. Thus $P_{i,a}(q) = 0$ for $a > \ell$, and

$$P_{i,\ell}(q) = \frac{e(J, M_{i,q})}{\ell!} = \frac{e(J)}{\ell!} \mathrm{rank}_{\mathcal{O}_{W,x}} M_{i,q}.$$

As $q \ge q_0$ is arbitrary, $P_{i,a} = 0$ for $a > \ell$. By (5.5), $\deg P_{i,\ell} \le n - 1 - \ell = k - 1$. Since J is $\mathfrak{m}_{W,x}$-primary, $e(J) > 0$. Let ρ_i be $\ell!/e(J)$ times the coefficient of u_1^{k-1} in $P_{i,\ell}$. Then

$$\lim_{q \to +\infty} \frac{\mathrm{rank}_{\mathcal{O}_{W,x}} M_{i,q}}{q^{k-1}} = \rho_i, \quad 0 \le i \le r-1. \tag{5.6}$$

Since $\dim \mathcal{O}_{X,w} = k$, by Definition 2.2(2) with $\xi = w$,

$$\mathrm{vol}(E) = \lim_{m \to +\infty} \frac{k! \cdot \mathrm{length}(\mathcal{O}_{X,w}/\mathfrak{a}_m(v_E))}{m^k}.$$

The ring $\mathcal{O}_{X,w}$ is the localization of $\mathcal{O}_{X,x}$ at the prime ideal I_1, and $\mathfrak{a}_m(v_E) = I_m \mathcal{O}_{X,w}$: if $f = g/g'$ with $g \in \mathcal{O}_{X,x}$, $g' \in \mathcal{O}_{X,x} \backslash I_1$, and $f \in \mathfrak{a}_m(v_E)$, then $\mathrm{mult}_E \mathrm{div}(g') = 0$, so $g \in I_m$. Since $\mathcal{O}_{X,w}/I_1\mathcal{O}_{X,w}$ is the fraction field of $\mathcal{O}_{W,x}$, $I_j\mathcal{O}_{X,w}/I_{j+1}\mathcal{O}_{X,w}$ is a vector space over it of dimension $\mathrm{rank}_{\mathcal{O}_{W,x}} \mathrm{gr}_j$. Thus, by (5.6),

$$\begin{aligned}\mathrm{vol}(E) &= \lim_{m \to +\infty} \frac{k!}{m^k} \sum_{j=0}^{m-1} \mathrm{rank}_{\mathcal{O}_{W,x}} \mathrm{gr}_j = \lim_{m \to +\infty} \frac{k!}{m^k} \sum_{i=0}^{r-1} \sum_{0 \le q < (m-i)/r} \mathrm{rank}_{\mathcal{O}_{W,x}} M_{i,q} \\ &= \frac{(k-1)!}{r^k} \sum_{i=0}^{r-1} \rho_i.\end{aligned}$$

Step 5: the Riemann sum. For $a_1, a_2 \in \mathbb{N}$ let $\gamma_{i,a_1,a_2} \in \mathbb{Q}$ be the coefficient of $u_1^{a_1} u_2^{a_2}$ in P_i. By **Steps 3 and 4**, $\gamma_{i,a_1,a_2} = 0$ if $a_1 + a_2 \ge n$ or $a_2 > \ell$, and

$$\gamma_{i,k-1,\ell} = \frac{e(J)\rho_i}{\ell!}.$$

Fix $\beta > 0$. For $m \in \mathbb{N}$, $0 \le i \le r-1$, and $q \in \mathbb{N}$ such that $rq + i \le m$, let $t_{i,q} := \lceil (m - rq - i)/\beta \rceil$. Then

$$\sum_{j=0}^{m} \mathrm{length}\left(\mathrm{gr}_j / J^{\lceil (m-j)/\beta \rceil} \mathrm{gr}_j \right) = \sum_{i=0}^{r-1} \sum_{0 \le q \le (m-i)/r} \psi_i(q, t_{i,q}).$$

We divide the pairs (i, q) on the right into three classes.

(1) $q < q_0$. For fixed q, $\psi_i(q,t) = O(t^\ell)$ by **Step 4**, and $t_{i,q} \le m/\beta + 1$. Since $\ell \le n-1$, the pairs in (1) contribute $O(m^{n-1})$.

(2) $q \geq q_0$ and $t_{i,q} < t_0$. Then $m - rq - i < \beta t_0$, so $\#\{q \geq q_0 \mid t_{i,q} < t_0\} \leq \beta t_0/r + 1$ for each i. Since $\psi_i(q,t)$ is increasing in t, $\psi_i(q, t_{i,q}) \leq \psi_i(q, t_0) = P_i(q, t_0) = O(m^{n-1})$ by (5.5) and $q \leq m$. Thus the pairs in (2) contribute $O(m^{n-1})$.

(3) $q \geq q_0$ and $t_{i,q} \geq t_0$. Then $\psi_i(q, t_{i,q}) = P_i(q, t_{i,q})$. Since $|t_{i,q} - (m - rq)/\beta| \leq r/\beta + 1$ and $0 \leq q \leq m/r$, by (5.5),

$$P_i(q, t_{i,q}) = \sum_{a_1+a_2=n-1} \gamma_{i,a_1,a_2} q^{a_1} \Big(\frac{m - rq}{\beta}\Big)^{a_2} + O(m^{n-2}).$$

For each i, the pairs in (3) are all (i,q) with $0 \leq q \leq m/r$ except at most $q_0 + \beta t_0/r + 2$ values of q, at which the sum on the right is $O(m^{n-1})$.

Let $a_1, a_2 \in \mathbb{N}$ with $a_1 + a_2 = n - 1$. The function $\tau^{a_1}(m - r\tau)^{a_2}$ of $\tau \in [0, m/r]$ is bounded by m^{n-1}, increasing on $[0, \tau_0]$ and decreasing on $[\tau_0, m/r]$ for some $\tau_0 \in [0, m/r]$. Thus, by the substitution $\tau = m\tau'/r$ and

$$\int_0^1 \tau'^{a_1}(1 - \tau')^{a_2} d\tau' = \frac{a_1! a_2!}{n!},$$

we have

$$\begin{aligned} \sum_{0 \leq q \leq m/r} q^{a_1}(m - rq)^{a_2} &= \int_0^{m/r} \tau^{a_1}(m - r\tau)^{a_2} d\tau + O(m^{n-1}) \\ &= \frac{a_1! a_2!}{r^{a_1+1} n!} m^n + O(m^{n-1}). \end{aligned}$$

Therefore the limit $\phi_J(\beta)$ exists, and

$$\phi_J(\beta) = \sum_{i=0}^{r-1} \sum_{a_1+a_2=n-1} \gamma_{i,a_1,a_2} \frac{a_1! a_2!}{r^{a_1+1} \beta^{a_2}}.$$

Since $\gamma_{i,a_1,a_2} = 0$ for $a_2 > \ell$, by the value of $\gamma_{i,k-1,\ell}$ and **Step 4**,

$$\beta^\ell \phi_J(\beta) = \sum_{i=0}^{r-1} \frac{(k-1)! e(J) \rho_i}{r^k} + \sum_{i=0}^{r-1} \sum_{\substack{a_1+a_2=n-1 \\ a_2 < \ell}} \gamma_{i,a_1,a_2} \frac{a_1! a_2!}{r^{a_1+1}} \beta^{\ell - a_2},$$

and the first sum on the right is $\mathrm{vol}(E) \cdot e(J)$. The double sum tends to 0 as $\beta \to 0^+$, so

$$\lim_{\beta \to 0^+} \beta^\ell \phi_J(\beta) = \mathrm{vol}(E) \cdot e(J). \quad \square$$

5.3. **Valuations near an lc place.**

Lemma 5.5. *Notation and conditions as in Set-up 2.8, and assume that $\ell \geq 1$. Let E be an lc place of (X, B) with $\mathrm{center}_X E = W$. Let P be a prime divisor over W with $\mathrm{center}_W P = \{x\}$. Then there exist a real number $b \geq 0$ and valuations v_λ on X with $\mathrm{center}_X(v_\lambda) = x$, $\lambda \in \mathbb{R}_{>0}$, such that*

(1) $a(v_\lambda, X, B) = \lambda \cdot a(P, W, \Delta_W, \mathbf{M})$ and $cv_\lambda(D) = a(E, X, \Delta) + \lambda b$ for any $\lambda > 0$, and
(2) the valuations v_λ satisfy

$$\limsup_{\lambda \to 0^+} \lambda^\ell \,\mathrm{vol}(v_\lambda) \leq \mathrm{vol}(E) \cdot \mathrm{vol}(P).$$

Proof. Let

$$t := a(P, W, \Delta_W, \mathbf{M}).$$

Then $t > 0$, as $(W, \Delta_W, \mathbf{M})$ is generalized klt by Definition-Theorem 2.7(1). We divide the proof into several steps.

Step 1. In this step we construct a log resolution of (X, B) on which E is a prime divisor mapping to a variety carrying P.

Let $W' \to W$ be a projective birational morphism from a smooth variety W' on which P is a prime divisor. Let $h : Y \to X$ be a log resolution of (X, B) on which E is a prime divisor, and let S be the prime divisor E on Y. Let

$$K_Y + B_Y := h^*(K_X + B) \quad \text{and} \quad K_S + B_S := (K_Y + B_Y)|_S.$$

Since (X, B) is lc and $\operatorname{mult}_S B_Y = 1$, $B_S = (B_Y - S)|_S$ is a simple normal crossing divisor with coefficients ≤ 1.

Let $\mu : S' \to S$ be a projective birational morphism from a smooth variety S' such that the induced map $h_{S'} : S' \dashrightarrow W'$ is a morphism and

$$\operatorname{Exc}(\mu) \cup \mu_*^{-1} \operatorname{Supp} B_S \cup \operatorname{Supp} h_{S'}^* P$$

is a simple normal crossing divisor. Let $K_{S'} + B_{S'} := \mu^*(K_S + B_S)$. Then $\operatorname{Supp} B_{S'} \subset \operatorname{Exc}(\mu) \cup \mu_*^{-1} \operatorname{Supp} B_S$ and all coefficients of $B_{S'}$ are ≤ 1.

Since S is quasi-projective, μ is the blow-up of a coherent ideal sheaf $\mathfrak{a} \subset \mathcal{O}_S$. Let $\mathfrak{b} \subset \mathcal{O}_Y$ be the preimage of $\mathfrak{a}$ under $\mathcal{O}_Y \to \mathcal{O}_S$, and let $\rho_1 : Y_1 \to Y$ be the blow-up of $\mathfrak{b}$. Since $\mathfrak{b}\mathcal{O}_S = \mathfrak{a}$, $\rho_{1*}^{-1} S$ is the blow-up of $\mathfrak{a}$, that is, S'. Let $Y_2 \to Y_1$ be a projective birational morphism such that the composition $h_2 : Y_2 \to X$ is a log resolution of (X, B), let $\rho_2 : Y_2 \to Y$ be the composition, and let $S_2 := \rho_{2*}^{-1} S$. Then $Y_2 \to Y_1$ restricts to a birational morphism $\mu_2 : S_2 \to S'$. Let

$$K_{Y_2} + B_{Y_2} := h_2^*(K_X + B), \quad K_{S_2} + B_{S_2} := (K_{Y_2} + B_{Y_2})|_{S_2}, \quad \text{and} \quad h_{S_2} := h_{S'} \circ \mu_2 : S_2 \to W'.$$

Since $K_{Y_2} + B_{Y_2}$ is the pullback of $K_Y + B_Y$, $K_{S_2} + B_{S_2}$ is the pullback of $K_S + B_S$, that is,

$$K_{S_2} + B_{S_2} = \mu_2^*(K_{S'} + B_{S'}).$$

Step 2. In this step we find a prime divisor Q on S' which computes t.

Let

$$\mathcal{Q} := \{Q \mid Q \text{ is a prime divisor on } S' \text{ with } h_{S'}(Q) = P\}.$$

Since $h(S) = W$, $h_{S'}$ is surjective. Since P is Cartier on W', $\operatorname{Supp} h_{S'}^* P = h_{S'}^{-1}(P)$. Since $h_{S'}$ is surjective, some irreducible component of $h_{S'}^{-1}(P)$ maps onto P, so $\mathcal{Q}$ is a non-empty finite set and $\operatorname{mult}_Q h_{S'}^* P \in \mathbb{N}^+$ for any $Q \in \mathcal{Q}$.

For any $\tau \in \mathbb{R}$,

$$K_{S_2} + B_{S_2} + \tau h_{S_2}^* P = \mu_2^*(K_{S'} + B_{S'} + \tau h_{S'}^* P),$$

and $(S', B_{S'} + \tau h_{S'}^* P)$ is a simple normal crossing sub-pair. A prime divisor on S' whose image in W' contains the generic point of P either belongs to $\mathcal{Q}$ or dominates W', and if it dominates W', then the coefficient of $B_{S'} + \tau h_{S'}^* P$ along it is ≤ 1. Thus Definition-Theorem 2.7(2) for h_2, S_2, and h_{S_2} implies

$$t = \min_{Q \in \mathcal{Q}} \frac{1 - \operatorname{mult}_Q B_{S'}}{\operatorname{mult}_Q h_{S'}^* P}.$$

We fix $Q \in \mathcal{Q}$ attaining the minimum, and let $r := \operatorname{mult}_Q h_{S'}^* P \in \mathbb{N}^+$. Then

$$\operatorname{mult}_Q B_{S'} = 1 - rt.$$

Step 3. In this step we construct v_λ and prove (1).

Since S' is normal, μ_2 is an isomorphism over the generic point of Q by Zariski's main theorem. Let $Q_2 := \mu_{2*}^{-1} Q$. Then $\operatorname{mult}_{Q_2} B_{S_2} = \operatorname{mult}_Q B_{S'} = 1 - rt$. Since $\operatorname{Supp} B_{Y_2} \cup \operatorname{Exc}(h_2)$ is a simple normal crossing divisor containing S_2 and $\operatorname{codim}_{Y_2} Q_2 = 2$, at most one prime divisor $G \neq S_2$ in $\operatorname{Supp} B_{Y_2} \cup \operatorname{Exc}(h_2)$ contains Q_2. We let $G := 0$ if no such prime divisor exists. Since $B_{S_2} = (B_{Y_2} - S_2)|_{S_2}$, near the generic point of Q_2 we have

$$B_{Y_2} = S_2 + (1 - rt)G.$$

Let $\sigma : Y_3 \to Y_2$ be the blow-up of Q_2 and F the exceptional prime divisor of σ which dominates Q_2. Since $S_2 + G$ is a simple normal crossing divisor near the generic point of Q_2 and $\operatorname{codim}_{Y_2} Q_2 = 2$,

$$a(F, X, B) = a(S_2, X, B) + 1 - (1 - rt) = rt.$$

Let $U \subset Y_3$ be the open subset on which Y_3 is smooth and $\sigma_*^{-1}(\operatorname{Supp} B_{Y_2} \cup \operatorname{Exc}(h_2)) \cup \operatorname{Exc}(\sigma)$ is a simple normal crossing divisor. By the local description near the generic point of Q_2 above, U contains the generic point of the irreducible component of $\sigma_*^{-1}S_2 \cap F$ which dominates Q_2. By [Sza94, Resolution Lemma], there exists a projective birational morphism $\kappa : X' \to Y_3$ which is an isomorphism over U and such that the composition $h' : X' \to X$ is a log resolution of (X, B). Let $E' := (\sigma \circ \kappa)_*^{-1}S_2$, which is E on X', and keep the letter F for $\kappa_*^{-1}F$. Let η be the generic point of the irreducible component of $E' \cap F$ which dominates Q_2, and let $K_{X'} + B' := h'^*(K_X + B)$. Since S_2 and G intersect transversally along Q_2, $\eta \notin \sigma_*^{-1}G$. Since any other component of $\operatorname{Supp} B_{Y_2} \cup \operatorname{Exc}(h_2)$ does not contain Q_2, near η we have

$$\operatorname{Supp} B' \cup \operatorname{Exc}(h') = E' \cup F, \quad \operatorname{mult}_{E'} B' = 1, \quad \text{and} \quad \operatorname{mult}_F B' = 1 - rt.$$

Thus $\mathcal{O}_{X',\eta}$ is a regular local ring with $\dim \mathcal{O}_{X',\eta} = 2$, and we let z, y be local equations of E', F at η respectively.

For any $\lambda > 0$, let v_λ be the quasi-monomial valuation of Definition 2.2(6) given by h', η, the regular system of parameters z, y, and the weights $(1, \lambda/r)$, that is,

$$v_\lambda(\varphi) = \min\left\{\beta_1 + \frac{\lambda}{r}\beta_2 \;\middle|\; c_\beta \neq 0\right\} \quad \text{for} \quad \varphi \in \mathcal{O}_{X',\eta}\backslash\{0\},$$

where

$$\varphi = \sum_{\beta \in \mathbb{N}^2} c_\beta z^{\beta_1} y^{\beta_2}$$

in the completion of $\mathcal{O}_{X',\eta}$ with all c_β in a coefficient field, as in Definition 2.2(6). The point η maps to the generic point of Q_2, hence to the generic point of Q, hence to the generic point of P, hence to x, as $\operatorname{center}_W P = \{x\}$. Thus $h'(\eta) = x$, and $\mathcal{O}_{X,x} \subset \mathcal{O}_{X',\eta}$ with $\mathfrak{m}_x \subset \mathfrak{m}_\eta$. Since $v_\lambda \geq 0$ on $\mathcal{O}_{X',\eta}$ and $v_\lambda \geq \min\{1, \lambda/r\} > 0$ on $\mathfrak{m}_\eta$, we have $\operatorname{center}_X(v_\lambda) = x$.

Since E' and F are prime divisors on X', [JM12, Proposition 5.1] implies

$$A_{X'}(v_\lambda) = 1 + \frac{\lambda}{r}.$$

Since the local equations at η of the components of B' other than E' and F are units, Definition 2.2(4) implies

$$a(v_\lambda, X, B) = a(E, X, B) + \frac{\lambda}{r}a(F, X, B) = 0 + \frac{\lambda}{r} \cdot rt = \lambda t.$$

Since $\operatorname{Supp} D \subset \operatorname{Supp} B$, the support of h'^*D near η is contained in $E' \cup F$, so

$$cv_\lambda(D) = c \operatorname{mult}_{E'} h'^*D + \frac{\lambda}{r} c \operatorname{mult}_F h'^*D.$$

Since D is $\mathbb{Q}$-Cartier,

$$c \operatorname{mult}_{E'} h'^*D = a(E, X, \Delta) - a(E, X, B) = a(E, X, \Delta).$$

Let

$$b := c \operatorname{mult}_F h'^*D/r.$$

Then $b \geq 0$ as $D \geq 0$, and (1) holds.

Step 4. In this step we bound v_λ from below on $\mathcal{O}_{X,x}$.

Let I_j be as in Definition-Proposition 5.4, and for $f \in \mathcal{O}_{X,x}$ let $\overline{f}$ be the image of f in $\mathcal{O}_{W,x}$. We show that for any $\lambda > 0$:

(a) $v_\lambda(g) \geq j$ for any integer $j \geq 0$ and any $g \in I_j$;

(b) for any $f \in \mathcal{O}_{X,x}$ with $\overline{f} \neq 0$,

$$v_\lambda(f) \geq \min\{1, \lambda \operatorname{mult}_P \operatorname{div}(\overline{f})\},$$

and $v_\lambda(f) \geq 1$ for any $f \in \mathcal{O}_{X,x}$ with $\overline{f} = 0$.

For (a), let $g \in I_j \backslash \{0\}$. Since $\mathcal{O}_{X',\eta}$ is a regular local ring and $E' = \operatorname{div}(z)$ near η, $z^{\operatorname{mult}_E \operatorname{div}(g)}$ divides $h'^* g$ in $\mathcal{O}_{X',\eta}$. Since $v_\lambda \geq 0$ on $\mathcal{O}_{X',\eta}$ and $v_\lambda(z) = 1$,

$$v_\lambda(g) \geq \operatorname{mult}_E \operatorname{div}(g) \geq j.$$

For (b), if $\overline{f} = 0$, then $f \in I_1$ by Definition-Proposition 5.4, and $v_\lambda(f) \geq 1$ by (a). Assume that $\overline{f} \neq 0$, and let $\alpha := \operatorname{mult}_P \operatorname{div}(\overline{f})$, where $\operatorname{div}(\overline{f})$ is taken on W'. Then $\alpha \in \mathbb{N}$ as $\overline{f}$ is regular near x. Let $\pi : E' \to W'$ be the composition $E' \to S_2 \xrightarrow{h_{S_2}} W'$. Near η, $X' \to Y_3$ is an isomorphism, $\sigma_*^{-1} S_2 \to S_2$ is an isomorphism as Q_2 is a Cartier divisor on S_2 near the generic point of Q_2, and μ_2 is an isomorphism. Hence $E' \to S'$ is an isomorphism near η and maps η to the generic point of Q. Since $h_{S'}(Q) = P$ and $\operatorname{Supp}(\operatorname{div}(\overline{f}) - \alpha P)$ does not contain P, the order of $\pi^* \overline{f}$ in the DVR $\mathcal{O}_{E',\eta}$ is

$$\operatorname{mult}_Q h_{S'}^* \operatorname{div}(\overline{f}) = \alpha \operatorname{mult}_Q h_{S'}^* P = r\alpha.$$

Since E' and F intersect transversally at η, the image of y is a uniformizer of $\mathcal{O}_{E',\eta} = \mathcal{O}_{X',\eta}/(z)$, and the image of $h'^* f$ in $\mathcal{O}_{E',\eta}$ is $\pi^* \overline{f}$. Thus

$$h'^* f = u y^{r\alpha} + z f_1$$

for some unit $u \in \mathcal{O}_{X',\eta}$ and some $f_1 \in \mathcal{O}_{X',\eta}$. Since $v_\lambda(u y^{r\alpha}) = \lambda\alpha$ and $v_\lambda(z f_1) \geq 1$, we have $v_\lambda(f) \geq \min\{1, \lambda\alpha\}$.

Step 5. In this step we prove (2).

For any integer $i \geq 1$, let

$$J_i := \{\psi \in \mathcal{O}_{W,x} \mid \operatorname{mult}_P \operatorname{div}(\psi) \geq i\},$$

and let $\widetilde{J}_i \subset \mathcal{O}_{X,x}$ be the preimage of J_i. Since W is normal by Definition-Theorem 2.7 and $\operatorname{center}_W P = \{x\}$, v_P has center x on W, and $J_i = \mathfrak{a}_i(v_P)$. Since $\dim \mathcal{O}_{W,x} = \ell$, Lemma 2.3 implies that J_i is $\mathfrak{m}_{W,x}$-primary and

$$\lim_{i \to +\infty} \frac{e(J_i)}{i^\ell} = \operatorname{vol}(P). \tag{5.7}$$

Fix an integer $i \geq 1$ and a real number $\lambda \in (0, 1/i]$. By **Step 4**(b), since $\lambda i \leq 1$,

$$v_\lambda(f) \geq \lambda i \quad \text{for any} \quad f \in \widetilde{J}_i.$$

Let m be a positive integer. By **Step 4**(a), $I_m \subset \mathfrak{a}_m(v_\lambda)$, so

$$\operatorname{length}\big(\mathcal{O}_{X,x}/\mathfrak{a}_m(v_\lambda)\big) = \sum_{j=0}^{m-1} \operatorname{length}\Big(\big(I_j + \mathfrak{a}_m(v_\lambda)\big)/\big(I_{j+1} + \mathfrak{a}_m(v_\lambda)\big)\Big).$$

For $0 \leq j \leq m-1$, the j-th summand is the length of a quotient of $\operatorname{gr}_j = I_j/I_{j+1}$. Let $N_j := \lceil (m-j)/(\lambda i) \rceil$. For any $g \in I_j$ and $f_1, \ldots, f_{N_j} \in \widetilde{J}_i$,

$$v_\lambda(f_1 \cdots f_{N_j} g) \geq N_j \lambda i + j \geq m,$$

so $J_i^{N_j}$ annihilates

$$\big(I_j + \mathfrak{a}_m(v_\lambda)\big)/\big(I_{j+1} + \mathfrak{a}_m(v_\lambda)\big),$$

which is thus a quotient of $\operatorname{gr}_j / J_i^{N_j} \operatorname{gr}_j$. Since the term $j = m$ below is 0, we obtain

$$\operatorname{length}\big(\mathcal{O}_{X,x}/\mathfrak{a}_m(v_\lambda)\big) \leq \sum_{j=0}^{m} \operatorname{length}\Big(\operatorname{gr}_j / J_i^{\lceil (m-j)/(\lambda i) \rceil} \operatorname{gr}_j\Big). \tag{5.8}$$

Multiply (5.8) by $n!/m^n$ and let $m \to +\infty$ with i and λ fixed. By Definition 2.2(2) and Lemma 2.3, the left-hand side tends to $\operatorname{vol}(v_\lambda)$, as $\dim \mathcal{O}_{X,x} = n$. By Definition-Proposition 5.4 with $J = J_i$ and $\beta = \lambda i$, the right-hand side tends to $\phi_{J_i}(\lambda i)$. Thus

$$\lambda^\ell \operatorname{vol}(v_\lambda) \leq \frac{1}{i^\ell} (\lambda i)^\ell \phi_{J_i}(\lambda i) \quad \text{for any} \quad \lambda \in (0, 1/i].$$

Let $\lambda \to 0^+$ with i fixed. Definition-Proposition 5.4 implies

$$\limsup_{\lambda \to 0^+} \lambda^\ell \operatorname{vol}(v_\lambda) \le \frac{\operatorname{vol}(E) \cdot e(J_i)}{i^\ell}.$$

The valuations v_λ do not depend on i. Letting $i \to +\infty$, (5.7) implies (2). The lemma follows. □

Proof of Theorem 5.1. Let E be an lc place of (X, B) with $\operatorname{center}_X E = W$ and P a prime divisor over W with $\operatorname{center}_W P = \{x\}$. Let

$$a_E := a(E, X, \Delta) \quad \text{and} \quad t := a(P, W, \Delta_W, \mathbf{M}).$$

Then $a_E > 0$ as (X, Δ) is klt, and $t > 0$ by Definition-Theorem 2.7(1). By Lemma 5.5, there exist a real number $b \ge 0$ and valuations v_λ, $\lambda \in \mathbb{R}_{>0}$, with $\operatorname{center}_X(v_\lambda) = x$ which satisfy Lemma 5.5(1)(2).

Let $\epsilon \in (0, 1]$. Since $B_\epsilon = B - \epsilon c D$, Definition 2.2(4) implies $a(v, X, B_\epsilon) = a(v, X, B) + \epsilon c v(D)$ for any valuation v with $\operatorname{center}_X v = x$. By Lemma 5.5(1),

$$a(v_\lambda, X, B_\epsilon) = \lambda t + \epsilon(a_E + \lambda b) \quad \text{for any} \quad \lambda > 0.$$

Let

$$\lambda(\epsilon) := \frac{\ell a_E \epsilon}{k(t + \epsilon b)}.$$

Then $\lambda(\epsilon)(t + \epsilon b) = \ell a_E \epsilon / k$, so

$$a(v_{\lambda(\epsilon)}, X, B_\epsilon) = \frac{\ell a_E \epsilon}{k} + a_E \epsilon = \frac{n a_E \epsilon}{k}.$$

By Lemma 2.9, (X, B_ϵ) is klt, so by Definition 2.2(5) and $\epsilon = k\lambda(\epsilon)(t + \epsilon b)/(\ell a_E)$,

$$\frac{\widehat{\operatorname{vol}}(X \ni x, B_\epsilon)}{n^n \epsilon^k} \le \frac{(n a_E \epsilon / k)^n \operatorname{vol}(v_{\lambda(\epsilon)})}{n^n \epsilon^k} = \frac{a_E^n \epsilon^\ell \operatorname{vol}(v_{\lambda(\epsilon)})}{k^n} = \frac{a_E^k (t + \epsilon b)^\ell}{k^k \ell^\ell} \lambda(\epsilon)^\ell \operatorname{vol}(v_{\lambda(\epsilon)}).$$

Since $0 < \lambda(\epsilon) \le \ell a_E \epsilon / (kt)$, $\lambda(\epsilon) \to 0^+$ as $\epsilon \to 0^+$. Thus Lemma 5.5(2) implies

$$\limsup_{\epsilon \to 0^+} \frac{\widehat{\operatorname{vol}}(X \ni x, B_\epsilon)}{n^n \epsilon^k} \le \frac{a_E^k \operatorname{vol}(E) \cdot t^\ell \operatorname{vol}(P)}{k^k \ell^\ell}.$$

The left-hand side depends neither on E nor on P. Since $(W, \Delta_W, \mathbf{M})$ is generalized klt and $\dim W = \ell$, Lemma 4.2 implies that the infimum of $a(P, W, \Delta_W, \mathbf{M})^\ell \operatorname{vol}(P)$ over all prime divisors P over W with $\operatorname{center}_W P = \{x\}$ is $\widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M})$. By Set-up 2.8, the infimum of $a(E, X, \Delta)^k \operatorname{vol}(E)$ over all lc places E of (X, B) with $\operatorname{center}_X E = W$ is $\widehat{\operatorname{vol}}^B(X \ni w, \Delta)$. Taking the infimum over P and then over E, the theorem follows. □

6. Values of sections on projective klt families

Let F be a field and $y_1, \dots, y_n$ variables of positive integral degrees $\deg y_i$. The degree of a monomial $y^\alpha := \prod_{i=1}^n y_i^{\alpha_i}$, $\alpha \in \mathbb{N}^n$, is $\sum_{i=1}^n \alpha_i \deg y_i$. Fix a monomial order on $F[y_1, \dots, y_n]$. For an ideal $I \subset F[y_1, \dots, y_n]$, let $\langle \operatorname{LT}(I) \rangle$ be the ideal generated by the leading terms of the non-zero elements of I. A *standard monomial* with respect to I is a monomial y^α such that $y^\alpha \notin \langle \operatorname{LT}(I) \rangle$.

Theorem 6.1. *Let T, t, $\pi : X \to T$, $\mathcal{L}$, B, ψ, and $\mathcal{B}_m$ be as follows.*

(1) T is a variety with $\dim T \ge 1$, and $t \in T$ is a closed point.
(2) $\pi : X \to T$ is a projective surjective morphism from a normal variety X whose generic fiber is geometrically integral of dimension $d \ge 1$.
(3) $\mathcal{L}$ is a line bundle on X, and $B \ge 0$ is an $\mathbb{R}$-divisor on X such that (X, B) is klt over a neighborhood of t.

(4) ψ is a surjective homomorphism

$$\mathcal{O}_{T,t}[y_1,\dots,y_n] \to \bigoplus_{m\geq 0}(\pi_*\mathcal{L}^{\otimes m})_t$$

of graded $\mathcal{O}_{T,t}$-algebras with $\deg y_i \geq 1$ for every i.

(5) $\mathcal{B}_m$ is the set of standard monomials of degree m with respect to $\ker(\psi)\otimes_{\mathcal{O}_{T,t}}\mathbb{C}(T)$ and a fixed lexicographic monomial order.

Then there exist an integer $m_0 \geq 1$ and a real number $C > 0$ satisfying the following. For any integer $m \geq m_0$, any rational functions b_β, $\beta \in \mathcal{B}_m$, on T which are not all zero, any divisorial valuation v on X with $\pi(\operatorname{center}_X v) = t$, and any local generator e of $\mathcal{L}$ at $\operatorname{center}_X v$,

$$v\Big(\sum_{\beta\in\mathcal{B}_m} b_\beta\psi(\beta)/e^m\Big) \leq \min_{\beta\in\mathcal{B}_m} v(b_\beta) + Cma(v,X,B).$$

The proof of Theorem 6.1 is given in Subsection 6.3.

6.1. **Standard monomials.**

Lemma 6.2. *Let S be a domain, $R = \bigoplus_{m\geq 0} R_m$ a graded S-algebra which is torsion-free over S, and $\psi : S[y_1,\dots,y_n] \to R$ a surjective homomorphism of graded S-algebras with $\deg y_i \geq 1$ for every i. Fix a monomial order which is lexicographic, or which compares two monomials first by their total degrees in finitely many disjoint sets of variables, taken in a fixed order, and then by a lexicographic order. Let $\mathcal{B}_m$ be the set of standard monomials of degree m with respect to $\ker(\psi)\otimes_S \operatorname{Frac}(S)$. Then there exist $g \in S\setminus\{0\}$ and an integer $N \geq 1$ such that for every $m \geq 0$,*

$$R_m \subset \bigoplus_{\beta\in\mathcal{B}_m} g^{-Nm}S\cdot\psi(\beta) \quad \text{in } R_m\otimes_S \operatorname{Frac}(S),$$

where the sum is direct.

Proof. Let $F := \operatorname{Frac}(S)$ and $I := \ker(\psi)\otimes_S F$. Since localization is exact, I is the kernel of the graded homomorphism $\psi\otimes_S F : F[y_1,\dots,y_n] \to R\otimes_S F$, so I is a homogeneous ideal. Since R is torsion-free over S, $R_m \to R_m\otimes_S F$ is injective, so $R_m \subset R_m\otimes_S F$ for any m.

Step 1. In this step we fix a Gröbner basis of I and clear the denominators of the elements of $\mathcal{G}$. Since $F[y_1,\dots,y_n]$ is Noetherian, there exists a finite subset $\mathcal{G} \subset I\setminus\{0\}$ such that $\langle \operatorname{LT}(I)\rangle$ is generated by $\{\operatorname{LT}(q)\}_{q\in\mathcal{G}}$, that is, $\mathcal{G}$ is a Gröbner basis of I [CLO15, Chapter 2, §5, Definition 5]. For $q \in \mathcal{G}$, let q' be the homogeneous component of q containing $\operatorname{LT}(q)$. Then $q' \in I$ since I is homogeneous, and $\operatorname{LT}(q') = \operatorname{LT}(q)$ since the monomials of q' are monomials of q. Possibly replacing each q with q' divided by the coefficient of $\operatorname{LT}(q)$, we may assume that any $q \in \mathcal{G}$ is homogeneous and

$$q = y^{\alpha(q)} - r_q, \tag{6.1}$$

where $y^{\alpha(q)} = \operatorname{LT}(q)$ and r_q is an F-linear combination of monomials $y^{\alpha'} < y^{\alpha(q)}$ of degree $\deg y^{\alpha(q)}$. Let $g \in S\setminus\{0\}$ be an element such that $gr_q \in S[y_1,\dots,y_n]$ for any $q \in \mathcal{G}$.

Step 2. Let Σ be the finite set of pairs $(\alpha(q),\alpha')$, where $q \in \mathcal{G}$ and $y^{\alpha'}$ is a monomial of r_q. In this step we construct $\omega \in (\mathbb{N}^+)^n$ such that

$$\omega\cdot\alpha \geq \omega\cdot\alpha' + 1 \quad \text{for any } (\alpha,\alpha') \in \Sigma. \tag{6.2}$$

Let

$$N' := \max\big(\{1\}\cup\{\alpha_i,\alpha_i' \mid (\alpha,\alpha')\in\Sigma,\ 1\leq i\leq n\}\big).$$

Assume first that the order is lexicographic. Possibly reordering indices, we may assume that the order compares the exponents of y_1 first, then those of y_2, and so on. Let $\omega_n := 1$ and

$\omega_i := 1 + N' \sum_{j=i+1}^{n} \omega_j$ for $i = n-1, \dots, 1$. Let $\alpha, \alpha' \in \{0, 1, \dots, N'\}^n$ such that $y^\alpha > y^{\alpha'}$. Let i be the smallest index such that $\alpha_i \neq \alpha'_i$. Then $\alpha_i \geq \alpha'_i + 1$, so

$$\omega \cdot (\alpha - \alpha') \geq \omega_i - N' \sum_{j=i+1}^{n} \omega_j = 1. \tag{6.3}$$

In particular, (6.2) holds.

Assume now that the order compares first the total degrees in Y_1, then those in Y_2, and so on up to Y_k, for disjoint sets of variables $Y_1, \dots, Y_k$, and then a lexicographic order. Let $\chi_j \in \{0,1\}^n$ be the indicator vector of Y_j, so that $\chi_j \cdot \alpha$ is the total degree of y^α in Y_j, and let ω^{lex} be the vector constructed above for the lexicographic order. Let $c_k, c_{k-1}, \dots, c_1$ be positive integers, taken successively for $j = k, k-1, \dots, 1$, such that

$$c_j > \max_{(\alpha,\alpha') \in \Sigma} \Big(|\omega^{\mathrm{lex}} \cdot (\alpha - \alpha')| + \sum_{i=j+1}^{k} c_i |\chi_i \cdot (\alpha - \alpha')| \Big),$$

and let $\omega := \omega^{\mathrm{lex}} + \sum_{j=1}^{k} c_j \chi_j$. Let $(\alpha, \alpha') \in \Sigma$. If $\chi_j \cdot \alpha = \chi_j \cdot \alpha'$ for any j, then $y^\alpha > y^{\alpha'}$ in the lexicographic order, and (6.2) follows from (6.3). Otherwise, let j be the smallest index such that $\chi_j \cdot \alpha \neq \chi_j \cdot \alpha'$. Then $\chi_j \cdot (\alpha - \alpha') \geq 1$, so

$$\omega \cdot (\alpha - \alpha') \geq c_j - |\omega^{\mathrm{lex}} \cdot (\alpha - \alpha')| - \sum_{i=j+1}^{k} c_i |\chi_i \cdot (\alpha - \alpha')| \geq 1.$$

Step 3. Let $N := \max_i \omega_i$. In this step we show that for any $m \geq 0$ and any monomial y^α of degree m, there exist $b_\beta \in g^{-\omega \cdot \alpha} S$, $\beta \in \mathcal{B}_m$, such that

$$y^\alpha - \sum_{\beta \in \mathcal{B}_m} b_\beta \beta \in I. \tag{6.4}$$

We argue by induction on $\omega \cdot \alpha \in \mathbb{N}$. If $y^\alpha \in \mathcal{B}_m$, then we let $b_{y^\alpha} := 1$ and $b_\beta := 0$ for $\beta \neq y^\alpha$. Otherwise, y^α belongs to the monomial ideal $\langle \mathrm{LT}(I) \rangle$, which is generated by $\{y^{\alpha(q)}\}_{q \in \mathcal{G}}$, so $y^\alpha = y^\gamma y^{\alpha(q)}$ for some $q \in \mathcal{G}$ and $\gamma \in \mathbb{N}^n$. By (6.1),

$$y^\alpha - y^\gamma r_q = y^\gamma q \in I, \qquad y^\gamma r_q = \sum_{\alpha'} e_{\alpha'} y^{\gamma + \alpha'},$$

where α' runs through the exponents of the monomials of r_q and $e_{\alpha'} \in g^{-1} S$. Since q is homogeneous, $\deg y^{\gamma+\alpha'} = m$, and by (6.2), $\omega \cdot (\gamma + \alpha') \leq \omega \cdot \alpha - 1$. By induction, any $y^{\gamma+\alpha'}$ satisfies (6.4) with coefficients in $g^{-(\omega \cdot \alpha - 1)} S$. Multiplying by $e_{\alpha'} \in g^{-1} S$ and adding up, y^α satisfies (6.4) with coefficients in $g^{-\omega \cdot \alpha} S$. Since $\deg y_i \geq 1$ for any i, $\sum_i \alpha_i \leq m$, so $\omega \cdot \alpha \leq Nm$ and $b_\beta \in g^{-Nm} S$.

Step 4. In this step we prove the inclusion of Lemma 6.2 and that the sum is direct. The monomials in $\mathcal{B}_m$ are linearly independent modulo I [CLO15, Chapter 5, §3, Proposition 1]. Since $I = \ker(\psi \otimes_S F)$, $\{\psi(\beta)\}_{\beta \in \mathcal{B}_m}$ is linearly independent over F in $R_m \otimes_S F$. Thus the sum of the modules $g^{-Nm} S \cdot \psi(\beta)$, $\beta \in \mathcal{B}_m$, is direct. Since ψ is surjective and graded, R_m is generated over S by the elements $\psi(y^\alpha)$ with $\deg y^\alpha = m$. Applying $\psi \otimes_S F$ to (6.4), **Step 3** implies

$$\psi(y^\alpha) = \sum_{\beta \in \mathcal{B}_m} b_\beta \psi(\beta) \in \bigoplus_{\beta \in \mathcal{B}_m} g^{-Nm} S \cdot \psi(\beta).$$

The lemma follows. □

6.2. **Families of nodal curves.** For a dominant rational map $g : Y \dashrightarrow Y'$ between varieties and a valuation v on Y such that the map $g_*v : \phi \to v(\phi \circ g)$ on the non-zero rational functions ϕ on Y' is not identically zero, g_*v is a valuation on Y', called the push-forward of v.

Lemma 6.3. *Let $g : Y \to Y'$ be a dominant morphism between varieties, and v a divisorial valuation on Y with center ξ on Y such that $g(\xi)$ is not the generic point of Y'. Then g_*v is a divisorial valuation on Y' with center $g(\xi)$ on Y'.*

Proof. Let $\phi \in \mathfrak{m}_{g(\xi)} \backslash \{0\}$. Since g is dominant, $\phi \circ g \in \mathfrak{m}_\xi \backslash \{0\}$, so $v(\phi \circ g) > 0$. Thus g_*v is a valuation on Y'. Since $v \geq 0$ on $\mathcal{O}_{Y,\xi}$, $v > 0$ on $\mathfrak{m}_\xi$, and $\mathcal{O}_{Y',g(\xi)} \to \mathcal{O}_{Y,\xi}$ is a local homomorphism, g_*v has center $g(\xi)$ on Y'. Let E be a prime divisor over Y and $c > 0$ a real number such that $v = cv_E$. The value group of g_*v is a non-zero subgroup of $c\mathbb{Z}$. The residue field of v_E is the field of rational functions on E, of transcendence degree $\dim Y - 1$. By the Abhyankar inequality [Bou72, Chapter VI, §10.3, Corollary 1] for the extension of fields of rational functions given by g, of transcendence degree $\dim Y - \dim Y'$, the residue field of g_*v has transcendence degree at least $\dim Y' - 1$, and at most $\dim Y' - 1$ by [Bou72, Chapter VI, §10.3, Corollary 1] for the field of rational functions on Y' since the value group of g_*v is non-zero. Thus g_*v is a divisorial valuation. □

Lemma 6.4. *Let T be a normal variety such that $(T, 0)$ is klt. Let $f : X \to T$ be a projective flat morphism whose fibers are reduced curves with only nodes as singularities and whose generic fiber is smooth and geometrically integral. Then:*

(1) X is normal, and $\omega_{X/T}$ is invertible and commutes with base change;
(2) for any canonical divisor K_T of T and any Cartier divisor $K_{X/T}$ on X with $\mathcal{O}_X(K_{X/T}) \cong \omega_{X/T}$, the divisor $K_{X/T} + f^*K_T$ is a canonical divisor of X;
(3) $(X, 0)$ is klt, and

$$a(f_*v, T, 0) \leq a(v, X, 0)$$

for every divisorial valuation v on X with a center ξ on X such that $f(\xi)$ is not the generic point of T.

Proof. **Step 1.** In this step we show that X is a normal variety.

Let η be the generic point of T and X_η the generic fiber of f. Since f is flat, f is open, so the generic point of any irreducible component of X maps to η. Since X_η is integral, X is irreducible.

Since $(T, 0)$ is klt, T is Cohen–Macaulay. Let $\xi \in X$ be a closed point. The fiber $X_{f(\xi)}$ is a reduced curve, hence Cohen–Macaulay, and $\mathcal{O}_{T,f(\xi)} \to \mathcal{O}_{X,\xi}$ is flat, so $\mathcal{O}_{X,\xi}$ is Cohen–Macaulay [Mat89, Theorem 23.3, Corollary]. Thus X is Cohen–Macaulay, and in particular X satisfies S_2.

Let $\xi \in X$ be a point such that $\dim \mathcal{O}_{X,\xi} = 1$. Since f is flat,

$$\dim \mathcal{O}_{X,\xi} = \dim \mathcal{O}_{T,f(\xi)} + \dim \mathcal{O}_{X_{f(\xi)},\xi},$$

so $\dim \mathcal{O}_{T,f(\xi)} \leq 1$. If $f(\xi) = \eta$, then $\mathcal{O}_{X,\xi}$ is a local ring of X_η, which is smooth, so $\mathcal{O}_{X,\xi}$ is regular. Otherwise, $\dim \mathcal{O}_{T,f(\xi)} = 1$, so $\mathcal{O}_{T,f(\xi)}$ is a DVR since T is normal, and $\dim \mathcal{O}_{X_{f(\xi)},\xi} = 0$, that is, ξ is the generic point of an irreducible component of the curve $X_{f(\xi)}$. Since $X_{f(\xi)}$ is reduced, $\mathcal{O}_{X_{f(\xi)},\xi} = \mathcal{O}_{X,\xi}/\mathfrak{m}_{f(\xi)}\mathcal{O}_{X,\xi}$ is a field, that is, $\mathfrak{m}_\xi = \mathfrak{m}_{f(\xi)}\mathcal{O}_{X,\xi}$ is principal. Thus the 1-dimensional local ring $\mathcal{O}_{X,\xi}$ is a DVR. Therefore X satisfies R_1, and X is normal by Serre's criterion.

Step 2. In this step we prove the rest of (1), and (2). Since f is flat with Cohen–Macaulay fibers, $\omega_{X/T}$ is flat over T and commutes with base change [Kol23, 2.68.2]. The sheaf $\omega_{X/T}$ is invertible since the fibers of f are nodal curves, which are Gorenstein. Let K_T be a canonical divisor of T, and $K_{X/T}$ a Cartier divisor such that $\mathcal{O}_X(K_{X/T}) \cong \omega_{X/T}$. Since $(T, 0)$ is klt, K_T is $\mathbb{Q}$-Cartier, so f^*K_T is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. Since T is normal and f is flat,

$$\operatorname{codim}_X f^{-1}(T \backslash T_{\mathrm{reg}}) \geq 2,$$

and over the non-singular locus T_{reg}, $\omega_{X/T} \otimes f^*\omega_T$ is the dualizing sheaf of $f^{-1}(T_{\mathrm{reg}})$. Thus

$$K_X := K_{X/T} + f^*K_T \tag{6.5}$$

is a canonical divisor of X. In particular, K_X is $\mathbb{Q}$-Cartier and $(X, 0)$ is a pair.

Step 3. Let E be a prime divisor over X such that $f(\mathrm{center}_X E)$ is not dense in T. In this step we construct a normal variety T', a projective birational morphism $\mu : T' \to T$, and a prime divisor P on T' such that E maps onto P, and we show that, with $X' := X \times_T T'$ and $f' : X' \to T'$ the projection,

$$\mathrm{mult}_E \,\mathrm{div}(\phi \circ f) = \mathrm{mult}_E(f'^*P) \cdot \mathrm{mult}_P \,\mathrm{div}(\phi \circ \mu) \tag{6.6}$$

for any non-zero rational function ϕ on T, $\mathrm{mult}_E \, f'^*P > 0$, and

$$a(E, X, 0) \geq a(P, T, 0) \,\mathrm{mult}_E \, f'^*P. \tag{6.7}$$

Let ξ be the generic point of $\mathrm{center}_X E$. Since $f(\xi)$ is not the generic point of T, Lemma 6.3 for f and v_E shows that $\nu := f_* v_E$ is a divisorial valuation with center $f(\xi)$, that is, there exist a normal variety T', a projective birational morphism $\mu : T' \to T$, and a prime divisor P on T' such that ν is a positive multiple of the valuation $\phi \to \mathrm{mult}_P \,\mathrm{div}(\phi \circ \mu)$.

Let $X' := X \times_T T'$, and let $f' : X' \to T'$ and $\mu' : X' \to X$ be the projections. The morphism f' is projective and flat, the fibers of f' are base changes of fibers of f, and the generic fiber of f' is X_η. By the first paragraph of **Step 1**, X' is irreducible, and μ' is birational. Since μ' is projective, E has a center on X'. Let ξ' be the generic point of $\mathrm{center}_{X'} E$. Since $\mathrm{mult}_E \,\mathrm{div}(\zeta) \geq 0$ for any $\zeta \in \mathcal{O}_{X',\xi'} \backslash \{0\}$ and $\mathrm{mult}_E \,\mathrm{div}(\zeta) > 0$ for any $\zeta \in \mathfrak{m}_{\xi'} \backslash \{0\}$, $f'(\xi')$ is the center of ν on T', that is, the generic point η_P of P. Thus E maps onto P.

Possibly shrinking T' to a neighborhood of η_P, we may assume that T' and P are smooth, $P = \mathrm{div}(q)$ for a regular function q on T', and

$$K_{T'} - \mu^* K_T = (a(P, T, 0) - 1)P, \tag{6.8}$$

where $K_{T'}$ is a canonical divisor of T' such that $\mu_* K_{T'} = K_T$. Since T' is smooth, **Step 1** and **Step 2** applied to f' imply that X' is a normal variety and $K_{X'} = K_{X'/T'} + f'^* K_{T'}$ is a canonical divisor of X' for any Cartier divisor $K_{X'/T'}$ such that $\mathcal{O}_{X'}(K_{X'/T'}) \cong \omega_{X'/T'}$. Since $\omega_{X/T}$ commutes with base change, we may let $K_{X'/T'} := \mu'^* K_{X/T}$. Since $\mu'^* f^* K_T = f'^* \mu^* K_T$, (6.5) and (6.8) imply

$$K_{X'} = \mu'^* K_{X/T} + f'^* K_{T'} = \mu'^* K_X + f'^*(K_{T'} - \mu^* K_T) = \mu'^* K_X + (a(P, T, 0) - 1) f'^* P.$$

Since μ' is birational,

$$a(E, X, 0) = a(E, X', 0) + (a(P, T, 0) - 1) \,\mathrm{mult}_E \, f'^* P. \tag{6.9}$$

We show that (X', f'^*P) is lc near ξ'. Since $f'(\xi') = \eta_P$, ξ' is a point of the fiber X'_{η_P}. If X'_{η_P} is smooth at ξ', then f' is smooth at ξ', so near ξ', X' is smooth and $f'^*P = \mathrm{div}(q \circ f')$ is a smooth divisor, and (X', f'^*P) is lc near ξ'. Otherwise, ξ' is a node of X'_{η_P}. By [Sta, Tag 0CBY], there exist an étale morphism $V \to T'$ from an affine variety V, a point $\eta_V \in V$ over η_P, a regular function θ on V, and étale morphisms $U \to X'$ and $U \to W := \mathrm{Spec}\, \mathcal{O}(V)[x, y]/(xy - \theta)$ over T' such that ξ' is the image of a point $\xi_U \in U$ whose image in V is η_V. Since $V \to T'$ is étale, $\mathcal{O}_{V,\eta_V}$ is a DVR with uniformizer q, and $\theta \neq 0$ in $\mathcal{O}_{V,\eta_V}$ since the generic fiber of f' is smooth, so $\theta = uq^N$ for a unit $u \in \mathcal{O}_{V,\eta_V}$ and an integer $N \geq 0$, and $N \geq 1$ since ξ' is a node of X'_{η_P}. Possibly shrinking V to a neighborhood of η_V and replacing x with $u^{-1}x$, we may assume that $\theta = q^N$ and that q is a member of a system of local coordinates at any point of the zero locus of q on V. Then W is étale locally isomorphic to $\{xy = z^N\} \times \mathbb{A}^{\dim T - 1}$, and the pullback of f'^*P to W corresponds to $\mathrm{div}(z) = \{x = z = 0\} + \{y = z = 0\}$, the sum of the torus invariant prime divisors of the toric surface $\{xy = z^N\} \subset \mathbb{A}^3$ times $\mathbb{A}^{\dim T - 1}$. Thus the pullback of (X', f'^*P) to U is étale locally a toric variety with a reduced torus invariant boundary, which is lc, and (X', f'^*P) is lc near ξ'.

Since f'^*P is Cartier near ξ', $a(E,X',f'^*P) = a(E,X',0) - \mathrm{mult}_E\, f'^*P \geq 0$. By (6.9), we obtain (6.7).

Finally, let ϕ be a non-zero rational function on T and $k := \mathrm{mult}_P \operatorname{div}(\phi\circ\mu)$. Then $\phi\circ\mu = u'q^k$ for a unit $u' \in \mathcal{O}_{T',\eta_P}$. Since $f'(\xi') = \eta_P$, $u'\circ f'$ is a unit at ξ', so

$$\mathrm{mult}_E \operatorname{div}(\phi\circ f) = \mathrm{mult}_E \operatorname{div}\left((\phi\circ\mu)\circ f'\right) = k\,\mathrm{mult}_E \operatorname{div}(q\circ f') = k\,\mathrm{mult}_E\, f'^*P,$$

which is (6.6). Since $q\circ f' \in \mathfrak{m}_{\xi'}$, $\mathrm{mult}_E\, f'^*P > 0$.

Step 4. In this step we prove (3). Let E be a prime divisor over X. If $f(\mathrm{center}_X E)$ is dense in T, then the generic point of $\mathrm{center}_X E$ is a point of X_η, which is smooth, so X is smooth near the generic point of $\mathrm{center}_X E$ and $a(E,X,0) \geq 1$. Otherwise, since $(T,0)$ is klt, **Step 3** implies

$$a(E,X,0) \geq a(P,T,0)\,\mathrm{mult}_E\, f'^*P > 0.$$

Thus $(X,0)$ is klt.

Let E be a prime divisor over X and $c > 0$ a real number such that $v = cv_E$, and let $\xi := \mathrm{center}_X v$. Assume that $f(\xi) \neq \eta$. Then $f(\mathrm{center}_X E) = \overline{\{f(\xi)\}}$ is not dense in T. By (6.6),

$$f_*v(\phi) = c\,\mathrm{mult}_E(f'^*P)\,\mathrm{mult}_P \operatorname{div}(\phi\circ\mu)$$

for any non-zero rational function ϕ on T, so by Definition 2.1 and (6.7),

$$a(f_*v,T,0) = c\,\mathrm{mult}_E(f'^*P)a(P,T,0) \leq ca(E,X,0) = a(v,X,0).$$

The lemma follows. □

Lemma 6.5. *Let T be a normal variety such that $(T,0)$ is klt. Let $f : X \to T$ be a projective flat morphism whose fibers are reduced curves with only nodes as singularities and whose generic fiber is smooth and geometrically integral. Let $\mathcal{L}$ be a line bundle on X which is very ample over T. Then there exist an integer $m_0 \geq 1$ and a real number $C > 0$ satisfying the following. Let $m \geq m_0$ be an integer. Then $f_*\mathcal{L}^{\otimes m}$ is locally free. Let v be a divisorial valuation on X with a center ξ on X such that $f(\xi)$ is not the generic point of T, e a local generator of $\mathcal{L}$ at ξ, $(s_\alpha)_\alpha$ a basis of $f_*\mathcal{L}^{\otimes m}$ near $f(\xi)$, and $s = \sum_\alpha b_\alpha s_\alpha \neq 0$ with rational functions b_α on T. Then*

$$v(s/e^m) \leq \min_\alpha v(b_\alpha) + Cma(v,X,0).$$

Proof. By Lemma 6.4, X is normal and $(X,0)$ is klt. In particular, $a(v,X,0)$ is defined. Let $\mathrm{Sing}(f) \subset X$ be the closed subset of points at which f is not smooth. Let η be the generic point of T, and X_η the generic fiber of f.

Step 1. In this step we fix m_0 and C.

By relative Serre vanishing and cohomology and base change, there exists an integer $m_0 \geq 1$ such that for any integer $m \geq m_0$, $R^1 f_*\mathcal{L}^{\otimes m} = 0$, and $f_*\mathcal{L}^{\otimes m}$ is locally free and commutes with arbitrary base change.

Since $\mathrm{Sing}(f)$ is quasi-compact, by [Sta, Tag 0CBY] there exist a finite set I and, for any $i \in I$,

- an étale morphism $\widehat{T}_i \to T$ from a connected affine variety $\widehat{T}_i$, and a regular function $\widehat{\phi}_i$ on $\widehat{T}_i$,
- étale morphisms of $\widehat{T}_i$-schemes $U_i \to \widehat{X}_i := X \times_T \widehat{T}_i$ and $U_i \to \operatorname{Spec} \mathcal{O}(\widehat{T}_i)[x,y]/(xy - \widehat{\phi}_i)$,

such that the images of the U_i in X cover $\mathrm{Sing}(f)$. Since f is flat and X_η is geometrically integral, $\widehat{X}_i$ is a variety. Since X_η is smooth, $\widehat{\phi}_i \neq 0$. Let $\widehat{f}_i : \widehat{X}_i \to \widehat{T}_i$ be the projection. Since $\widehat{X}_i \to X$ is étale, $(\widehat{X}_i, 0)$ is klt [Kol13, 2.14]. Thus there exists a real number $\beta > 0$ such that

$$\left(\widehat{X}_i, \beta\widehat{f}_i^*\operatorname{div}(\widehat{\phi}_i)\right) \text{ is lc for any } i \in I.$$

Since $\mathrm{Sing}(f) \to T$ is proper and quasi-finite, hence finite, there exists a positive integer n_0 such that any fiber of $\mathrm{Sing}(f) \to T$ has at most n_0 geometric points.

We let

$$C := \deg(\mathcal{L}|_{X_\eta})\Big(1 + \frac{3n_0}{2\beta}\Big).$$

Fix an integer $m \geq m_0$, and v, e, $(s_\alpha)_\alpha$, and $s = \sum_\alpha b_\alpha s_\alpha$ as in the lemma. Let $\xi := \operatorname{center}_X v$, $\tau := f(\xi)$, and $X_\tau := X \times_T \operatorname{Spec} \kappa(\tau)$.

Step 2. In this step we extend the base so that the nodes of X_τ become split, and we bound their thickness.

By Lemma 6.3 for f, f_*v is a divisorial valuation on T with center τ, so $f_*v = \gamma v_P$ for a real number $\gamma > 0$ and a prime divisor P on a normal variety T' with a projective birational morphism $\mu : T' \to T$ such that $\operatorname{center}_T P$ is the closure of τ. Let $R := \mathcal{O}_{T',P}$, and let $q \in R$ be a uniformizer.

Let R'' be a discrete valuation ring which dominates R, is unramified over R and is essentially of finite type over $\mathbb{C}$. Let κ'' be the residue field of R'' and $X_0'' := X_\tau \times_{\kappa(\tau)} \kappa''$. We choose R'' satisfying the following condition:

($\star$) for any singular point p of X_0'', there exist $i \in I$, a morphism $\operatorname{Spec} R'' \to \widehat{T}_i$ over T, and a κ''-rational point of $U_i \times_{\widehat{T}_i} \operatorname{Spec} R''$ which maps to p.

We show that such an R'' exists. Let p be a singular point of $X_\tau \times_{\kappa(\tau)} \kappa(P)$ and $i \in I$ an index such that the image of p in X is contained in the image of U_i. A point p' of $U_i \times_T \operatorname{Spec} R$ over p maps to a closed point of the étale R-scheme $\widehat{T}_i \times_T \operatorname{Spec} R$, whose local ring R_1 is unramified over R. As p is a singular point of the fiber, p' maps to the point $x = y = 0$ of the closed fiber of $\operatorname{Spec} R_1[x,y]/(xy - \widehat{\phi}_i)$, so $\kappa(p')$ is finite and separable over the residue field of R_1, and an unramified extension of R_1 with residue field $\kappa(p')$ satisfies ($\star$) for one point over p. Since ($\star$) for a point is preserved by further unramified extensions, each such extension increases the number of singular points of X_0'' satisfying ($\star$), which is at most n_0 for any discrete valuation ring R'' which dominates R and is unramified over R, so finitely many such extensions yield R''.

Let $\Pi = \{p_1, \dots, p_{\#\Pi}\}$ be the set of singular points of X_0''. Thus $\#\Pi \leq n_0$. For $1 \leq k \leq \#\Pi$, let $i(k) \in I$ and $\operatorname{Spec} R'' \to \widehat{T}_{i(k)}$ be as in ($\star$) for p_k, and $\phi_k \in R''$ the image of $\widehat{\phi}_{i(k)}$. By ($\star$), $X \times_T \operatorname{Spec} R''$ at p_k and $\operatorname{Spec} R''[x,y]/(xy - \phi_k)$ at the point $x = y = 0$ of the closed fiber have a common étale neighborhood with residue field κ''. Thus p_k is κ''-rational, and both branches of X_0'' through p_k are defined over κ''. Moreover $\phi_k \neq 0$ as $\operatorname{Spec} R'' \to \widehat{T}_{i(k)}$ maps the generic point to the generic point. Since R'' is unramified over R, q is a uniformizer of R''. Let N_k be the integer such that

$$\phi_k \in q^{N_k} \cdot (R'')^\times.$$

Then $N_k \geq 1$ as p_k is singular.

Let T'' be the normalization of T' in the fraction field of R'', $\mu'' : T'' \to T$ the induced morphism, P'' the prime divisor on T'' such that $\mathcal{O}_{T'',P''} = R''$, η'' the generic point of P'', and $X'' := X \times_T T''$. Then X'' is a variety, $\kappa(P'') = \kappa''$, X_0'' is the fiber of X'' over η'', and

$$\operatorname{mult}_{P''} \operatorname{div}(\phi \circ \mu'') = \operatorname{mult}_P \operatorname{div}(\phi \circ \mu) \quad \text{for any non-zero rational function } \phi \text{ on } T.$$

Let $\mathcal{O}_v$ be the valuation ring of v. Since R'' is flat over R, $\mathcal{O}_v \otimes_R R'' \subset \mathbb{C}(X'')$, and since the residue fields of $\mathcal{O}_v$ and R'' are extensions of $\kappa(P)$, there exists a prime ideal $\mathfrak{q}$ of $\mathcal{O}_v \otimes_R R''$ containing the maximal ideals of $\mathcal{O}_v$ and R''. Let v'' be a valuation on X'' whose valuation ring dominates $(\mathcal{O}_v \otimes_R R'')_{\mathfrak{q}}$, and let $g'' : X'' \to X$ and $f'' : X'' \to T''$ be the projections. Since valuation rings are maximal for domination, g''_*v'' and f''_*v'' are positive multiples of v and of $v_{P''}$ respectively. We normalize v'' so that $g''_*v'' = v$. Then for any non-zero rational function ϕ on T,

$$(f''_*v'')(\phi \circ \mu'') = (f_*v)(\phi) = \gamma \operatorname{mult}_P \operatorname{div}(\phi \circ \mu) = \gamma \operatorname{mult}_{P''} \operatorname{div}(\phi \circ \mu''),$$

so

$$(f''_*v'')(\phi) = \gamma \operatorname{mult}_{P''} \operatorname{div}(\phi) \quad \text{for any non-zero rational function } \phi \text{ on } T'',$$

and v'' has a center on X'' over η''.

Let $1 \le k \le \#\Pi$ and $i := i(k)$. The morphism $\operatorname{Spec} R'' \to \widehat{T}_i$ over T induces a dominant rational map $X'' \dashrightarrow \widehat{X}_i$ over X. Let $\widetilde{v}$ be the push-forward of v'' under $X'' \dashrightarrow \widehat{X}_i$. Then the push-forward of $\widetilde{v}$ along $\widehat{X}_i \to X$ is v, $\widetilde{v}$ is a divisorial valuation [Bou72, Chapter VI, §10.3, Corollary 1], and the center of $\widetilde{v}$ on $\widehat{X}_i$ is the image of the center of v'' under $X \times_T \operatorname{Spec} R'' \to \widehat{X}_i$. Since $\widehat{X}_i \to X$ is étale, $a(\widetilde{v}, \widehat{X}_i, 0) = a(v, X, 0)$ [Kol13, 2.14]. Since $(\widehat{X}_i, \beta \widehat{f}_i^* \operatorname{div}(\widehat{\phi}_i))$ is lc,

$$\gamma N_k = v''(\phi_k) = \widetilde{v}(\widehat{f}_i^* \widehat{\phi}_i) \le \frac{1}{\beta} a(\widetilde{v}, \widehat{X}_i, 0) = \frac{1}{\beta} a(v, X, 0) \quad \text{for any } 1 \le k \le \#\Pi. \tag{6.10}$$

Step 3. In this step we construct a smooth variety $\widetilde{Z}$ over the double cover $t^2 = q$, and compute the log discrepancy of an extension $\widetilde{w}$ of v.

Let $T^\circ \subset T''$ be an affine open subset containing η'', $P^\circ := P'' \cap T^\circ$, $X^\circ := X \times_T T^\circ$, and $f^\circ : X^\circ \to T^\circ$ the projection. Let $i_0 := \min_\alpha \operatorname{mult}_P \operatorname{div}(b_\alpha \circ \mu)$. Possibly shrinking T° to a neighborhood of η'', we may assume the following:

(a) T° and P° are smooth, and $\operatorname{div}(q) = P^\circ$ on T°;
(b) for any $1 \le k \le \#\Pi$, $\operatorname{Spec} R'' \to \widehat{T}_{i(k)}$ extends to a morphism $T^\circ \to \widehat{T}_{i(k)}$, and ϕ_k / q^{N_k} is a unit of $\mathcal{O}(T^\circ)$;
(c) the reduced subscheme of $\operatorname{Sing}(f^\circ)$ is the disjoint union of the images of sections $\sigma_k : P^\circ \to X^\circ$, $1 \le k \le \#\Pi$, with $\sigma_k(\eta'') = p_k$, and étale locally along $\sigma_k(P^\circ)$, X° is isomorphic over T° to $\operatorname{Spec} \mathcal{O}(T^\circ)[x, y]/(xy - \phi_k)$;
(d) T° maps into an open subset of T over which $(s_\alpha)_\alpha$ is a basis of $f_* \mathcal{L}^{\otimes m}$, and $q^{-i_0} b_\alpha \in \mathcal{O}(T^\circ)$ for any α.

Here (c) holds near η'' by ($\star$) and (b), since $\operatorname{Sing}(f^\circ) \to T^\circ$ is finite and the points of Π are κ''-rational.

Let $\widetilde{T} \subset T^\circ \times \mathbb{A}^1$ be the subvariety defined by $t^2 = q$, $\widetilde{P} \subset \widetilde{T}$ the prime divisor defined by $t = 0$, and $\psi : \widetilde{T} \to T$ the induced morphism. By (a), $dq \ne 0$ along P°, so $\widetilde{T}$ is smooth, and $\widetilde{T} \to T^\circ$ is a double cover, étale over $T^\circ \backslash P^\circ$, which maps $\widetilde{P}$ isomorphically onto P°. Let $Z := X^\circ \times_{T^\circ} \widetilde{T}$ with projections $\widetilde{f} : Z \to \widetilde{T}$ and $g : Z \to X$. The morphism $\widetilde{f}$ satisfies the hypotheses of Lemma 6.4, and Z is normal. By (c), Z is smooth away from the images of the sections $\widetilde{\sigma}_k : \widetilde{P} \to Z$ induced by the σ_k, and étale locally along $\widetilde{\sigma}_k(\widetilde{P})$, Z is the product of a smooth variety with the surface

$$\{xy = t^{2N_k}\} \subset \mathbb{A}^3,$$

since t is part of a system of étale coordinates of $\widetilde{T}$ along $\widetilde{P}$ and we may replace x by $q^{N_k} x / \phi_k$. Since $2N_k \ge 2$, the reduced singular locus of Z is $\bigcup_{k=1}^{\#\Pi} \widetilde{\sigma}_k(\widetilde{P})$.

Let $h : \widetilde{Z} \to Z$ be the composition of $\max_{1 \le k \le \#\Pi} N_k$ blow-ups, each of which has center the reduced singular locus ($h = \mathrm{id}_Z$ if $\Pi = \emptyset$). The blow-up of $\{xy = t^{2N}\}$ at the origin is smooth if $N = 1$, and if $N \ge 2$ it has exactly one singular point, at which it is of the form $xy = t^{2N-2}$. Blow-ups commute with étale base change and with products by smooth varieties. Thus $\widetilde{Z}$ is smooth, and étale locally along $\widetilde{\sigma}_k(\widetilde{P})$, h is the product of a smooth variety with the minimal resolution of $\{xy = t^{2N_k}\}$. The minimal resolution of $\{xy = t^{2N_k}\}$ is toric and is covered by the charts with coordinates (u_ρ, u'_ρ), $0 \le \rho \le 2N_k - 1$, on which

$$t = u_\rho u'_\rho, \quad x = u_\rho^{\rho+1} {u'_\rho}^{\rho}, \quad y = u_\rho^{2N_k - \rho - 1} {u'_\rho}^{2N_k - \rho}, \quad \frac{dx}{x} \wedge dt = du_\rho \wedge du'_\rho.$$

Since $\frac{dx}{x} \wedge dt$ generates the dualizing sheaf of $\{xy = t^{2N_k}\}$, h is crepant:

$$K_{\widetilde{Z}} = h^* K_Z.$$

Since

$$\operatorname{div}(t) = \operatorname{div}(u_\rho) + \operatorname{div}(u'_\rho)$$

on each chart, $(\widetilde{f}\circ h)^*\widetilde{P}$ is a reduced snc divisor, and over $\widetilde{\sigma}_k(\widetilde{P})$ it contains a chain of $2N_k-1$ exceptional prime divisors joining the strict transforms of the branches $\{x=t=0\}$ and $\{y=t=0\}$.

Write $(\widetilde{f}\circ h)^*\widetilde{P}=\sum_{j\in J}D_j$ with prime divisors D_j, let $\widetilde{\kappa}$ be the residue field of $\widetilde{T}$ at the generic point of $\widetilde{P}$, and let $\Gamma_j:=D_j\times_{\widetilde{P}}\operatorname{Spec}\widetilde{\kappa}$. Then $\widetilde{\kappa}\cong\kappa''$, and the Γ_j are smooth projective curves over $\widetilde{\kappa}$: the strict transforms of the irreducible components of $Z\times_{\widetilde{T}}\operatorname{Spec}\widetilde{\kappa}\cong X_0''$, and the exceptional curves, each $\cong\mathbb{P}^1_{\widetilde{\kappa}}$. Possibly shrinking T° to a neighborhood of η'', we may assume that D_j is smooth for any $j\in J$.

The morphism $\widetilde{Z}\to X''$ is generically finite of degree 2, as h is birational and $Z\to X^\circ$ is the base change of the double cover $\widetilde{T}\to T^\circ$. Let $\widetilde{w}$ be a valuation on $\widetilde{Z}$ whose push-forward along $\widetilde{Z}\to X''$ is v''. The push-forward of $\widetilde{w}$ along the generically finite morphism $\widetilde{Z}\to X$ is $g''_*v''=v$, so $\widetilde{w}$ is a divisorial valuation by [Bou72, Chapter VI, §10.3, Corollary 1] for $\widetilde{Z}\to X$. Since $\widetilde{P}$ is the only prime divisor on $\widetilde{T}$ over P° and $\operatorname{mult}_{\widetilde{P}}\operatorname{div}(q)=2$, we have

$$\big((\widetilde{f}\circ h)_*\widetilde{w}\big)(\phi)=\frac{\gamma}{2}\operatorname{mult}_{\widetilde{P}}\operatorname{div}(\phi)\quad\text{for any non-zero rational function }\phi\text{ on }\widetilde{T}.$$

In particular,

$$\widetilde{w}(t)=\frac{\gamma}{2},$$

and the center $\widetilde{c}$ of $\widetilde{w}$ on $\widetilde{Z}$ maps to the generic point of $\widetilde{P}$.

Let $R(\psi):=K_{\widetilde{T}}-\psi^*K_T$. Possibly shrinking T° to a neighborhood of η'', we may assume that $\operatorname{Supp}R(\psi)\subset\widetilde{P}$. By Lemma 6.4(1)(2) for f and for $\widetilde{f}$, we may take $K_X=K_{X/T}+f^*K_T$, $K_{Z/\widetilde{T}}=g^*K_{X/T}$, and $K_Z=K_{Z/\widetilde{T}}+\widetilde{f}^*K_{\widetilde{T}}$. Let $\pi:=g\circ h:\widetilde{Z}\to X$. Since $\widetilde{f}^*\psi^*K_T=g^*f^*K_T$, the definition of $R(\psi)$ and the crepancy of h give

$$K_{\widetilde{Z}}=\pi^*K_X+(\widetilde{f}\circ h)^*R(\psi).$$

Since $\pi_*\widetilde{w}=v$ and

$$\operatorname{mult}_{\widetilde{P}}\operatorname{div}(\phi\circ\psi)=2\operatorname{mult}_P\operatorname{div}(\phi\circ\mu)\quad\text{for any non-zero rational function }\phi\text{ on }T,$$

the ramification formula for log discrepancies implies

$$a\big(\widetilde{w},\widetilde{Z},-(\widetilde{f}\circ h)^*R(\psi)\big)=a(v,X,0)\quad\text{and}\quad 1+\operatorname{mult}_{\widetilde{P}}R(\psi)=a(\widetilde{P},\widetilde{T},-R(\psi))=2a(P,T,0).$$

Since

$$\widetilde{w}\big((\widetilde{f}\circ h)^*R(\psi)\big)=\frac{\gamma}{2}\operatorname{mult}_{\widetilde{P}}R(\psi),$$

we obtain

$$a(\widetilde{w},\widetilde{Z},0)=a(v,X,0)-\frac{\gamma}{2}\operatorname{mult}_{\widetilde{P}}R(\psi)=a(v,X,0)-\gamma a(P,T,0)+\frac{\gamma}{2}.\tag{6.11}$$

Step 4. In this step we bound the vertical part of the divisor of the section.

Let $\widetilde{R}:=\mathcal{O}_{\widetilde{T},\widetilde{P}}$ and $\widetilde{Z}_{\widetilde{R}}:=\widetilde{Z}\times_{\widetilde{T}}\operatorname{Spec}\widetilde{R}$. Then $\widetilde{Z}_{\widetilde{R}}\to\operatorname{Spec}\widetilde{R}$ is an arithmetic surface, that is, a regular fibered surface over a Dedekind scheme of dimension 1, with closed fiber

$$F=\sum_{j\in J}\Gamma_j.$$

Intersection numbers on $\widetilde{Z}_{\widetilde{R}}$ are taken over $\widetilde{\kappa}$. Since $\widetilde{Z}$ is integral, $\widetilde{T}$ is normal and the generic fiber of $\widetilde{f}\circ h$ is geometrically integral, $(\widetilde{f}\circ h)_*\mathcal{O}_{\widetilde{Z}}=\mathcal{O}_{\widetilde{T}}$, so F is connected by Zariski's connectedness theorem.

Let G_0 be the graph whose vertices are the irreducible components of X_0'', with one edge for each $p_k\in\Pi$ joining the components which contain the branches through p_k (a loop if both branches are contained in one component). Let G be the dual graph of F: the vertex set is J, and there exist $\#(\Gamma_j\cap\Gamma_{j'})$ edges between distinct $j,j'\in J$. By **Step 3**, G is the graph G_0

with the edge of any $p_k \in \Pi$ replaced by a path with $2N_k$ edges, and distinct Γ_j, $\Gamma_{j'}$ intersect transversally at $\widetilde{\kappa}$-rational points, so

$$\Gamma_j \cdot \Gamma_{j'} = \#(\Gamma_j \cap \Gamma_{j'}) \quad \text{for any distinct } j, j' \in J.$$

Since F is connected, G is connected, and since G has exactly $\sum_{k=1}^{\#\Pi} 2N_k$ edges and a shortest path in G passes through any edge at most once,

$$\operatorname{dist}_G(j, j') \le 2 \sum_{k=1}^{\#\Pi} N_k \quad \text{for any } j, j' \in J. \tag{6.12}$$

Let $s_0 := q^{-i_0} s$, $\widetilde{\mathcal{L}} := \pi^* \mathcal{L}$, and $\widetilde{s}_0 := \pi^* s_0$, which is a regular section of $\widetilde{\mathcal{L}}^{\otimes m}$ by (d). By (d) and $m \ge m_0$, the restrictions of the s_α to X_0'' form a basis of $H^0(X_0'', \mathcal{L}^{\otimes m}|_{X_0''})$. Since $q^{-i_0} b_\alpha$ is a unit of R'' for some α, $s_0|_{X_0''} \ne 0$. Possibly shrinking T° to a neighborhood of η'', we may write

$$\operatorname{div}(\widetilde{s}_0) = H + \sum_{j \in J} a_j D_j, \tag{6.13}$$

where $a_j \in \mathbb{N}$ and $H \ge 0$ is a divisor with no component contained in $(\widetilde{f} \circ h)^{-1}(\widetilde{P})$. Since

$$Z \times_{\widetilde{T}} \widetilde{P} \cong X^\circ \times_{T^\circ} P^\circ$$

is reduced, s_0 does not vanish along some irreducible component of X_0'', and h is an isomorphism at the generic point of the strict transform of that component, we have $a_j = 0$ for some $j \in J$.

Since $\Gamma_j \cdot F = 0$ for any $j \in J$ [Liu02, Proposition 9.1.21(a)],

$$\Gamma_j \cdot \Gamma_j = - \sum_{j' \in J \setminus \{j\}} \Gamma_j \cdot \Gamma_{j'}.$$

Intersecting (6.13) with Γ_j on $\widetilde{Z}_{\widetilde{R}}$, we obtain

$$H \cdot \Gamma_j + \sum_{j' \in J \setminus \{j\}} (a_{j'} - a_j) \Gamma_j \cdot \Gamma_{j'} = m \deg(\widetilde{\mathcal{L}}|_{\Gamma_j}) \quad \text{for any } j \in J. \tag{6.14}$$

Here $H \cdot \Gamma_j = \deg(H|_{\Gamma_j}) \ge 0$ and $\deg(\widetilde{\mathcal{L}}|_{\Gamma_j}) \ge 0$. Moreover $\deg(\widetilde{\mathcal{L}}|_{\Gamma_j}) = 0$ if Γ_j is exceptional over Z, and the other Γ_j are birational to the irreducible components of X_0''. Since f is flat,

$$\sum_{j \in J} \deg(\widetilde{\mathcal{L}}|_{\Gamma_j}) = \deg(\mathcal{L}|_{X_0''}) = \deg(\mathcal{L}|_{X_\eta}).$$

The sum of (6.14) over $j \in J$ is $\sum_{j \in J} H \cdot \Gamma_j = m \deg(\mathcal{L}|_{X_\eta})$, hence

$$H \cdot \Gamma_j \le m \deg(\mathcal{L}|_{X_\eta}) \quad \text{for any } j \in J. \tag{6.15}$$

Let $j, j' \in J$ such that $\Gamma_j \cap \Gamma_{j'} \ne \emptyset$ and $a_j > a_{j'}$, and let $J_+ := \{i \in J \mid a_i \ge a_j\}$. In the sum of (6.14) over $i \in J_+$, the terms of pairs in J_+ cancel, and we obtain

$$\sum_{i \in J_+} \sum_{i' \in J \setminus J_+} (a_i - a_{i'}) \Gamma_i \cdot \Gamma_{i'} = \sum_{i \in J_+} \left(H \cdot \Gamma_i - m \deg(\widetilde{\mathcal{L}}|_{\Gamma_i}) \right) \le \sum_{i \in J} H \cdot \Gamma_i = m \deg(\mathcal{L}|_{X_\eta}).$$

Any term on the left is non-negative, as $a_i \ge a_j > a_{i'}$, and the term of $(i, i') = (j, j')$ is $(a_j - a_{j'}) \Gamma_j \cdot \Gamma_{j'} \ge a_j - a_{j'}$. Thus

$$|a_j - a_{j'}| \le m \deg(\mathcal{L}|_{X_\eta}) \quad \text{for any } j, j' \in J \text{ such that } \Gamma_j \cap \Gamma_{j'} \ne \emptyset.$$

Since $a_j = 0$ for some $j \in J$, (6.12) implies

$$\max_{j \in J} a_j \le 2m \deg(\mathcal{L}|_{X_\eta}) \sum_{k=1}^{\#\Pi} N_k. \tag{6.16}$$

Since $\sum_{j\in J} D_j = \operatorname{div}(t)$ on $\widetilde{Z}$ and $\widetilde{w}(t) = \gamma/2$,

$$\widetilde{w}\Big(\sum_{j\in J} a_j D_j\Big) \le \Big(\max_{j\in J} a_j\Big)\widetilde{w}(t) \le \gamma m \deg(\mathcal{L}|_{X_\eta}) \sum_{k=1}^{\#\Pi} N_k. \tag{6.17}$$

Step 5. In this step we bound the horizontal part of the divisor of the section and conclude.

By (6.15), for any $j \in J$, any closed point of Γ_j has coefficient at most $m\deg(\mathcal{L}|_{X_\eta})$ in $H|_{\Gamma_j}$. Possibly shrinking T° to a neighborhood of η'', we may assume that for any $j \in J$, the components of $H|_{D_j}$ are smooth, pairwise disjoint and dominate $\widetilde{P}$. Then they are the closures of the points of $\operatorname{Supp} H|_{\Gamma_j}$, and for any component H' of $H|_{D_j}$, $\operatorname{mult}_{H'}(H|_{D_j})$ is the coefficient of the generic point of H' in $H|_{\Gamma_j}$, so

$$\Big(D_j, \frac{1}{m\deg(\mathcal{L}|_{X_\eta})} H|_{D_j}\Big) \text{ is lc for any } j \in J.$$

Since $\widetilde{Z}$ is smooth, D_j is a smooth prime divisor and $D_j \not\subset \operatorname{Supp} H$, [EMY03, Theorem 0.1] implies

$$\operatorname{mld}\Big(V; \widetilde{Z}, D_j + \frac{1}{m\deg(\mathcal{L}|_{X_\eta})} H\Big) = \operatorname{mld}\Big(V; D_j, \frac{1}{m\deg(\mathcal{L}|_{X_\eta})} H|_{D_j}\Big) \ge 0$$

for any proper closed subset $V \subset D_j$, where $\operatorname{mld}(V; Y, \Delta)$ denotes the infimum of $a(E, Y, \Delta)$ over the prime divisors E over Y with $\operatorname{center}_Y E \subset V$ [EMY03, Definition 1.1]. Since $\widetilde{w}(t) > 0$, $\widetilde{c} \in D_j$ for some $j \in J$. As $\widetilde{w}$ is a divisorial valuation,

$$a\Big(\widetilde{w}, \widetilde{Z}, D_j + \frac{1}{m\deg(\mathcal{L}|_{X_\eta})} H\Big) \ge 0.$$

If $\widetilde{c}$ is the generic point of D_j, this holds since $D_j \not\subset \operatorname{Supp} H$ and then

$$\widetilde{w}(\zeta) = \frac{\gamma}{2} \operatorname{mult}_{D_j} \operatorname{div}(\zeta) \quad \text{for any non-zero rational function } \zeta \text{ on } \widetilde{Z}.$$

Thus

$$\widetilde{w}(H) \le m\deg(\mathcal{L}|_{X_\eta})\big(a(\widetilde{w}, \widetilde{Z}, 0) - \widetilde{w}(D_j)\big). \tag{6.18}$$

Since $\pi^* e$ is a local generator of $\widetilde{\mathcal{L}}$ at $\widetilde{c}$, $\pi_* \widetilde{w} = v$, $s = q^{i_0} s_0$, $v(q) = \gamma$, and

$$v(b_\alpha) = \gamma \operatorname{mult}_P \operatorname{div}(b_\alpha \circ \mu) \quad \text{for any } \alpha,$$

(6.13) implies

$$v(s/e^m) = \min_\alpha v(b_\alpha) + \widetilde{w}(H) + \widetilde{w}\Big(\sum_{j\in J} a_j D_j\Big). \tag{6.19}$$

Suppose that $\Pi \ne \emptyset$. Then $\sum_{k=1}^{\#\Pi} N_k \ge 1$. By (6.18) with $\widetilde{w}(D_j) \ge 0$, (6.11), (6.17), $a(P, T, 0) > 0$, (6.10), and $\#\Pi \le n_0$,

$$\begin{aligned}
\widetilde{w}(H) + \widetilde{w}\Big(\sum_{j\in J} a_j D_j\Big) &\le m\deg(\mathcal{L}|_{X_\eta})\Big(a(v, X, 0) + \frac{\gamma}{2} + \gamma \sum_{k=1}^{\#\Pi} N_k\Big) \\
&\le m\deg(\mathcal{L}|_{X_\eta})\Big(a(v, X, 0) + \frac{3}{2} \sum_{k=1}^{\#\Pi} \gamma N_k\Big) \\
&\le m\deg(\mathcal{L}|_{X_\eta})\Big(1 + \frac{3n_0}{2\beta}\Big) a(v, X, 0) = C m a(v, X, 0).
\end{aligned}$$

Suppose that $\Pi = \emptyset$. Then $h = \mathrm{id}_Z$ and F is smooth and connected, hence irreducible. Thus $J = \{j\}$, $D_j = \operatorname{div}(t)$, $a_j = 0$, and $\widetilde{w}(D_j) = \gamma/2$. By (6.18) and (6.11),

$$\widetilde{w}(H) \le m\deg(\mathcal{L}|_{X_\eta})\big(a(v, X, 0) - \gamma a(P, T, 0)\big) \le C m a(v, X, 0).$$

In both cases, (6.19) implies

$$v(s/e^m) \le \min_\alpha v(b_\alpha) + Cma(v, X, 0).$$

The lemma follows. □

6.3. **Proof of Theorem 6.1.**

Proof of Theorem 6.1. Possibly shrinking T to an affine neighborhood of t, we may assume that (X, B) is klt. We fix a non-zero rational section ς of $\mathcal{L}$. Let v be a divisorial valuation on X such that $\pi(\mathrm{center}_X\, v) = t$, and let $\xi := \mathrm{center}_X\, v$.

Step 1. In this step we fix a tower of nodal curve families over a smooth base which dominates X, and we bound the log discrepancies of v along the tower by $a(v, X, B)$.

By [dJ97, Theorem 5.9], with the trivial group and the empty closed subset of X, and forgetting the group action and the sections of the pluri nodal fibration [dJ97, Situation 5.8], there exist varieties $X_0, \dots, X_d$ and morphisms

$$X_d \xrightarrow{f_d} X_{d-1} \xrightarrow{f_{d-1}} \cdots \xrightarrow{f_1} X_0 = T', \qquad \mu : T' \to T, \qquad h : X_d \to X,$$

satisfying the following.

(1) μ is projective, generically finite and surjective.
(2) For any $1 \le i \le d$, f_i is projective and flat, the fibers of f_i are reduced curves with only nodes as singularities, and the generic fiber $X_{i,\eta}$ of f_i is smooth and geometrically integral.
(3) h is projective, generically finite and surjective, and $\pi \circ h = \mu \circ f_1 \circ \cdots \circ f_d$.

In (2), the generic fiber of f_i is smooth since f_i is smooth over the complement of a proper closed subset of X_{i-1} [dJ97, Situation 5.8], and in (3), h is projective since $\pi \circ h = \mu \circ f_1 \circ \cdots \circ f_d$ is projective and π is separated. Since flatness and families of nodal curves are preserved by base change, and a flat morphism to a variety whose generic fiber is integral has integral total space, possibly replacing T' with a resolution, $X_1, \dots, X_d$ with the base changes, and h with the composition, we may assume that T' is smooth. Each X_i is projective over T, so the inverse image Σ_i of t in X_i is proper. For $0 \le i \le d-1$, we fix a finite set $\mathcal{U}_i$ of affine open subsets of X_i covering Σ_i.

We show by induction on i that $(X_i, 0)$ is klt. Since $X_0 = T'$ is smooth, $(X_0, 0)$ is klt. Assume that $(X_{i-1}, 0)$ is klt for some $1 \le i \le d$. By Lemma 6.4, X_i is normal, $(X_i, 0)$ is klt, and

$$a((f_i)_* v', X_{i-1}, 0) \le a(v', X_i, 0) \tag{6.20}$$

for any divisorial valuation v' on X_i with a center ξ' on X_i such that $f_i(\xi')$ is not the generic point of X_{i-1}.

Let (Z, Δ_Z) be a klt pair, $\Omega \subset Z$ an open subset, and $g \in \mathcal{O}(\Omega)\backslash\{0\}$. A log resolution of $(\Omega, \Delta_Z|_\Omega + \mathrm{div}(g))$ shows that $(\Omega, \Delta_Z|_\Omega + \lambda\, \mathrm{div}(g))$ is klt for some real number $\lambda > 0$. Since

$$a(v', \Omega, \Delta_Z|_\Omega + \lambda\, \mathrm{div}(g)) = a(v', Z, \Delta_Z) - \lambda v'(g),$$

there exists a real number $C_g > 0$ such that

$$0 \le v'(g) \le C_g a(v', Z, \Delta_Z) \tag{6.21}$$

for any divisorial valuation v' on Z with center in Ω.

Next we lift v to the tower. Write $v = \lambda_E v_E$, where $\lambda_E > 0$ and E is a prime divisor over X. Let $Y \to X$ be a projective birational morphism from a normal variety Y such that E is a prime divisor on Y. Let Y_d be the normalization of Y in $\mathbb{C}(X_d)$. Since $Y_d \to Y$ is finite and surjective, there exists a prime divisor E_d on Y_d mapping onto E, and there exists a positive integer r_E such that

$$\mathrm{mult}_{E_d}\, \mathrm{div}(\phi \circ h) = r_E\, \mathrm{mult}_E\, \mathrm{div}(\phi)$$

for any non-zero rational function ϕ on X. Let $v_d := \frac{\lambda_E}{r_E} v_{E_d}$. Then $h_* v_d = v$. Since h is proper, v_d has a center ξ_d on X_d, and $h(\xi_d) = \xi$. For $0 \le i \le d$, let ξ_i be the image of ξ_d in X_i. By condition (3), ξ_i maps to t, so ξ_i is not the generic point of X_i as $\dim T \ge 1$, and by Lemma 6.3,

$$v_i := (f_{i+1} \circ \cdots \circ f_d)_* v_d$$

is a divisorial valuation on X_i with center ξ_i. Moreover $\xi_i \in \Sigma_i$. Since $(f_i)_* v_i = v_{i-1}$ and $f_i(\xi_i) = \xi_{i-1}$ is not the generic point of X_{i-1} for $1 \le i \le d$, (6.20) implies

$$a(v_0, X_0, 0) \le a(v_1, X_1, 0) \le \cdots \le a(v_d, X_d, 0). \tag{6.22}$$

We let B_d be the $\mathbb{R}$-divisor on X_d such that $K_{X_d} + B_d = h^*(K_X + B)$. The inequality $B_d \ge 0$ possibly fails. Let $\widetilde{X}$ be the normalization of X in $\mathbb{C}(X_d)$. Then h factors through a projective birational morphism $X_d \to \widetilde{X}$ and the finite morphism $\widetilde{h} : \widetilde{X} \to X$. Let $K_{\widetilde{X}} + \widetilde{B} := \widetilde{h}^*(K_X + B)$. Let v' be a divisorial valuation on X_d with a center on X_d, and write $v' = \lambda' v_{P'}$, where $\lambda' > 0$ and P' is a prime divisor over X_d. Since h is generically finite, only the generic point of X_d maps to the generic point of X, so by Lemma 6.3 for h, there exist a prime divisor P over X and a positive integer r such that

$$\operatorname{mult}_{P'} \operatorname{div}(\phi \circ h) = r \operatorname{mult}_P \operatorname{div}(\phi)$$

for any non-zero rational function ϕ on X. By the ramification formula for the finite morphism $\widetilde{h}$, $a(P', \widetilde{X}, \widetilde{B}) = r a(P, X, B)$. Since $K_{X_d} + B_d$ is the pullback of $K_{\widetilde{X}} + \widetilde{B}$,

$$a(v', X_d, B_d) = \lambda' a(P', \widetilde{X}, \widetilde{B}) = \lambda' r a(P, X, B) = a(h_* v', X, B). \tag{6.23}$$

In particular, $a(v_d, X_d, B_d) = a(v, X, B)$.

Since $(X_d, 0)$ is klt and (X_d, B_d) is sub-klt by (6.23), comparing both log discrepancies on a log resolution of (X_d, B_d) implies that there exists a real number $C_0 > 0$ such that $a(v', X_d, 0) \le C_0 a(v', X_d, B_d)$ for any divisorial valuation v' on X_d with a center on X_d. For $v' = v_d$, together with (6.23) and (6.22), we obtain

$$a(v_i, X_i, 0) \le C_0 a(v, X, B) \quad \text{for any } 0 \le i \le d. \tag{6.24}$$

Step 2. In this step we construct subrings $\mathcal{R}_i \subset \mathbb{C}(X_i)$ with bases over $\mathbb{C}(X_0)$ adapted to the tower, and we compute the standard monomials of the homogenization of each level.

For $1 \le k \le n$, let φ_k be the rational function $h^*(\psi(y_k)/\varsigma^{\deg y_k})$ on X_d.

Claim 6.6. *Let $1 \le i \le d$ and $\mathcal{S}$ a finite set of rational functions on X_i. Then there exist rational functions u_i and $\zeta_{i,j}$, $j \in \Lambda_i$, on X_i, where Λ_i is a finite set containing 0 and $\zeta_{i,0} = 1$, such that u_i is transcendental over $\mathbb{C}(X_{i-1})$ and*

$$\mathcal{R}'_i := \bigoplus_{j \in \Lambda_i} \mathbb{C}(X_{i-1})[u_i]\zeta_{i,j}$$

is a direct sum and a subring of $\mathbb{C}(X_i)$ containing $\mathcal{S}$.

Proof. The curve $X_{i,\eta}$ is projective over $\mathbb{C}(X_{i-1})$ with function field $\mathbb{C}(X_i)$. Let $D \ge 0$ be a non-zero divisor on $X_{i,\eta}$ whose support contains the poles of all elements of $\mathcal{S}$. By Serre vanishing, for $N \gg 0$ the restriction

$$H^0(X_{i,\eta}, \mathcal{O}(ND)) \to \bigoplus_{P \in \operatorname{Supp} D} \mathcal{O}(ND) \otimes \kappa(P)$$

is surjective, so there exists a rational function u_i on X_i whose pole divisor on $X_{i,\eta}$ is ND. Since $D \ne 0$, u_i is not constant on the geometrically integral curve $X_{i,\eta}$, so u_i is transcendental over $\mathbb{C}(X_{i-1})$. Then u_i defines a finite morphism $X_{i,\eta} \to \mathbb{P}^1_{\mathbb{C}(X_{i-1})}$, and $X_{i,\eta} \setminus \operatorname{Supp} D$ is the inverse image of $\mathbb{A}^1_{\mathbb{C}(X_{i-1})}$. Thus $X_{i,\eta} \setminus \operatorname{Supp} D = \operatorname{Spec} \mathcal{R}'_i$, where $\mathcal{R}'_i$ is the integral closure of $\mathbb{C}(X_{i-1})[u_i]$ in $\mathbb{C}(X_i)$, and $\mathcal{S} \subset \mathcal{R}'_i$. Since $\mathcal{R}'_i$ is a finitely generated torsion-free module over the principal ideal domain $\mathbb{C}(X_{i-1})[u_i]$, $\mathcal{R}'_i$ is free. Moreover, $\mathcal{R}'_i / \mathbb{C}(X_{i-1})[u_i]$ is torsion-free, since an element of

$\mathcal{R}_i' \cap \mathbb{C}(X_{i-1})(u_i)$ belongs to the integrally closed ring $\mathbb{C}(X_{i-1})[u_i]$. Thus $\mathcal{R}_i'/\mathbb{C}(X_{i-1})[u_i]$ is free, and lifts of a basis of $\mathcal{R}_i'/\mathbb{C}(X_{i-1})[u_i]$ together with $\zeta_{i,0} := 1$ form a basis $\{\zeta_{i,j}\}_{j\in\Lambda_i}$ of $\mathcal{R}_i'$. □

Let $\mathcal{S}_d := \{\varphi_1, \dots, \varphi_n\}$. For $i = d, d-1, \dots, 1$ in turn, by Claim 6.6 we fix rational functions u_i and $\zeta_{i,j}$, $j \in \Lambda_i$, on X_i which satisfy the conclusion of Claim 6.6 for $\mathcal{S} = \mathcal{S}_i$, and we let $\mathcal{S}_{i-1} \subset \mathbb{C}(X_{i-1})$ be the finite set of coefficients of the elements of $\mathcal{S}_i$ and of $\zeta_{i,j}\zeta_{i,j'}$, $j, j' \in \Lambda_i$, in the $\mathbb{C}(X_{i-1})$-basis $\{u_i^q\zeta_{i,j}\}_{q\in\mathbb{N},j\in\Lambda_i}$ of $\mathcal{R}_i'$. Let $\mathcal{R}_0 := \mathbb{C}(X_0)$, and for $1 \le i \le d$ let

$$\mathcal{R}_i := \bigoplus_{j\in\Lambda_i} \mathcal{R}_{i-1}[u_i]\zeta_{i,j} \subset \mathbb{C}(X_i).$$

By induction on i, $\mathcal{R}_i$ is a subring of $\mathbb{C}(X_i)$ containing $\mathcal{S}_i$: if $\mathcal{S}_{i-1} \subset \mathcal{R}_{i-1}$, then the products $\zeta_{i,j}\zeta_{i,j'}$ and the elements of $\mathcal{S}_i$ have coefficients in $\mathcal{R}_{i-1}$, and the sum is direct since $\{u_i^q\zeta_{i,j}\}_{q,j}$ is linearly independent over $\mathbb{C}(X_{i-1})$. In particular $\varphi_k \in \mathcal{R}_d$ for $1 \le k \le n$. Let $\mathcal{T}_i$ be the set of tower monomials

$$\tau = \prod_{i'=1}^{i} u_{i'}^{q_{i'}}\zeta_{i',j_{i'}}, \qquad q_{i'} \in \mathbb{N},\ j_{i'} \in \Lambda_{i'}.$$

By induction on i, $\mathcal{T}_i$ is a $\mathbb{C}(X_0)$-basis of $\mathcal{R}_i$.

We fix weights from the bottom up. Assume that integers $M_1, \dots, M_{i-1} \ge 1$ are fixed. For $\tau = \prod_{i'} u_{i'}^{q_{i'}}\zeta_{i',j_{i'}}$ in $\mathcal{T}_{i-1}$, let

$$\mathrm{wt}(\tau) := \sum_{i'=1}^{i-1}(q_{i'} + \mathrm{wt}(\zeta_{i',j_{i'}})),$$

where $\mathrm{wt}(\zeta_{i',0}) := 0$ and $\mathrm{wt}(\zeta_{i',j}) := M_{i'}$ for $j \ne 0$, and let $\mathcal{R}_{i-1,\le m}$ be the $\mathbb{C}(X_0)$-span of the elements $\tau \in \mathcal{T}_{i-1}$ with $\mathrm{wt}(\tau) \le m$. Since $\mathcal{S}_{i-1} \subset \mathcal{R}_{i-1}$ is finite, we may fix an integer $M_i \ge 1$ such that for any $j, j' \in \Lambda_i\backslash\{0\}$,

$$\zeta_{i,j}\zeta_{i,j'} \in \sum_{0\le q<M_i,\ j''\in\Lambda_i} \mathcal{R}_{i-1,\le M_i-1-q}u_i^q\zeta_{i,j''}. \tag{6.25}$$

After $M_1, \dots, M_d$ are fixed, the definitions of wt and $\mathcal{R}_{i-1,\le m}$ above define $\mathrm{wt}(\tau)$ for $\tau \in \mathcal{T}_i$ and $\mathcal{R}_{i,\le m}$ for any $0 \le i \le d$. In particular $\mathcal{R}_{0,\le m} = \mathbb{C}(X_0)$ for any $m \in \mathbb{N}$. By induction on i,

$$\mathcal{R}_{i,\le m} \cdot \mathcal{R}_{i,\le m'} \subset \mathcal{R}_{i,\le m+m'} \quad \text{for any } m, m' \in \mathbb{N}. \tag{6.26}$$

For the inductive step, the space $\mathcal{R}_{i,\le m}$ is spanned by the elements $Gu_i^q\zeta_{i,j}$ with

$$G \in \mathcal{R}_{i-1,\le m-q-\mathrm{wt}(\zeta_{i,j})}.$$

For $G'u_i^{q'}\zeta_{i,j'}$ with

$$G' \in \mathcal{R}_{i-1,\le m'-q'-\mathrm{wt}(\zeta_{i,j'})},$$

if $j = 0$ or $j' = 0$, then the product belongs to $\mathcal{R}_{i,\le m+m'}$ by (6.26) for $i-1$. If $j, j' \ne 0$, then by (6.25) the product is a sum of terms $GG'\gamma u_i^{q+q'+q''}\zeta_{i,j''}$ with $\gamma \in \mathcal{R}_{i-1,\le M_i-1-q''}$, and by (6.26) for $i-1$ each term has weight at most

$$(m-q-M_i) + (m'-q'-M_i) + (M_i-1-q'') + (q+q'+q'') + M_i = m + m' - 1.$$

Let z be a variable. By (6.26),

$$\mathcal{A} := \bigoplus_{m\ge0} \mathcal{R}_{d,\le m}z^m \subset \mathbb{C}(X_d)[z]$$

is a graded $\mathbb{C}(X_0)$-algebra.

Fix $1 \le i \le d$. Let

$$P_i := \mathbb{C}(X_{i-1})[z, U_i, V_{i,j} \mid j \in \Lambda_i\backslash\{0\}],$$

graded by $\deg z = \deg U_i = 1$ and $\deg V_{i,j} = M_i$, and let $V_{i,0} := 1$. Let $\mathcal{A}_i$ be the image and J_i the kernel of the homomorphism $P_i \to \mathbb{C}(X_i)[z]$ sending z, U_i, and $V_{i,j}$ to z, u_iz, and $\zeta_{i,j}z^{M_i}$.

We order the monomials of P_i first by the total degree in $\{V_{i,j}\}_{j\neq 0}$ and then by a lexicographic order. Then for any $m \in \mathbb{N}$:

(6.27) the standard monomials of degree m with respect to J_i are the monomials $z^{m-q-\deg V_{i,j}} U_i^q V_{i,j}$ with $q \in \mathbb{N}$, $j \in \Lambda_i$, and $q + \deg V_{i,j} \leq m$, and the elements $u_i^q \zeta_{i,j} z^m$, which are the images of the standard monomials of degree m, form a $\mathbb{C}(X_{i-1})$-basis of $\mathcal{A}_{i,m}$.

By (6.25), for $j, j' \in \Lambda_i \backslash \{0\}$ there exist $\gamma_{q,j''} \in \mathbb{C}(X_{i-1})$ such that

$$V_{i,j}V_{i,j'} - \sum_{0\leq q<M_i,\ j''\in\Lambda_i} \gamma_{q,j''} z^{2M_i-q-\deg V_{i,j''}} U_i^q V_{i,j''} \in J_i,$$

where $2M_i - q - \deg V_{i,j''} > 0$ as $q < M_i$. The leading monomial of the left hand side is $V_{i,j}V_{i,j'}$, since the other terms have degree at most 1 in $\{V_{i,j}\}_{j\neq 0}$. Thus any standard monomial with respect to J_i belongs to the set $\mathcal{M}_i$ of monomials $z^a U_i^q V_{i,j}$, $a, q \in \mathbb{N}$, $j \in \Lambda_i$. The image of $z^a U_i^q V_{i,j}$ is $u_i^q \zeta_{i,j} z^{a+q+\deg V_{i,j}}$, so distinct elements of $\mathcal{M}_i$ have distinct images, and by Claim 6.6 the images of $\mathcal{M}_i$ are linearly independent over $\mathbb{C}(X_{i-1})$. On the other hand, the images of the standard monomials form a basis of $P_i/J_i \cong \mathcal{A}_i$ by [CLO15, Chapter 5, §3, Proposition 4]. An element of $\mathcal{M}_i$ which is not standard would then be a linear combination of other elements of $\mathcal{M}_i$ modulo J_i. Thus the standard monomials are exactly the elements of $\mathcal{M}_i$, which implies (6.27).

Step 3. In this step we prove a value estimate on each level which is linear in the degree, and we iterate the estimate along the tower.

Fix $1 \leq i \leq d$. Since X_i is projective over the affine variety T, there exist a line bundle $\mathcal{H}_i$ on X_i which is very ample over T and $\sigma_i \in H^0(X_i, \mathcal{H}_i)\backslash\{0\}$ such that $u_i\sigma_i$ and $\zeta_{i,j}\sigma_i$, $j \in \Lambda_i$, are regular sections of $\mathcal{H}_i$: let $\mathcal{H}_i$ be a sufficiently high power of an ample line bundle such that $\mathcal{I} \otimes \mathcal{H}_i$ is globally generated, where $\mathcal{I} \subset \mathcal{O}_{X_i}$ is the non-zero ideal sheaf of local functions g such that gu_i and $g\zeta_{i,j}$ are regular for any j. Then $\mathcal{H}_i$ is very ample over X_{i-1}, and since $M_i \geq 1$, $\zeta_{i,j}\sigma_i^{M_i}$ is regular. The homomorphism

$$\mathcal{A}_i \to \bigoplus_{m\geq 0} H^0(X_{i,\eta}, \mathcal{H}_i^{\otimes m})$$

sending Gz^m to $G\sigma_i^m$ is an injective homomorphism of graded rings.

Let $\Omega \in \mathcal{U}_{i-1}$ and

$$R_\Omega := \bigoplus_{m\geq 0} H^0(f_i^{-1}(\Omega), \mathcal{H}_i^{\otimes m}).$$

Then R_Ω is torsion-free over the Noetherian domain $\mathcal{O}(\Omega)$ as X_i is integral, $R_{\Omega,0} = \mathcal{O}(\Omega)$ as $(f_i)_*\mathcal{O}_{X_i} = \mathcal{O}_{X_{i-1}}$, X_{i-1} being normal and $X_{i,\eta}$ geometrically integral, and R_Ω is a finitely generated $\mathcal{O}(\Omega)$-algebra as $\mathcal{H}_i$ is very ample over X_{i-1}. Thus there exists a surjective homomorphism of graded $\mathcal{O}(\Omega)$-algebras

$$\psi_\Omega : \mathcal{O}(\Omega)[z, U_i, V_{i,j}, Q_1, \ldots, Q_{n_\Omega} \mid j \in \Lambda_i\backslash\{0\}] \to R_\Omega$$

sending z, U_i, and $V_{i,j}$ to σ_i, $u_i\sigma_i$, and $\zeta_{i,j}\sigma_i^{M_i}$, with $\deg Q_1, \ldots, \deg Q_{n_\Omega} \geq 1$. We order the monomials first by the total degree in $Q_1, \ldots, Q_{n_\Omega}$, then by the total degree in $\{V_{i,j}\}_{j\neq 0}$, and then by the lexicographic order extending the one of **Step 2**. Let

$$K_\Omega := \ker(\psi_\Omega) \otimes_{\mathcal{O}(\Omega)} \mathbb{C}(X_{i-1}),$$

and let $\mathcal{B}_{\Omega,m}$ be the set of standard monomials of degree m with respect to K_Ω. By Lemma 6.2 with $\mathcal{O}(\Omega)$ in place of S, there exist $g_{i,\Omega} \in \mathcal{O}(\Omega)\backslash\{0\}$ and an integer $N_{i,\Omega} \geq 1$ such that for any $m \geq 0$,

$$R_{\Omega,m} \subset \bigoplus_{\beta\in\mathcal{B}_{\Omega,m}} g_{i,\Omega}^{-N_{i,\Omega}m} \mathcal{O}(\Omega) \cdot \psi_\Omega(\beta). \tag{6.28}$$

Since the homomorphism

$$P_i \to R_\Omega \otimes_{\mathcal{O}(\Omega)} \mathbb{C}(X_{i-1}) = \bigoplus_m H^0(X_{i,\eta}, \mathcal{H}_i^{\otimes m})$$

is the composition of $P_i \to \mathcal{A}_i$ and the injective homomorphism above, $K_\Omega \cap P_i = J_i$. The order compares the total degree in $Q_1, \dots, Q_{n_\Omega}$ first, so the order is of n_Ω-elimination type, and by [CLO15, Chapter 3, §1, Exercise 5], $\mathcal{G} \cap P_i$ is a Gröbner basis of J_i for a Gröbner basis $\mathcal{G}$ of K_Ω. An element of $\mathcal{G}$ whose leading monomial does not involve $Q_1, \dots, Q_{n_\Omega}$ belongs to P_i, so a monomial in $z, U_i, V_{i,j}$ is standard with respect to K_Ω if and only if it is standard with respect to J_i. By (6.27),

$$z^{m-q-\deg V_{i,j}} U_i^q V_{i,j} \in \mathcal{B}_{\Omega,m} \quad \text{for any } q \in \mathbb{N} \text{ and } j \in \Lambda_i \text{ such that } q + \deg V_{i,j} \le m. \tag{6.29}$$

By Lemma 6.5 for f_i, with X_{i-1} and $\mathcal{H}_i$ in place of T and $\mathcal{L}$, there exist an integer $m_i' \ge 1$ and a real number $C_i' > 0$ satisfying the conclusion of Lemma 6.5. The hypotheses hold since $(X_{i-1}, 0)$ is klt by **Step 1**, and f_i satisfies condition (2). Let $\Omega \in \mathcal{U}_{i-1}$ contain ξ_{i-1}. Let $m \ge m_i'$ be an integer, and let $b_{q,j}$, for $q \in \mathbb{N}$ and $j \in \Lambda_i$ with $q + \deg V_{i,j} \le m$, be rational functions on X_{i-1}, not all zero. Let

$$F_i := \sum_{q,j} b_{q,j} u_i^q \zeta_{i,j} \quad \text{and} \quad s := F_i \sigma_i^m = \sum_{q,j} b_{q,j} \psi_\Omega(z^{m-q-\deg V_{i,j}} U_i^q V_{i,j}).$$

Let e_i be a local generator of $\mathcal{H}_i$ at ξ_i. By Lemma 6.5, $(f_i)_* \mathcal{H}_i^{\otimes m}$ has a basis $(s_\alpha)_\alpha$ near ξ_{i-1}, and writing $s = \sum_\alpha b_\alpha s_\alpha$ with rational functions b_α on X_{i-1},

$$v_i(s/e_i^m) \le \min_\alpha v_{i-1}(b_\alpha) + C_i' m a(v_i, X_i, 0).$$

By (6.28) at ξ_{i-1},

$$s_\alpha = \sum_{\beta \in \mathcal{B}_{\Omega,m}} c_{\alpha,\beta} \psi_\Omega(\beta), \qquad c_{\alpha,\beta} \in g_{i,\Omega}^{-N_{i,\Omega} m} \mathcal{O}_{X_{i-1},\xi_{i-1}}.$$

The elements $\psi_\Omega(\beta)$, $\beta \in \mathcal{B}_{\Omega,m}$, are linearly independent over $\mathbb{C}(X_{i-1})$ [CLO15, Chapter 5, §3, Proposition 1(ii)], so by (6.29), $b_{q,j} = \sum_\alpha b_\alpha c_{\alpha,(q,j)}$, where $c_{\alpha,(q,j)}$ is the coefficient of $z^{m-q-\deg V_{i,j}} U_i^q V_{i,j}$. Since $v_{i-1} \ge 0$ on $\mathcal{O}_{X_{i-1},\xi_{i-1}}$,

$$\min_{q,j} v_{i-1}(b_{q,j}) \ge \min_\alpha v_{i-1}(b_\alpha) - N_{i,\Omega} m v_{i-1}(g_{i,\Omega}).$$

Since σ_i / e_i is regular at ξ_i, $v_i(F_i) \le v_i(s/e_i^m)$. By (6.21) for $g_{i,\Omega}$ on X_{i-1} and by (6.24),

$$v_i(F_i) \le \min_{q,j} v_{i-1}(b_{q,j}) + (N_{i,\Omega} C_{g_{i,\Omega}} + C_i') C_0 m a(v, X, B). \tag{6.30}$$

Let

$$C_i := m_i' C_0 \max_{\Omega \in \mathcal{U}_{i-1}} (N_{i,\Omega} C_{g_{i,\Omega}} + C_i').$$

For any integer $m \ge 1$, a sum $\sum_{q+\deg V_{i,j} \le m} b_{q,j} u_i^q \zeta_{i,j}$ is also a sum over $q + \deg V_{i,j} \le m_i' m$, so (6.30) holds with $m_i' m$ in place of m. Thus for any integer $m \ge 1$ and any $b_{q,j} \in \mathbb{C}(X_{i-1})$,

$$v_i\Big(\sum_{q+\deg V_{i,j} \le m} b_{q,j} u_i^q \zeta_{i,j} \Big) \le \min_{q,j} v_{i-1}(b_{q,j}) + C_i m a(v, X, B). \tag{6.31}$$

Now let $m \ge 1$. We show by induction on i that for any $G \in \mathcal{R}_{i,\le m}$ with coefficients γ_τ' in $\mathcal{T}_i$,

$$v_i(G) \le \min_\tau v_0(\gamma_\tau') + (C_1 + \cdots + C_i) m a(v, X, B).$$

The case $i = 0$ is trivial. For $i \ge 1$, write $G = \sum_{q,j} G_{q,j} u_i^q \zeta_{i,j}$ with

$$G_{q,j} \in \mathcal{R}_{i-1,\le m-q-\mathrm{wt}(\zeta_{i,j})} \subset \mathcal{R}_{i-1,\le m}.$$

The coefficients of the $G_{q,j}$ in $\mathcal{T}_{i-1}$ are exactly the γ'_τ, divided according to (q,j), since $\mathcal{T}_i$ is a basis. Thus (6.31) for $b_{q,j} = G_{q,j}$ and the inductive hypothesis for the $G_{q,j}$ imply the inequality for G. For $i = d$ we obtain, for any $G \in \mathcal{R}_{d,\le m}$ with coefficients γ_τ in $\mathcal{T}_d$,

$$v_d(G) \le \min_\tau v_0(\gamma_\tau) + (C_1 + \cdots + C_d) m a(v, X, B). \tag{6.32}$$

Step 4. In this step we return to the standard monomials $\mathcal{B}_m$ and conclude.

Let $I := \ker(\psi) \otimes_{\mathcal{O}_{T,t}} \mathbb{C}(T)$ and X_η the generic fiber of π, so that

$$\mathbb{C}(T)[y_1, \dots, y_n]/I \cong \bigoplus_m H^0(X_\eta, \mathcal{L}^{\otimes m}).$$

Since $\varphi_k \in \mathcal{R}_d$, we may fix an integer $N_0 \ge 1$ such that $\varphi_k \in \mathcal{R}_{d,\le N_0 \deg y_k}$ for $1 \le k \le n$. Let

$$\Phi : \mathbb{C}(T')[y_1, \dots, y_n] \to \mathcal{A}$$

be the homomorphism sending y_k to $\varphi_k z^{N_0 \deg y_k}$, which is well defined by (6.26). For any monomial β of degree m, $\Phi(\beta) = h^*(\psi(\beta)/\varsigma^m) z^{N_0 m}$. We show that $\ker(\Phi) = I \otimes_{\mathbb{C}(T)} \mathbb{C}(T')$. The homomorphism Φ factors through

$$\bigoplus_m H^0(X_\eta, \mathcal{L}^{\otimes m}) \otimes_{\mathbb{C}(T)} \mathbb{C}(T') = \bigoplus_m H^0(X'_\eta, \mathcal{L}^{\otimes m}), \qquad X'_\eta := X_\eta \times_{\operatorname{Spec} \mathbb{C}(T)} \operatorname{Spec} \mathbb{C}(T'),$$

and X'_η is integral since X_η is geometrically integral. By condition (3), h induces a proper morphism from the generic fiber of $X_d \to T'$ to X'_η which is generically finite onto the image, and both have dimension d, so the morphism is surjective, hence dominant. Thus the pullback of a non-zero rational function on X'_η is a non-zero rational function on X_d, so the second homomorphism, sending s to $h^*(s/\varsigma^m) z^{N_0 m}$, is injective. By Buchberger's criterion [CLO15, Chapter 2, §6, Theorem 6], which only involves divisions in $\mathbb{C}(T)[y_1, \dots, y_n]$ and does not depend on the coefficient field, the reduced Gröbner basis of I with respect to the given lexicographic order is a Gröbner basis of $I \otimes_{\mathbb{C}(T)} \mathbb{C}(T')$. Thus $\mathcal{B}_m$ is the set of standard monomials of degree m with respect to $\ker(\Phi)$.

Let $\Omega \in \mathcal{U}_0$. Let R'_Ω be the $\mathcal{O}(\Omega)$-subalgebra of $\mathcal{A}$ generated by z, $u_i z$, $\zeta_{i,j} z^{M_i}$, and $\Phi(y_k)$, for $1 \le i \le d$, $j \in \Lambda_i \backslash \{0\}$, and $1 \le k \le n$ respectively. Then R'_Ω is a graded $\mathcal{O}(\Omega)$-algebra which is torsion-free as $\mathcal{A}$ is a $\mathbb{C}(X_0)$-vector space, and we let

$$\psi'_\Omega : \mathcal{O}(\Omega)[z, U_i, V_{i,j}, y_1, \dots, y_n \mid 1 \le i \le d,\ j \in \Lambda_i \backslash \{0\}] \to R'_\Omega$$

be the surjective homomorphism sending z, U_i, $V_{i,j}$, and y_k to z, $u_i z$, $\zeta_{i,j} z^{M_i}$, and $\Phi(y_k)$, where $\deg z = \deg U_i = 1$, $\deg V_{i,j} = M_i$, and y_k has degree $N_0 \deg y_k$. We order the monomials first by the total degree in the tower variables $z, U_i, V_{i,j}$, and then by a lexicographic order which compares $y_1, \dots, y_n$ as the given lexicographic order. Let

$$K'_\Omega := \ker(\psi'_\Omega) \otimes_{\mathcal{O}(\Omega)} \mathbb{C}(T'),$$

and let $\mathcal{B}'_{\Omega,m}$ be the set of standard monomials of degree m with respect to K'_Ω. By Lemma 6.2 with $\mathcal{O}(\Omega)$ in place of S, there exist $g_\Omega \in \mathcal{O}(\Omega) \backslash \{0\}$ and an integer $N_\Omega \ge 1$ such that for any $m \ge 0$,

$$R'_{\Omega,m} \subset \bigoplus_{\beta' \in \mathcal{B}'_{\Omega,m}} g_\Omega^{-N_\Omega m} \mathcal{O}(\Omega) \cdot \psi'_\Omega(\beta'). \tag{6.33}$$

The order compares the total degree in the tower variables first, so the order is of elimination type for the tower variables, and by [CLO15, Chapter 3, §1, Exercise 5], a monomial in $y_1, \dots, y_n$ is standard with respect to K'_Ω if and only if it is standard with respect to $K'_\Omega \cap \mathbb{C}(T')[y_1, \dots, y_n] = \ker(\Phi)$. Thus $\mathcal{B}_m \subset \mathcal{B}'_{\Omega, N_0 m}$.

Let $m \ge 1$ be an integer, b_β for $\beta \in \mathcal{B}_m$ rational functions on T, not all zero, and

$$s := \sum_{\beta \in \mathcal{B}_m} b_\beta \psi(\beta) \quad \text{and} \quad F := h^*(s/\varsigma^m) = \sum_{\beta \in \mathcal{B}_m} b_\beta h^*(\psi(\beta)/\varsigma^m).$$

By (6.26) and the choice of N_0, $F \in \mathcal{R}_{d,\le N_0 m}$. Write

$$F = \sum_{\tau \in \mathcal{T}_d,\ \mathrm{wt}(\tau) \le N_0 m} \gamma_\tau \tau,$$

where the γ_τ are rational functions on X_0. Let $\Omega \in \mathcal{U}_0$ contain ξ_0. For $\tau = \prod_i u_i^{q_i} \zeta_{i,j_i}$,

$$\tau z^{N_0 m} = \psi'_\Omega\Big(z^{N_0 m - \mathrm{wt}(\tau)} \prod_i U_i^{q_i} V_{i,j_i} \Big) \in R'_{\Omega, N_0 m}.$$

By (6.33),

$$\tau z^{N_0 m} = \sum_{\beta'} c_{\tau,\beta'} \psi'_\Omega(\beta'), \qquad c_{\tau,\beta'} \in g_\Omega^{-N_\Omega N_0 m} \mathcal{O}(\Omega).$$

On the other hand,

$$F z^{N_0 m} = \sum_{\beta \in \mathcal{B}_m} b_\beta \Phi(\beta) = \sum_{\beta \in \mathcal{B}_m} b_\beta \psi'_\Omega(\beta).$$

Since the elements $\psi'_\Omega(\beta')$, $\beta' \in \mathcal{B}'_{\Omega, N_0 m}$, are linearly independent over $\mathbb{C}(T')$ [CLO15, Chapter 5, §3, Proposition 1(ii)] and $\mathcal{B}_m \subset \mathcal{B}'_{\Omega, N_0 m}$, we have $b_\beta = \sum_\tau \gamma_\tau c_{\tau,\beta}$ for any $\beta \in \mathcal{B}_m$. Since $\xi_0 \in \Omega$, $v_0 \ge 0$ on $\mathcal{O}(\Omega)$, and $v_0(b_\beta) = v(b_\beta)$. Let

$$C_{d+1} := C_0 \max_{\Omega \in \mathcal{U}_0} N_\Omega C_{g_\Omega}.$$

By (6.21) for g_Ω on X_0 and by (6.24),

$$\min_\tau v_0(\gamma_\tau) \le \min_{\beta \in \mathcal{B}_m} v(b_\beta) + N_\Omega N_0 m v_0(g_\Omega) \le \min_{\beta \in \mathcal{B}_m} v(b_\beta) + C_{d+1} N_0 m a(v, X, B). \tag{6.34}$$

Since $F = h^*(s/\varsigma^m)$ and $v_d(h^*\phi) = v(\phi)$ for any non-zero rational function ϕ on X (**Step 1**), (6.34) and (6.32) for $G = F$ with $N_0 m$ in place of m imply

$$v(s/\varsigma^m) = v_d(F) \le \min_{\beta \in \mathcal{B}_m} v(b_\beta) + N_0(C_1 + \cdots + C_{d+1}) m a(v, X, B). \tag{6.35}$$

Finally we replace ς by a local generator e of $\mathcal{L}$ at ξ. Let $\mathcal{U}$ be a finite set of affine open subsets of X covering X such that $\mathcal{L}|_\Omega$ has a generator e_Ω for any $\Omega \in \mathcal{U}$, and write $\varsigma/e_\Omega = g_{\Omega,1}/g_{\Omega,2}$ with $g_{\Omega,1}, g_{\Omega,2} \in \mathcal{O}(\Omega)\backslash\{0\}$. By (6.21) for (X, B) and $g_{\Omega,1}, g_{\Omega,2}$,

$$0 \le v(g_{\Omega,1}), v(g_{\Omega,2}) \le C_{d+2} a(v, X, B) \quad \text{whenever } \xi \in \Omega,$$

where

$$C_{d+2} := \max_{\Omega \in \mathcal{U}} \max\{C_{g_{\Omega,1}}, C_{g_{\Omega,2}}\}.$$

For $\Omega \in \mathcal{U}$ containing ξ, $v(e/e_\Omega) = 0$, so

$$|v(\varsigma/e)| = |v(g_{\Omega,1}) - v(g_{\Omega,2})| \le C_{d+2} a(v, X, B).$$

Since $v(s/e^m) = v(s/\varsigma^m) + m v(\varsigma/e)$, (6.35) implies

$$v\Big(\sum_{\beta \in \mathcal{B}_m} b_\beta \psi(\beta)/e^m \Big) \le \min_{\beta \in \mathcal{B}_m} v(b_\beta) + C m a(v, X, B), \qquad C := N_0(C_1 + \cdots + C_{d+1}) + C_{d+2},$$

for any integer $m \ge 1$.

The integer N_0 and the constants $C_0, \ldots, C_{d+2}$ do not depend on v, since $C_1, \ldots, C_{d+2}$ are maxima over the finite sets $\mathcal{U}_0, \ldots, \mathcal{U}_{d-1}$ and $\mathcal{U}$. Thus the theorem holds with $m_0 := 1$ and the constant C above. □

7. The lower bound

Theorem 7.1. *Notation and conditions as in Set-up 2.8, and assume that $\ell \ge 1$. Then*

$$\liminf_{\epsilon \to 0^+} \frac{\widehat{\mathrm{vol}}(X \ni x, B_\epsilon)}{n^n \epsilon^k} \ge \frac{\widehat{\mathrm{vol}}(X \ni w, \Delta) \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{k^k \ell^\ell}.$$

We prove Theorem 7.1 in Subsection 7.5.

7.1. **Gauss valuations along a stratum.**

Set-up 7.2. Notation and conditions as in Set-up 2.8. Let h, B_Y, E, $E_1, \dots, E_m$, and S be as follows.

(1) $\ell \geq 1$.
(2) $h : Y \to X$ is a log resolution of $(X, B + D)$ such that $\mathcal{I}_W \mathcal{O}_Y$ is invertible, and $K_Y + B_Y := h^*(K_X + B)$.
(3) $E = \sum_j E_j$ is the reduced divisor on Y with
$$\operatorname{Supp} E = \operatorname{Exc}(h) \cup \operatorname{Supp} h_*^{-1}(B + D) \cup \operatorname{Supp}(\mathcal{O}_Y/\mathcal{I}_W \mathcal{O}_Y),$$
and E is snc.
(4) $m \geq 1$, and $E_1, \dots, E_m$ are components of E with $\operatorname{mult}_{E_j} B_Y = 1$ for $1 \leq j \leq m$.
(5) S is an irreducible component of $E_1 \cap \cdots \cap E_m$ with $h(S) = W$.

Let $K_S + B_S := (K_Y + B_Y)|_S$.

Since E is snc, S is smooth and $\dim S = n - m$. Since $h(S) = W$, $n - m \geq \ell$, that is, $m \leq k$. Since h is projective, S is projective over W.

Construction-Proposition 7.3. Let X be a normal variety of dimension n, $h : Y \to X$ a projective birational morphism from a smooth variety Y, $E = \sum_j E_j$ a reduced snc divisor on Y, $m \geq 1$, $E_1, \dots, E_m$ components of E, and S an irreducible component of $E_1 \cap \cdots \cap E_m$. Let $Y^\circ \subset Y$ be an affine open subset with $S \cap Y^\circ \neq \emptyset$, and $y_1, \dots, y_{n-m}, z_1, \dots, z_m \in \mathcal{O}(Y^\circ)$ functions defining an étale morphism $Y^\circ \to \mathbb{A}^n$ such that
$$E_j \cap Y^\circ = \{z_j = 0\} \quad \text{for } 1 \leq j \leq m \quad \text{and} \quad S^\circ := S \cap Y^\circ = \{z_1 = \cdots = z_m = 0\}.$$
Then:

(1) there exists a unique homomorphism ι from $\mathcal{O}(S^\circ)$ to the completion of $\mathcal{O}(Y^\circ)$ along S° which lifts the identity of $\mathcal{O}(S^\circ)$ and maps $y_i|_{S^\circ}$ to y_i for $1 \leq i \leq n - m$;
(2) every $f \in \mathcal{O}(Y^\circ)$ has a unique expansion $f = \sum_{\beta \in \mathbb{N}^m} \iota(f_\beta) z^\beta$ with $f_\beta \in \mathcal{O}(S^\circ)$;
(3) for every valuation θ on S with a center on S° and every $\boldsymbol{\alpha} \in \mathbb{R}^m_{>0}$, there exist valuations $\mu_{\theta,\boldsymbol{\alpha}}$ and $u_{\boldsymbol{\alpha}}$ on X with
$$\mu_{\theta,\boldsymbol{\alpha}}(f) = \min_\beta \{\theta(f_\beta) + \boldsymbol{\alpha} \cdot \beta\} \quad \text{and} \quad u_{\boldsymbol{\alpha}}(f) = \min\{\boldsymbol{\alpha} \cdot \beta \mid f_\beta \neq 0\} \quad \text{for every } f \in \mathcal{O}(Y^\circ) \backslash \{0\};$$
(4) for every $g \in \mathcal{O}(S^\circ)$, $\iota(g)$ is a root of a non-zero polynomial in one variable with coefficients in $\mathbb{C}[y_1, \dots, y_{n-m}]$;
(5) $u_{\boldsymbol{\alpha}}$ is the quasi-monomial valuation (Definition 2.2(6)) at the generic point of S in the regular system of parameters $z_1, \dots, z_m$ with weights $\boldsymbol{\alpha}$; in particular, $u_{\boldsymbol{\alpha}}$ does not depend on the choice of Y°, $y_1, \dots, y_{n-m}$, and $z_1, \dots, z_m$.

Proof. Let $I \subset \mathcal{O}(Y^\circ)$ be the ideal of S°, $\widehat{\mathcal{O}(Y^\circ)} := \varprojlim_N \mathcal{O}(Y^\circ)/I^N$ the completion of $\mathcal{O}(Y^\circ)$ along S°, and $y := (y_1, \dots, y_{n-m})$. Since $(y, z_1, \dots, z_m)$ is étale, the subscheme $\{z_1 = \cdots = z_m = 0\}$ of Y° is smooth, so $I = (z_1, \dots, z_m)$, $z_1, \dots, z_m$ is a regular sequence, and $y|_{S^\circ} : S^\circ \to \mathbb{A}^{n-m}$ is étale.

(1) Since $y|_{S^\circ}$ is étale, it is formally étale [Gro67, Définition 17.1.1, Définition 17.3.1]. Thus for any $N \geq 1$, any $\mathbb{C}[y]$-algebra homomorphism $\mathcal{O}(S^\circ) \to \mathcal{O}(Y^\circ)/I^N$ lifts uniquely to a $\mathbb{C}[y]$-algebra homomorphism $\mathcal{O}(S^\circ) \to \mathcal{O}(Y^\circ)/I^{N+1}$, as $(I^N/I^{N+1})^2 = 0$. Starting from the identity of $\mathcal{O}(S^\circ) = \mathcal{O}(Y^\circ)/I$, the lifts define ι, and ι is unique.

(2) Since $z_1, \dots, z_m$ is a regular sequence generating I, I^N/I^{N+1} is a free $\mathcal{O}(S^\circ)$-module with basis
$$\{z^\beta \mid \beta \in \mathbb{N}^m,\ \beta_1 + \cdots + \beta_m = N\} \quad \text{for any } N \geq 0.$$
Since ι lifts the identity of $\mathcal{O}(S^\circ)$, the homomorphism
$$\mathcal{O}(S^\circ)[[z_1, \dots, z_m]] \to \widehat{\mathcal{O}(Y^\circ)}, \qquad g \to \iota(g), \quad z_j \to z_j, \tag{7.1}$$

induces isomorphisms between the N-th graded pieces of the $(z_1,\dots,z_m)$-adic filtration and of the I-adic filtration for any $N\geq 0$. Thus (7.1) is an isomorphism, which is the existence and uniqueness of the expansion.

(3) Let $f\in\mathcal{O}(Y^\circ)\backslash\{0\}$ and $\beta_0\in\mathbb{N}^m$ with $f_{\beta_0}\neq 0$. Since $\theta\geq 0$ on $\mathcal{O}(S^\circ)$ and $\boldsymbol{\alpha}\in\mathbb{R}^m_{>0}$, only finitely many β satisfy

$$\theta(f_\beta)+\boldsymbol{\alpha}\cdot\beta\leq\theta(f_{\beta_0})+\boldsymbol{\alpha}\cdot\beta_0,$$

so the minimum defining $\mu_{\theta,\boldsymbol{\alpha}}(f)$ is attained. By (7.1), the expansion of a product is the product of the power series. We call $\theta(f_\beta)+\boldsymbol{\alpha}\cdot\beta$ the *weight* of the term $\iota(f_\beta)z^\beta$. Let

$$\operatorname{gr}_\theta\mathcal{O}(S^\circ):=\bigoplus_{\lambda\in\mathbb{R}}\{g\in\mathcal{O}(S^\circ)\mid\theta(g)\geq\lambda\}/\{g\in\mathcal{O}(S^\circ)\mid\theta(g)>\lambda\},$$

a domain as θ is a valuation, and let $Z_1,\dots,Z_m$ be variables, Z_j of degree α_j. The sum of the terms of minimal weight of f defines a non-zero homogeneous element of the domain $\operatorname{gr}_\theta\mathcal{O}(S^\circ)[Z_1,\dots,Z_m]$, in which the term $\iota(f_\beta)z^\beta$ corresponds to the image of f_β in degree $\theta(f_\beta)$ times Z^β. The element defined by ff' is the product of the elements defined by f and f', and hence

$$\mu_{\theta,\boldsymbol{\alpha}}(ff')=\mu_{\theta,\boldsymbol{\alpha}}(f)+\mu_{\theta,\boldsymbol{\alpha}}(f')\quad\text{for any } f,f'\in\mathcal{O}(Y^\circ)\backslash\{0\}.$$

The inequality

$$\mu_{\theta,\boldsymbol{\alpha}}(f+f')\geq\min\{\mu_{\theta,\boldsymbol{\alpha}}(f),\mu_{\theta,\boldsymbol{\alpha}}(f')\}$$

holds term by term. Thus $\mu_{\theta,\boldsymbol{\alpha}}$ extends to a valuation on X. Replacing $\theta(f_\beta)$ with 0 for $f_\beta\neq 0$ in the proof for $\mu_{\theta,\boldsymbol{\alpha}}$ shows that $u_{\boldsymbol{\alpha}}$ extends to a valuation on X.

(4) Since $y|_{S^\circ}:S^\circ\to\mathbb{A}^{n-m}$ is étale, $\dim S^\circ=n-m$, so the image of the morphism $S^\circ\to\mathbb{A}^{n-m+1}$ given by $(y|_{S^\circ},g)$ is not dense, and there exists a polynomial $\Phi_g\in\mathbb{C}[y_1,\dots,y_{n-m}][U]\backslash\{0\}$ with $\Phi_g(y|_{S^\circ},g)=0$. As ι is a ring homomorphism with $\iota(y_i|_{S^\circ})=y_i$,

$$\Phi_g(y_1,\dots,y_{n-m},\iota(g))=0,\quad\text{where }\Phi_g(y_1,\dots,y_{n-m},U)\neq 0\text{ in }\mathcal{O}(Y^\circ)[U].\tag{7.2}$$

(5) Let $\eta_S\in S^\circ$ be the generic point of S. Then $z_1,\dots,z_m$ is a regular system of parameters of $\mathcal{O}_{Y,\eta_S}$, and ι extends to the residue field of $\mathcal{O}_{Y,\eta_S}$, with image a coefficient field of $\widehat{\mathcal{O}}_{Y,\eta_S}$, so the expansion in (2) shows that $u_{\boldsymbol{\alpha}}$ satisfies Definition 2.2(6) with $X'=Y$ and $\eta=\eta_S$. The quasi-monomial valuation depends only on $E_1,\dots,E_m$ at η_S and on $\boldsymbol{\alpha}$, and not on the choice of a coefficient field or of local equations of $E_1,\dots,E_m$ [JM12, Proposition 3.1]. □

To obtain the upper bound for v in Proposition 7.5(1), we need the following lemma.

Lemma 7.4. *Let X be a smooth variety, E a reduced snc divisor on X, and v a valuation on X with $\operatorname{center}_X v=\xi$ and $a(v,X,E)<+\infty$. Then:*

(1) v extends uniquely to a map

$$v:\widehat{\mathcal{O}}_{X,\xi}\backslash\{0\}\to\mathbb{R}_{\geq 0}$$

with $v(fg)=v(f)+v(g)$ and $v(f+g)\geq\min\{v(f),v(g)\}$ for any $f,g\in\widehat{\mathcal{O}}_{X,\xi}\backslash\{0\}$ with $f+g\neq 0$, again denoted by v;

(2) $v(\partial f)\geq v(f)-a(v,X,E)$ for any $f\in\widehat{\mathcal{O}}_{X,\xi}$ and any $\mathbb{C}$-derivation ∂ of $\widehat{\mathcal{O}}_{X,\xi}$ such that $\partial(g)\in g\cdot\widehat{\mathcal{O}}_{X,\xi}$ for any local equation g of a component of E through ξ.

Proof. Our argument follows [OAI26, Proof of Proposition 4.2]. Since $a(v,X,E)<+\infty$, the log discrepancy of v in the sense of [JM12, Section 5] is finite, so v extends uniquely to $\widehat{\mathcal{O}}_{X,\xi}$ [JM12, Corollary 5.11]. The extension is ≥ 0 on $\widehat{\mathcal{O}}_{X,\xi}$. Let $\delta>0$ be a real number with $v\geq\delta$ on $\mathfrak{m}_\xi$. Then $v\geq N'\delta$ on $\mathfrak{m}_\xi^{N'}\widehat{\mathcal{O}}_{X,\xi}$ for any $N'\geq 1$.

Let $d:=\dim X$. Possibly shrinking X to a neighborhood of ξ, we may assume that there exist an étale morphism $(x_1,\dots,x_d):X\to\mathbb{A}^d$ and an integer $0\leq N\leq d$ such that $\{x_1=0\},\dots,\{x_N=0\}$ are the components of E and all of them pass through ξ. Let $\partial_{x_1},\dots,\partial_{x_d}$ be

the derivations of $\mathcal{O}_X$ dual to the basis $dx_1, \dots, dx_d$ of Ω_X, $L_i := x_i\partial_{x_i}$ for $1 \le i \le N$, and $L_i := \partial_{x_i}$ for $N < i \le d$. By the Leibniz rule,

$$\partial(\mathfrak{m}_\xi^{N'}\widehat{\mathcal{O}}_{X,\xi}) \subset \mathfrak{m}_\xi^{N'-1}\widehat{\mathcal{O}}_{X,\xi} \quad \text{for any } N' \ge 1,$$

so ∂ is continuous for the $\mathfrak{m}_\xi$-adic topology and is determined by

$$\partial|_{\mathcal{O}_{X,\xi}} = \sum_{i=1}^{d} \partial(x_i)\partial_{x_i}.$$

Since $\partial(x_i) \in x_i\widehat{\mathcal{O}}_{X,\xi}$ for $1 \le i \le N$, there exist $\phi_1, \dots, \phi_d \in \widehat{\mathcal{O}}_{X,\xi}$ with $\partial = \sum_{i=1}^d \phi_i L_i$, where L_i also denotes the continuous extension of L_i to $\widehat{\mathcal{O}}_{X,\xi}$. Since $v(\phi_i) \ge 0$, it suffices to show

$$v(L_i f) \ge v(f) - a(v, X, E) \tag{7.3}$$

for any $f \in \widehat{\mathcal{O}}_{X,\xi}\backslash\{0\}$ and $1 \le i \le d$.

Let $f \in \widehat{\mathcal{O}}_{X,\xi}\backslash\{0\}$, N' a positive integer with $(N'-1)\delta > v(f)$, and $f' \in \mathcal{O}_{X,\xi}$ with $f - f' \in \mathfrak{m}_\xi^{N'}\widehat{\mathcal{O}}_{X,\xi}$. Then $v(f') = v(f)$ and

$$v(L_i f - L_i f') \ge (N'-1)\delta > v(f) \ge v(f) - a(v, X, E),$$

where $a(v, X, E) \ge 0$ by Lemma 2.4. Thus (7.3) for f' implies (7.3) for f, and we only need to prove (7.3) for $f \in \mathcal{O}_{X,\xi}\backslash\{0\}$.

First assume that there exist a real number $q > 0$, a projective birational morphism $\pi : X' \to X$ from a smooth variety X', and a prime divisor P on X' such that $v(f) = q\,\mathrm{mult}_P \mathrm{div}(f)$ for any non-zero rational function f on X, where $\mathrm{div}(f)$ is taken on X'. We show (7.3) for any non-zero rational function f on X. Let $\lambda_i := dx_i/x_i$ for $1 \le i \le N$, $\lambda_i := dx_i$ for $N < i \le d$, and $\omega := \lambda_1 \wedge \cdots \wedge \lambda_d$. Since ω is a nowhere vanishing section of $\Omega_X^d(\log E)$,

$$\mathrm{mult}_P \mathrm{div}(\pi^*\omega) = a(P, X, E) - 1.$$

Since $df/f = \sum_{i=1}^d (L_i f/f)\lambda_i$, replacing λ_i with df/f in ω yields

$$\lambda_1 \wedge \cdots \wedge \lambda_{i-1} \wedge \frac{df}{f} \wedge \lambda_{i+1} \wedge \cdots \wedge \lambda_d = \frac{L_i f}{f}\omega.$$

At the generic point of P, $\pi^*(df/f)$ and $\pi^*\lambda_1, \dots, \pi^*\lambda_d$ are sections of $\Omega^1_{X'}(\log P)$, so the left-hand side pulls back to a section of $\Omega^d_{X'}(\log P) = \Omega^d_{X'}(P)$ there. Thus

$$\mathrm{mult}_P \mathrm{div}(L_i f) - \mathrm{mult}_P \mathrm{div}(f) + a(P, X, E) - 1 \ge -1,$$

that is,

$$\mathrm{mult}_P \mathrm{div}(L_i f) \ge \mathrm{mult}_P \mathrm{div}(f) - a(P, X, E).$$

Since $a(v, X, E) = qa(P, X, E)$, (7.3) holds for v.

Now let v be arbitrary and $f \in \mathcal{O}_{X,\xi}\backslash\{0\}$. Then $L_i f \in \mathcal{O}_{X,\xi}$, and we may assume that $L_i f \ne 0$, as otherwise (7.3) is trivial. Possibly shrinking X, we may assume that f and $L_i f$ are regular on X. Let (X', D') be a log smooth pair over X [JM12, Definition 3.2] such that D' contains the supports of the pullbacks of E, $\mathrm{div}(f)$, and $\mathrm{div}(L_i f)$, let η' be the generic point of the smallest stratum of D' containing $\mathrm{center}_{X'} v$, and let $v' := r_{X',D'}(v) \in \mathrm{QM}_{\eta'}(X', D')$ be the retraction of v onto the quasi-monomial valuations of (X', D') [JM12, Section 4]. By [JM12, Lemma 4.7, Corollary 5.4],

$$v'(f) = v(f), \quad v'(L_i f) = v(L_i f), \quad v'(g) = v(g), \quad \text{and} \quad A_X(v') \le A_X(v)$$

for any local equation g of a component of E, so $a(v', X, E) \le a(v, X, E)$. By [JM12, Lemma 4.5, Remark 3.9], v' is the limit of divisorial valuations w in $\mathrm{QM}_{\eta'}(X', D')$, with $w(f) \to v'(f)$, $w(L_i f) \to v'(L_i f)$, and $a(w, X, E) \to a(v', X, E)$, since A_X is linear on $\mathrm{QM}_{\eta'}(X', D')$ [JM12,

Proposition 5.1]. The centers on X of v' and of all such w are generizations of ξ. The divisorial case for w and the limit over w imply

$$v(L_i f) = v'(L_i f) \geq v'(f) - a(v', X, E) \geq v(f) - a(v, X, E).$$

The lemma follows. □

Proposition 7.5. *Notation and conditions as in Set-up 7.2, and as in Construction-Proposition 7.3 with h, E, $E_1, \dots, E_m$, and S from Set-up 7.2. Assume that each component of E meeting Y° other than $E_1, \dots, E_m$ is $\{y_i = 0\}$ for some i. Let v be a divisorial valuation on X whose center on Y is a point $\xi \in S^\circ$ with $h(\xi) = x$, let v also denote the extension of v to $\widehat{\mathcal{O}}_{Y,\xi}$ (Lemma 7.4), let θ be the valuation on S with $\theta(g) = v(\iota(g))$ for every $g \in \mathcal{O}(S^\circ)$, where $v(\iota(g))$ is the value of v at the image of $\iota(g)$ under the natural map from the completion of $\mathcal{O}(Y^\circ)$ along S° to $\widehat{\mathcal{O}}_{Y,\xi}$, and let $\boldsymbol{\alpha} := (v(z_1), \dots, v(z_m))$. Then θ is a divisorial valuation on S with $\operatorname{center}_S(\theta) = \xi$, and:*

(1) for every $f \in \mathcal{O}_{Y,\xi}$,

$$\mu_{\theta,\boldsymbol{\alpha}}(f) \leq v(f) \leq \bigl(1 + ma(v, Y, E) / \min_{1 \leq j \leq m} v(z_j)\bigr) \cdot \mu_{\theta,\boldsymbol{\alpha}}(f);$$

(2) $a(\mu_{\theta,\boldsymbol{\alpha}}, X, B) = a(\theta, S, B_S) \leq a(v, X, B)$;

(3) $\mu_{\theta,\boldsymbol{\alpha}}(g) = v(g)$ for every local equation g at ξ of a component of E.

Proof. Since v is a divisorial valuation and Y is smooth, $a(v, Y, E) < +\infty$, so the extension of v to $\widehat{\mathcal{O}}_{Y,\xi}$ exists and is unique by Lemma 7.4(1). Let $\delta > 0$ be a real number with $v \geq \delta$ on $\mathfrak{m}_\xi$. Then $v \geq N\delta$ on $\mathfrak{m}_\xi^N \widehat{\mathcal{O}}_{Y,\xi}$ for any $N \geq 1$. This implies:

(a) if a series $\sum_{i \geq 1} c_i$ converges in $\widehat{\mathcal{O}}_{Y,\xi}$, then $v(\sum_{i \geq 1} c_i) \geq \inf_i v(c_i)$;

(b) for any $f \in \widehat{\mathcal{O}}_{Y,\xi} \backslash \{0\}$, there exists $f' \in \mathcal{O}_{Y,\xi}$ with $v(f') = v(f)$, so $v(\widehat{\mathcal{O}}_{Y,\xi} \backslash \{0\})$ is contained in the value group of v, which is a discrete subgroup of $\mathbb{R}$ as v is a divisorial valuation.

Completing (7.1) at the maximal ideal of ξ, we have

$$\widehat{\mathcal{O}}_{Y,\xi} = \widehat{\mathcal{O}}_{S,\xi}[[z_1, \dots, z_m]],$$

under which $\iota(g)$ corresponds to g for $g \in \mathcal{O}(S^\circ)$ and $f \in \mathcal{O}(Y^\circ)$ corresponds to $\sum_\beta f_\beta z^\beta$. We use the identification above in the rest of the proof. Thus θ is the restriction of v to $\mathcal{O}(S^\circ) \subset \widehat{\mathcal{O}}_{S,\xi}$, $\theta(g) < +\infty$ for $g \neq 0$, and θ is a valuation on S with $\operatorname{center}_S(\theta) = \xi$. Moreover, $v(g) = \theta(g)$ for any non-zero rational function g on S, viewed in $\operatorname{Frac} \widehat{\mathcal{O}}_{S,\xi}$.

θ is a divisorial valuation. Since $h(\xi) = x$ and $\dim W = \ell \geq 1$, ξ is not the generic point of S, so θ is non-trivial. By (b), the value group of θ is contained in the value group of v, hence discrete. Let

$$\mathbb{K} := \mathbb{C}(X)(\iota(\mathbb{C}(S))) \subset \operatorname{Frac} \widehat{\mathcal{O}}_{Y,\xi}.$$

By Construction-Proposition 7.3(4), $\mathbb{K}$ is algebraic over $\mathbb{C}(X)$, so

$$\operatorname{trdeg}_{\mathbb{C}} \mathbb{K} = n \quad \text{and} \quad \operatorname{trdeg}_{\iota(\mathbb{C}(S))} \mathbb{K} = n - \dim S = m.$$

The residue field $\kappa_{v|_{\mathbb{K}}}$ of $v|_{\mathbb{K}}$ contains the residue field of $v|_{\mathbb{C}(X)}$, so $\operatorname{trdeg}_{\mathbb{C}} \kappa_{v|_{\mathbb{K}}} \geq n - 1$. By Abhyankar's inequality [Bou72, Chapter VI, §10.3, Corollary 1] for the extension $\mathbb{K}/\iota(\mathbb{C}(S))$, the residue field κ_θ of θ satisfies

$$\operatorname{trdeg}_{\mathbb{C}} \kappa_\theta \geq (n - 1) - m = \dim S - 1.$$

Thus θ is a divisorial valuation.

(c) Units. Let $g \in \mathcal{O}(Y^\circ)$ with $g(\xi) \neq 0$. Then $g_0 = g|_{S^\circ}$ is a unit of $\mathcal{O}_{S,\xi}$, so $\theta(g_0) = 0$, and since $\mu_{\theta,\boldsymbol{\alpha}} \geq 0$ on $\mathcal{O}(Y^\circ)$, $0 \leq \mu_{\theta,\boldsymbol{\alpha}}(g) \leq \theta(g_0) = 0$. Since any unit of $\mathcal{O}_{Y,\xi}$ is g/g' with $g, g' \in \mathcal{O}(Y^\circ)$ and $g(\xi)g'(\xi) \neq 0$, $\mu_{\theta,\boldsymbol{\alpha}} = v = 0$ on the units of $\mathcal{O}_{Y,\xi}$.

(3). Any local equation at ξ of a component of E is the product of a unit of $\mathcal{O}_{Y,\xi}$ and one of 1, z_j for some $1 \le j \le m$, or y_i for some i. By (c), we only need to consider z_j and y_i. The expansion of z_j is z_j, so $\mu_{\theta,\boldsymbol{\alpha}}(z_j) = \alpha_j = v(z_j)$. Since $y_i = \iota(y_i|_{S^\circ})$, $\mu_{\theta,\boldsymbol{\alpha}}(y_i) = \theta(y_i|_{S^\circ}) = v(y_i)$.

(1). Let $f \in \mathcal{O}_{Y,\xi}\backslash\{0\}$. Write $f = g/g'$ with $g, g' \in \mathcal{O}(Y^\circ)$ and $g'(\xi) \neq 0$. By (c), $\mu_{\theta,\boldsymbol{\alpha}}(f) = \mu_{\theta,\boldsymbol{\alpha}}(g)$ and $v(f) = v(g)$, so we may assume that $f = g \in \mathcal{O}(Y^\circ)$. Since $v(g_\beta z^\beta) = \theta(g_\beta) + \boldsymbol{\alpha}\cdot\beta$ for any β, (a) implies $v(g) \ge \mu_{\theta,\boldsymbol{\alpha}}(g)$.

Let $t := \mu_{\theta,\boldsymbol{\alpha}}(g)$ and $\beta \in \mathbb{N}^m$ with $\theta(g_\beta) + \boldsymbol{\alpha}\cdot\beta = t$. For $1 \le j \le m$, let $N_j := \lfloor t/\alpha_j \rfloor$. Then $\beta_j \le N_j$, as $\alpha_j\beta_j \le t$. Let ∂_{z_j} be the partial derivative of $\widehat{\mathcal{O}}_{S,\xi}[[z_1,\dots,z_m]]$ with respect to z_j. Since $z_j\partial_{z_j}(cz^{\beta'}) = \beta'_j cz^{\beta'}$ for any $c \in \widehat{\mathcal{O}}_{S,\xi}$ and $\beta' \in \mathbb{N}^m$,

$$\prod_{j=1}^{m}\prod_{\gamma\in\{0,\dots,N_j\}\backslash\{\beta_j\}}(z_j\partial_{z_j}-\gamma)g = \prod_{j=1}^{m}\prod_{\gamma\in\{0,\dots,N_j\}\backslash\{\beta_j\}}(\beta_j-\gamma)\cdot g_\beta z^\beta + R, \tag{7.4}$$

where R is the sum of

$$\prod_{j=1}^{m}\prod_{\gamma\in\{0,\dots,N_j\}\backslash\{\beta_j\}}(\beta'_j-\gamma)\cdot g_{\beta'} z^{\beta'}$$

over all $\beta' \in \mathbb{N}^m$ with $\beta'_j > N_j$ for some j. The terms with $\beta' \neq \beta$ and $\beta'_j \le N_j$ for any j vanish. The first term on the right-hand side has value t, as $\prod_j\prod_\gamma(\beta_j-\gamma) \neq 0$. A term of R with $\beta'_j > N_j$ has value at least $\boldsymbol{\alpha}\cdot\beta' \ge \alpha_j(N_j+1) > t$, as $\theta \ge 0$ on $\mathcal{O}(S^\circ)$. By (a), $v(R) \ge \min_{1\le j\le m}\alpha_j(N_j+1) > t$. Thus the left-hand side has value t.

Each $z_j\partial_{z_j}$ is a $\mathbb{C}$-derivation of $\widehat{\mathcal{O}}_{Y,\xi}$ satisfying the hypothesis of Lemma 7.4 for (Y,E): by the description in (3), it suffices to note $z_j\partial_{z_j}(z_{j'}) \in z_{j'}\widehat{\mathcal{O}}_{Y,\xi}$ for $1 \le j' \le m$ and $z_j\partial_{z_j}(y_i) = 0$, as $y_i \in \widehat{\mathcal{O}}_{S,\xi}$. By Lemma 7.4(2) and since $a(v,Y,E) \ge 0$ (Lemma 2.4), for any $f' \in \widehat{\mathcal{O}}_{Y,\xi}$ and $\gamma \in \mathbb{Z}$,

$$v\big((z_j\partial_{z_j}-\gamma)f'\big) \ge \min\{v(f') - a(v,Y,E), v(f')\} = v(f') - a(v,Y,E).$$

The operator on the left-hand side of (7.4) is a product of $\sum_{j=1}^m N_j$ such factors, so $t \ge v(g) - \sum_{j=1}^m N_j\cdot a(v,Y,E)$. Since $N_j \le t/\alpha_j \le t/\min_{1\le j'\le m}\alpha_{j'}$,

$$v(g) \le t + \sum_{j=1}^{m} N_j\cdot a(v,Y,E) \le \Big(1 + \frac{ma(v,Y,E)}{\min_{1\le j\le m} v(z_j)}\Big)\mu_{\theta,\boldsymbol{\alpha}}(g).$$

(2). Since θ is a divisorial valuation, there exist a real number $q > 0$, a projective birational morphism $\pi : S' \to S$ from a smooth variety S' which is an isomorphism outside a reduced snc divisor Q on S', and a component P of Q, such that $\theta(g) = q\,\mathrm{mult}_P\,\mathrm{div}(g)$ for any non-zero rational function g on S, where $\mathrm{div}(g)$ is taken on S'. Let $t_P \in \mathcal{O}_{S',P}$ be a local equation of P. Let

$$\widehat{S} := \operatorname{Spec}\widehat{\mathcal{O}}_{S,\xi}, \quad \widehat{Y} := \operatorname{Spec}\widehat{\mathcal{O}}_{Y,\xi}, \quad \widehat{S}' := S'\times_S\widehat{S}, \quad \widehat{Y}' := S'\times_S\widehat{Y} = \widehat{S}'\times_{\widehat{S}}\widehat{Y},$$

where $\widehat{Y} \to \widehat{S}$ is induced by $\widehat{\mathcal{O}}_{S,\xi} \subset \widehat{\mathcal{O}}_{S,\xi}[[z_1,\dots,z_m]]$. Let $A_{\widehat{Y}}$ be the log discrepancy on $\widehat{Y}$ of [JM12, Section 5].

The morphism $\widehat{Y} \to S\times\mathbb{A}^m$ induced by

$$\mathcal{O}_{S,\xi}[z_1,\dots,z_m] \to \widehat{\mathcal{O}}_{S,\xi}[[z_1,\dots,z_m]]$$

is regular, as it is the composite of a localization and the completion of a local ring essentially of finite type over $\mathbb{C}$ [Mat89, Theorem 32.1(i) and Corollary to Theorem 32.6], and $\widehat{Y}'$ is the base change of $S'\times\mathbb{A}^m$ along it. Since

$$\Big(S'\times\mathbb{A}^m, Q\times\mathbb{A}^m + \sum_{j=1}^{m} S'\times\{z_j=0\}\Big)$$

is log smooth, $\widehat{Y}'$ is regular and $\widehat{G} := Q_{\widehat{Y}'} + \sum_{j=1}^m \widehat{Z}_j$ is snc [Sta, Tags 07C1, 0H7S], where $Q_{\widehat{Y}'}$ is the pullback of Q and $\widehat{Z}_j := \operatorname{div}(z_j)$ on $\widehat{Y}'$. Moreover, $\widehat{Y}' \to \widehat{Y}$ is projective birational and an isomorphism outside $Q_{\widehat{Y}'}$, so $(\widehat{Y}', \widehat{G})$ is a log smooth pair over $\widehat{Y}$ [JM12, Definition 3.2].

Let η be the center of v on $\widehat{Y}'$, which exists as $\widehat{Y}' \to \widehat{Y}$ is projective. Since $v(g) = \theta(g)$ for any non-zero rational function g on S, that is, on S', and $\operatorname{center}_{S'}(\theta)$ is the generic point of P, η maps to the generic point of P. Since $v(z_j) = \alpha_j > 0$, $\eta \in \bigcap_{j=1}^m \widehat{Z}_j$. A component of Q through the generic point of P is P itself, so the components of $\widehat{G}$ through η are $\widehat{Z}_1, \dots, \widehat{Z}_m$ and the component $\widehat{P}_1$ of the pullback of P through η, whose local equation at η is t_P. Let η_1 be the generic point of the component of $\widehat{P}_1 \cap \widehat{Z}_1 \cap \cdots \cap \widehat{Z}_m$ containing η, and η_0 the image of η_1 in $\widehat{S}'$. Since $\widehat{Y}' = \widehat{S}' \times_{\widehat{S}} \widehat{Y}$ and $\widehat{Y} = \operatorname{Spec} \widehat{\mathcal{O}}_{S,\xi}[[z_1, \dots, z_m]]$,

$$\widehat{\mathcal{O}}_{\widehat{Y}',\eta_1} = \widehat{\mathcal{O}}_{\widehat{S}',\eta_0}[[z_1, \dots, z_m]]. \tag{7.5}$$

Since η_0 is the generic point of a component of the pullback of P to $\widehat{S}'$, $\mathcal{O}_{\widehat{S}',\eta_0}$ is a discrete valuation ring with uniformizer t_P. Let $\widehat{\mu}$ be the retraction of v onto the quasi-monomial valuations of $(\widehat{Y}', \widehat{G})$ [JM12, Section 4], that is, the monomial valuation at η_1 in the regular system of parameters $(t_P, z_1, \dots, z_m)$ of $\mathcal{O}_{\widehat{Y}',\eta_1}$ with weights $v(t_P) = \theta(t_P) = q$ and $v(z_j) = \alpha_j$.

We show that $\widehat{\mu}(f) = \mu_{\theta,\boldsymbol{\alpha}}(f)$ for any non-zero rational function f on X. Let $f \in \mathcal{O}(Y^\circ)\backslash\{0\}$, so $f = \sum_\beta f_\beta z^\beta$ in $\widehat{\mathcal{O}}_{\widehat{S}',\eta_0}[[z_1, \dots, z_m]]$ with $f_\beta \in \mathcal{O}(S^\circ)$. For any β with $f_\beta \neq 0$, $\theta(f_\beta) = q \operatorname{mult}_P \operatorname{div}(f_\beta)$ and $t_P^{-\operatorname{mult}_P \operatorname{div}(f_\beta)} f_\beta$ is a unit of $\mathcal{O}_{S',P}$, hence of $\widehat{\mathcal{O}}_{\widehat{S}',\eta_0}$. Thus

$$\widehat{\mu}(f) = \min_\beta \{q \operatorname{mult}_P \operatorname{div}(f_\beta) + \boldsymbol{\alpha} \cdot \beta\} = \mu_{\theta,\boldsymbol{\alpha}}(f).$$

For a prime divisor F over $\widehat{Y}$ (resp. S), let $A_{\widehat{Y}}(F)$ (resp. $A_S(F)$) be the value of $A_{\widehat{Y}}$ (resp. A_S) at the valuation $\operatorname{mult}_F \operatorname{div}(\cdot)$. By [JM12, Proposition 5.1, (5.1)],

$$A_{\widehat{Y}}(\widehat{\mu}) = qA_{\widehat{Y}}(\widehat{P}_1) + \sum_{j=1}^m \alpha_j A_{\widehat{Y}}(\widehat{Z}_j) = qA_S(P) + \sum_{j=1}^m \alpha_j = A_S(\theta) + \sum_{j=1}^m \alpha_j.$$

Here $A_{\widehat{Y}}(\widehat{Z}_j) = 1$ as $\widehat{Y}' \to \widehat{Y}$ is an isomorphism at the generic point of $\widehat{Z}_j$, and

$$A_{\widehat{Y}}(\widehat{P}_1) = 1 + \operatorname{mult}_{\widehat{P}_1} K_{\widehat{Y}'/\widehat{Y}} = 1 + \operatorname{mult}_P K_{S'/S} = A_S(P),$$

since $K_{\widehat{Y}'/\widehat{Y}}$, the divisor of $\operatorname{Fitt}_0(\Omega_{\widehat{Y}'/\widehat{Y}})$ [JM12, Section 1.3], is the pullback of $K_{S'/S}$ under the flat morphism $\widehat{Y}' \to S'$ by [JM12, Lemma 1.2(ii)], as $\widehat{Y} \to S$ is regular, and t_P is a local equation of both P and $\widehat{P}_1$. By [JM12, Proposition 5.13], $A_Y(\mu_{\theta,\boldsymbol{\alpha}}) = A_{\widehat{Y}}(\widehat{\mu})$ and $A_Y(v) = A_{\widehat{Y}}(v)$, and by [JM12, Corollary 5.4], $A_{\widehat{Y}}(\widehat{\mu}) \leq A_{\widehat{Y}}(v)$. Thus

$$A_Y(\mu_{\theta,\boldsymbol{\alpha}}) = A_S(\theta) + \sum_{j=1}^m \alpha_j \leq A_Y(v). \tag{7.6}$$

Since $\operatorname{Supp} B_Y \subset \operatorname{Supp} E$, we write $B_Y = \sum_j b_j E_j$. Then $b_j = 1$ for $1 \leq j \leq m$. Since h is a log resolution of (X, B), Definition 2.2(4) implies

$$a(\mu_{\theta,\boldsymbol{\alpha}}, X, B) = A_Y(\mu_{\theta,\boldsymbol{\alpha}}) - \sum_j b_j \mu_{\theta,\boldsymbol{\alpha}}(E_j) \quad \text{and} \quad a(v, X, B) = A_Y(v) - \sum_j b_j v(E_j).$$

Since $(K_Y + E_1 + \cdots + E_m)|_S = K_S$, $B_S = \sum_{j>m} b_j E_j|_S$. For $j > m$ with $\xi \in E_j$, $E_j \cap Y^\circ = \{y_i = 0\}$ for some i, and $y_i|_{S^\circ}$ is a local equation of $E_j|_S$ at ξ with $\theta(y_i|_{S^\circ}) = \mu_{\theta,\boldsymbol{\alpha}}(y_i)$. Thus

$$a(\theta, S, B_S) = A_S(\theta) - \sum_{j>m} b_j \mu_{\theta,\boldsymbol{\alpha}}(E_j).$$

By (3), $\mu_{\theta,\boldsymbol{\alpha}}(E_j) = v(E_j)$ for any j, and $\mu_{\theta,\boldsymbol{\alpha}}(E_j) = \alpha_j$ for $1 \le j \le m$. Combining the last three equalities with (7.6),

$$\begin{aligned} a(\mu_{\theta,\boldsymbol{\alpha}}, X, B) &= A_S(\theta) - \sum_{j>m} b_j \mu_{\theta,\boldsymbol{\alpha}}(E_j) = a(\theta, S, B_S) \\ &\le A_Y(v) - \sum_{j=1}^{m} \alpha_j - \sum_{j>m} b_j v(E_j) = a(v, X, B). \end{aligned}$$

This proves (2). □

7.2. **Thresholds on strata and the Stein factorization.**

Lemma 7.6. *Let (X, B) be an lc pair, W a minimal lc center of (X, B), and $(W, \Delta_W, \mathbf{M})$ the generalized pair induced by adjunction from (X, B) to W (Definition-Theorem 2.7). Let $h : Y \to X$ be a log resolution of (X, B), $K_Y + B_Y := h^*(K_X + B)$, $m \ge 1$, $E_1, \dots, E_m$ distinct prime divisors on Y with $\operatorname{mult}_{E_j} B_Y = 1$ for $1 \le j \le m$, S an irreducible component of $E_1 \cap \cdots \cap E_m$ with $h(S) = W$, and $K_S + B_S := (K_Y + B_Y)|_S$. Let θ be a divisorial valuation on S with a center ξ on S such that $h(\xi)$ is not the generic point of W. Then $(h|_S)_*\theta$ is a divisorial valuation on W with center $h(\xi)$ on W, and*

$$a((h|_S)_*\theta, W, \Delta_W, \mathbf{M}) \le a(\theta, S, B_S).$$

Proof. By Lemma 6.3 for $h|_S$, $(h|_S)_*\theta$ is a divisorial valuation on W with center $h(\xi)$.

Let $W' \to W$ be a projective birational morphism from a normal variety W' such that the center of $(h|_S)_*\theta$ on W' is the generic point of a prime divisor P on W'. Then there exists a real number $e > 0$ such that $((h|_S)_*\theta)(f) = e \operatorname{mult}_P \operatorname{div}(f)$ for any non-zero rational function f on W, and

$$a((h|_S)_*\theta, W, \Delta_W, \mathbf{M}) = e a(P, W, \Delta_W, \mathbf{M}).$$

We show that we may assume that the induced map $h_S : S \dashrightarrow W'$ is a morphism. Since W is quasi-projective, W' is the blow-up of W along a non-zero ideal sheaf $\mathfrak{J} \subset \mathcal{O}_W$. Let $\widetilde{\mathfrak{J}} \subset \mathcal{O}_X$ be the preimage of $\mathfrak{J}$ under $\mathcal{O}_X \to \mathcal{O}_W$. Since $h(S) = W \not\subset V(\mathfrak{J})$, there exists a projective birational morphism $g : Y' \to Y$ such that $h \circ g$ is a log resolution of (X, B), $\widetilde{\mathfrak{J}} \cdot \mathcal{O}_{Y'}$ is invertible, and g is an isomorphism over the generic point of S. Let $S' := g_*^{-1}S$, $E'_j := g_*^{-1}E_j$, and $K_{Y'} + B_{Y'} := g^*(K_Y + B_Y)$. Then S' is an irreducible component of $E'_1 \cap \cdots \cap E'_m$, $\operatorname{mult}_{E'_j} B_{Y'} = 1$ for $1 \le j \le m$, $(h \circ g)(S') = W$, and $(K_{Y'} + B_{Y'})|_{S'}$ is the pullback of $K_S + B_S$ under $S' \to S$, so $a(\theta, S, B_S)$ does not change if S is replaced by S'. Write $\widetilde{\mathfrak{J}} \cdot \mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-R)$. Since $(h \circ g)(S') = W \not\subset V(\mathfrak{J})$, $S' \not\subset \operatorname{Supp} R$, so $\mathfrak{J} \cdot \mathcal{O}_{S'} = \mathcal{O}_{S'}(-R|_{S'})$ is invertible and $S' \to W$ factors through W'. Possibly replacing h, E_j, and S with $h \circ g$, E'_j, and S', we may assume that $h_S : S \to W'$ is a morphism.

If $m = 1$, let $Y_1 := Y$, $F := S$, $g_F := \mathrm{id}_S$, and $H := 0$. If $m \ge 2$, let $\pi_1 : Y_1 \to Y$ be the blow-up of S, F the exceptional prime divisor, $g_F : F \to S$ the induced $\mathbb{P}^{m-1}$-bundle, $H_j := F \cap (\pi_1)_*^{-1}E_j$ for $1 \le j \le m$, and $H := \sum_{j=1}^m H_j$. The restriction of H_j to any fiber of g_F is a hyperplane. Since $\operatorname{Exc}(h) \cup \operatorname{Supp} h_*^{-1}B$ is snc, no prime divisor on Y other than $E_1, \dots, E_m$ with non-zero coefficient in B_Y contains S, so

$$a(F, X, B) = m - \sum_{j=1}^{m} \operatorname{mult}_{E_j} B_Y = 0.$$

Thus in both cases F is an lc place of (X, B) with $\operatorname{center}_X F = h(S) = W$, and F is a prime divisor on the log resolution Y_1 of (X, B). Let $K_{Y_1} + B_{Y_1}$ be the pullback of $K_Y + B_Y$ and $K_F + B_F := (K_{Y_1} + B_{Y_1})|_F$. Then

$$K_F + B_F = g_F^*(K_S + B_S) \quad \text{and} \quad B_F = H + g_F^* B_S.$$

Let $Q := h_S^*P$, which is defined over a neighborhood of the generic point of P. By Definition-Theorem 2.7(2) for the lc place F and the morphism $h_S \circ g_F : F \to W'$,

$$a(P, W, \Delta_W, \mathbf{M}) = \sup\{t \in \mathbb{R} \mid (F, B_F + tg_F^*Q) \text{ is sub-lc over the generic point of } P\}.$$

Write $\theta(f) = \lambda \operatorname{mult}_G \operatorname{div}(f)$ for any non-zero rational function f on S, where $\lambda > 0$ is a real number and G is a prime divisor on a smooth variety S'' with a projective birational morphism $\phi : S'' \to S$. Let $F'' := F \times_S S''$, let $g_F'' : F'' \to S''$ and $\phi_F : F'' \to F$ be the projections, and let $G_F := g_F''^{-1}(G)$, which is a prime divisor on F''. The center of θ on W' is the center of $(h|_S)_*\theta$, that is, the generic point of P, so $h_S(\phi(G))$ is dense in P, and so is the image of G_F in W'. Since g_F is smooth, $K_{F''} - \phi_F^*K_F = g_F''^*(K_{S''} - \phi^*K_S)$. Together with $K_F + H = g_F^*K_S$, which follows from $K_F + B_F = g_F^*(K_S + B_S)$ and $B_F = H + g_F^*B_S$, this implies that for any real number t,

$$\phi_F^*(K_F + B_F + tg_F^*Q) - K_{F''} = g_F''^*\big(\phi^*(K_S + B_S + tQ) - K_{S''}\big) + \phi_F^*H$$

near G_F. Since $G_F \not\subset \operatorname{Supp} \phi_F^*H$,

$$a(G_F, F, B_F + tg_F^*Q) = a(G, S, B_S + tQ) = a(G, S, B_S) - t \operatorname{mult}_G \phi^*Q.$$

Thus if $(F, B_F + tg_F^*Q)$ is sub-lc over the generic point of P, then $a(G, S, B_S) - t \operatorname{mult}_G \phi^*Q \geq 0$. Since $\lambda \operatorname{mult}_G \phi^*Q$ is the value of θ at a local equation of P, $\lambda \operatorname{mult}_G \phi^*Q = e$. Therefore $te \leq \lambda a(G, S, B_S) = a(\theta, S, B_S)$ for any such t, and

$$a((h|_S)_*\theta, W, \Delta_W, \mathbf{M}) = ea(P, W, \Delta_W, \mathbf{M}) \leq a(\theta, S, B_S).$$

The lemma follows. □

Construction 7.7. Notation and conditions as in Set-up 7.2. Let $B_S = \sum_i b_{S,i}S_i$ with distinct prime divisors S_i on S, and $B'_S := \sum_i b'_{S,i}S_i$, where $b'_{S,i} := 1/2$ if $b_{S,i} = 1$ and $b'_{S,i} := \max\{b_{S,i}, 0\}$ otherwise. Let $S \xrightarrow{\pi_S} T \xrightarrow{\tau} W$ be the Stein factorization of $h|_S$, and $r := \deg \tau$. Let $W'' \to W$ be a projective birational morphism such that $\mathbf{M}$ descends to W''. Let T'' be the normalization of the irreducible component of $W'' \times_W T$ dominating T, and $\tau^*\mathbf{M}$ the $\boldsymbol{b}$-divisor of T which descends to T'' with trace the pullback of $\mathbf{M}_{W''}$. It does not depend on the choice of W''. Let Δ_T be the $\mathbb{R}$-divisor on T with $K_T + \Delta_T + (\tau^*\mathbf{M})_T = \tau^*(K_W + \Delta_W + \mathbf{M}_W)$.

Proposition 7.8. *Notation and conditions as in Construction 7.7, and assume that $k \geq 2$, Δ is a $\mathbb{Q}$-divisor, and $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Then, possibly shrinking X to a neighborhood of x:*

(1) $\Delta_T \geq 0$, $(T, \Delta_T, \tau^*\mathbf{M})$ is a generalized klt pair, and $a(u, T, \Delta_T, \tau^*\mathbf{M}) = a(\tau_*u, W, \Delta_W, \mathbf{M})$ for every divisorial valuation u on T with a center on T;

(2) there exists a finite morphism $\tau_1 : T_1 \to T$ from a normal variety T_1 such that $\tau \circ \tau_1$ is Galois and the $\mathbb{Q}$-divisor Δ_{T_1} with

$$K_{T_1} + \Delta_{T_1} + ((\tau \circ \tau_1)^*\mathbf{M})_{T_1} = (\tau \circ \tau_1)^*(K_W + \Delta_W + \mathbf{M}_W),$$

where $(\tau \circ \tau_1)^*\mathbf{M}$ is defined as $\tau^*\mathbf{M}$ in Construction 7.7, satisfies $\Delta_{T_1} \geq 0$.

Proof. Since Δ is a $\mathbb{Q}$-divisor and $K_X + \Delta$ and D are $\mathbb{Q}$-Cartier, $c \in \mathbb{Q}$. Thus B is a $\mathbb{Q}$-divisor and $K_X + B$ is $\mathbb{Q}$-Cartier, so Δ_W is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is a $\mathbb{Q}$-$\boldsymbol{b}$-divisor by Definition-Theorem 2.7(3).

Step 1. We construct a crepant dlt modification over X and a morphism from an lc place over W to T. If $m \geq 2$, let F be the exceptional prime divisor $\mathbb{P}(N_{S/Y})$ of the blow-up of Y along S. If $m = 1$, let $F := S$. Since E is snc, no component of E other than $E_1, \dots, E_m$ contains S, so $a(F, X, B) = m - \sum_{j=1}^m \operatorname{mult}_{E_j} B_Y = 0$. Thus F is an lc place of (X, B) with $\operatorname{center}_X F = h(S) = W$. By Lemma 5.2, possibly shrinking X to a neighborhood of x, there exists a projective birational morphism $f : Y_F \to X$ from a normal variety Y_F such that F is a $\mathbb{Q}$-Cartier prime divisor on Y_F, $f^{-1}(W) = F$ as sets, and F is the only f-exceptional prime divisor, as $k \geq 2$. Then

$$K_{Y_F} + F + f_*^{-1}B = f^*(K_X + B),$$

so $(Y_F, F + f_*^{-1}B)$ is lc. Let $(Y', B_{Y'})$ be a crepant dlt modification of $(Y_F, F + f_*^{-1}B)$ [Fuj11, Theorem 10.4]. Then $B_{Y'}$ is a $\mathbb{Q}$-divisor and $K_{Y'} + B_{Y'}$ is the pullback of $K_X + B$. For any lc

center Z' of $(Y', B_{Y'})$, let $K_{Z'} + B_{Z'} := (K_{Y'} + B_{Y'})|_{Z'}$. Then Z' is normal and $B_{Z'} \geq 0$ [Fuj11, Lemma 10.2]. Let $q : Y' \to Y_F$ be the induced morphism and $F' := q_*^{-1}F$. Since $\operatorname{mult}_{F'} B_{Y'} = 1$, F' is an lc center of $(Y', B_{Y'})$, and the image of F' in X is W.

Let $F \to S$ be the projection if $m \geq 2$ and id_S if $m = 1$. Since F' and T are normal and τ is finite, the rational map $F' \dashrightarrow T$ induced by $F \to S \xrightarrow{\pi_S} T$ is a morphism, so the morphism $F' \to W$ factors through τ.

Step 2. Let $\sigma : T' \to W$ be a finite surjective morphism from a normal variety T'. Define $\sigma^*\mathbf{M}$ and $\Delta_{T'}$ as $\tau^*\mathbf{M}$ and Δ_T in Construction 7.7, with σ in place of τ. For a valuation u on T' with a center on T', let

$$a(u, T', \Delta_{T'}, \sigma^*\mathbf{M}) := a(u, T'', \Delta_{T''}),$$

where $g : T'' \to T'$ is a projective birational morphism from a normal variety T'' to which $\sigma^*\mathbf{M}$ descends,

$$K_{T''} + \Delta_{T''} + (\sigma^*\mathbf{M})_{T''} := g^*(K_{T'} + \Delta_{T'} + (\sigma^*\mathbf{M})_{T'}),$$

and the right-hand side is the log discrepancy of Definition 2.2(4) for the sub-pair $(T'', \Delta_{T''})$. By Definition 2.2(4) and since $\sigma^*\mathbf{M}$ descends to T'', it does not depend on g, and it is the generalized log discrepancy of Definition 2.6(3) once $\Delta_{T'} \geq 0$. We prove (a) in this step and (b) in **Step 3**:

(a) for any divisorial valuation u on T' with a center on T', $\sigma_* u$ is a divisorial valuation on W with a center on W, and

$$a(u, T', \Delta_{T'}, \sigma^*\mathbf{M}) = a(\sigma_* u, W, \Delta_W, \mathbf{M});$$

(b) if there exist an lc center Z' of $(Y', B_{Y'})$ with image W in X and a morphism $Z' \to T'$ whose composite with σ is the morphism $Z' \to W$ induced by $Y' \to X$, then $\Delta_{T'} \geq 0$.

Let u be a divisorial valuation on T' with a center ξ on T'. Since u is non-trivial, ξ is not the generic point of T', and since σ is finite, $\sigma(\xi)$ is not the generic point of W. By Lemma 6.3, $\sigma_* u$ is a divisorial valuation on W with center $\sigma(\xi)$.

Let $W'' \to W$ be a projective birational morphism from a smooth variety W'' such that $\mathbf{M}$ descends to W'' and the center of $\sigma_* u$ on W'' is the generic point of a prime divisor P on W''. Let T'' be the normalization of the irreducible component of $W'' \times_W T'$ dominating T', $\sigma'' : T'' \to W''$ the induced finite morphism, and $g : T'' \to T'$ the induced projective birational morphism. Then $\sigma^*\mathbf{M}$ descends to T'' with trace $\sigma''^*\mathbf{M}_{W''}$. Let $K_{W''} + \Delta_{W''} + \mathbf{M}_{W''}$ be the pullback of $K_W + \Delta_W + \mathbf{M}_W$. Then

$$K_{T''} + \Delta_{T''} = \sigma''^*(K_{W''} + \Delta_{W''}).$$

The center of u on T'' maps to the generic point of P and σ'' is finite, so the center of u on T'' is the generic point of a prime divisor Q on T'', and there exists a real number $\lambda > 0$ such that $u(\zeta) = \lambda \operatorname{mult}_Q \operatorname{div}(\zeta)$ for any non-zero rational function ζ on T'. Let $e_Q := \operatorname{mult}_Q \sigma''^* P$. Then $\sigma_* u = \lambda e_Q v_P$, and $\operatorname{mult}_Q(K_{T''} - \sigma''^* K_{W''}) = e_Q - 1$ by the Hurwitz formula. Thus $\operatorname{mult}_Q \Delta_{T''} = e_Q \operatorname{mult}_P \Delta_{W''} - (e_Q - 1)$, and

$$a(u, T', \Delta_{T'}, \sigma^*\mathbf{M}) = \lambda(1 - \operatorname{mult}_Q \Delta_{T''}) = \lambda e_Q(1 - \operatorname{mult}_P \Delta_{W''}) = a(\sigma_* u, W, \Delta_W, \mathbf{M}).$$

This proves (a).

Step 3. We prove (b). Let Z' and $Z' \to T'$ be as in (b), and let P be a prime divisor on T'. Since $Z' \to W$ is proper and surjective and σ is finite, $Z' \to T'$ is proper and surjective. Since T' is normal, P is Cartier near the generic point of P, so the preimage of P in Z' has an irreducible component Q which is a prime divisor on Z' mapping onto P. Let $e \geq 1$ be the multiplicity along Q of the pullback of P.

By [Fuj11, Definition 10.1], there exists a resolution $\psi : Y'' \to Y'$ such that $\operatorname{Exc}(\psi)$ and $\operatorname{Exc}(\psi) \cup \operatorname{Supp} \psi_*^{-1} B_{Y'}$ are snc divisors and $a(G, Y', B_{Y'}) > 0$ for every ψ-exceptional prime divisor G. Let E'' be the reduced divisor with support $\operatorname{Exc}(\psi) \cup \operatorname{Supp} \psi_*^{-1} B_{Y'}$, and $K_{Y''} + B_{Y''} := \psi^*(K_{Y'} + B_{Y'})$, which is also the pullback of $K_X + B$. Then $\operatorname{Supp} B_{Y''} \subset E''$, the coefficients of $B_{Y''}$ are ≤ 1, and $\operatorname{mult}_G B_{Y''} = 1 - a(G, Y', B_{Y'}) < 1$ for every ψ-exceptional prime divisor G.

Let G' be an lc place of $(Y', B_{Y'})$ with $\operatorname{center}_{Y'} G' = Z'$, and $v := v_{G'}$. If the center of v on Y'' were contained in a ψ-exceptional prime divisor G, then Lemma 2.4 for Y'' and E'' would give

$$0 = a(v, Y'', B_{Y''}) \geq (1 - \operatorname{mult}_G B_{Y''})v(G) > 0.$$

Thus the center of v on Y'' is not contained in $\operatorname{Exc}(\psi)$, so ψ is an isomorphism over the generic point of Z' by Zariski's main theorem. Since $(Y', B_{Y'})$ is snc near the generic point of Z', the strict transform Z'' of Z' on Y'' is an irreducible component of $E''_1 \cap \cdots \cap E''_N$ for some integer $N \geq 1$ and distinct prime divisors $E''_1, \dots, E''_N$ on Y'' with $\operatorname{mult}_{E''_j} B_{Y''} = 1$ for $1 \leq j \leq N$, the only components of E'' through the generic point of Z'' are $E''_1, \dots, E''_N$, and $(K_{Y''} + B_{Y''})|_{Z''}$ is the pullback of $K_{Z'} + B_{Z'}$ under $Z'' \to Z'$.

By [Kol07, Theorem 3.26] for an ideal sheaf on Y'' with cosupport the exceptional locus of $Y'' \to X$, there exists a composite $\rho : Y''' \to Y''$ of blow-ups of smooth centers having snc with the total transform of E'' [Kol07, Definition 3.25] such that the composite $h''' : Y''' \to X$ of ρ, ψ, and $Y' \to X$ is a log resolution of (X, B) whose exceptional divisors and $h'''^{-1}_* B$ are contained in the total transform E''' of E'', as E''' contains $\rho_*^{-1} E''$ and every ρ-exceptional prime divisor, and every exceptional prime divisor of $Y'' \to X$ is ψ-exceptional or the strict transform of F' or of a q-exceptional prime divisor, which have coefficient 1 in $B_{Y'}$ [Fuj11, Theorem 10.4], and so is contained in E''. Let $K_{Y'''} + B_{Y'''} := h'''^*(K_X + B)$. Since the components of E'' through the generic point of Z'' have coefficient 1 in $B_{Y''}$, the exceptional divisor of the blow-up along a center of ρ through the generic point of the current transform of Z'' has coefficient 1, so there exist an irreducible component Z''' of $E'''_1 \cap \cdots \cap E'''_N$ for distinct prime divisors E'''_j with $\operatorname{mult}_{E'''_j} B_{Y'''} = 1$ and a birational morphism $Z''' \to Z'$ such that $K_{Z'''} + B_{Z'''} := (K_{Y'''} + B_{Y'''})|_{Z'''}$ is the pullback of $K_{Z'} + B_{Z'}$. In particular, $h'''(Z''') = W$.

Let $\theta := v_Q$, which is a divisorial valuation on Z''' with a center on Z''' as $Z''' \to Z'$ is proper and birational. Then

$$a(\theta, Z''', B_{Z'''}) = a(Q, Z', B_{Z'}) = 1 - \operatorname{mult}_Q B_{Z'} \leq 1,$$

as $B_{Z'} \geq 0$. Since Q maps onto P, the push-forward u of θ along $Z' \to T'$ satisfies $u(\zeta) = e \operatorname{mult}_P \operatorname{div}(\zeta)$ for any non-zero rational function ζ on T'. Since $\sigma(P) \neq W$, the center of θ on Z''' does not map to the generic point of W, and the push-forward of θ to W is $\sigma_* u$. By (a) and Lemma 7.6 for h''', $E'''_1, \dots, E'''_N$, and Z''',

$$ea(P, T', \Delta_{T'}, \sigma^* \mathbf{M}) = a(\sigma_* u, W, \Delta_W, \mathbf{M}) \leq a(\theta, Z''', B_{Z'''}) \leq 1.$$

Since T' is normal, any g as in **Step 2** is an isomorphism over the generic point of P, so

$$a(P, T', \Delta_{T'}, \sigma^* \mathbf{M}) = 1 - \operatorname{mult}_P \Delta_{T'}.$$

Thus $\operatorname{mult}_P \Delta_{T'} \geq 1 - 1/e \geq 0$. This proves (b).

Step 4. We prove (1). By **Step 1**, F' is an lc center of $(Y', B_{Y'})$ with image W, and the morphism $F' \to W$ factors through τ. By (b) for $\sigma = \tau$ and $Z' = F'$, $\Delta_T \geq 0$, and (a) for $\sigma = \tau$ is the equality in (1). The variety T is quasi-projective as τ is finite, $\tau^* \mathbf{M}$ descends to T'' of Construction 7.7 with trace the pullback of $\mathbf{M}_{W''}$, which is nef/W and hence nef/T, and $K_T + \Delta_T + (\tau^* \mathbf{M})_T = \tau^*(K_W + \Delta_W + \mathbf{M}_W)$ is $\mathbb{R}$-Cartier. Thus $(T, \Delta_T, \tau^* \mathbf{M})$ is a generalized pair. For any prime divisor E' over T, (a) for $u = v_{E'}$ implies that

$$a(E', T, \Delta_T, \tau^* \mathbf{M}) = a(\tau_* v_{E'}, W, \Delta_W, \mathbf{M}) > 0,$$

since $\tau_* v_{E'}$ is a divisorial valuation on W with a center on W and $(W, \Delta_W, \mathbf{M})$ is generalized klt by Definition-Theorem 2.7(1). Thus $(T, \Delta_T, \tau^* \mathbf{M})$ is generalized klt.

Step 5. We prove (2). Since $(Y', B_{Y'})$ has finitely many lc centers [Fuj11, Theorem 9.1(1)], there exists an lc center $Z \subset F'$ of $(Y', B_{Y'})$ with image W which is minimal among the lc centers contained in F' with image W. Any lc center contained in Z is contained in F', so Z is minimal among all lc centers of $(Y', B_{Y'})$ with image W, that is, Z is a source of W [Kol16, Definition 18]. Let $Z \to T_1 \to W$ be the Stein factorization of $Z \to W$. Then T_1 is normal as Z is normal, and T_1 is the spring of W [Kol16, Definition 18]. Since $(Y', B_{Y'}) \to X$ is crepant and

$B_{Y'}$ is a $\mathbb{Q}$-divisor, $T_1 \to W$ is Galois [Kol16, Proposition 19]. The morphism $Z \to T$ obtained by restricting $F' \to T$ factors through a morphism $\tau_1 : T_1 \to T$ by the universal property of the Stein factorization, and τ_1 is finite as T_1 is finite over W. Thus $\tau \circ \tau_1 : T_1 \to W$ is Galois. By (b) for $\sigma = \tau \circ \tau_1$ and $Z' = Z$, $\Delta_{T_1} \geq 0$, and Δ_{T_1} is a $\mathbb{Q}$-divisor since Δ_W is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is a $\mathbb{Q}$-$\boldsymbol{b}$-divisor. □

7.3. **Valuations with small normalized volume.**

Lemma 7.9. *Notation and conditions as in Set-up 2.8, and assume that Δ is a $\mathbb{Q}$-divisor, and $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Then there exist real numbers $\alpha, \gamma > 0$ such that*

$$a(v, X, B) \leq \gamma\epsilon \quad \text{and} \quad v(f) \geq \alpha \text{ for every } f \in \mathcal{I}_W \cdot \mathcal{O}_{X,x}$$

for any rational number $\epsilon \in (0, 1]$ and any valuation v on X with $\operatorname{center}_X v = x$, $cv(D) = 1$, and $a(v, X, B_\epsilon)^n \operatorname{vol}(v) \leq 2\widehat{\operatorname{vol}}(X \ni x, B_\epsilon)$.

Proof. Since Δ is a $\mathbb{Q}$-divisor and $K_X + \Delta$ and D are $\mathbb{Q}$-Cartier, $c \in \mathbb{Q}$. Let $\kappa_n > 0$ be the constant of [Zhu24, Lemma 3.4], which depends only on n, and let $\gamma := 2/\kappa_n$. Let F be an lc place of (X, B) with $\operatorname{center}_X F = W$, and let $g_1, \dots, g_N$ be generators of the ideal $\mathcal{I}_W \cdot \mathcal{O}_{X,x}$. Since (X, Δ) is klt,

$$c \operatorname{mult}_F D = a(F, X, \Delta) - a(F, X, B) = a(F, X, \Delta) > 0,$$

and $\operatorname{mult}_F \operatorname{div}(g_j) > 0$ for $1 \leq j \leq N$ as $\operatorname{center}_X F = W$. Let

$$\alpha := \frac{\kappa_n}{2c \operatorname{mult}_F D} \min_{1 \leq j \leq N} \operatorname{mult}_F \operatorname{div}(g_j) > 0.$$

Let $\epsilon \in (0, 1]$ be a rational number and v a valuation on X as in the lemma. Then B_ϵ is a $\mathbb{Q}$-divisor and $K_X + B_\epsilon$ is $\mathbb{Q}$-Cartier. By Lemma 2.9, (X, B_ϵ) is klt, so $\widehat{\operatorname{vol}}(X \ni x, B_\epsilon) > 0$ [LLX20, Theorem 2.5]. By the definition of $\widehat{\operatorname{vol}}(X \ni x, B_\epsilon)$ and the assumption on v,

$$0 < \widehat{\operatorname{vol}}(X \ni x, B_\epsilon) \leq a(v, X, B_\epsilon)^n \operatorname{vol}(v) \leq 2\widehat{\operatorname{vol}}(X \ni x, B_\epsilon) < +\infty.$$

By Definition 2.2(4) and Lemma 2.4 on a log resolution of (X, B), $a(v, X, B) \geq 0$ as (X, B) is lc. Since $B_\epsilon = B - \epsilon c D$ and $cv(D) = 1$,

$$a(v, X, B_\epsilon) = a(v, X, B) + \epsilon \geq \epsilon. \tag{7.7}$$

By [Zhu24, Lemma 3.4], for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D' \geq 0$ on a neighborhood of x with $v(D') > 0$,

$$\operatorname{lct}(X \ni x, B_\epsilon; D') \geq \kappa_n \frac{\widehat{\operatorname{vol}}(X \ni x, B_\epsilon)}{a(v, X, B_\epsilon)^n \operatorname{vol}(v)} \cdot \frac{a(v, X, B_\epsilon)}{v(D')} \geq \frac{\kappa_n a(v, X, B_\epsilon)}{2v(D')}. \tag{7.8}$$

By Lemma 2.9, $\operatorname{lct}(X \ni x, B_\epsilon; D) = \epsilon c$. Since $v(D) = 1/c$, (7.8) for $D' = D$ implies $a(v, X, B_\epsilon) \leq 2\epsilon/\kappa_n$. By (7.7),

$$a(v, X, B) \leq a(v, X, B_\epsilon) \leq \gamma\epsilon.$$

Let $1 \leq j \leq N$. Since $\operatorname{center}_X v = x \in W$, $v(g_j) > 0$. Since $x \in \operatorname{center}_X F$, the pair $(X, B_\epsilon + t \operatorname{div}(g_j))$ is not lc near x if $a(F, X, B_\epsilon + t \operatorname{div}(g_j)) < 0$. As

$$a(F, X, B_\epsilon) = a(F, X, B) + \epsilon c \operatorname{mult}_F D = \epsilon c \operatorname{mult}_F D,$$

it follows that

$$\operatorname{lct}(X \ni x, B_\epsilon; \operatorname{div}(g_j)) \leq \frac{a(F, X, B_\epsilon)}{\operatorname{mult}_F \operatorname{div}(g_j)} = \frac{\epsilon c \operatorname{mult}_F D}{\operatorname{mult}_F \operatorname{div}(g_j)}.$$

Thus (7.8) for $D' = \operatorname{div}(g_j)$ and (7.7) imply

$$v(g_j) \geq \frac{\kappa_n a(v, X, B_\epsilon) \operatorname{mult}_F \operatorname{div}(g_j)}{2\epsilon c \operatorname{mult}_F D} \geq \frac{\kappa_n \operatorname{mult}_F \operatorname{div}(g_j)}{2c \operatorname{mult}_F D} \geq \alpha.$$

Any $f \in \mathcal{I}_W \cdot \mathcal{O}_{X,x}$ is an $\mathcal{O}_{X,x}$-linear combination of $g_1, \dots, g_N$, and $v \geq 0$ on $\mathcal{O}_{X,x}$ as $\operatorname{center}_X v = x$, so $v(f) \geq \min_j v(g_j) \geq \alpha$. The lemma follows. □

7.4. **Graded algebras along the stratum.**

Proposition 7.10. *Notation and conditions as in Construction 7.7, and as in Construction-Proposition 7.3 with h, E, $E_1,\dots,E_m$, and S from Set-up 7.2, and assume that $k\geq 2$. Let $\mathbf{b}\in\mathbb{R}^m_{>0}$ have $\mathbb{Q}$-linearly independent coordinates. Then for every $\zeta\in(0,1)$ there exist a real number $C>0$ and a continuous function $\varphi:\mathbb{R}^m_{>0}\to\mathbb{R}_{>0}$ with $r\cdot\varphi(\mathbf{b})\geq(1-\zeta)\operatorname{vol}(u_{\mathbf{b}})$ such that*

$$\operatorname{vol}(\mu_{\theta,\mathbf{b}})\geq\varphi\big(\mathbf{b}+Ca(\theta,S,B'_S)\cdot(1,\dots,1)\big)\cdot\operatorname{vol}((\pi_S)_*\theta)$$

for every divisorial valuation θ on S with $\operatorname{center}_S(\theta)\in S^\circ\cap h^{-1}(x)$.

Proof. For $\alpha\in\mathbb{R}^m$ we let $|\alpha|$ be the sum of the coordinates of α. Let $Z_1,\dots,Z_m$ be variables.

Step 1. In this step we compare $\mu_{\theta,\mathbf{b}}$ with the graded algebra of $u_{\mathbf{b}}$ on $\mathcal{O}_{X,x}$, for any valuation θ on S with $\operatorname{center}_S(\theta)\in S^\circ\cap h^{-1}(x)$.

Since the coordinates of $\mathbf{b}$ are $\mathbb{Q}$-linearly independent, $\mathbf{b}\cdot\alpha\neq\mathbf{b}\cdot\alpha'$ for any distinct $\alpha,\alpha'\in\mathbb{N}^m$. Thus for any $f\in\mathcal{O}(Y^\circ)\backslash\{0\}$ there exists exactly one $\alpha\in\mathbb{N}^m$ with $f_\alpha\neq 0$ and $\mathbf{b}\cdot\alpha=u_{\mathbf{b}}(f)$, and we let

$$\operatorname{in}(f):=f_\alpha Z^\alpha\in\mathbb{C}(S)[Z_1,\dots,Z_m]$$

be the leading form of f. Since ι is a ring homomorphism and $\mathcal{O}(S^\circ)$ is a domain, for any $f,g\in\mathcal{O}(Y^\circ)\backslash\{0\}$ and $\alpha''\in\mathbb{N}^m$,

$$(fg)_{\alpha''}=\sum_{\alpha+\alpha'=\alpha''}f_\alpha g_{\alpha'}\quad\text{and}\quad\operatorname{in}(fg)=\operatorname{in}(f)\operatorname{in}(g).$$

Hence

$$\operatorname{in}:\mathbb{C}(X)^\times=\mathbb{C}(Y)^\times\to\mathbb{C}(S)(Z_1,\dots,Z_m)^\times,\quad\operatorname{in}(g_1/g_2):=\frac{\operatorname{in}(g_1)}{\operatorname{in}(g_2)}\ \text{ for } g_1,g_2\in\mathcal{O}(Y^\circ)\backslash\{0\},$$

is a well-defined multiplicative map.

Let $F\in\mathcal{O}_{X,w}\backslash\{0\}$. As $h(S)=W$, F is regular at the generic point of S, so $F=g_1/g_2$ with $g_1,g_2\in\mathcal{O}(Y^\circ)$ and $g_2|_{S^\circ}\neq 0$. If moreover $F\in\mathcal{O}_{X,x}$, then F is regular at $\operatorname{center}_S(\theta)$ as $h(\operatorname{center}_S(\theta))=x$, and we may take g_2 such that $g_2|_{S^\circ}$ does not vanish at $\operatorname{center}_S(\theta)$. Since $(g_2)_0=g_2|_{S^\circ}\neq 0$, we have $u_{\mathbf{b}}(g_2)=0$ and $\operatorname{in}(g_2)=g_2|_{S^\circ}$. Thus there exists $\alpha\in\mathbb{N}^m$ with

$$u_{\mathbf{b}}(F)=u_{\mathbf{b}}(g_1)=\mathbf{b}\cdot\alpha\quad\text{and}\quad\operatorname{in}(F)=\frac{(g_1)_\alpha}{g_2|_{S^\circ}}Z^\alpha.$$

For $R\in\{\mathcal{O}_{X,w},\mathcal{O}_{X,x}\}$ and $\alpha\in\mathbb{N}^m$, let

$$\operatorname{gr}_\alpha R:=\{F\in R\mid u_{\mathbf{b}}(F)\geq\mathbf{b}\cdot\alpha\}/\{F\in R\mid u_{\mathbf{b}}(F)>\mathbf{b}\cdot\alpha\},\quad\operatorname{gr}_{u_{\mathbf{b}}}R:=\bigoplus_{\alpha\in\mathbb{N}^m}\operatorname{gr}_\alpha R,$$

and let $\mathcal{G}_w:=\operatorname{gr}_{u_{\mathbf{b}}}\mathcal{O}_{X,w}$ and $\mathcal{G}_x:=\operatorname{gr}_{u_{\mathbf{b}}}\mathcal{O}_{X,x}$. Writing two elements of R with a common denominator $g_2\in\mathcal{O}(Y^\circ)$ such that $g_2|_{S^\circ}\neq 0$, and such that $g_2|_{S^\circ}$ does not vanish at $\operatorname{center}_S(\theta)$ if $R=\mathcal{O}_{X,x}$, we see that $F\to\operatorname{in}(F)$ induces injective homomorphisms of $\mathbb{N}^m$-graded rings, by which we regard

$$\mathcal{G}_x\subset\mathcal{G}_w\subset\mathbb{C}(S)[Z_1,\dots,Z_m].$$

We have:

(1) For $F\in R$, $u_{\mathbf{b}}(F)>0$ if and only if $(h^*F)|_S=0$, if and only if $F|_W=0$ as $h(S)=W$. Thus $\{F\in R\mid u_{\mathbf{b}}(F)>0\}=\mathcal{I}_WR$ and $\operatorname{gr}_0R=R/\mathcal{I}_WR$, so $(\mathcal{G}_w)_0=\mathbb{C}(W)$ and $(\mathcal{G}_x)_0=\mathcal{O}_{W,x}$. Any $F\in\mathcal{O}_{X,w}$ is F_1/F_2 with $F_1,F_2\in\mathcal{O}_{X,x}$ and $F_2\notin\mathcal{I}_W\mathcal{O}_{X,x}$, and $\operatorname{in}(F)=\operatorname{in}(F_1)/(F_2|_W)$, so $\mathcal{G}_w$ is the localization of $\mathcal{G}_x$ at $\mathcal{O}_{W,x}\backslash\{0\}$.

(2) The ratios of homogeneous elements of $\mathcal{G}_x$ generate the field $\mathbb{C}(S)(Z_1,\dots,Z_m)$, since $Z_j=\operatorname{in}(z_j)$, any $f\in\mathcal{O}(S^\circ)\backslash\{0\}$ is $\operatorname{in}(g)$ for any $g\in\mathcal{O}(Y^\circ)$ with $g|_{S^\circ}=f$, and any element of $\mathbb{C}(X)^\times$ is F_1/F_2 with $F_1,F_2\in\mathcal{O}_{X,x}\backslash\{0\}$.

(3) For any $F \in \mathcal{O}_{X,x}\backslash\{0\}$ with $\mathrm{in}(F) = c_F Z^\alpha$,

$$\theta(c_F) \geq 0 \quad \text{and} \quad \mathbf{b}\cdot\alpha = u_{\mathbf{b}}(F) \leq \mu_{\theta,\mathbf{b}}(F) \leq \mathbf{b}\cdot\alpha + \theta(c_F). \tag{7.9}$$

Moreover, $\mathrm{center}_X(\mu_{\theta,\mathbf{b}}) = x$. Write $F = g_1/g_2$ with $g_1, g_2 \in \mathcal{O}(Y^\circ)$ and $g_2|_{S^\circ}$ not vanishing at $\mathrm{center}_S(\theta)$. Then $\theta(g_2|_{S^\circ}) = 0$ and $\theta \geq 0$ on $\mathcal{O}(S^\circ)$, so $\mu_{\theta,\mathbf{b}}(g_2) = 0$ and

$$\mu_{\theta,\mathbf{b}}(F) = \mu_{\theta,\mathbf{b}}(g_1) = \min_{\alpha'}\{\theta((g_1)_{\alpha'}) + \mathbf{b}\cdot\alpha'\}.$$

The last minimum satisfies

$$\mathbf{b}\cdot\alpha = \min\{\mathbf{b}\cdot\alpha' \mid (g_1)_{\alpha'} \neq 0\} \leq \min_{\alpha'}\{\theta((g_1)_{\alpha'}) + \mathbf{b}\cdot\alpha'\} \leq \theta((g_1)_\alpha) + \mathbf{b}\cdot\alpha = \theta(c_F) + \mathbf{b}\cdot\alpha.$$

If $F \in \mathfrak{m}_x$, then $(g_1)_0/g_2|_{S^\circ} = (h^*F)|_{S^\circ}$ vanishes at $\mathrm{center}_S(\theta)$, so $\mu_{\theta,\mathbf{b}}(F) > 0$. Thus $\mathrm{center}_X(\mu_{\theta,\mathbf{b}}) = x$.

Let $\omega_{\theta,\mathbf{b}}$ be the valuation on $\mathbb{C}(S)(Z_1,\dots,Z_m)$ with

$$\omega_{\theta,\mathbf{b}}\Big(\sum_\alpha f_\alpha Z^\alpha\Big) = \min_\alpha\{\mathbf{b}\cdot\alpha + \theta(f_\alpha)\} \quad \text{for } f_\alpha \in \mathbb{C}(S),$$

and for $\lambda \in \mathbb{R}$ let $\{\omega_{\theta,\mathbf{b}} \geq \lambda\}$ be the set of elements of $\mathbb{C}(S)(Z_1,\dots,Z_m)$ with value at least λ. By (3), $\omega_{\theta,\mathbf{b}} \geq 0$ on $\mathcal{G}_x$, and $\omega_{\theta,\mathbf{b}} \geq \mathbf{b}\cdot\alpha$ on $(\mathcal{G}_x)_\alpha$. As $\omega_{\theta,\mathbf{b}}$ of a polynomial is the minimum over the homogeneous components, $\mathcal{G}_x \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda\}$ is a homogeneous ideal of $\mathcal{G}_x$ containing $(\mathcal{G}_x)_\alpha$ whenever $\mathbf{b}\cdot\alpha \geq \lambda$.

Let $\lambda > 0$. For $\alpha \in \mathbb{N}^m$, let

$$\mathrm{in}_\alpha(\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})) \subset (\mathcal{G}_x)_\alpha$$

be the image of the set of $F \in \mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})$ with $u_{\mathbf{b}}(F) \geq \mathbf{b}\cdot\alpha$. By (7.9), any $F \in \mathcal{O}_{X,x}$ with $u_{\mathbf{b}}(F) \geq \lambda$ belongs to $\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})$, and the values of $u_{\mathbf{b}}$ on $\mathcal{O}_{X,x}\backslash\{0\}$ belong to the discrete set $\{\mathbf{b}\cdot\alpha \mid \alpha \in \mathbb{N}^m\}$. Thus filtering $\mathcal{O}_{X,x}/\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})$ by the images of $\{u_{\mathbf{b}} \geq \mathbf{b}\cdot\alpha\}$ implies

$$\mathrm{length}(\mathcal{O}_{X,x}/\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})) = \sum_{\mathbf{b}\cdot\alpha<\lambda} \mathrm{length}\left((\mathcal{G}_x)_\alpha/\,\mathrm{in}_\alpha(\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}}))\right). \tag{7.10}$$

If $F \in \mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})$ and $\mathrm{in}(F) = c_F Z^\alpha$, then by (7.9),

$$\omega_{\theta,\mathbf{b}}(c_F Z^\alpha) = \mathbf{b}\cdot\alpha + \theta(c_F) \geq \mu_{\theta,\mathbf{b}}(F) \geq \lambda.$$

Hence

$$\mathrm{in}_\alpha(\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})) \subset (\mathcal{G}_x)_\alpha \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda\}.$$

The $\mathcal{O}_{X,x}$-modules $(\mathcal{G}_x)_\alpha$ are $\mathcal{O}_{W,x}$-modules, and all residue fields are $\mathbb{C}$, so lengths of finite length modules are dimensions over $\mathbb{C}$. Therefore

$$\mathrm{length}(\mathcal{O}_{X,x}/\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})) \geq \sum_{\mathbf{b}\cdot\alpha<\lambda} \dim_{\mathbb{C}} \frac{(\mathcal{G}_x)_\alpha}{(\mathcal{G}_x)_\alpha \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda\}} = \dim_{\mathbb{C}} \frac{\mathcal{G}_x}{\mathcal{G}_x \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda\}}. \tag{7.11}$$

Step 2. In this step we construct $\mathcal{C}_x$, $\overline{\mathcal{C}}$, and φ, which do not depend on θ.

Since $u_{\mathbf{b}} \geq 0$ on $\mathcal{O}_{X,w}$ and $u_{\mathbf{b}} > 0$ on $\mathfrak{m}_w = \mathcal{I}_W\mathcal{O}_{X,w}$ by (1), $\mathrm{center}_X u_{\mathbf{b}} = w$. As in (7.10), filtering $\mathcal{O}_{X,w}/\mathfrak{a}_\lambda(u_{\mathbf{b}})$ by the images of $\{u_{\mathbf{b}} \geq \mathbf{b}\cdot\alpha\}$, and using that the image of $\mathfrak{a}_\lambda(u_{\mathbf{b}})$ in $(\mathcal{G}_w)_\alpha$ is 0 for $\mathbf{b}\cdot\alpha < \lambda$, we obtain

$$\mathrm{length}(\mathcal{O}_{X,w}/\mathfrak{a}_\lambda(u_{\mathbf{b}})) = \sum_{\mathbf{b}\cdot\alpha<\lambda} \dim_{\mathbb{C}(W)}(\mathcal{G}_w)_\alpha.$$

As $\dim \mathcal{O}_{X,w} = k$, Definition 2.2 implies

$$\lim_{\lambda\to+\infty} \frac{k!}{\lambda^k} \sum_{\mathbf{b}\cdot\alpha<\lambda} \dim_{\mathbb{C}(W)}(\mathcal{G}_w)_\alpha = \mathrm{vol}(u_{\mathbf{b}}).$$

We regard $\mathbb{C}(W) \subset \mathbb{C}(T) \subset \mathbb{C}(S)$, and $[\mathbb{C}(T) : \mathbb{C}(W)] = \deg\tau = r$. Let $\mathcal{G}_w[\mathbb{C}(T)]$ be the subring of $\mathbb{C}(S)[Z_1,\dots,Z_m]$ generated by $\mathcal{G}_w$ and $\mathbb{C}(T)$. Let $\kappa \in \mathbb{C}(T)$ with $\mathbb{C}(T) = \mathbb{C}(W)(\kappa)$. By

(2), $\kappa = g_1/g_2$ with $g_1, g_2 \in \mathcal{G}_w\backslash\{0\}$. Comparing the homogeneous components of $\kappa g_2 = g_1$, $\kappa = (g_1)_\beta/(g_2)_\beta$ for any β with $(g_2)_\beta \neq 0$. Let $g := (g_1)_\beta$ and $g_0 := (g_2)_\beta$. Since

$$\mathcal{G}_w[\mathbb{C}(T)] = \sum_{0\le j<r} \kappa^j \mathcal{G}_w,$$

we have $g_0^{r-1}\mathcal{G}_w[\mathbb{C}(T)] \subset \mathcal{G}_w$. Let $\alpha_0 := \deg g_0^{r-1}$. Then

$$\dim_{\mathbb{C}(W)}(\mathcal{G}_w)_\alpha \le r \dim_{\mathbb{C}(T)} \mathcal{G}_w[\mathbb{C}(T)]_\alpha \le \dim_{\mathbb{C}(W)}(\mathcal{G}_w)_{\alpha+\alpha_0}$$

for any $\alpha \in \mathbb{N}^m$, so, since $\mathbf{b}\cdot\alpha$ and $\mathbf{b}\cdot(\alpha+\alpha_0)$ differ by the constant $\mathbf{b}\cdot\alpha_0$ and $(\lambda+\mathbf{b}\cdot\alpha_0)/\lambda \to 1$,

$$\lim_{\lambda\to+\infty} \frac{k!}{\lambda^k} \sum_{\mathbf{b}\cdot\alpha<\lambda} \dim_{\mathbb{C}(T)} \mathcal{G}_w[\mathbb{C}(T)]_\alpha = \frac{\mathrm{vol}(u_{\mathbf{b}})}{r}. \tag{7.12}$$

Let $\mathcal{R} \subset \mathbb{C}(S)[Z_1, \dots, Z_m]$ be an $\mathbb{N}^m$-graded $\mathbb{C}(T)$-subalgebra satisfying the following.

(i) $\mathcal{R}_0 = \mathbb{C}(T)$.
(ii) The ratios of homogeneous elements of $\mathcal{R}$ generate the field $\mathbb{C}(S)(Z_1, \dots, Z_m)$.
(iii) There exists $\alpha_{\mathcal{R}} \in \mathbb{N}^m$ such that

$$\dim_{\mathbb{C}(T)} \mathcal{R}_\alpha \le \dim_{\mathbb{C}(T)} \mathcal{G}_w[\mathbb{C}(T)]_{\alpha+\alpha_{\mathcal{R}}} \quad \text{for any } \alpha \in \mathbb{N}^m.$$

For $\mathbf{b}' \in \mathbb{R}^m_{>0}$ and $\lambda > 0$ let

$$H_{\mathcal{R},\mathbf{b}'}(\lambda) := \sum_{\mathbf{b}'\cdot\alpha<\lambda} \dim_{\mathbb{C}(T)} \mathcal{R}_\alpha.$$

We show:

(A) for any $\mathbf{b}' \in \mathbb{R}^m_{>0}$, the limit

$$\varphi_{\mathcal{R}}(\mathbf{b}') := \lim_{\lambda\to+\infty} k!\lambda^{-k} H_{\mathcal{R},\mathbf{b}'}(\lambda)$$

exists and is positive, and $\varphi_{\mathcal{R}}$ is continuous;
(B) for any $\zeta \in (0,1)$, there exists a $\mathbb{C}(T)$-subalgebra $\mathcal{R}' \subset \mathcal{R}$ generated by finitely many homogeneous elements, satisfying (i)–(iii), such that

$$\varphi_{\mathcal{R}'}(\mathbf{b}) \ge (1-\zeta)\varphi_{\mathcal{R}}(\mathbf{b}).$$

Since $\pi_{S*}\mathcal{O}_S = \mathcal{O}_T$, $\mathbb{C}(T)$ is algebraically closed in $\mathbb{C}(S)$, so $\mathbb{C}(S) \otimes_{\mathbb{C}(T)} \overline{\mathbb{C}(T)}$ is a domain. Let $\mathbb{L}$ be the fraction field of $\mathbb{C}(S) \otimes_{\mathbb{C}(T)} \overline{\mathbb{C}(T)}$. Then

$$\mathrm{trdeg}_{\overline{\mathbb{C}(T)}} \mathbb{L} = \dim S - \dim T = k - m.$$

Fix a regular system of parameters $\varpi_1, \dots, \varpi_{k-m}$ at a smooth closed point of a variety over $\overline{\mathbb{C}(T)}$ with function field $\mathbb{L}$, and let $v_0 : \mathbb{L}^\times \to \mathbb{Z}^{k-m}$ be the valuation sending a regular function at the point to the lexicographically smallest exponent in the expansion of the function in $\overline{\mathbb{C}(T)}[[\varpi_1, \dots, \varpi_{k-m}]]$, where $\mathbb{Z}^{k-m}$ is ordered lexicographically. Then

$$\dim_{\overline{\mathbb{C}(T)}} \big(\{v_0 \ge v\}/\{v_0 > v\}\big) \le 1 \quad \text{for any } v \in \mathbb{Z}^{k-m}. \tag{7.13}$$

Let

$$\Gamma := \{(\alpha, v_0(f)) \mid \alpha \in \mathbb{N}^m,\ f \in \mathbb{L}^\times,\ fZ^\alpha \in \mathcal{R} \otimes_{\mathbb{C}(T)} \overline{\mathbb{C}(T)}\} \subset \mathbb{N}^m \times \mathbb{Z}^{k-m},$$

a semigroup. For $\alpha \in \mathbb{N}^m$, by (7.13), filtering $\mathcal{R}_\alpha \otimes_{\mathbb{C}(T)} \overline{\mathbb{C}(T)}$ by the values of v_0 shows that its dimension over $\overline{\mathbb{C}(T)}$ equals the number of values $v_0(f)$ with fZ^α a non-zero element of $\mathcal{R}_\alpha \otimes_{\mathbb{C}(T)} \overline{\mathbb{C}(T)}$. Thus

$$\#\big(\Gamma \cap (\{\alpha\} \times \mathbb{Z}^{k-m})\big) = \dim_{\mathbb{C}(T)} \mathcal{R}_\alpha \quad \text{and} \quad H_{\mathcal{R},\mathbf{b}'}(\lambda) = \#\{(\alpha, v) \in \Gamma \mid \mathbf{b}'\cdot\alpha < \lambda\}. \tag{7.14}$$

We bound the growth of Γ. Since

$$\mathbf{b}\cdot\alpha \le |\alpha| \max\{\mathbf{b}\cdot\alpha' \mid \alpha' \in \mathbb{R}^m_{\ge 0},\ |\alpha'| = 1\} \quad \text{for } \alpha \in \mathbb{N}^m,$$

(iii) and (7.12) imply $\sum_{|\alpha|\le i}\dim\mathcal{R}_\alpha = O(i^k)$. Since $\mathbb{C}(S)(Z_1,\dots,Z_m)$ has transcendence degree k over $\mathbb{C}(T)$, (ii) implies that there exist homogeneous elements of $\mathcal{R}$ algebraically independent over $\mathbb{C}(T)$, of degrees $\alpha^{(1)},\dots,\alpha^{(k)}$, and $\alpha^{(j)}\neq 0$ for any j by (i). Multiplication by the j'-th power of the first element, $0\le j'<i$, injects $\bigoplus_{|\alpha|=i}\mathcal{R}_\alpha$ into $\bigoplus_{|\alpha|=i+j'|\alpha^{(1)}|}\mathcal{R}_\alpha$, and the total degrees $i+j'|\alpha^{(1)}|$ are distinct and less than $i(1+|\alpha^{(1)}|)$. Thus

$$i\cdot\#\{(\alpha,\upsilon)\in\Gamma\mid|\alpha|=i\}\le\sum_{|\alpha|<i(1+|\alpha^{(1)}|)}\dim\mathcal{R}_\alpha=O(i^k).$$

Moreover, distinct monomials in the k algebraically independent elements are linearly independent over $\mathbb{C}(T)$, so

$$H_{\mathcal{R},\mathbf{b}'}(\lambda)\ge\#\Big\{(i_1,\dots,i_k)\in\mathbb{N}^k\;\Big|\;\sum_j i_j\mathbf{b}'\cdot\alpha^{(j)}<\lambda\Big\}\ge\frac{1}{k!}\Big(\frac{\lambda}{\max_j\mathbf{b}'\cdot\alpha^{(j)}}\Big)^k \tag{7.15}$$

for any $\lambda\ge\max_j\mathbf{b}'\cdot\alpha^{(j)}$.

Let $\mathrm{Cone}(\Gamma)\subset\mathbb{R}^k$ be the closed convex cone generated by Γ, where $\mathbb{R}^k=\mathbb{R}^m\times\mathbb{R}^{k-m}$, and $\mathbb{Z}\Gamma$ the group generated by Γ. By (ii), the fraction field of $\mathcal{R}\otimes_{\mathbb{C}(T)}\overline{\mathbb{C}(T)}$ is $\mathbb{L}(Z_1,\dots,Z_m)$, and if $fZ^\alpha=g_1/g_2$ with $f\in\mathbb{L}^\times$, $\alpha\in\mathbb{Z}^m$, and g_1,g_2 in $\mathcal{R}\otimes_{\mathbb{C}(T)}\overline{\mathbb{C}(T)}$, then $fZ^\alpha=(g_1)_{\alpha+\beta}/(g_2)_\beta$ for any β with $(g_2)_\beta\neq 0$, so $(\alpha,v_0(f))\in\mathbb{Z}\Gamma$, as it is the difference of the elements of Γ defined by $(g_1)_{\alpha+\beta}$ and $(g_2)_\beta$. Taking $f\in\{1,\varpi_1,\dots,\varpi_{k-m}\}$ and any $\alpha\in\mathbb{Z}^m$, as $v_0(\varpi_j)$ is the j-th unit vector of $\mathbb{Z}^{k-m}$, we obtain $\mathbb{Z}\Gamma=\mathbb{Z}^k$.

By (i), $\Gamma\cap(\{0\}\times\mathbb{Z}^{k-m})=\{0\}$. Since $\#\{(\alpha,\upsilon)\in\Gamma\mid|\alpha|=i\}=O(i^{k-1})$, $(\Gamma,\{|\alpha|\ge 0\})$ is an admissible pair in the sense of [KK12, Definition 1.9], as $\Gamma\subset\{|\alpha|\ge 0\}$, a rational half-space of $\mathbb{R}^k$, and Γ has bounded growth with respect to $\{|\alpha|\ge 0\}$ in the sense of [KK12, Definition 1.17], so the pair is strongly admissible by [KK12, Theorem 1.18], that is, $\mathrm{Cone}(\Gamma)$ is strictly convex and $\mathrm{Cone}(\Gamma)\cap\{|\alpha|=0\}=\{0\}$. As $\Gamma\subset\mathbb{N}^m\times\mathbb{Z}^{k-m}$, the cone $\mathrm{Cone}(\Gamma)$ is contained in $\mathbb{R}^m_{\ge 0}\times\mathbb{R}^{k-m}$, so for any $\mathbf{b}'\in\mathbb{R}^m_{>0}$ the function $\mathbf{b}'\cdot\alpha$ has positive upper and lower bounds on the compact set $\mathrm{Cone}(\Gamma)\cap\{|\alpha|=1\}$. Hence for any closed convex cone $\Sigma\subset\mathrm{Cone}(\Gamma)$,

$$\Sigma_{\mathbf{b}'}:=\{(\alpha,\upsilon)\in\Sigma\mid\mathbf{b}'\cdot\alpha<1\}$$

is a bounded convex set, and $\lambda\Sigma_{\mathbf{b}'}$ is the set of $(\alpha,\upsilon)\in\Sigma$ with $\mathbf{b}'\cdot\alpha<\lambda$. By (7.14),

$$H_{\mathcal{R},\mathbf{b}'}(\lambda)\le\#(\mathbb{Z}\Gamma\cap\lambda\,\mathrm{Cone}(\Gamma)_{\mathbf{b}'}).$$

As $\mathbb{Z}\Gamma=\mathbb{Z}^k$, $\mathrm{Cone}(\Gamma)$ has non-empty interior. Counting lattice points in dilates of bounded convex sets,

$$\limsup_{\lambda\to+\infty}\frac{H_{\mathcal{R},\mathbf{b}'}(\lambda)}{\lambda^k}\le\mathrm{vol}_{\mathbb{R}^k}(\mathrm{Cone}(\Gamma)_{\mathbf{b}'}),$$

where $\mathrm{vol}_{\mathbb{R}^k}$ is the Lebesgue measure on $\mathbb{R}^k$. Let Σ be a closed convex cone with $\Sigma\backslash\{0\}$ contained in the interior of $\mathrm{Cone}(\Gamma)$. By [KK12, Theorem 1.6], all but finitely many points of $\mathbb{Z}\Gamma\cap\Sigma$ belong to Γ, so

$$H_{\mathcal{R},\mathbf{b}'}(\lambda)\ge\#(\mathbb{Z}\Gamma\cap\lambda\Sigma_{\mathbf{b}'})-\#((\mathbb{Z}\Gamma\cap\Sigma)\backslash\Gamma).$$

As the boundary of the convex set $\mathrm{Cone}(\Gamma)_{\mathbf{b}'}$ has measure zero, the supremum of $\mathrm{vol}_{\mathbb{R}^k}(\Sigma_{\mathbf{b}'})$ over such Σ is $\mathrm{vol}_{\mathbb{R}^k}(\mathrm{Cone}(\Gamma)_{\mathbf{b}'})$. Thus the limit in (A) exists, and

$$\varphi_{\mathcal{R}}(\mathbf{b}')=k!\,\mathrm{vol}_{\mathbb{R}^k}(\mathrm{Cone}(\Gamma)_{\mathbf{b}'})>0, \tag{7.16}$$

where positivity follows from (7.15). If $\delta>0$ and $\mathbf{b}''\le(1+\delta)\mathbf{b}'$ coordinatewise, then $\mathbf{b}''\cdot\alpha\le(1+\delta)\mathbf{b}'\cdot\alpha$ for $\alpha\in\mathbb{N}^m$, so

$$H_{\mathcal{R},\mathbf{b}''}(\lambda)\ge H_{\mathcal{R},\mathbf{b}'}(\lambda/(1+\delta))\quad\text{and}\quad\varphi_{\mathcal{R}}(\mathbf{b}'')\ge(1+\delta)^{-k}\varphi_{\mathcal{R}}(\mathbf{b}').$$

For $\mathbf{b}''$ close to $\mathbf{b}'$, both $\mathbf{b}''\le(1+\delta)\mathbf{b}'$ and $\mathbf{b}'\le(1+\delta)\mathbf{b}''$ hold with $\delta\to 0$, so $\varphi_{\mathcal{R}}$ is continuous. This proves (A).

For (B), choose rational polyhedral closed convex cones $\Sigma \subset \Sigma'$ such that $\Sigma\backslash\{0\}$ is contained in the interior of Σ', $\Sigma'\backslash\{0\}$ is contained in the interior of $\mathrm{Cone}(\Gamma)$, and

$$\mathrm{vol}_{\mathbb{R}^k}(\Sigma_{\mathbf{b}}) \geq (1-\zeta)\,\mathrm{vol}_{\mathbb{R}^k}(\mathrm{Cone}(\Gamma)_{\mathbf{b}}).$$

Since $\mathbb{Z}\Gamma = \mathbb{Z}^k$, each extremal ray of Σ' contains points of $\mathbb{Z}\Gamma$ arbitrarily far from the origin, so by [KK12, Theorem 1.6] each extremal ray of Σ' contains a point of Γ. Let $\Gamma' \subset \Gamma$ be the subsemigroup generated by one such point on each extremal ray of Σ' and by finitely many elements of Γ generating the group $\mathbb{Z}\Gamma$. Then $\mathbb{Z}\Gamma' = \mathbb{Z}\Gamma$ and $\Sigma' \subset \mathrm{Cone}(\Gamma')$, so by [KK12, Theorem 1.6] all but finitely many points of $\mathbb{Z}\Gamma \cap \Sigma$ belong to Γ'. For each generator (α, υ) of Γ', choose an element fZ^α of $\mathcal{R}_\alpha \otimes_{\mathbb{C}(T)} \overline{\mathbb{C}(T)}$ with $v_0(f) = \upsilon$, and write fZ^α as a finite sum $\sum_j \eta_j \otimes g_j$ with $\eta_j \in \overline{\mathbb{C}(T)}$ and $g_j \in \mathcal{R}_\alpha$. Let $\mathcal{R}'$ be the $\mathbb{C}(T)$-subalgebra of $\mathcal{R}$ generated by the elements g_j for all generators (α, υ) of Γ' and by finitely many homogeneous elements of $\mathcal{R}$ whose ratios generate $\mathbb{C}(S)(Z_1, \dots, Z_m)$. Then $\mathcal{R}'$ satisfies (i)–(iii), and $\mathcal{R}' \otimes_{\mathbb{C}(T)} \overline{\mathbb{C}(T)}$ contains the elements fZ^α, so the semigroup of $\mathcal{R}'$ contains Γ'. By (7.14) for $\mathcal{R}'$,

$$H_{\mathcal{R}',\mathbf{b}}(\lambda) \geq \#(\mathbb{Z}\Gamma \cap \lambda\Sigma_{\mathbf{b}}) - \#((\mathbb{Z}\Gamma \cap \Sigma)\backslash\Gamma'),$$

so by (7.16),

$$\varphi_{\mathcal{R}'}(\mathbf{b}) \geq k!\,\mathrm{vol}_{\mathbb{R}^k}(\Sigma_{\mathbf{b}}) \geq (1-\zeta)\varphi_{\mathcal{R}}(\mathbf{b}).$$

This proves (B).

By (1) and (2), $\mathcal{R} = \mathcal{G}_w[\mathbb{C}(T)]$ satisfies (i)–(iii) with $\alpha_{\mathcal{R}} = 0$, and by (7.12),

$$\varphi_{\mathcal{G}_w[\mathbb{C}(T)]}(\mathbf{b}) = \mathrm{vol}(u_{\mathbf{b}})/r.$$

Let $\mathcal{R}' \subset \mathcal{G}_w[\mathbb{C}(T)]$ be as in (B). By (1), $\mathcal{G}_w[\mathbb{C}(T)] = \mathbb{C}(T)\cdot\mathcal{G}_x$, so there exist finitely many homogeneous elements of $\mathcal{G}_x$ whose $\mathbb{C}(T)$-span contains the generators of $\mathcal{R}'$. Let $\mathcal{C}_x$ be the $\mathcal{O}_{W,x}$-subalgebra of $\mathcal{G}_x$ generated by the homogeneous elements above and by finitely many homogeneous elements of $\mathcal{G}_x$ whose ratios generate $\mathbb{C}(S)(Z_1, \dots, Z_m)$, which exist by (2). Then $\mathcal{C}_x$ is a finitely generated $\mathbb{N}^m$-graded $\mathcal{O}_{W,x}$-subalgebra of $\mathcal{G}_x$ with $(\mathcal{C}_x)_0 = \mathcal{O}_{W,x}$, the fraction field of $\mathcal{C}_x$ is $\mathbb{C}(S)(Z_1, \dots, Z_m)$, and $\mathcal{R}' \subset \mathbb{C}(T)\cdot\mathcal{C}_x$.

Let $\overline{\mathcal{C}}$ be the integral closure of $\mathcal{C}_x$ in $\mathbb{C}(S)(Z_1, \dots, Z_m)$. We have:

- $\overline{\mathcal{C}}$ is an $\mathbb{N}^m$-graded subring of $\mathbb{C}(S)[Z_1, \dots, Z_m]$. Since $\mathbb{C}(S)[Z_1, \dots, Z_m]$ is normal, it contains $\overline{\mathcal{C}}$. The automorphisms of $\mathbb{C}(S)[Z_1, \dots, Z_m]$ multiplying each Z_j by an element of $\mathbb{C}^\times$ preserve $\mathcal{C}_x$, hence $\overline{\mathcal{C}}$, and the homogeneous components of any $g \in \mathbb{C}(S)[Z_1, \dots, Z_m]$ are $\mathbb{C}$-linear combinations of the images of g under finitely many such automorphisms.
- $\overline{\mathcal{C}}$ is a finite $\mathcal{C}_x$-module, since $\mathcal{C}_x$ is a localization of a finitely generated $\mathbb{C}$-algebra.
- $\overline{\mathcal{C}}_0 = (\tau_*\mathcal{O}_T)_x$. If $g \in \overline{\mathcal{C}}_0$, then the degree 0 part of a monic equation of g over $\mathcal{C}_x$ is a monic equation over $\mathcal{O}_{W,x}$; thus $g \in \mathbb{C}(S)$ is regular on $(h|_S)^{-1}(U)$ for a neighborhood U of x in W, as S is normal, and $g \in ((h|_S)_*\mathcal{O}_S)_x = (\tau_*\mathcal{O}_T)_x$. Conversely, $(\tau_*\mathcal{O}_T)_x \subset \mathbb{C}(S)$ is finite over $\mathcal{O}_{W,x}$.
- There exists a homogeneous $\mathfrak{f} \in \mathcal{C}_x\backslash\{0\}$ with $\mathfrak{f}\overline{\mathcal{C}} \subset \mathcal{C}_x$. If $g \in \overline{\mathcal{C}}$ is homogeneous and $g = g_1/g_2$ with $g_1, g_2 \in \mathcal{C}_x$, then $g = (g_1)_{\alpha+\deg g}/(g_2)_\alpha$ for any α with $(g_2)_\alpha \neq 0$. We take $\mathfrak{f}$ to be the product of such denominators $(g_2)_\alpha$ for finitely many homogeneous generators of the $\mathcal{C}_x$-module $\overline{\mathcal{C}}$.

Here and below $\otimes\mathbb{C}(T)$ means $\otimes_{(\tau_*\mathcal{O}_T)_x}\mathbb{C}(T)$, and we identify $\overline{\mathcal{C}} \otimes \mathbb{C}(T)$ with the subring $\mathbb{C}(T)\cdot\overline{\mathcal{C}}$ of $\mathbb{C}(S)[Z_1, \dots, Z_m]$. Then $\mathcal{R} = \overline{\mathcal{C}} \otimes \mathbb{C}(T)$ satisfies (i)–(iii): (i) holds as $\mathbb{C}(T)$ is the fraction field of $(\tau_*\mathcal{O}_T)_x$, (ii) holds as $\mathcal{C}_x \subset \overline{\mathcal{C}}$, and (iii) holds with $\alpha_{\mathcal{R}} = \deg\mathfrak{f}$, since

$$\mathfrak{f}\cdot(\overline{\mathcal{C}} \otimes \mathbb{C}(T)) \subset \mathbb{C}(T)\cdot\mathcal{G}_x = \mathcal{G}_w[\mathbb{C}(T)].$$

Let $H_{\mathbf{b}'} := H_{\mathcal{R},\mathbf{b}'}$ and $\varphi := \varphi_{\mathcal{R}}$ for $\mathcal{R} = \overline{\mathcal{C}} \otimes \mathbb{C}(T)$. By (A), $\varphi : \mathbb{R}^m_{>0} \to \mathbb{R}_{>0}$ is well-defined and continuous. Since $\mathcal{R}' \subset \mathbb{C}(T)\cdot\mathcal{C}_x$ and $\mathbb{C}(T)\cdot\mathcal{C}_x \subset \overline{\mathcal{C}} \otimes \mathbb{C}(T)$, (B) implies

$$r\cdot\varphi(\mathbf{b}) \geq r\cdot\varphi_{\mathcal{R}'}(\mathbf{b}) \geq (1-\zeta)\,\mathrm{vol}(u_{\mathbf{b}}). \tag{7.17}$$

Step 3. In this step we embed $\overline{\mathcal{C}}$ into a section ring of $S \times \mathbb{P}^{m-1}$, fix C by Theorem 6.1, and prove the inequality of the proposition.

Since (X, B) is lc and E is snc, the coefficients of B_S are ≤ 1, $\operatorname{Supp} B'_S$ is snc, and B'_S has coefficients in $[0, 1)$. Thus (S, B'_S) is klt. Let $f_j Z^{\alpha_j}$, $1 \leq j \leq N_1$, be homogeneous generators of the $(\tau_* \mathcal{O}_T)_x$-algebra $\overline{\mathcal{C}}$ with $\alpha_j \neq 0$, which exist as $\overline{\mathcal{C}}_0 = (\tau_* \mathcal{O}_T)_x$. Let $R_S \geq 0$ be a divisor on S such that $\operatorname{div}(f_j) + R_S \geq 0$ for any j. Since S is quasi-projective, there exist a line bundle $\mathcal{H}_S$ on S which is very ample/T and a non-zero section $\sigma \in H^0(S, \mathcal{H}_S)$ with $\operatorname{div}(\sigma) \geq R_S$: take $\mathcal{H}_S = \mathcal{O}_S(R_S) \otimes \mathcal{A}$ for a sufficiently high power $\mathcal{A}$ of an ample line bundle and σ the product of the canonical section of $\mathcal{O}_S(R_S)$ and a non-zero section of $\mathcal{A}$. As $|\alpha_j| \geq 1$,

$$\operatorname{div}(f_j \sigma^{|\alpha_j|}) \geq \operatorname{div}(f_j) + R_S \geq 0.$$

We regard $Z_1, \ldots, Z_m$ as the homogeneous coordinates of $\mathbb{P}^{m-1}$. Let $\operatorname{pr}_S : S \times \mathbb{P}^{m-1} \to S$ be the projection and $\mathcal{L} := \mathcal{H}_S \boxtimes \mathcal{O}(1)$, which is very ample/$T$ via $\pi_S \circ \operatorname{pr}_S$. Let $t \in \tau^{-1}(x)$ and $\overline{\mathcal{C}}_t := \overline{\mathcal{C}} \otimes_{(\tau_* \mathcal{O}_T)_x} \mathcal{O}_{T,t}$, with $(\overline{\mathcal{C}}_t)_0 = \mathcal{O}_{T,t}$. The section ring

$$\bigoplus_{i \geq 0} ((\pi_S \circ \operatorname{pr}_S)_* \mathcal{L}^{\otimes i})_t$$

is $\mathbb{N}^m$-graded, with degree α piece $(\pi_{S*} \mathcal{H}_S^{\otimes |\alpha|})_t Z^\alpha$, and

$$c_\alpha Z^\alpha \to c_\alpha \sigma^{|\alpha|} Z^\alpha \tag{7.18}$$

defines an injective homomorphism of $\mathbb{N}^m$-graded $\mathcal{O}_{T,t}$-algebras from $\overline{\mathcal{C}}_t$ to the section ring. Choose a surjective homomorphism ψ_t of graded $\mathcal{O}_{T,t}$-algebras from $\mathcal{O}_{T,t}[\mathsf{y}_1, \ldots, \mathsf{y}_{N'}]$ onto the section ring, with each $\psi_t(\mathsf{y}_i)$ homogeneous of a degree in $\mathbb{N}^m \backslash \{0\}$, $\deg \mathsf{y}_i$ the total degree of $\psi_t(\mathsf{y}_i)$, and

$$\psi_t(\mathsf{y}_{N'-N_1+j}) = f_j \sigma^{|\alpha_j|} Z^{\alpha_j} \quad \text{for } 1 \leq j \leq N_1.$$

Fix the lexicographic order with $\mathsf{y}_1 > \cdots > \mathsf{y}_{N'}$. Let $I_t := \ker(\psi_t) \otimes_{\mathcal{O}_{T,t}} \mathbb{C}(T)$, let $\mathcal{B}_i$ be the set of standard monomials of degree i with respect to I_t (as defined before Lemma 6.2), and for $\alpha \in \mathbb{N}^m$ let $\mathcal{B}'_\alpha$ be the set of monomials in $\mathsf{y}_{N'-N_1+1}, \ldots, \mathsf{y}_{N'}$ of multidegree α belonging to $\mathcal{B}_{|\alpha|}$, where $\mathsf{y}_{N'-N_1+j}$ has multidegree α_j. Let

$$\rho : \mathbb{C}(T)[\mathsf{y}_{N'-N_1+1}, \ldots, \mathsf{y}_{N'}] \to \overline{\mathcal{C}} \otimes \mathbb{C}(T)$$

be the homomorphism with $\rho(\mathsf{y}_{N'-N_1+j}) = f_j Z^{\alpha_j}$. It maps monomials into $\overline{\mathcal{C}}$. Thus for a monomial β in $\mathsf{y}_{N'-N_1+1}, \ldots, \mathsf{y}_{N'}$, $\psi_t(\beta)$ is the image of $\rho(\beta)$ under (7.18).

As (7.18) is injective, the kernel of ρ is

$$I_t \cap \mathbb{C}(T)[\mathsf{y}_{N'-N_1+1}, \ldots, \mathsf{y}_{N'}].$$

Since the $f_j Z^{\alpha_j}$ generate the $(\tau_* \mathcal{O}_T)_x$-algebra $\overline{\mathcal{C}}$, ρ is surjective. Since the order is lexicographic with $\mathsf{y}_1, \ldots, \mathsf{y}_{N'-N_1}$ first, the elimination theorem [CLO15, Chapter 3, §1, Theorem 2] implies that a monomial in $\mathsf{y}_{N'-N_1+1}, \ldots, \mathsf{y}_{N'}$ is a standard monomial with respect to the kernel if and only if it is a standard monomial with respect to I_t. As the kernel is homogeneous for the multidegree, [CLO15, Chapter 5, §3, Proposition 1] implies that $\{\rho(\beta) \mid \beta \in \mathcal{B}'_\alpha\}$ is a $\mathbb{C}(T)$-basis of $\overline{\mathcal{C}}_\alpha \otimes \mathbb{C}(T)$. In particular,

$$M_\alpha := \bigoplus_{\beta \in \mathcal{B}'_\alpha} \mathcal{O}_{T,t} \rho(\beta) \subset (\overline{\mathcal{C}}_t)_\alpha$$

is a free $\mathcal{O}_{T,t}$-module of rank $\dim_{\mathbb{C}(T)}(\overline{\mathcal{C}}_\alpha \otimes \mathbb{C}(T))$.

The hypotheses of Theorem 6.1 hold for $\pi_S \circ \operatorname{pr}_S$, $\operatorname{pr}_S^* B'_S$, $\mathcal{H}_S \boxtimes \mathcal{O}(1)$, $k-1$, ψ_t in place of $\pi : X \to T$, B, $\mathcal{L}$, d, ψ: $\dim T = \ell \geq 1$, t is a closed point, $\pi_S \circ \operatorname{pr}_S$ is projective and surjective, as S is projective over W and τ is finite, $S \times \mathbb{P}^{m-1}$ is smooth, the generic fiber of $\pi_S \circ \operatorname{pr}_S$ is geometrically integral as $\mathbb{C}(T)$ is algebraically closed in $\mathbb{C}(S)$, of dimension $(k-m) + (m-1) = k - 1 \geq 1$ as $k \geq 2$, and $(S \times \mathbb{P}^{m-1}, \operatorname{pr}_S^* B'_S)$ is klt. By Theorem 6.1, we

obtain an integer $i_{0,t} \geq 1$ and a real number $C_t > 0$. As $\tau^{-1}(x)$ is finite, let $C := \max_{t\in\tau^{-1}(x)} C_t$ and $i_0 := \max_{t\in\tau^{-1}(x)} i_{0,t}$. Then C and φ do not depend on θ.

Now let θ be as in the statement, and $t \in \tau^{-1}(x)$ the image of $\mathrm{center}_S(\theta)$ in T. Since $\tau(t) = x$ and $\dim W \geq 1$, t is not the generic point of T, so $(\pi_S)_*\theta$ is a divisorial valuation on T with center t by Lemma 6.3. For $\lambda' \in \mathbb{R}$ let

$$\chi(\lambda') := \mathrm{length}\left(\mathcal{O}_{T,t}/\mathfrak{a}_{\lambda'}((\pi_S)_*\theta)\right), \quad \text{where } \mathfrak{a}_{\lambda'}((\pi_S)_*\theta) := \mathcal{O}_{T,t} \text{ if } \lambda' \leq 0.$$

Then χ is increasing. The function field of $S \times \mathbb{P}^{m-1}$ is $\mathbb{C}(S)(Z_2/Z_1, \dots, Z_m/Z_1)$. Let θ' be the weight 0 extension of θ: for a polynomial in $Z_2/Z_1, \dots, Z_m/Z_1$ with coefficients in $\mathbb{C}(S)$, θ' is the minimum of the θ-values of the coefficients. Let E' be a prime divisor on a log resolution $S' \to S$ of (S, B'_S) such that θ is a positive multiple of $v_{E'}$. Then θ' is the positive multiple of $v_{E'\times\mathbb{P}^{m-1}}$ that agrees with θ on non-zero rational functions on S, and $S' \times \mathbb{P}^{m-1}$ is a log resolution of $(S \times \mathbb{P}^{m-1}, \mathrm{pr}_S^* B'_S)$ with crepant boundary the pullback of the crepant boundary on S'. Hence

$$a(\theta', S \times \mathbb{P}^{m-1}, \mathrm{pr}_S^* B'_S) = a(\theta, S, B'_S) > 0.$$

The center of θ' maps to t. Let e_S be a local generator of $\mathcal{H}_S$ at $\mathrm{center}_S(\theta)$. Then $e := e_S Z_1$ is a local generator of $\mathcal{L}$ at the center of θ'. Let

$$\widetilde{\mathbf{b}} := \mathbf{b} + Ca(\theta, S, B'_S) \cdot (1, \dots, 1),$$

so that

$$\widetilde{\mathbf{b}} \cdot \alpha = \mathbf{b} \cdot \alpha + C|\alpha| a(\theta, S, B'_S) \quad \text{for } \alpha \in \mathbb{N}^m.$$

Let $\alpha \in \mathbb{N}^m$ with $|\alpha| \geq i_0$, $\lambda' \in \mathbb{R}$, and $g = \sum_{\beta\in\mathcal{B}'_\alpha} b_\beta \rho(\beta) \neq 0$ with $b_\beta \in \mathcal{O}_{T,t}$ and $\omega_{\theta,\mathbf{b}}(g) \geq \lambda'$. Write $g = c_g Z^\alpha$. Thus $\theta(c_g) \geq \lambda' - \mathbf{b} \cdot \alpha$. The image of g under (7.18) is $c_g \sigma^{|\alpha|} Z^\alpha$, and it equals $\psi_t(\sum_\beta b_\beta \beta)$. Since $\theta'(Z^\alpha/Z_1^{|\alpha|}) = 0$ and σ is regular at $\mathrm{center}_S(\theta)$, Theorem 6.1, with $b_\beta := 0$ for $\beta \in \mathcal{B}_{|\alpha|}\backslash\mathcal{B}'_\alpha$, implies

$$\begin{aligned} \lambda' - \mathbf{b} \cdot \alpha \leq \theta(c_g) \leq \theta(c_g) + |\alpha|\theta(\sigma/e_S) &= \theta'\Big(\frac{\psi_t(\sum_\beta b_\beta \beta)}{e^{|\alpha|}}\Big) \\ &\leq \min_{\beta\in\mathcal{B}'_\alpha} \theta(b_\beta) + C|\alpha| a(\theta, S, B'_S). \end{aligned} \tag{7.19}$$

Thus

$$b_\beta \in \mathfrak{a}_{\lambda' - \widetilde{\mathbf{b}}\cdot\alpha}((\pi_S)_*\theta) \quad \text{for any } \beta \in \mathcal{B}'_\alpha.$$

Hence the surjection

$$M_\alpha \to \bigoplus_{\beta\in\mathcal{B}'_\alpha} \mathcal{O}_{T,t}/\mathfrak{a}_{\lambda' - \widetilde{\mathbf{b}}\cdot\alpha}((\pi_S)_*\theta), \quad \sum_\beta b_\beta \rho(\beta) \to (b_\beta)_\beta,$$

vanishes on $M_\alpha \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda'\}$, and

$$\dim_{\mathbb{C}} \frac{M_\alpha}{M_\alpha \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda'\}} \geq \dim_{\mathbb{C}(T)} (\overline{\mathcal{C}}_\alpha \otimes \mathbb{C}(T)) \cdot \chi(\lambda' - \widetilde{\mathbf{b}} \cdot \alpha). \tag{7.20}$$

Let $\lambda > 0$ and $\lambda' := \lambda - \omega_{\theta,\mathbf{b}}(\mathfrak{f})$. We have:

(a) By (7.11) and $\mathcal{C}_x \subset \mathcal{G}_x$,

$$\mathrm{length}(\mathcal{O}_{X,x}/\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})) \geq \dim_{\mathbb{C}} \mathcal{C}_x/(\mathcal{C}_x \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda\}).$$

(b) $\omega_{\theta,\mathbf{b}} \geq 0$ on $\overline{\mathcal{C}}$, as $\overline{\mathcal{C}}$ is integral over $\mathcal{C}_x$. Thus $g \to \mathfrak{f}g$ induces an injection

$$\overline{\mathcal{C}}/(\overline{\mathcal{C}} \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda'\}) \to \mathcal{C}_x/(\mathcal{C}_x \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda\}).$$

(c) Let $\mathfrak{m}_t \subset (\tau_*\mathcal{O}_T)_x$ be the maximal ideal of t. The natural map

$$\overline{\mathcal{C}}/(\overline{\mathcal{C}} \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda'\}) \to \overline{\mathcal{C}}_t/(\overline{\mathcal{C}}_t \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda'\})$$

is bijective. Any $g' \in (\tau_*\mathcal{O}_T)_x\backslash\mathfrak{m}_t$ is a unit at $\mathrm{center}_S(\theta)$, so

$$\omega_{\theta,\mathbf{b}}(g/g') = \omega_{\theta,\mathbf{b}}(g) \quad \text{for } g \in \overline{\mathcal{C}},$$

which implies injectivity. For surjectivity, let $g/g' \in \overline{\mathcal{C}}_t$ with $g \in \overline{\mathcal{C}}$ and $g' \in (\tau_*\mathcal{O}_T)_x \backslash \mathfrak{m}_t$. For any integer $j \geq 1$, since $(\tau_*\mathcal{O}_T)_x/\mathfrak{m}_t^j = \mathcal{O}_{T,t}/\mathfrak{m}_{T,t}^j$, there exists $g'' \in (\tau_*\mathcal{O}_T)_x$ with $1 - g'g'' \in \mathfrak{m}_t^j$, and

$$\omega_{\theta,\mathbf{b}}(g/g' - g''g) = \omega_{\theta,\mathbf{b}}(g) + \theta(1 - g'g'') \geq j \min \theta(\mathfrak{m}_t) \geq \lambda'$$

for $j \gg 0$, where $\min \theta(\mathfrak{m}_t) > 0$ is the minimum of θ on finitely many generators of $\mathfrak{m}_t$.

(d) $\overline{\mathcal{C}}_t \cap \{\omega_{\theta,\mathbf{b}} \geq \lambda'\}$ is homogeneous, and $M_\alpha \subset (\overline{\mathcal{C}}_t)_\alpha$.

By (a)–(d) and (7.20), each quotient in (a)–(c) and (7.20) is finite-dimensional over $\mathbb{C}$, and

$$\operatorname{length}(\mathcal{O}_{X,x}/\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})) \geq \sum_{|\alpha| \geq i_0} \dim_{\mathbb{C}(T)}(\overline{\mathcal{C}}_\alpha \otimes \mathbb{C}(T)) \cdot \chi(\lambda' - \widetilde{\mathbf{b}} \cdot \alpha). \tag{7.21}$$

The finitely many α with $|\alpha| < i_0$ contribute at most

$$\sum_{|\alpha| < i_0} \dim_{\mathbb{C}(T)}(\overline{\mathcal{C}}_\alpha \otimes \mathbb{C}(T)) \cdot \chi(\lambda) = O(\lambda^\ell),$$

which is $o(\lambda^n)$ as $k \geq 1$. So we may sum over all $\alpha \in \mathbb{N}^m$ in (7.21) at the cost of $o(\lambda^n)$. Let $N \geq 1$ be an integer. Grouping the α with $(j-1)\lambda/N \leq \widetilde{\mathbf{b}} \cdot \alpha < j\lambda/N$, $1 \leq j \leq N$, and using that χ is increasing, we obtain

$$\sum_{\alpha \in \mathbb{N}^m} \dim_{\mathbb{C}(T)}(\overline{\mathcal{C}}_\alpha \otimes \mathbb{C}(T)) \cdot \chi(\lambda' - \widetilde{\mathbf{b}} \cdot \alpha) \geq \sum_{j=1}^{N} \Big(H_{\widetilde{\mathbf{b}}}\big(\tfrac{j\lambda}{N}\big) - H_{\widetilde{\mathbf{b}}}\big(\tfrac{(j-1)\lambda}{N}\big)\Big)\chi\big(\lambda' - \tfrac{j\lambda}{N}\big).$$

As $\dim \mathcal{O}_{X,x} = n$ and $\dim \mathcal{O}_{T,t} = \ell$, Definition 2.2 implies

$$\operatorname{length}(\mathcal{O}_{X,x}/\mathfrak{a}_\lambda(\mu_{\theta,\mathbf{b}})) \sim \frac{\operatorname{vol}(\mu_{\theta,\mathbf{b}})\lambda^n}{n!} \quad \text{and} \quad \chi(\lambda) \sim \frac{\operatorname{vol}((\pi_S)_*\theta)\lambda^\ell}{\ell!},$$

and, by the definition of φ in **Step 2**,

$$H_{\widetilde{\mathbf{b}}}(\lambda) \sim \frac{\varphi(\widetilde{\mathbf{b}})\lambda^k}{k!}.$$

Dividing by $\lambda^n/n!$ and letting $\lambda \to +\infty$, we obtain

$$\operatorname{vol}(\mu_{\theta,\mathbf{b}}) \geq \frac{n!}{k!\ell!}\varphi(\widetilde{\mathbf{b}}) \operatorname{vol}((\pi_S)_*\theta) \sum_{j=1}^{N} \Big(\big(\tfrac{j}{N}\big)^k - \big(\tfrac{j-1}{N}\big)^k\Big)\big(1 - \tfrac{j}{N}\big)^\ell.$$

As $N \to +\infty$, the sum converges to

$$\int_0^1 (1-y)^\ell d(y^k) = k\int_0^1 y^{k-1}(1-y)^\ell dy = \frac{k!\ell!}{(k+\ell)!} = \frac{k!\ell!}{n!}.$$

Therefore

$$\operatorname{vol}(\mu_{\theta,\mathbf{b}}) \geq \varphi\big(\mathbf{b} + Ca(\theta, S, B'_S) \cdot (1, \ldots, 1)\big) \cdot \operatorname{vol}((\pi_S)_*\theta).$$

Together with (7.17), the proposition follows. □

7.5. **Proof of Theorem 7.1.**

Proof of Theorem 7.1. Since $B - B_\epsilon = \epsilon cD$, for any $\epsilon \in [0,1]$ and any valuation v on X with a center on X,

$$a(v, X, B_\epsilon) = a(v, X, B) + \epsilon c v(D). \tag{7.22}$$

Step 1. Possibly shrinking X to a neighborhood of x, let Δ_j, δ_j, c_j, B_j, $B_{j,\epsilon}$, and $(W, \Delta_{W,j}, \mathbf{M}_j)$ be as in Construction-Proposition 3.1. By Construction-Proposition 3.1(1), (X, Δ_j, D, x) satisfies the conditions of Set-up 2.8 with minimal lc center W, so ℓ and k do not change when (X, Δ, D, x) is replaced by (X, Δ_j, D, x), while c, B, B_ϵ, and $(W, \Delta_W, \mathbf{M})$ are replaced by c_j, B_j, $B_{j,\epsilon}$, and

$(W, \Delta_{W,j}, \mathbf{M}_j)$. Assume that Theorem 7.1 holds for (X, Δ_j, D, x) for any j. By Corollary 3.2(1), Theorem 7.1 for (X, Δ_j, D, x), Corollary 3.2(2) with $\epsilon = 1$ and (3), and since $k + \ell = n$,

$$\begin{aligned}\liminf_{\epsilon\to 0^+} \frac{\widehat{\mathrm{vol}}(X \ni x, B_\epsilon)}{n^n\epsilon^k} &\geq (1+\delta_j)^{-n} \liminf_{\epsilon\to 0^+} \frac{\widehat{\mathrm{vol}}(X \ni x, B_{j,\epsilon})}{n^n\epsilon^k} \\ &\geq (1+\delta_j)^{-n} \cdot \frac{\widehat{\mathrm{vol}}(X \ni w, \Delta_j) \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_{W,j}, \mathbf{M}_j)}{k^k\ell^\ell} \\ &\geq \frac{(1-\delta_j)^n}{(1+\delta_j)^n} \cdot \frac{\widehat{\mathrm{vol}}(X \ni w, \Delta) \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{k^k\ell^\ell}\end{aligned}$$

for any j, and Theorem 7.1 follows by letting $j \to +\infty$. Possibly replacing (X, Δ, D, x) with (X, Δ_j, D, x), we may assume that Δ is a $\mathbb{Q}$-divisor, $K_X + \Delta$ is $\mathbb{Q}$-Cartier, and $c \in \mathbb{Q}$. Then B is a $\mathbb{Q}$-divisor and $K_X + B$ is $\mathbb{Q}$-Cartier, so Δ_W is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is a $\mathbb{Q}$-$\boldsymbol{b}$-divisor by Definition-Theorem 2.7(3).

Since (X, B) is lc, $a(v, X, B) \geq 0$ for any valuation v on X with $\mathrm{center}_X v = x$, so (7.22) implies

$$\widehat{\mathrm{vol}}(X \ni x, B_\epsilon) \leq \widehat{\mathrm{vol}}(X \ni x, B_{\epsilon'}) \leq (\epsilon'/\epsilon)^n \widehat{\mathrm{vol}}(X \ni x, B_\epsilon) \quad \text{for } 0 < \epsilon < \epsilon' \leq 1.$$

Thus, the case when the right side of (7.23) below is $+\infty$ being trivial, we may choose rational numbers $\epsilon_i \in (0, 1]$, $i \in \mathbb{N}^+$, with $\lim_{i\to+\infty} \epsilon_i = 0$ such that

$$\lim_{i\to+\infty} \frac{\widehat{\mathrm{vol}}(X \ni x, B_{\epsilon_i})}{\epsilon_i^k} = \liminf_{\epsilon\to 0^+} \frac{\widehat{\mathrm{vol}}(X \ni x, B_\epsilon)}{\epsilon^k}. \tag{7.23}$$

By Lemma 2.9, $(X \ni x, B_{\epsilon_i})$ is klt, so $\widehat{\mathrm{vol}}(X \ni x, B_{\epsilon_i}) > 0$ [LLX20, Theorem 2.5]. By Lemma 4.2, there exists a prime divisor Γ_i over X such that $\mathrm{center}_X \Gamma_i = \{x\}$ and

$$a(\Gamma_i, X, B_{\epsilon_i})^n \,\mathrm{vol}(\Gamma_i) \leq (1 + 1/i)\widehat{\mathrm{vol}}(X \ni x, B_{\epsilon_i}).$$

Since $x \in \mathrm{Supp}\, D$ and D is $\mathbb{Q}$-Cartier, $\mathrm{mult}_{\Gamma_i} D > 0$. Let $v_i := v_{\Gamma_i}/(c\,\mathrm{mult}_{\Gamma_i} D)$. Then $cv_i(D) = 1$. Since $a(\lambda v, X, B_{\epsilon_i})^n \,\mathrm{vol}(\lambda v)$ does not depend on $\lambda > 0$ for any valuation v on X with $\mathrm{center}_X v = x$, the first inequality in (7.24) below holds, and the equality there follows from (7.22):

$$\begin{gathered}a(v_i, X, B_{\epsilon_i})^n \,\mathrm{vol}(v_i) \leq (1 + 1/i)\widehat{\mathrm{vol}}(X \ni x, B_{\epsilon_i}) \quad \text{and} \\ a(v_i, X, B_{\epsilon_i}) = a(v_i, X, B) + \epsilon_i.\end{gathered} \tag{7.24}$$

Since $\epsilon_i \in \mathbb{Q}$ and $1 + 1/i \leq 2$, Lemma 7.9 implies that there exist real numbers $\alpha, \gamma > 0$ such that

$$a(v_i, X, B) \leq \gamma\epsilon_i \quad \text{and} \quad v_i(f) \geq \alpha \text{ for any } f \in \mathcal{I}_W \cdot \mathcal{O}_{X,x} \tag{7.25}$$

for any i. By (7.23) and (7.24), it suffices to show that

$$\liminf_{i\to+\infty} \frac{(a(v_i, X, B) + \epsilon_i)^n \,\mathrm{vol}(v_i)}{\epsilon_i^k} \geq \frac{n^n \cdot \widehat{\mathrm{vol}}(X \ni w, \Delta) \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{k^k\ell^\ell}. \tag{7.26}$$

Step 2. Let $h : Y \to X$, B_Y, and $E = \sum_j E_j$ be as in Set-up 7.2(2)(3), and let $K_Y + \Delta_Y := h^*(K_X + \Delta)$. Let ξ_i be the center of v_i on Y. Then $h(\xi_i) = x$. For any component E_j of E, let $v_i(E_j)$ be the value of v_i at a local equation of E_j at ξ_i, so $v_i(E_j) = 0$ if $\xi_i \notin E_j$. Since $\mathrm{Supp}\, B_Y \cup \mathrm{Supp}\, \Delta_Y \subset \mathrm{Supp}\, E$ and (Y, E) is lc, $a(v_i, Y, E) \geq 0$, and thus

$$a(v_i, X, B) \geq \sum_j (1 - \mathrm{mult}_{E_j} B_Y) v_i(E_j) \quad \text{and} \quad a(v_i, X, \Delta) \geq \sum_j (1 - \mathrm{mult}_{E_j} \Delta_Y) v_i(E_j),$$

where all terms are non-negative, as $\mathrm{mult}_{E_j} B_Y \leq 1$ and $\mathrm{mult}_{E_j} \Delta_Y < 1$ for any j. By (7.22) and (7.25), $a(v_i, X, B) \leq \gamma\epsilon_i$ and $a(v_i, X, \Delta) = a(v_i, X, B) + 1 \leq \gamma + 1$. Thus

$$v_i(E_j) \leq \frac{\gamma + 1}{1 - \mathrm{mult}_{E_j} \Delta_Y} \quad \text{for any } j, \quad \text{and} \quad v_i(E_j) \leq \frac{\gamma\epsilon_i}{1 - \mathrm{mult}_{E_j} B_Y} \quad \text{if } \mathrm{mult}_{E_j} B_Y < 1.$$

Passing to a subsequence, $\lim_{i\to+\infty} v_i(E_j)$ exists for any j, and it is 0 if $\mathrm{mult}_{E_j} B_Y < 1$. Possibly reordering indices, we may assume that $E_1, \dots, E_m$ are exactly the components E_j of E such that $\mathrm{mult}_{E_j} B_Y = 1$ and $\lim_i v_i(E_j) > 0$. Then

$$\lim_{i\to+\infty} v_i(E_j) = 0 \quad \text{for any } j > m, \tag{7.27}$$

and, possibly removing finitely many i, there exists a real number $\kappa > 0$ such that $v_i(E_j) \geq \kappa$ for any i and any $1 \leq j \leq m$.

Let $R \geq 0$ be the divisor on Y such that $\mathcal{I}_W \mathcal{O}_Y = \mathcal{O}_Y(-R)$. Then $\operatorname{Supp} R \subset \operatorname{Supp} E$. By (7.25), $\sum_j \mathrm{mult}_{E_j} R \cdot v_i(E_j) \geq \alpha$ for any i, so (7.27) implies $\mathrm{mult}_{E_{j_0}} R > 0$ for some $1 \leq j_0 \leq m$. In particular, $m \geq 1$ and $h(E_{j_0}) \subset W$. Since $v_i(E_j) > 0$ for $1 \leq j \leq m$, $\xi_i \in E_1 \cap \cdots \cap E_m$. Passing to a subsequence, there exists an irreducible component S of $E_1 \cap \cdots \cap E_m$ such that $\xi_i \in S$ for any i. Since E is snc, no component of E other than $E_1, \dots, E_m$ contains S. Let F be the exceptional prime divisor of the blow-up of S if $m \geq 2$, and $F := E_1$ if $m = 1$. Then

$$a(F, X, B) = m - \sum_{j=1}^{m} \mathrm{mult}_{E_j} B_Y = 0,$$

so $h(S) = \operatorname{center}_X F$ is an lc center of (X, B). Since $h(S) \subset h(E_{j_0}) \subset W$ and W is the minimal lc center of (X, B),

$$h(S) = W.$$

Thus h, B_Y, E, $E_1, \dots, E_m$, and S satisfy the conditions of Set-up 7.2. Let B_S, B'_S, T, τ, r, $\tau^*\mathbf{M}$, and Δ_T be as in Set-up 7.2 and Construction 7.7.

Since E is snc and $h^{-1}(x) \cap S$ is quasi-compact, finitely many affine open subsets of Y as in Construction-Proposition 7.3, on each of which every component of E meeting it other than $E_1, \dots, E_m$ is the zero locus of one of $y_1, \dots, y_{n-m}$, cover $h^{-1}(x) \cap S$. Passing to a subsequence, we fix one of them, Y°, together with $(y_1, \dots, y_{n-m}, z_1, \dots, z_m)$ and ι, such that $\xi_i \in Y^\circ$ for any i. Then $\xi_i \in S^\circ$, and $v_i(z_j) = v_i(E_j)$ for any i and any $1 \leq j \leq m$.

If $k = 1$, then $n - m = \dim S \geq \dim W = n - 1$, so $m = 1$ and $S = E_1$. Since E_1 is a prime divisor on Y with $h(E_1) = W$, $E_1 = h_*^{-1}W$, and $S \to W$ is birational. As W is normal by Definition-Theorem 2.7, $T = W$, $\tau = \mathrm{id}_W$, $r = 1$, $\Delta_T = \Delta_W$, and $\tau^*\mathbf{M} = \mathbf{M}$.

Step 3. Assume that $k \geq 2$. We prove that, possibly replacing X with an étale neighborhood of x,

$$\widehat{\mathrm{vol}}(T \ni p, \Delta_T, \tau^*\mathbf{M}) = r \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M}), \tag{7.28}$$

where $p \in T$ is the point over x fixed below.

By **Steps 1** and 2, the hypotheses of Proposition 7.8 hold. Possibly shrinking X to a neighborhood of x, Proposition 7.8(1) holds, and by Proposition 7.8(2) there exists a finite morphism $\tau_1 : T_1 \to T$ from a normal variety T_1 such that $\tau \circ \tau_1$ is Galois and $\Delta_{T_1} \geq 0$, where $(\tau \circ \tau_1)^*\mathbf{M}$ is defined as $\tau^*\mathbf{M}$ in Construction 7.7 and Δ_{T_1} is the $\mathbb{Q}$-divisor on T_1 with

$$K_{T_1} + \Delta_{T_1} + ((\tau \circ \tau_1)^*\mathbf{M})_{T_1} = (\tau \circ \tau_1)^*(K_W + \Delta_W + \mathbf{M}_W).$$

The image of ξ_i in T belongs to the finite set $\tau^{-1}(x)$. Passing to a subsequence, it is a point $p \in T$ independent of i. Let $q \in T_1$ be a point such that $\tau_1(q) = p$, let $G \subset \mathrm{Gal}(T_1/W)$ be the decomposition group of q, that is, the subgroup of elements fixing q, and let $\overline{q}$ be the image of q in T_1/G. By [GR71, Exposé V, Proposition 2.2], $T_1/G \to W$ is étale at $\overline{q}$. Let $W^\circ \subset T_1/G$ be an open neighborhood of $\overline{q}$ which is étale over W. By [Gro67, Proposition 18.1.1], possibly shrinking W°, there exist an étale morphism $\pi : X' \to X$ from an affine normal variety X' and a closed point $x' \in X'$ such that $\pi(x') = x$ and $W' := \pi^{-1}(W) \cong W^\circ$ over W, with x' corresponding to $\overline{q}$. Possibly shrinking X' to an affine neighborhood of x', we may assume that $\pi^{-1}(x) = \{x'\}$.

Since π is étale, X' is a normal quasi-projective variety, $\pi^*\Delta \geq 0$ is a $\mathbb{Q}$-divisor, $K_{X'} + \pi^*\Delta$ is $\mathbb{Q}$-Cartier, $(X', \pi^*\Delta)$ is klt, $\pi^*D \geq 0$ is a non-zero $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor with $x' \in \operatorname{Supp} \pi^*D$, and $\mathrm{lct}(X' \ni x', \pi^*\Delta; \pi^*D) = c$. By Lemma 4.4(2) for the étale morphism π, possibly shrinking X'

to a neighborhood of x', any lc center of (X', π^*B) contains x', W' is the minimal lc center of (X', π^*B), and

$$\widehat{\mathrm{vol}}(W' \ni x', \Delta_{W'}, \mathbf{M}') = \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M}), \tag{7.29}$$

where $(W', \Delta_{W'}, \mathbf{M}')$ is the generalized pair induced by adjunction from (X', π^*B) to W'. Thus $(X', \pi^*\Delta, \pi^*D, x')$ satisfies the conditions of Set-up 2.8 with minimal lc center W'.

For any i, Lemma 4.4(1) for the klt germ $(X \ni x, B_{\epsilon_i})$ implies that

$$\widehat{\mathrm{vol}}(X' \ni x', \pi^*B_{\epsilon_i}) = \widehat{\mathrm{vol}}(X \ni x, B_{\epsilon_i}),$$

and that there exists a divisorial valuation v_i' on X' such that $\pi_* v_i' = v_i$, $\mathrm{center}_{X'} v_i' = x'$, $a(v_i', X', \pi^*B_{\epsilon_i}) = a(v_i, X, B_{\epsilon_i})$, and $\mathrm{vol}(v_i') = \mathrm{vol}(v_i)$. Since $\pi_* v_i' = v_i$, $v_i'(\pi^*D) = v_i(D)$, so $cv_i'(\pi^*D) = 1$, and $a(v_i', X', \pi^*B) = a(v_i, X, B)$ by (7.22).

Let w' be the generic point of W'. By Theorem 1.3, there exist non-empty open subsets $U \subset W$ and $U' \subset W'$ such that

$$\widehat{\mathrm{vol}}(X \ni y, \Delta) = \frac{n^n}{k^k} \cdot \widehat{\mathrm{vol}}(X \ni w, \Delta) \quad \text{and} \quad \widehat{\mathrm{vol}}(X' \ni y', \pi^*\Delta) = \frac{n^n}{k^k} \cdot \widehat{\mathrm{vol}}(X' \ni w', \pi^*\Delta)$$

for any closed points $y \in U$ and $y' \in U'$. Since $W' \to W$ is étale, $U' \cap \pi^{-1}(U) \neq \emptyset$, and for a closed point $y' \in U' \cap \pi^{-1}(U)$, Lemma 4.4(1) implies $\widehat{\mathrm{vol}}(X' \ni y', \pi^*\Delta) = \widehat{\mathrm{vol}}(X \ni \pi(y'), \Delta)$. Thus

$$\widehat{\mathrm{vol}}(X' \ni w', \pi^*\Delta) = \widehat{\mathrm{vol}}(X \ni w, \Delta). \tag{7.30}$$

Let $h' : Y' := Y \times_X X' \to X'$ be the base change of h, and E', $B_{Y'}$ the pullbacks of E, B_Y. Then h' is a log resolution of $(X', \pi^*(B+D))$, $\mathcal{I}_{W'}\mathcal{O}_{Y'}$ is invertible, E' is snc with support as in Set-up 7.2(3), and $K_{Y'} + B_{Y'} = h'^*(K_{X'} + \pi^*B)$. The center ξ_i' of v_i' on Y' maps to ξ_i, and since $\pi_* v_i' = v_i$, the value of v_i' at a local equation at ξ_i' of the preimage of E_j is $v_i(E_j)$ for any j. The preimage of S in Y' is smooth. Passing to a subsequence, there exists an irreducible component S' of it such that $\xi_i' \in S'$ for any i. For $1 \le j \le m$, let E_j' be the component of the preimage of E_j containing S'. Then $\mathrm{mult}_{E_j'} B_{Y'} = 1$, S' is an irreducible component of $E_1' \cap \cdots \cap E_m'$, and, as in **Step 2**, $h'(S')$ is an lc center of (X', π^*B) contained in W', so $h'(S') = W'$. Passing to a subsequence, there exists an affine open subset $Y'^{\circ} \subset Y^{\circ} \times_X X'$ containing ξ_i' for any i on which the pullbacks of $y_1, \dots, y_{n-m}, z_1, \dots, z_m$ satisfy the conditions of Construction-Proposition 7.3 for h', E', $E_1', \dots, E_m'$, and S', and every component of E' meeting Y'° other than $E_1', \dots, E_m'$ is the zero locus of the pullback of one of $y_1, \dots, y_{n-m}$. Thus (7.27) and $v_i'(E_j') \ge \kappa$ for $1 \le j \le m$ hold for v_i', and $\xi_i' \in S' \cap Y'^{\circ}$ for any i.

Since $\pi^{-1}(W) = W'$, the preimage of S in Y' is $S \times_W W'$. Since $W' \to W$ is étale, $S \times_W W' \to T \times_W W' \to W'$ is the Stein factorization of $S \times_W W' \to W'$, and $T \times_W W'$ is normal. Let p' be the unique point of $T \times_W W'$ over p and x', and T' the irreducible component of $T \times_W W'$ containing p'. Then $S' \to T' \xrightarrow{\tau'} W'$ is the Stein factorization of $h'|_{S'}$, and p' is the image of ξ_i'. Let $r' := \deg \tau'$, and let $\tau'^*\mathbf{M}'$ and $\Delta_{T'}$ be as in Construction 7.7 for X'. Since h', $B_{Y'}$, E', $E_1', \dots, E_m'$, and S' satisfy the conditions of Set-up 7.2 for $(X', \pi^*\Delta, \pi^*D, x')$, Proposition 7.8(1) for X' implies that, possibly shrinking X' to a neighborhood of x', $\Delta_{T'} \ge 0$ and $(T', \Delta_{T'}, \tau'^*\mathbf{M}')$ is generalized klt near p'. Since $\Delta_{W'}$ and $\mathbf{M}'$ are $\mathbb{Q}$-coefficient by Definition-Theorem 2.7(3), so are $\Delta_{T'}$ and $\tau'^*\mathbf{M}'$.

Let $T_1^{\circ} \subset T_1$ be the preimage of $W^{\circ} \cong W'$. Since $T_1 \to T_1/G$ is the quotient by G and G fixes q, $T_1^{\circ} \to W'$ is a finite Galois morphism of degree $\#G$ whose fiber over x' is $\{q\}$. The morphism $T_1^{\circ} \to T \times_W W'$ induced by τ_1 and by $T_1^{\circ} \to W'$ maps q to p', hence maps T_1° onto T'. Since T' is normal, $T_1^{\circ} \to T'$ is the quotient by a subgroup $G' \subset G$. It is a finite Galois morphism of degree $\#G'$ whose fiber over p' is $\{q\}$, and $r' = \#G/\#G'$.

By Lemma 4.4(2), $\Delta_{W'}$ and $\mathbf{M}'$ are the pullbacks of Δ_W and $\mathbf{M}$. Therefore, the $\mathbb{Q}$-divisor B' of Proposition 4.1 for $T_1^{\circ} \to W'$ and $(W', \Delta_{W'}, \mathbf{M}')$ is $\Delta_{T_1}|_{T_1^{\circ}} \ge 0$, and so is the one for $T_1^{\circ} \to T'$

and $(T', \Delta_{T'}, \tau'^*\mathbf{M}')$. By Proposition 4.1 for $T_1^\circ \to W'$ and for $T_1^\circ \to T'$,

$$\#G \cdot \widehat{\mathrm{vol}}(W' \ni x', \Delta_{W'}, \mathbf{M}') = \widehat{\mathrm{vol}}(T_1^\circ \ni q, \Delta_{T_1}|_{T_1^\circ}, (\tau \circ \tau_1)^*\mathbf{M}) = \#G' \cdot \widehat{\mathrm{vol}}(T' \ni p', \Delta_{T'}, \tau'^*\mathbf{M}').$$

By (7.29) and $r' = \#G/\#G'$,

$$\widehat{\mathrm{vol}}(T' \ni p', \Delta_{T'}, \tau'^*\mathbf{M}') = r' \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M}).$$

By (7.30) and (7.29), possibly replacing (X, Δ, D, x), v_i, and the data of **Steps 2** and 3 with $(X', \pi^*\Delta, \pi^*D, x')$, v_i', and the primed data, we may assume that (7.28) holds.

Step 4. For any i, let θ_i be the valuation θ of Proposition 7.5 for $v = v_i$ and $\xi = \xi_i$, let $\boldsymbol{\alpha}_i := (v_i(z_1), \dots, v_i(z_m))$, and let $\mu_i := \mu_{\theta_i, \boldsymbol{\alpha}_i}$. By Proposition 7.5, θ_i is a divisorial valuation on S with $\mathrm{center}_S(\theta_i) = \xi_i$.

Since $E - B_Y \geq 0$,

$$a(v_i, Y, E) \leq a(v_i, Y, B_Y) = a(v_i, X, B) \leq \gamma\epsilon_i.$$

Since $v_i(z_j) \geq \kappa$ for $1 \leq j \leq m$, Proposition 7.5(1) implies

$$\mu_i(f) \leq v_i(f) \leq (1 + m\gamma\epsilon_i/\kappa)\mu_i(f) \quad \text{for any } f \in \mathcal{O}_{X,x}.$$

Thus $\mathrm{center}_X(\mu_i) = x$ and

$$\mathfrak{a}_t(v_i) \subset \mathfrak{a}_{t/(1+m\gamma\epsilon_i/\kappa)}(\mu_i) \quad \text{for any } t > 0,$$

so

$$\mathrm{vol}(v_i) \geq (1 + m\gamma\epsilon_i/\kappa)^{-n} \mathrm{vol}(\mu_i). \tag{7.31}$$

By Proposition 7.5(2),

$$a(\mu_i, X, B) = a(\theta_i, S, B_S) \leq a(v_i, X, B). \tag{7.32}$$

Since $h(\xi_i) = x$ and $\ell \geq 1$, x is not the generic point of W, so by Lemma 7.6 and (7.32), $(h|_S)_*\theta_i$ is a divisorial valuation on W with center x, and

$$a((h|_S)_*\theta_i, W, \Delta_W, \mathbf{M}) \leq a(\theta_i, S, B_S) = a(\mu_i, X, B). \tag{7.33}$$

Since $(W, \Delta_W, \mathbf{M})$ is generalized klt, $a(\mu_i, X, B) > 0$. If $k \geq 2$, then, by Lemma 6.3 for π_S, $(\pi_S)_*\theta_i$ is a divisorial valuation on T with center $\pi_S(\xi_i) = p$, as $\tau(p) = x$, and by Proposition 7.8(1) and (7.33),

$$a((\pi_S)_*\theta_i, T, \Delta_T, \tau^*\mathbf{M}) = a((h|_S)_*\theta_i, W, \Delta_W, \mathbf{M}) \leq a(\mu_i, X, B).$$

Since $(T, \Delta_T, \tau^*\mathbf{M})$ is generalized klt near p, Lemma 4.2 and (7.28) imply

$$a(\mu_i, X, B)^\ell \mathrm{vol}((\pi_S)_*\theta_i) \geq \widehat{\mathrm{vol}}(T \ni p, \Delta_T, \tau^*\mathbf{M}) = r \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M}). \tag{7.34}$$

If $k = 1$, then $T = W$, $\tau = \mathrm{id}_W$, $r = 1$, $\Delta_T = \Delta_W$, and $\tau^*\mathbf{M} = \mathbf{M}$ by **Step 2**. We let $p := x = h(\xi_i)$, the image of ξ_i in $T = W$. Then (7.34) follows from Lemma 4.2 for $(W, \Delta_W, \mathbf{M})$ and (7.33).

We have $B_S = \sum_{j>m} \mathrm{mult}_{E_j} B_Y \cdot E_j|_S$, and $E_j|_S$ is reduced for any $j > m$. Thus

$$B_S - B_S' \leq \frac{1}{2} \sum_{j>m,\ \mathrm{mult}_{E_j} B_Y = 1} E_j|_S.$$

For $j > m$, let $\theta_i(E_j|_S)$ be the value of θ_i at a local equation of $E_j|_S$ at ξ_i. By the choice of Y° in **Steps 2** and 3, Construction-Proposition 7.3(1), and the definition of θ_i in Proposition 7.5, $\theta_i(E_j|_S) = v_i(E_j)$. By (7.25), (7.27), and (7.32),

$$\begin{aligned} a(\theta_i, S, B_S') &\leq a(\theta_i, S, B_S) + \frac{1}{2} \sum_{j>m,\ \mathrm{mult}_{E_j} B_Y = 1} v_i(E_j) \\ &\leq \gamma\epsilon_i + \frac{1}{2} \sum_{j>m,\ \mathrm{mult}_{E_j} B_Y = 1} v_i(E_j) \to 0 \end{aligned} \tag{7.35}$$

as $i \to +\infty$. Since B'_S has coefficients in $[0,1)$ and snc support, (S, B'_S) is klt, so $a(\theta_i, S, B'_S) > 0$.

Step 5. Assume that $k \geq 2$.

By **Step 2**,

$$\boldsymbol{\alpha}_\infty := \lim_{i\to+\infty} \boldsymbol{\alpha}_i \in \mathbb{R}^m_{>0}.$$

Fix a real number $\rho \in (0,1)$ and a vector $\mathbf{b} \in \mathbb{R}^m_{>0}$ with $\mathbb{Q}$-linearly independent coordinates such that $(1-\rho)\boldsymbol{\alpha}_\infty < \mathbf{b} < (1+\rho)\boldsymbol{\alpha}_\infty$ coordinatewise. Then $(1-\rho)\boldsymbol{\alpha}_i \leq \mathbf{b} \leq (1+\rho)\boldsymbol{\alpha}_i$ coordinatewise for any $i \gg 0$. Since $\theta_i(f_\beta) \geq 0$ for any $f \in \mathcal{O}(Y^\circ)$ and any β, the definition of $\mu_{\theta,\boldsymbol{\alpha}}$ in Construction-Proposition 7.3(3) implies $(1-\rho)\mu_i \leq \mu_{\theta_i,\mathbf{b}} \leq (1+\rho)\mu_i$ on $\mathcal{O}(Y^\circ)$, hence on $\mathcal{O}_{Y,\xi_i} \supset \mathcal{O}_{X,x}$. Thus

$$\mathfrak{a}_t(\mu_i) \subset \mathfrak{a}_{(1-\rho)t}(\mu_{\theta_i,\mathbf{b}}) \quad \text{for any } t > 0,$$

and

$$\operatorname{vol}(\mu_i) \geq (1-\rho)^n \operatorname{vol}(\mu_{\theta_i,\mathbf{b}}) \quad \text{for any } i \gg 0. \tag{7.36}$$

Since $u_{\mathbf{b}}(f) > 0$ exactly when $f|_{S^\circ} = 0$, the center of $u_{\mathbf{b}}$ on Y is the generic point of S, so $\operatorname{center}_X u_{\mathbf{b}} = w$. Since $u_{\mathbf{b}}$ is the quasi-monomial valuation at the generic point of S with weights $\mathbf{b}$ along $E_1, \dots, E_m$ (Construction-Proposition 7.3(5)), $A_Y(u_{\mathbf{b}}) = \mathbf{b} \cdot (1, \dots, 1)$ [JM12, Proposition 5.1], so Definition 2.2(4) for h implies

$$a(u_{\mathbf{b}}, X, B) = \mathbf{b} \cdot (1 - \operatorname{mult}_{E_1} B_Y, \dots, 1 - \operatorname{mult}_{E_m} B_Y) = 0.$$

By (7.22) with $\epsilon = 1$,

$$a(u_{\mathbf{b}}, X, \Delta) = c u_{\mathbf{b}}(D) = \mathbf{b} \cdot \big(c \operatorname{mult}_{E_1} h^*D, \dots, c \operatorname{mult}_{E_m} h^*D\big).$$

Since $\operatorname{Supp} h^*D \subset \operatorname{Supp} E$, $1 = c v_i(D) = \sum_j c \operatorname{mult}_{E_j} h^*D \cdot v_i(E_j)$, so by (7.27),

$$\lim_{i\to+\infty} \boldsymbol{\alpha}_i \cdot (c \operatorname{mult}_{E_1} h^*D, \dots, c \operatorname{mult}_{E_m} h^*D) = 1.$$

Thus

$$1 - \rho \leq a(u_{\mathbf{b}}, X, \Delta) \leq 1 + \rho.$$

Since $u_{\mathbf{b}}$ is a valuation on X with center w, the definition of $\widehat{\operatorname{vol}}(X \ni w, \Delta)$ implies

$$\widehat{\operatorname{vol}}(X \ni w, \Delta) \leq a(u_{\mathbf{b}}, X, \Delta)^k \operatorname{vol}(u_{\mathbf{b}}) \leq (1+\rho)^k \operatorname{vol}(u_{\mathbf{b}}). \tag{7.37}$$

Step 6. We prove

$$\liminf_{i\to+\infty} a(\mu_i, X, B)^\ell \operatorname{vol}(\mu_i) \geq \widehat{\operatorname{vol}}(X \ni w, \Delta) \cdot \widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M}). \tag{7.38}$$

First assume that $k \geq 2$. Let ρ and $\mathbf{b}$ be as in **Step 5**, and $\zeta \in (0,1)$ a real number. Since $\operatorname{center}_S(\theta_i) = \xi_i \in S^\circ \cap h^{-1}(x)$ for any i, by Proposition 7.10 there exist a real number $C > 0$ and a continuous function $\varphi : \mathbb{R}^m_{>0} \to \mathbb{R}_{>0}$ with $r \cdot \varphi(\mathbf{b}) \geq (1-\zeta)\operatorname{vol}(u_{\mathbf{b}})$ such that

$$\operatorname{vol}(\mu_{\theta_i,\mathbf{b}}) \geq \varphi\big(\mathbf{b} + C a(\theta_i, S, B'_S) \cdot (1, \dots, 1)\big) \cdot \operatorname{vol}((\pi_S)_*\theta_i) \quad \text{for any } i.$$

Together with (7.36) and (7.34), this implies

$$a(\mu_i, X, B)^\ell \operatorname{vol}(\mu_i) \geq (1-\rho)^n \varphi\big(\mathbf{b} + C a(\theta_i, S, B'_S) \cdot (1, \dots, 1)\big) \cdot r \cdot \widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M})$$

for any $i \gg 0$. By (7.35) and the continuity of φ, letting $i \to +\infty$ and using $r\varphi(\mathbf{b}) \geq (1-\zeta)\operatorname{vol}(u_{\mathbf{b}})$ and (7.37),

$$\begin{aligned}\liminf_{i\to+\infty} a(\mu_i, X, B)^\ell \operatorname{vol}(\mu_i) &\geq (1-\rho)^n(1-\zeta)\operatorname{vol}(u_{\mathbf{b}}) \cdot \widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M}) \\ &\geq \frac{(1-\rho)^n(1-\zeta)}{(1+\rho)^k} \cdot \widehat{\operatorname{vol}}(X \ni w, \Delta) \cdot \widehat{\operatorname{vol}}(W \ni x, \Delta_W, \mathbf{M}).\end{aligned}$$

The left side does not depend on ρ, $\mathbf{b}$, and ζ. Letting $\zeta \to 0$ and $\rho \to 0$, we obtain (7.38).

Now assume that $k = 1$. By **Step 2**, $m = 1$, $S = E_1 = h_*^{-1}W$, and $S \to W$ is birational. Write $z := z_1$. Possibly shrinking X to an affine neighborhood of x, since $\mathcal{O}_{X,w}$ is a discrete

valuation ring, there exists $g \in \mathcal{O}(X)$ vanishing on W with $\operatorname{mult}_W g = 1$. Write $g = \sum_\beta \iota(g_\beta) z^\beta$ as in Construction-Proposition 7.3. Since $h(S) = W$, $g_0 = g|_{S^\circ} = 0$, and since $\operatorname{mult}_S \operatorname{div}(h^* g) = \operatorname{mult}_W g = 1$, g_1 is a non-zero regular function on S°. Since $(S^\circ, B'_S|_{S^\circ})$ is klt, there exists a real number $\eta_0 > 0$ such that $(S^\circ, B'_S|_{S^\circ} + \eta_0 \operatorname{div}(g_1))$ is lc. Thus

$$\theta(g_1) \le a(\theta, S, B'_S)/\eta_0$$

for any valuation θ on S with a center on S°. We show that, for any real number $b > 0$ and any divisorial valuation θ on S with $\operatorname{center}_S(\theta) \in S^\circ \cap h^{-1}(x)$,

$$\operatorname{vol}(\mu_{\theta,b}) \ge \frac{\operatorname{vol}((h|_S)_*\theta)}{b + a(\theta, S, B'_S)/\eta_0}. \tag{7.39}$$

Let $b_\theta := b + \theta(g_1)$, and fix a real number $t > 0$. For $f \in \mathcal{O}_{X,x}$ with $f \in \mathfrak{m}_x$, $f|_{S^\circ}$ vanishes at $\operatorname{center}_S(\theta)$, so $\mu_{\theta,b}(f) > 0$. Thus $\mathfrak{a}_t(\mu_{\theta,b})$ is $\mathfrak{m}_x$-primary, and any class in $\mathcal{O}_{X,x}/\mathfrak{a}_t(\mu_{\theta,b})$ is represented by an element of $\mathcal{O}(X)$. For a non-zero $f \in \mathcal{O}(X)$, write $f = \sum_\beta \iota(f_\beta) z^\beta$ as in Construction-Proposition 7.3. Then $\operatorname{mult}_S \operatorname{div}(h^* f) = \operatorname{mult}_W f$ is the smallest β such that $f_\beta \ne 0$, and we call $f_{\operatorname{mult}_W f}$ the *lowest-degree coefficient* of f. If $f \in \mathfrak{a}_t(\mu_{\theta,b})$ and $\operatorname{mult}_W f = \beta$, then

$$\theta(f_\beta) + b\beta \ge t.$$

For $\beta \in \mathbb{N}$, let $\mathcal{F}_\beta \subset \mathcal{O}_{X,x}/\mathfrak{a}_t(\mu_{\theta,b})$ be the image of $\{f \in \mathcal{O}(X) \mid f = 0 \text{ or } \operatorname{mult}_W f \ge \beta\}$. Since $\theta \ge 0$ on $\mathcal{O}(S^\circ)$, $\mu_{\theta,b}(f) \ge b \operatorname{mult}_W f$, so $\mathcal{F}_\beta = 0$ if $b\beta \ge t$, and

$$\operatorname{length}\left(\mathcal{O}_{X,x}/\mathfrak{a}_t(\mu_{\theta,b})\right) = \sum_{\beta \ge 0} \operatorname{length}(\mathcal{F}_\beta/\mathcal{F}_{\beta+1}).$$

For $f_W \in \mathcal{O}(W)$, let $\widetilde{f}_W \in \mathcal{O}(X)$ be a lift of f_W. Since

$$\mathcal{I}_W \cdot \{f \mid \operatorname{mult}_W f \ge \beta\} \subset \{f \mid \operatorname{mult}_W f \ge \beta + 1\},$$

the map $\varrho_\beta : \mathcal{O}(W) \to \mathcal{F}_\beta/\mathcal{F}_{\beta+1}$ sending f_W to the class of $\widetilde{f}_W g^\beta$ is well defined and $\mathcal{O}(W)$-linear. For a real number $\lambda \le 0$, let $\mathfrak{a}_\lambda((h|_S)_*\theta) := \mathcal{O}_{W,x}$. Since $g_0 = 0$, the lowest-degree coefficient of g^β is g_1^β, so for $f_W \ne 0$, the lowest-degree coefficient of $\widetilde{f}_W g^\beta$ and of $\widetilde{f}_W g^\beta - f'$, for any $f' \in \mathcal{O}(X)$ with $\operatorname{mult}_W f' \ge \beta + 1$, is $(f_W \circ h|_{S^\circ}) \cdot g_1^\beta \ne 0$. Hence if $f_W \in \ker \varrho_\beta$, then $\theta(f_W \circ h|_{S^\circ}) + \beta\theta(g_1) + b\beta \ge t$, that is, $f_W \in \mathfrak{a}_{t-\beta b_\theta}((h|_S)_*\theta)$. Since $\mathfrak{a}_{t-\beta b_\theta}((h|_S)_*\theta)$ is $\mathfrak{m}_x$-primary in $\mathcal{O}_{W,x}$ or equal to $\mathcal{O}_{W,x}$, we obtain

$$\begin{aligned}\operatorname{length}\left(\mathcal{O}_{X,x}/\mathfrak{a}_t(\mu_{\theta,b})\right) &\ge \sum_{\beta \ge 0} \operatorname{length}\left(\mathcal{O}_{W,x}/\mathfrak{a}_{t-\beta b_\theta}((h|_S)_*\theta)\right) \\ &\ge \frac{1}{b_\theta} \int_0^t \operatorname{length}\left(\mathcal{O}_{W,x}/\mathfrak{a}_{t'}((h|_S)_*\theta)\right) dt',\end{aligned}$$

where the last inequality holds since the function

$$t' \to \operatorname{length}(\mathcal{O}_{W,x}/\mathfrak{a}_{t'}((h|_S)_*\theta))$$

is increasing and vanishes for $t' \le 0$. Since

$$\operatorname{length}(\mathcal{O}_{W,x}/\mathfrak{a}_{t'}((h|_S)_*\theta)) = \operatorname{vol}((h|_S)_*\theta) t'^\ell/\ell! + o(t'^\ell)$$

and $n = \ell + 1$, letting $t \to +\infty$ we obtain

$$\operatorname{vol}(\mu_{\theta,b}) \ge \operatorname{vol}((h|_S)_*\theta)/b_\theta.$$

Since $b_\theta \le b + a(\theta, S, B'_S)/\eta_0$, (7.39) follows.

Since $m = 1$, $\mu_i = \mu_{\theta_i, v_i(z)}$. Since $\operatorname{mult}_{E_1} h^* D = \operatorname{mult}_W D$, the identity

$$1 = c v_i(D) = \sum_j c \operatorname{mult}_{E_j} h^* D \cdot v_i(E_j)$$

and (7.27) imply $c v_i(z) \operatorname{mult}_W D \to 1$. Since W is an lc center of (X, B), $\operatorname{mult}_W B = 1$, so $a(W, X, \Delta) = 1 - \operatorname{mult}_W \Delta = c \operatorname{mult}_W D$. Since $\mathcal{O}_{X,w}$ is a discrete valuation ring, any valuation

on X with center w is a positive multiple of v_W, so $\widehat{\mathrm{vol}}(X \ni w, \Delta) = a(W, X, \Delta) = c \operatorname{mult}_W D$. Thus $1/v_i(z) \to \widehat{\mathrm{vol}}(X \ni w, \Delta)$. By (7.39) for $\theta = \theta_i$ and $b = v_i(z)$, and by (7.34),

$$a(\mu_i, X, B)^\ell \operatorname{vol}(\mu_i) \geq \frac{a(\mu_i, X, B)^\ell \operatorname{vol}((h|_S)_* \theta_i)}{v_i(z) + a(\theta_i, S, B'_S)/\eta_0} \geq \frac{\widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{v_i(z) + a(\theta_i, S, B'_S)/\eta_0}.$$

By (7.35), the right side tends to $\widehat{\mathrm{vol}}(X \ni w, \Delta) \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})$, which implies (7.38).

Step 7. For any real numbers $\lambda, \epsilon > 0$,

$$\frac{(\lambda + \epsilon)^n}{\lambda^\ell} \geq \frac{n^n \epsilon^k}{k^k \ell^\ell}, \tag{7.40}$$

since $\lambda \to (\lambda + \epsilon)^n / \lambda^\ell$ attains the minimum $n^n \epsilon^k / (k^k \ell^\ell)$ on $(0, +\infty)$ at $\lambda = \ell\epsilon/k$. By **Step 4**, $a(\mu_i, X, B) > 0$. By (7.32), (7.31), and (7.40) with $\lambda = a(\mu_i, X, B)$ and $\epsilon = \epsilon_i$,

$$\begin{aligned} \frac{(a(v_i, X, B) + \epsilon_i)^n \operatorname{vol}(v_i)}{\epsilon_i^k} &\geq \frac{(a(\mu_i, X, B) + \epsilon_i)^n \operatorname{vol}(\mu_i)}{(1 + m\gamma\epsilon_i/\kappa)^n \epsilon_i^k} \\ &\geq \frac{n^n}{(1 + m\gamma\epsilon_i/\kappa)^n k^k \ell^\ell} \cdot a(\mu_i, X, B)^\ell \operatorname{vol}(\mu_i). \end{aligned}$$

Since $\epsilon_i \to 0$, (7.38) implies (7.26). □

8. The limit at the generic point of the minimal lc center

Proposition 8.1. *Notation and conditions as in Set-up 2.8, and assume that $k \geq 2$. Then*

$$\lim_{\sigma \to 0^+} \frac{\widehat{\mathrm{vol}}(X \ni w, B_\sigma)}{\sigma^k} = \widehat{\mathrm{vol}}^B(X \ni w, \Delta) \in (0, +\infty).$$

Proof. The argument in **Step 3** showing that prime divisors computing lc thresholds are lc places of (X, B) is similar to [LLX20, Proof of Proposition 6.9]. Possibly shrinking X to a neighborhood of x, we let Δ_j, δ_j, c_j, B_j, and $B_{j,\epsilon}$ be as in Construction-Proposition 3.1.

Step 1. For any prime divisor E over X and any $\sigma \in [0, 1]$,

$$a(E, X, B_\sigma) = a(E, X, B) + \sigma c \operatorname{mult}_E D = \sigma a(E, X, \Delta) + (1 - \sigma) a(E, X, B). \tag{8.1}$$

By Lemma 2.9, (X, B_σ) is klt, so $\widehat{\mathrm{vol}}(X \ni w, B_\sigma)$ is defined for any $\sigma \in (0, 1]$. We show that for any $0 < \sigma \leq \sigma' \leq 1$,

$$\begin{aligned} \widehat{\mathrm{vol}}(X \ni w, B_\sigma) &\leq \widehat{\mathrm{vol}}(X \ni w, B_{\sigma'}), \\ \widehat{\mathrm{vol}}(X \ni w, B_\sigma) &\leq \sigma^k \widehat{\mathrm{vol}}^B(X \ni w, \Delta) < +\infty. \end{aligned} \tag{8.2}$$

Since $B_\sigma \geq B_{\sigma'}$, $a(v, X, B_\sigma) \leq a(v, X, B_{\sigma'})$ for any valuation v on X with $\operatorname{center}_X v = w$. Taking the infimum over v implies the first inequality of (8.2).

Let E be an lc place of (X, B) with $\operatorname{center}_X E = W$. Since $a(E, X, B) = 0$, (8.1) implies $a(E, X, B_\sigma) = \sigma a(E, X, \Delta)$. The valuation of E has center w, so

$$\widehat{\mathrm{vol}}(X \ni w, B_\sigma) \leq a(E, X, B_\sigma)^k \operatorname{vol}(E) = \sigma^k a(E, X, \Delta)^k \operatorname{vol}(E).$$

Taking the infimum over E implies $\widehat{\mathrm{vol}}(X \ni w, B_\sigma) \leq \sigma^k \widehat{\mathrm{vol}}^B(X \ni w, \Delta)$. Since W is an lc center of (X, B), there exists an lc place E of (X, B) with $\operatorname{center}_X E = W$, so $\widehat{\mathrm{vol}}^B(X \ni w, \Delta) < +\infty$. This proves (8.2).

Step 2. We show that $\widehat{\mathrm{vol}}(X \ni w, \Delta) > 0$. This implies $\widehat{\mathrm{vol}}^B(X \ni w, \Delta) > 0$, since $\widehat{\mathrm{vol}}(X \ni w, \Delta) \leq \widehat{\mathrm{vol}}^B(X \ni w, \Delta)$ by the second inequality of (8.2) with $\sigma = 1$, for which $B_\sigma = \Delta$.

First assume that Δ is a $\mathbb{Q}$-divisor and $K_X+\Delta$ is $\mathbb{Q}$-Cartier. By Theorem 1.3 with $\xi = w$ and [LLX20, Theorem 2.5],

$$\widehat{\mathrm{vol}}(X \ni w, \Delta) = \frac{k^k}{n^n}\widehat{\mathrm{vol}}(X \ni x', \Delta) > 0$$

for a general closed point x' of W.

In general, fix j. By Construction-Proposition 3.1(1), Δ_j is a $\mathbb{Q}$-divisor, $K_X+\Delta_j$ is $\mathbb{Q}$-Cartier, and (X,Δ_j) is klt, so $\widehat{\mathrm{vol}}(X \ni w, \Delta_j) > 0$ by the previous paragraph. By Corollary 3.2(2) with $\epsilon = 1$,

$$\widehat{\mathrm{vol}}(X \ni w, \Delta) \geq (1+\delta_j)^{-k}\widehat{\mathrm{vol}}(X \ni w, \Delta_j) > 0.$$

Step 3. In this step we assume that Δ is a $\mathbb{Q}$-divisor and $K_X+\Delta$ is $\mathbb{Q}$-Cartier, and we show that there exists a real number $\sigma_0 \in (0,1)$ such that

$$\widehat{\mathrm{vol}}(X \ni w, B_\sigma) = \sigma^k \widehat{\mathrm{vol}}^B(X \ni w, \Delta) \quad \text{for any } \sigma \in (0,\sigma_0). \tag{8.3}$$

In particular, the limit in the proposition holds when Δ is a $\mathbb{Q}$-divisor and $K_X+\Delta$ is $\mathbb{Q}$-Cartier.

Since Δ is a $\mathbb{Q}$-divisor and $K_X+\Delta$ and D are $\mathbb{Q}$-Cartier, $c \in \mathbb{Q}$. Let $I \subset [0,1]$ be the finite set consisting of 1 and the coefficients of Δ, and let $J \subset \mathbb{R}_{>0}$ be the finite set of the non-zero coefficients of cD. Let $\mathrm{LCT}_n(I,J)$ be the set of the numbers $\mathrm{lct}(V,\Gamma;M)$, where (V,Γ) is an lc pair of dimension n with the coefficients of Γ in I, and M is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on V with the coefficients of M in J. By [HMX14, Theorem 1.1], $\mathrm{LCT}_n(I,J)$ satisfies the ACC. Thus there exists a real number $\sigma_0 \in (0,1)$ such that

$$\mathrm{LCT}_n(I,J) \cap (1-\sigma_0, 1) = \emptyset, \tag{8.4}$$

as otherwise $\mathrm{LCT}_n(I,J)$ contains a strictly increasing sequence converging to 1.

Let $\sigma \in (0,\sigma_0) \cap \mathbb{Q}$. Then B_σ is a $\mathbb{Q}$-divisor and K_X+B_σ is $\mathbb{Q}$-Cartier. For an $\mathfrak{m}_w$-primary ideal $\mathfrak{a} \subset \mathcal{O}_{X,w}$ and a prime divisor E over X with $W \subset \mathrm{center}_X E$, let

$$\mathrm{mult}_E \mathfrak{a} := \min\{\mathrm{mult}_E \mathrm{div}(f) \mid f \in \mathfrak{a}\},$$

and let

$$\mathrm{lct}(X \ni w, B_\sigma; \mathfrak{a}) := \inf\left\{ \frac{a(E,X,B_\sigma)}{\mathrm{mult}_E \mathfrak{a}} \;\middle|\; E \text{ a prime divisor over } X,\ \mathrm{center}_X E = W \right\}.$$

We say that E *computes* $\mathrm{lct}(X \ni w, B_\sigma; \mathfrak{a})$ if $\mathrm{center}_X E = W$ and E attains the infimum. Since $\mathfrak{a}$ is $\mathfrak{m}_w$-primary, a prime divisor E over X with $W \subset \mathrm{center}_X E$ and $\mathrm{mult}_E \mathfrak{a} > 0$ satisfies $\mathrm{center}_X E = W$, so $\mathrm{lct}(X \ni w, B_\sigma; \mathfrak{a})$ is the infimum of $a(E,X,B_\sigma)/\mathrm{mult}_E \mathfrak{a}$ over the prime divisors E over X with $W \subset \mathrm{center}_X E$, which is the threshold of [Kim26, Lemma 2.2]. By [Kim26, Lemma 2.2],

$$\widehat{\mathrm{vol}}(X \ni w, B_\sigma) = \inf\{\mathrm{lct}(X \ni w, B_\sigma; \mathfrak{a})^k \cdot e(\mathfrak{a}) \mid \mathfrak{a} \subset \mathcal{O}_{X,w} \text{ an } \mathfrak{m}_w\text{-primary ideal}\}. \tag{8.5}$$

Fix an $\mathfrak{m}_w$-primary ideal $\mathfrak{a} \subset \mathcal{O}_{X,w}$. There exist an open neighborhood $X^\circ \subset X$ of w and a coherent ideal sheaf $\mathfrak{a}^\circ \subset \mathcal{O}_{X^\circ}$ such that $\mathfrak{a}^\circ_w = \mathfrak{a}$ and $\mathrm{Supp}(\mathcal{O}_{X^\circ}/\mathfrak{a}^\circ) = W \cap X^\circ$. We keep the notation Δ, D, B, B_σ for the restrictions of Δ, D, B, B_σ to X°. Let $X' \to X^\circ$ be a log resolution of (X°, B_σ) and $\mathfrak{a}^\circ$, where $\mathfrak{a}^\circ \cdot \mathcal{O}_{X'} = \mathcal{O}_{X'}(-G)$, let $\mathrm{mult}_E \mathfrak{a}^\circ := \mathrm{mult}_E G$ for a prime divisor E over X°, and let $\mathrm{lct}(X^\circ, B_\sigma; \mathfrak{a}^\circ)$ be the infimum of $a(E,X^\circ,B_\sigma)/\mathrm{mult}_E \mathfrak{a}^\circ$ over the prime divisors E over X° with $\mathrm{mult}_E \mathfrak{a}^\circ > 0$. This is the lc threshold of [Xu21, Definition 2.8], which a prime divisor computes if it attains the infimum. The prime divisors P on X' with $\mathrm{mult}_P \mathfrak{a}^\circ > 0$ are finitely many, and $\mathrm{center}_{X^\circ} P \subset W \cap X^\circ$ for each of them. Then

$$\mathrm{lct}(X^\circ, B_\sigma; \mathfrak{a}^\circ) = \min_P \frac{a(P,X^\circ,B_\sigma)}{\mathrm{mult}_P \mathfrak{a}^\circ},$$

$$\mathrm{lct}(X \ni w, B_\sigma; \mathfrak{a}) = \min_P \frac{a(P,X^\circ,B_\sigma)}{\mathrm{mult}_P \mathfrak{a}^\circ},$$

where P runs over the prime divisors on X' with $\operatorname{mult}_P \mathfrak{a}^\circ > 0$, with $\operatorname{center}_{X^\circ} P = W \cap X^\circ$ in the second minimum. Possibly replacing X° with the complement of the centers $\operatorname{center}_{X^\circ} P \subsetneq W \cap X^\circ$, which do not contain w, we may assume that $\operatorname{center}_{X^\circ} P = W \cap X^\circ$ for any prime divisor P on X' with $\operatorname{mult}_P \mathfrak{a}^\circ > 0$. Then

$$\operatorname{lct}(X^\circ, B_\sigma; \mathfrak{a}^\circ) = \operatorname{lct}(X \ni w, B_\sigma; \mathfrak{a}).$$

Let E be a prime divisor over X computing $\operatorname{lct}(X \ni w, B_\sigma; \mathfrak{a})$, which exists as the minimum above is attained by some P. Then E computes $\operatorname{lct}(X^\circ, B_\sigma; \mathfrak{a}^\circ)$.

Since (X°, B_σ) is a quasi-projective klt pair, by [Xu21, Lemma 2.11] there exists a projective birational morphism $\rho : Y \to X^\circ$ such that $\operatorname{Exc}(\rho) = E$, $-E$ is ample over X°, and, with $\Delta_Y := \rho_*^{-1}\Delta$ and $D_Y := \rho_*^{-1}D$,

$$(Y, E + \Delta_Y + (1-\sigma)cD_Y) \text{ is lc.} \tag{8.6}$$

Here E is exceptional over X° as $\operatorname{codim}_X W = k \geq 2$, and E is $\mathbb{Q}$-Cartier as $-E$ is ample over X°. Since $D_Y = \rho^* D - (\operatorname{mult}_E D)E$, D_Y is $\mathbb{Q}$-Cartier, so $K_Y + E + \Delta_Y$ is $\mathbb{Q}$-Cartier and, as $D_Y \geq 0$, $(Y, E + \Delta_Y)$ is lc.

Let F be an lc place of (X, B) with $\operatorname{center}_X F = W$. By (8.1) with $\sigma = 1$,

$$c \operatorname{mult}_F D = a(F, X, \Delta) > 0,$$

so $c > 0$ and $W \subset \operatorname{Supp} D$, and thus $cD_Y \neq 0$. Let

$$\tau := \operatorname{lct}(Y, E + \Delta_Y; cD_Y).$$

Since $\dim Y = n$, the coefficients of $E + \Delta_Y$ are in I, and the coefficients of cD_Y are in J, we have $\tau \in \operatorname{LCT}_n(I, J)$. By (8.6), $\tau \geq 1 - \sigma > 1 - \sigma_0$, so $\tau \geq 1$ by (8.4). Thus

$$(Y, E + \Delta_Y + cD_Y) \text{ is lc.} \tag{8.7}$$

Since $\operatorname{Exc}(\rho) = E$ and $\rho(E) = W \cap X^\circ$, $\rho^{-1}(W \cap X^\circ) = E$ by Zariski's main theorem. Thus $\operatorname{center}_Y F \subset E$, so $\operatorname{mult}_F E > 0$. Since $\rho_*^{-1}B = \Delta_Y + cD_Y$ and $\operatorname{Exc}(\rho) = E$,

$$K_Y + E + \Delta_Y + cD_Y = \rho^*(K_{X^\circ} + B) + a(E, X, B)E.$$

By (8.7) and $a(F, X, B) = 0$,

$$0 \leq a(F, Y, E + \Delta_Y + cD_Y) = a(F, X, B) - a(E, X, B) \operatorname{mult}_F E = -a(E, X, B) \operatorname{mult}_F E.$$

Since $\operatorname{mult}_F E > 0$ and (X, B) is lc, $a(E, X, B) = 0$. Thus E is an lc place of (X, B) with $\operatorname{center}_X E = W$.

For any $m \in \mathbb{N}^+$,

$$\mathfrak{a}^m \subset \{f \in \mathcal{O}_{X,w} \mid \operatorname{mult}_E \operatorname{div}(f) \geq m \operatorname{mult}_E \mathfrak{a}\},$$

so, multiplying the lengths of the quotients of $\mathcal{O}_{X,w}$ by $k!/m^k$ and letting $m \to +\infty$, Definition 2.2(2)(3) implies $e(\mathfrak{a}) \geq (\operatorname{mult}_E \mathfrak{a})^k \operatorname{vol}(E)$. Since $a(E, X, B) = 0$, (8.1) implies $a(E, X, B_\sigma) = \sigma a(E, X, \Delta)$. As E computes $\operatorname{lct}(X \ni w, B_\sigma; \mathfrak{a})$,

$$\begin{aligned}\operatorname{lct}(X \ni w, B_\sigma; \mathfrak{a})^k \cdot e(\mathfrak{a}) &\geq \frac{a(E, X, B_\sigma)^k}{(\operatorname{mult}_E \mathfrak{a})^k} \cdot (\operatorname{mult}_E \mathfrak{a})^k \operatorname{vol}(E) \\ &= \sigma^k a(E, X, \Delta)^k \operatorname{vol}(E) \geq \sigma^k \widehat{\operatorname{vol}}^B(X \ni w, \Delta).\end{aligned}$$

Since $\mathfrak{a}$ is arbitrary, (8.5) implies $\widehat{\operatorname{vol}}(X \ni w, B_\sigma) \geq \sigma^k \widehat{\operatorname{vol}}^B(X \ni w, \Delta)$. With (8.2), (8.3) holds for any $\sigma \in (0, \sigma_0) \cap \mathbb{Q}$.

Since $\sigma \to \widehat{\operatorname{vol}}(X \ni w, B_\sigma)$ is increasing by (8.2), (8.3) extends from $(0, \sigma_0) \cap \mathbb{Q}$ to $(0, \sigma_0)$.

Step 4. We prove the limit in general. By (8.2), we only need to show that

$$\liminf_{\sigma \to 0^+} \frac{\widehat{\operatorname{vol}}(X \ni w, B_\sigma)}{\sigma^k} \geq \widehat{\operatorname{vol}}^B(X \ni w, \Delta).$$

Fix j. By Construction-Proposition 3.1(1), Δ_j is a $\mathbb{Q}$-divisor, $K_X + \Delta_j$ is $\mathbb{Q}$-Cartier, and (X, Δ_j, D, x), together with W, satisfies the conditions of Set-up 2.8, as the lc places of (X, B_j) are

exactly the lc places of (X,B). Let $\widehat{\mathrm{vol}}^{B_j}(X\ni w,\Delta_j)$ be defined as in Set-up 2.8 for (X,Δ_j,D,x). By **Step 3**,

$$\widehat{\mathrm{vol}}(X\ni w,B_{j,\sigma})=\sigma^k\widehat{\mathrm{vol}}^{B_j}(X\ni w,\Delta_j)$$

for any $0<\sigma\ll 1$, where the bound on σ depends on j. Since the lc places of (X,B_j) are exactly the lc places of (X,B), and $a(E,X,\Delta_j)\geq(1-\delta_j)a(E,X,\Delta)$ for any prime divisor E over X by Construction-Proposition 3.1(2) with $\epsilon=1$, we have

$$\widehat{\mathrm{vol}}^{B_j}(X\ni w,\Delta_j)\geq(1-\delta_j)^k\widehat{\mathrm{vol}}^{B}(X\ni w,\Delta).$$

By Corollary 3.2(2) with $\epsilon=\sigma$, for any $0<\sigma\ll 1$,

$$\frac{\widehat{\mathrm{vol}}(X\ni w,B_\sigma)}{\sigma^k}\geq\frac{\widehat{\mathrm{vol}}(X\ni w,B_{j,\sigma})}{(1+\delta_j)^k\sigma^k}=\frac{\widehat{\mathrm{vol}}^{B_j}(X\ni w,\Delta_j)}{(1+\delta_j)^k}\geq\left(\frac{1-\delta_j}{1+\delta_j}\right)^k\widehat{\mathrm{vol}}^{B}(X\ni w,\Delta).$$

Thus

$$\liminf_{\sigma\to 0^+}\frac{\widehat{\mathrm{vol}}(X\ni w,B_\sigma)}{\sigma^k}\geq\left(\frac{1-\delta_j}{1+\delta_j}\right)^k\widehat{\mathrm{vol}}^{B}(X\ni w,\Delta)$$

for any j. Since $\lim_{j\to+\infty}\delta_j=0$, the proposition follows. □

9. Proof of the main theorem

Proof of Theorem 1.4. Since every lc center of (X,B) contains x, the minimal lc center of (X,B) containing x is the minimal lc center of (X,B), so X, x, Δ, D, c, B, and W satisfy Set-up 2.8, and w, ℓ, k, $(W,\Delta_W,\mathbf{M})$, and $\widehat{\mathrm{vol}}^{B}(X\ni w,\Delta)$ are those of Set-up 2.8. Moreover, $\Delta+(1-\epsilon)cD=B_\epsilon$ for any $\epsilon\in[0,1]$. Let

$$\underline{L}:=\liminf_{\epsilon\to 0^+}\frac{\widehat{\mathrm{vol}}(X\ni x,B_\epsilon)}{n^n\epsilon^k}\quad\text{and}\quad\overline{L}:=\limsup_{\epsilon\to 0^+}\frac{\widehat{\mathrm{vol}}(X\ni x,B_\epsilon)}{n^n\epsilon^k}.$$

Possibly shrinking X to a neighborhood of x, we let Δ_j, δ_j, c_j, B_j, and $B_{j,\epsilon}$ be as in Construction-Proposition 3.1.

Case 1. $\ell\geq 1$.

We first show that

$$0<\widehat{\mathrm{vol}}(W\ni x,\Delta_W,\mathbf{M})<+\infty. \tag{9.1}$$

Fix j, and let $(W,\Delta_{W,j},\mathbf{M}_j)$ be as in Construction-Proposition 3.1(3). By Construction-Proposition 3.1(1), (X,B_j) is lc and W is the minimal lc center of (X,B_j). Since $c_j\in\mathbb{Q}$, $B_j=\Delta_j+c_jD$ is a $\mathbb{Q}$-divisor and $K_X+B_j=(K_X+\Delta_j)+c_jD$ is $\mathbb{Q}$-Cartier. Thus, by Definition-Theorem 2.7(1)(3), $(W,\Delta_{W,j},\mathbf{M}_j)$ is generalized klt, $\Delta_{W,j}$ is a $\mathbb{Q}$-divisor, and $\mathbf{M}_j$ is a $\mathbb{Q}$-$\boldsymbol{b}$-divisor. Since $\dim W=\ell\geq 1$, Proposition 4.6 implies that

$$0<\widehat{\mathrm{vol}}(W\ni x,\Delta_{W,j},\mathbf{M}_j)<+\infty.$$

Since $0<\delta_j<\frac{1}{2}$, Corollary 3.2(3) implies that

$$\frac{\widehat{\mathrm{vol}}(W\ni x,\Delta_{W,j},\mathbf{M}_j)}{(1+\delta_j)^\ell}\leq\widehat{\mathrm{vol}}(W\ni x,\Delta_W,\mathbf{M})\leq\frac{\widehat{\mathrm{vol}}(W\ni x,\Delta_{W,j},\mathbf{M}_j)}{(1-\delta_j)^\ell},$$

which implies (9.1).

Next we show that

$$\lim_{\sigma\to 0^+}\frac{\widehat{\mathrm{vol}}(X\ni w,B_\sigma)}{\sigma^k}=\widehat{\mathrm{vol}}^{B}(X\ni w,\Delta)\in(0,+\infty). \tag{9.2}$$

If $k\geq 2$, then (9.2) is Proposition 8.1. If $k=1$, then $\mathcal{O}_{X,w}$ is a discrete valuation ring, so W is the only prime divisor over X with center W, and every valuation on X with center w is a

positive multiple of v_W, and $\mathrm{vol}(v_W) = 1$. Thus $a(W, X, B) = 0$ as W is an lc center of (X, B), and since $B_\sigma = \sigma\Delta + (1-\sigma)B$, for any $\sigma \in (0, 1]$,

$$\frac{\widehat{\mathrm{vol}}(X \ni w, B_\sigma)}{\sigma} = \frac{a(W, X, B_\sigma)}{\sigma} = a(W, X, \Delta) = \widehat{\mathrm{vol}}^B(X \ni w, \Delta) = \widehat{\mathrm{vol}}(X \ni w, \Delta), \tag{9.3}$$

where the last equality follows from the case $\sigma = 1$. Moreover, $a(W, X, \Delta) \in (0, 1]$ as (X, Δ) is klt and $\Delta \geq 0$. This implies (9.2).

Now fix $\sigma \in (0, 1]$. By Lemma 2.9, (X, B_σ) is klt and $\mathrm{lct}(X \ni x, B_\sigma; D) = \sigma c$. Thus X, x, B_σ, D, σc, $B_\sigma + \sigma c D = B$, and W satisfy Set-up 2.8 in the roles of X, x, Δ, D, c, B, and W. Since $B_\sigma + \sigma c D = B$, the generic point w, the dimensions ℓ and k, and the generalized pair $(W, \Delta_W, \mathbf{M})$ do not change, and B_ϵ is replaced by

$$B_\sigma + (1-\epsilon)\sigma c D = \Delta + (1 - \sigma\epsilon)cD = B_{\sigma\epsilon}, \quad \epsilon \in [0, 1].$$

Thus Theorem 7.1 implies that

$$\liminf_{\epsilon \to 0^+} \frac{\widehat{\mathrm{vol}}(X \ni x, B_{\sigma\epsilon})}{n^n \epsilon^k} \geq \frac{\widehat{\mathrm{vol}}(X \ni w, B_\sigma) \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{k^k \ell^\ell}.$$

The left-hand side is $\sigma^k \underline{L}$. Thus, for any $\sigma \in (0, 1]$,

$$\underline{L} \geq \frac{\widehat{\mathrm{vol}}(X \ni w, B_\sigma)}{\sigma^k} \cdot \frac{\widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{k^k \ell^\ell}. \tag{9.4}$$

The left-hand side of (9.4) does not depend on σ. Letting $\sigma \to 0^+$ in (9.4), by (9.1) and (9.2),

$$\underline{L} \geq \frac{\widehat{\mathrm{vol}}^B(X \ni w, \Delta) \cdot \widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{k^k \ell^\ell}.$$

By Theorem 5.1, $\overline{L}$ is at most the right-hand side. Since $\underline{L} \leq \overline{L}$, the limit in Theorem 1.4(1) exists and

$$\underline{L} = \overline{L} = \frac{\widehat{\mathrm{vol}}^B(X \ni w, \Delta)}{k^k} \cdot \frac{\widehat{\mathrm{vol}}(W \ni x, \Delta_W, \mathbf{M})}{\ell^\ell},$$

which is positive and finite by (9.1) and (9.2). Together with (9.3) when $k = 1$, this proves (1) and (2). By Theorem 7.1, (3) follows.

Case 2. $\ell = 0$, that is, $W = \{x\}$.

Then $k = n$ and $w = x$. Fix j. By Construction-Proposition 3.1(1), (X, Δ_j) is klt, $K_X + \Delta_j$ is $\mathbb{Q}$-Cartier, $c_j D \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor as $c_j \in \mathbb{Q}$, $(X, \Delta_j + c_j D) = (X, B_j)$ is lc, and $\{x\} = W$ is the minimal lc center of (X, B_j). By Theorem 1.2 for $(X \ni x, \Delta_j)$ and $c_j D$ in place of D, since $\Delta_j + (1-\epsilon)c_j D = B_{j,\epsilon}$, there exists a quasi-monomial valuation v_j on X with $\mathrm{center}_X v_j = x$ such that

$$\widehat{\mathrm{vol}}(X \ni x, B_{j,\epsilon}) = a(v_j, X, \Delta_j)^n \,\mathrm{vol}(v_j) \cdot \epsilon^n \quad \text{for any } 0 < \epsilon \ll 1.$$

Let $L_j := a(v_j, X, \Delta_j)^n \,\mathrm{vol}(v_j)/n^n$. Then

$$\lim_{\epsilon \to 0^+} \frac{\widehat{\mathrm{vol}}(X \ni x, B_{j,\epsilon})}{n^n \epsilon^n} = L_j.$$

Since v_j is quasi-monomial, $a(v_j, X, \Delta_j) < +\infty$ [JM12, Proposition 5.1], so $L_j < +\infty$. Since $\widehat{\mathrm{vol}}(X \ni x, \Delta_j) \leq a(v_j, X, \Delta_j)^n \,\mathrm{vol}(v_j)$, [LLX20, Theorem 2.5] implies

$$0 < \frac{\widehat{\mathrm{vol}}(X \ni x, \Delta_j)}{n^n} \leq L_j < +\infty. \tag{9.5}$$

By Corollary 3.2(1), dividing by $n^n \epsilon^n$ and letting $\epsilon \to 0^+$, since $0 < \delta_j < \frac{1}{2}$, we have

$$\frac{L_j}{(1+\delta_j)^n} \leq \underline{L} \leq \overline{L} \leq \frac{L_j}{(1-\delta_j)^n}. \tag{9.6}$$

By (9.5) and (9.6), $\overline{L} < +\infty$ and

$$\overline{L} \leq \left(\frac{1+\delta_j}{1-\delta_j}\right)^n \underline{L}$$

for any j. Since $\lim_{j\to+\infty} \delta_j = 0$, $\overline{L} \leq \underline{L}$. Thus the limit in Theorem 1.4(1) exists and is equal to $\underline{L} = \overline{L} < +\infty$. By Corollary 3.2(1) with $\epsilon = 1$,

$$\widehat{\mathrm{vol}}(X \ni x, \Delta_j) \geq (1-\delta_j)^n \widehat{\mathrm{vol}}(X \ni x, \Delta).$$

Thus, by (9.6) and (9.5),

$$\underline{L} \geq \frac{L_j}{(1+\delta_j)^n} \geq \frac{\widehat{\mathrm{vol}}(X \ni x, \Delta_j)}{n^n(1+\delta_j)^n} \geq \left(\frac{1-\delta_j}{1+\delta_j}\right)^n \frac{\widehat{\mathrm{vol}}(X \ni x, \Delta)}{n^n}$$

for any j. Since $\widehat{\mathrm{vol}}(X \ni x, \Delta_j) > 0$ by (9.5), $\underline{L} > 0$, which proves (1). Letting $j \to +\infty$,

$$\underline{L} \geq \frac{\widehat{\mathrm{vol}}(X \ni x, \Delta)}{n^n}.$$

Since $w = x$ and $k = n$, this is (3). Statement (2) is empty when $\ell = 0$. □

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